

DE GRUYTER

*Rishabha Malviya, Shristy Verma,
Sonali Sundram, Harshil Shah*

DIGITAL BLOCKCHAIN

BIG DATA, ARTIFICIAL INTELLIGENCE, AND
VIRTUAL REALITY IN HEALTHCARE ECOSYSTEM

DE GRUYTER SERIES ON THE APPLICATIONS
OF MATHEMATICS IN ENGINEERING AND
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Digital Blockchain

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Rishabha Malviya, Shristy Verma, Sonali Sundram, Harshil Shah

Digital Blockchain



Big Data, Artificial Intelligence, and Virtual Reality in Healthcare
Ecosystem

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Preface

The rapid advancement of technology is reshaping the healthcare and pharmaceutical industries in unforeseen manners. The combination of blockchain, big data, and artificial intelligence (AI) has created new opportunities in drug research, clinical trials, and patient care. The book “Digital Blockchain- Big Data, Artificial Intelligence and Virtual Reality in Healthcare Ecosystem” explores the revolutionary potential of these technologies in the pharmaceutical and healthcare industries. Blockchain ensures data integrity, confidentiality, and transparency, whereas big data and AI allow for new advances in drug development, clinical trials, and personalized medicine. Virtual reality is being more widely used in medical training, patient rehabilitation, and treatments, producing immersive environments that improve patient outcomes and healthcare professional capacities.

Beginning with the basis of artificial intelligence (AI) and big data in pharmaceutical research, the book educates readers through the complex relations between these technologies and their uses in healthcare. It investigates blockchain’s rising importance in pharmaceutical supply chain security, as well as the advancements of AI-powered decision-making systems. These digital technologies have a significant impact on the entire lifecycle of medication research and delivery, as well as ethical, regulatory, and intellectual property considerations.

The future of healthcare depends on developing collaborations among pharmaceutical companies, startups, and academic institutions to fully

realize the possibilities of AI, big data, and blockchain. This book provides an orientation for healthcare practitioners, technicians, industrial experts, and academicians, as a forward-thinking resource with incorporates disruptive technology into the healthcare ecosystem.

Foreword

The healthcare sector is undergoing an evolutionary change as a result of the widespread adoption of digital technologies such as blockchain, big data, artificial intelligence (AI), and virtual reality. These technologies not only enhance pharmaceutical research, drug development, and clinical trials, but they also transform patient care, safety, and data management. The book, *Digital Blockchain: Big Data, Artificial Intelligence and Virtual Reality in Healthcare Ecosystem*, authored by Dr. Rishabha Malviya and his team, explores various applications of these technologies in the healthcare and pharmaceutical sector.

The pharmaceutical sector has long played an important role in global healthcare systems, but issues such as difficulties in drug development research, data management complications, and regulatory compliance remain. Blockchain and AI are tackling these concerns by increasing transparency and accelerating drug discovery. The emergence of technologies opens up new possibilities for better, more efficient healthcare systems.

Each chapter provides new insights into how blockchain's transparency and security, AI's computing power, and big data's analytical skills are transforming every stage of the healthcare value chain. This book thoroughly covers the ground to demonstrate the way technology is not only enhancing but also significantly altering every aspect of healthcare, whether it be smart pharma, drug production, clinical trials, or post-market surveillance.

The book emphasizes the value of collaboration among industry stakeholders, including research institutions, pharmaceutical corporations, technological entrepreneurs, and regulatory organizations. The combination of these organizations promises a future in which patient outcomes improve, systems are more transparent, and healthcare is more personalized and accessible.

The content is technically advanced and forward-thinking, taking readers through current developments while forecasting future trends. As AI, blockchain, big data, and Virtual reality merge to form the emerging standard in healthcare, this book is an essential resource for healthcare practitioners, technologists, industrial experts, and researchers.

Happy Reading and Best Wishes.



Dr. Dhruv Galgotia

CEO, Galgotias University,

Greater Noida, India

Chapter 1 Introduction to smart pharma: foundation of AI and big data

Abstract

Due to its potential to completely transform the pharmaceutical business, artificial intelligence (AI) has emerged as a potent instrument in the medical community. However, the potential isn't fully realized yet. This chapter provides information about AI and big data in the pharma and other healthcare industries. Nevertheless, AI techniques such as big data and natural language processing (NLP) can expedite and rehabilitate the process by facilitating a more precise and efficient analysis of large datasets. To achieve this, the test administration may be simplified. The application of AI in the field of chemistry has recently garnered significant attention due to its potential to provide valuable insights into developing efficient drug delivery systems. AI significantly influences the medication manufacturing process and the drug repurposing strategy. Despite its numerous obstacles, AI has the potential to enhance medical imaging services and help diagnose patients. However, healthcare systems are responsible for informing patients about the benefits and drawbacks of AI-supported healthcare systems. A conclusion of the recent state of the field for AI applications in the healthcare industry is provided in the chapter, taking into account the rapid advancements in AI combined with AI-integrated medical systems.

Keywords: pharma, AI, big data, deep learning, machine learning, NLP, digital health,

1.1 Introduction

The pharmaceutical industry has been significantly impacted by the recent surge in interest in the application of AI in medicinal chemistry. The objective of the lengthy and complex drug discovery process, which has historically been reliant on resource-intensive methods such as high-throughput screening and experimental testing, is to identify and develop new pharmaceuticals. However, by enabling more precise and effective analysis of big datasets, ML and NLP (natural language processing) methods in AI offer an opportunity to speed up and improve the process. Researchers have recently shown that deep learning (DL) may be successfully applied to precisely forecast the efficacy of medicinal molecules. In recent years, there has been a significant increase in interest in the implementation of AI in the field of medicine, as it has the potential to revolutionize the pharmaceutical industry. The long and intricate drug discovery process has historically relied on resource-intensive techniques such as high-throughput screening and trial-and-error testing to identify and propagate novel pharmaceuticals. Therefore, by enabling more precise and effective analysis of big datasets, ML along with NLP techniques in AI provides a way to speed up and improve the process. Researchers have recently shown that DL may be successfully applied to precisely forecast the efficacy of medicinal molecules [[→1](#), [→2](#), [→3](#)]. Medicinal chemistry is currently characterized by the use of a trial-and-error approach and

intensive screening procedures. One of these techniques is to analyze an abundance of possible therapeutic molecules to find those that meet the requirements. It has been observed, nevertheless, that these methods are costly, time-consuming, and frequently produce less accurate results. Furthermore, the scope of their application may be restricted by the availability of appropriate chemicals as well as the challenges associated with accurately predicting their physiological reactions. Numerous AI algorithms, such as unsupervised learning strategies, reinforcement learning, and evolution or rule-based algorithms, may be of great assistance in solving these issues [[→4](#), [→5](#), [→6](#)].

AI has brought about significant changes in the area of drug research and development (R&D). It involves researching proteins and small compounds, helping to plan clinical trials, and monitoring things after putting them on the market [[→7](#)]. AI includes various methodological areas, such as information representation, reasoning, finding solutions, and, most importantly, machine learning (ML) as a foundational paradigm. Algorithms used in ML can identify patterns within a specially classified dataset. Deep learning (DL) is a specific subfield of ML that focuses on the application of artificial neural networks (ANNs). The aforementioned elements combine to create a network of linked, highly developed computing units known as “perceptions,” which are designed to resemble actual neurons in the human brain and transmit electrical impulses in a comparable way [[→8](#), [→9](#)]. NLP algorithms can extract relevant details from unstructured data that can be found in a variety of books, articles, and

patents. Additional study of the extracted information may reveal the hidden relationships of knowledge in several papers, hence relieving the workload of researchers who would otherwise need to examine every document [[→10](#), [→11](#)]. Furthermore, algorithms that are used to generate virtual screens based on their capacity to synthesize workable substances include nearest-neighbor classifiers, random forest (RF), extreme learning machines, support vector machines (SVMs), and deep neural networks (DNNs) [[→12](#), [→13](#)]. These algorithms also can forecast toxicity and in vivo activity. To provide a complete framework that would aid in the identification and development of treatments in areas like immunoncology and cardiovascular illnesses, a group of biopharmaceutical firms has partnered with IT companies [[→14](#)]. Based only on a molecule's chemical structure, the AI/ML models may also anticipate possible biological interactions of a molecule that hasn't been tested or synthesized. Scientists in the absorption, distribution, metabolism, and excretion (ADME) sector use AI/ML models to identify small molecules with the best potential for synthesis, testing, and process development based on their ADME profiles [[→15](#), [→16](#)]. Artificial intelligence (AI) facilitates the best possible operation of medication delivery systems and is a useful tool in the discovery and repurposing of pharmaceuticals. It collects information from several sources, evaluates pharmacokinetic, patient, and molecular data, and contrasts unknown compounds with recognized ones. The accuracy with which drug delivery methods are chosen, taking patient needs and illness circumstances into account, determines the efficacy [[→17](#), [→18](#)].

AI algorithms can forecast the behavior of drugs on various levels by examining intricate connections between pharmacological properties, formulation components, and physiological parameters. They enable the prediction of the physicochemical characteristics, stability, and in vitro release profile of the medicinal compound, thereby facilitating the development of a more comprehensive understanding of efficient drug delivery systems and medication delivery processes. The methodology previously described is also employed to conduct in vivo-in vitro correlation studies and to enhance the evaluation of in vivo pharmacokinetic parameters and drug distribution. By applying appropriate AI techniques, researchers can identify potential risks and impediments related to drug delivery systems (DDS) early in the development phase, enabling proactive alterations and modifications to reduce risks and enhance the effectiveness of medication. AI and computer modeling help to lessen the likelihood of unexpected outcomes by reducing reliance on costly and time-consuming trial-and-error trials [[→19](#), [→20](#)]. Even though AI applications in pharmacy seem promising, there are still many unsolved research issues. Long-term effects like medication adherence, ethical concerns about data privacy and potential biases, technical difficulties with integrating data and system maintenance, and patient and chemist usability are a few of these. The satisfaction of patients with the device is another factor. To enhance the understanding of the implications of integrating AI into the pharmacological system, scientists may address these knowledge

gaps, develop solutions that optimize its benefits, mitigate its drawbacks, and ensure its ethical use [[→ 21](#)].

1.2 Big data in the pharmaceutical landscape

Healthcare data has been digitized with the help of the growth and development of big data. A new domain called “big data” was introduced because of the data’s rapid expansion over the past few years. Big data analytics and other intelligent healthcare solutions, which also save costs and boost efficiency, are making healthcare more mobile and smarter. By employing predictive analytics to forecast pharmacological outcomes and pinpoint the patients who would benefit most from pharmaceutical treatments, chemists can get a deeper understanding of the dangers and adverse effects of drugs [[→ 22](#), [→ 23](#), [→ 24](#)]. Big data’s potential benefits in the medical and health sectors have made it a prominent study topic in recent years. Utilizing big data-driven clinical decision support systems has shown promise in treating ailments like Alzheimer’s disease. Utilizing big data techniques, electronic medical record optimization has also been accomplished. Data about chemical characteristics, pharmacokinetics, clinical behaviors, and drug toxicity have been created in the fields of toxicology, pharmaceuticals, and pharmacology. High-throughput screening (HTS) and next-generation sequencing (NGS) have significantly increased data output and correspondingly increased the amount of data [[→ 25](#), [→ 26](#)].

Digital health care will be enabled by big data analysis, which will reduce the requirement for in-person visits, enhance patient outcomes, and guarantee that clinicians have immediate access to all patient medical records. Big data analysis will become a vital tool for healthcare organizations via digital medicine with aggregated real-time data, offering more chances for proactive action and a more comprehensive picture of patients' circumstances [[→27](#), [→28](#)]. Data streams are constantly expanding in the modern era as a result of the utilization of electronic medical records, the vast amounts of data transmitted by diverse types of sensors, and the information that patients post on social media. The healthcare sector has always generated vast amounts of data. The medical sector also generates substantial quantities of data, such as genetic information, clinical notes, imaging data, and health-related behavior. Healthcare organizations can assist in disease surveillance, medical decision-making, and public health management by effectively utilizing the data [[→29](#)]. The problem can be addressed with the use of electronic health records (EHRs), which are electronic patient health data compilations in digital format. Increasingly, EHRs are being implemented in poor countries. Through a strong network connection, these records may be exchanged between different healthcare settings. Digitally recorded health records are less likely to include duplicate data as they can be shared, updated, and altered throughout different healthcare systems, reducing the chance of missing any necessary paperwork. Physicians, patients, and hospital management all immediately benefit from EHR initiatives [[→30](#)]. The

electronic medical record is a critical element of e-health utilization, as it contains information regarding a patient's health. An electronic medical record is comprised of legal documents that are generated in hospital contexts. Consequently, these data serve as the electronic health record's principal information source. The term "e-health" was first coined in the early twenty-first century and denotes the integration of contemporary communication and information technologies into the provision of healthcare services, according to a researcher. Effective management of electronic health and the ability to share medical information over greater geographic distances require a multidisciplinary team with experience in computer science, instrumentation, and communications [[→31](#), [→32](#)]. In big analytics, it is one of the crucial steps, consisting of five stages: data production, acquisition, preservation, analytics, and visualization [→fig 1.1](#). The main source of data production is the Internet, which is followed by data acquisition, which entails gathering, sorting, and exchanging enormous volumes of data. Large infrastructure is used to store data efficiently to guarantee its use in the future. Utilizing ML, computational intelligence, and data mining techniques, data analytics can extract data from vast volumes of data, guaranteeing effective use and improved use in the future [[→33](#)].

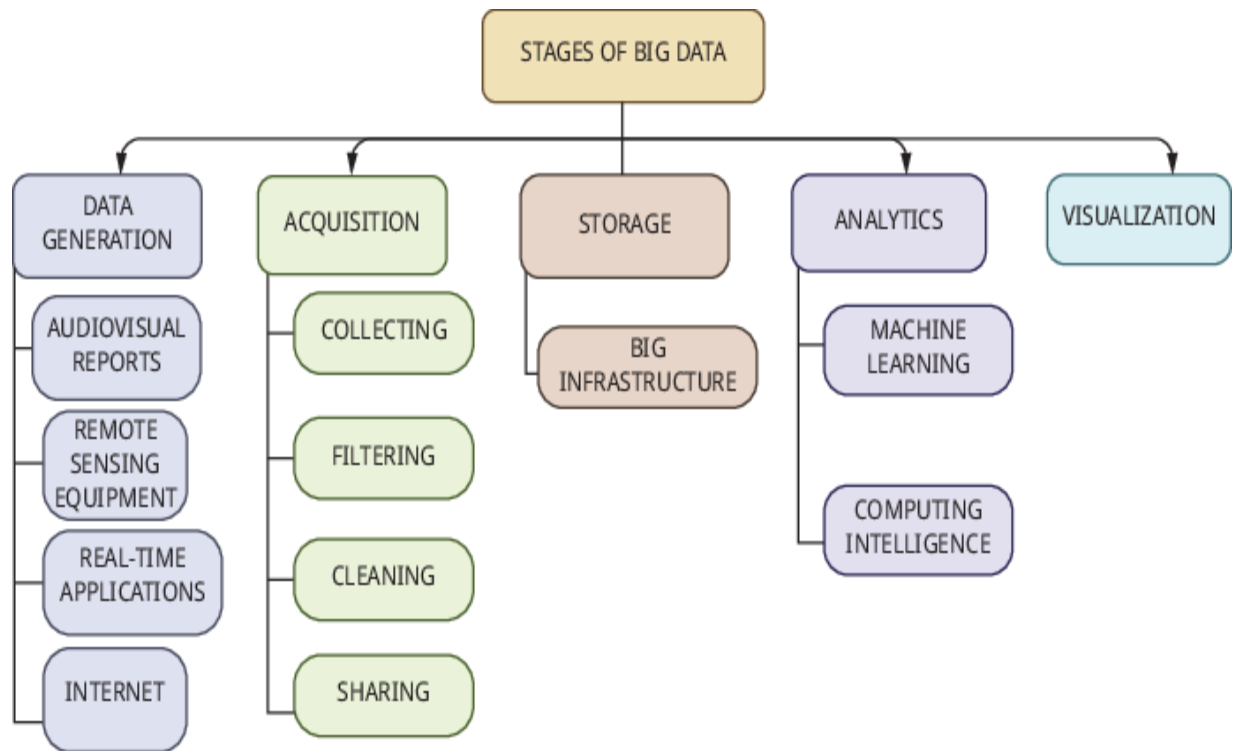


Fig. 1.1: Different stages of big data along with classification.

Through the use of massive volumes of data, big data offers clinical trial projects real-time monitoring, enabling organizations to efficiently plan trial stages and manage patient involvement. Big data analytics is very helpful in providing intelligent information to avoid or address potential problems as well as find new opportunities to boost efficiency and save costs. Big data analytics, for instance, is utilized in the rare illness sector to predict the risks associated with new publications, research discoveries, or drug interactions. Big data applies predictive analytics to procedures like segmentation and sales force focusing on pharmaceutical products sales as well as supply chain operations, measures sentiment, aids in marketing and sales decision-making, and analyzes high-frequency product transactions [[→ 34](#), [→ 35](#)]. There has been an increase in the use of portable sensors in

the monitoring and diagnosis of various medical conditions, including diabetes and other illnesses. These sensors allow patients to receive accurate assessments and recommendations regarding biological factors found in wound fluids, such as pH, temperature, bacterial substances, and sweat biomarkers. A real-time monitoring method for analyte concentrations in wound fluid is provided by a multitude of sensors that are embedded in bandages or dressings. Inkjet-printed sensors are employed in smart bandages to detect intermittent hemorrhage, pH fluctuations, and air pressure changes at the wound site. Moreover, smart bandage detectors can be linked to microneedle biosensors to detect cancer biomarkers like tyrosinase (TYR) and screen for skin melanoma [[→ 36](#), [→ 37](#), [→ 38](#), [→ 39](#), [→ 40](#), [→ 41](#), [→ 42](#), [→ 43](#), [→ 44](#), [→ 45](#)]. Furthermore, by enhancing public health assessment and surveillance, predictive analytics and big data can increase public health precision. Therefore, from the standpoint of public health, large-scale data collection is an invaluable tool for epidemiological studies, needs for population health assessments, population-based treatment evaluation, and well-informed policy-making [[→ 46](#), [→ 47](#)]. By assisting in the identification of the most effective ways for treating each patient uniquely, the use of big data analytics for healthcare can support individualized care, which may influence lowering medical costs, improving living standards, and reducing healthcare resource waste. Scientists can use inductive reasoning to uncover patterns in exploratory data that are unrelated to any specific hypothesis by analyzing the data using techniques like data mining. Thus, it becomes feasible to do

predictive analysis and real-time analysis, which helps medical professionals start treatments earlier and reduces their risk of morbidity and mortality [[→ 48](#)].

1.3 Integration of big data and AI in pharmaceutical research

The process of finding new drugs is a tribute to biological complexity and human inventiveness in the context of advancements in medicine and pharmaceuticals. Big data and AI are two formidable forces that have emerged as leaders in a few industries, including healthcare and pharmaceuticals, as all continue to advance into the information age [[→ 49](#), [→ 50](#)]. The process of finding new drugs could be significantly changed by combining AI with big data. The combination may provide the solution to several issues that have previously plagued the sector, including accelerating drug pipelines, increasing the precision of treatment outcome forecasts, and optimizing R&D expenditures [[→ 51](#), [→ 52](#)]. The methodologies employed to accelerate the drug development process have also evolved in tandem with the advancement of AI and big data. Medication candidates can now be developed in a more organized, cost-effective manner and at a significantly faster pace with the assistance of the AI method. Computing power and algorithms that utilize current information to provide superior analytics and classification are implemented throughout the drug development process, from the discovery

of a potential medication to the pharmaceutical industry's production process [[→ 53](#), [→ 54](#), [→ 55](#)].

Big data, or an enormous collection of extensive data from many domains, is increasingly being used in drug development. To analyze the heterogeneous data – which ranges from clinical to bioassay – advanced computer strategies and processing capacity are required. Big data analysis is moving toward high-performance computers, cloud services, and GPUs due to the expanding volume of big data. The intricacy of the data must be resolved in part by AI, underscoring the necessity of sophisticated computing skills in the pharmaceutical sector [[→ 56](#), [→ 57](#)]. Drug research and discovery depend heavily on multidimensional data, such as target structures, chemical assays, chemical frameworks, and clinical data. There are now more opportunities for the successful use of AI in medication development due to the quick growth of these data sources. Advances in hardware have also boosted data availability and processing capability, opening new modeling methodologies such as large-scale networks. Because biocompatible nanomaterials, a kind of modern nanotechnology, offer flexible and dependable drug delivery systems for systemic disorders like cancer, they have had a substantial influence on drug development. The development of biocompatible nanomaterials, AI, and hardware is opening the door to a more successful and efficient drug discovery method [[→ 58](#), [→ 59](#)]. Molecular dynamic (MD) simulations served as the foundation for early attempts to apply AI to nano simulation for drug development. Later, a similar method was employed to assess the carbon nanotubes' sensitivity

to organic substances [[→ 60](#), [→ 61](#)]. AI algorithms, especially ML models, depend on data. The more data they are given, the more accurate, predictive, and adaptive they become. AI evolved as a tool for large data interpretation, analysis, and application in the drug development industry. Furthermore, big data offers a platform for comprehensive patient analysis. The increasing acceptance of personalized medicine underscores the importance of comprehending the impact of an individual's genetic composition on their response to medication. The customization of drugs to match individual patient profiles, which improves treatment efficacy and reduces adverse effects, was made feasible by AI-driven analysis of large genetic data [[→ 62](#), [→ 63](#), [→ 64](#)]. Big data produced during the drug discovery process and made available through open databases may prove to be very beneficial in lowering drug attrition throughout the pipeline. To determine the preferred order of ranking for drug candidates with specified medicinal actions, conventional machine learning techniques like QSAR modeling can be employed. The technique may be employed as well to weed out substances that are inappropriate or have known side effects. Advances in AI have enabled CADD research to leverage the robust architecture of big data studies, thereby taking the field to new heights. These advancements include the creation of several deep learning algorithms and notable boosts in processing power [[→ 65](#), [→ 66](#)]. An important connection between chemical frameworks and their characteristics and activities can still be made with the help of the effective method of QSAR. When QSAR modeling was first used in drug

development, it was based on modest data sets and straightforward linear regression techniques. ML is used in drug research and discovery for several reasons other than QSAR. For example, the numerical outline for chemical pattern-matching investigations was used for de novo drug creation using generative models [[→67](#), [→68](#), [→69](#), [→70](#), [→71](#)]. These propagative models can produce synthetic molecules with the right characteristics and functionality. As a result, by using ML modeling for a range of data types, including public databases, CADD may produce substantial value in the big data age [[→72](#), [→73](#)].

1.4 Role of stakeholders

Stakeholder analysis, which identifies and ranks stakeholders according to the information obtained, is an essential step in the design of health innovations. Because stakeholder selection criteria in participatory research are sometimes ambiguous, stakeholder results must be taken into account while developing participatory planning procedures. Stakeholder analysis aids in understanding the setting in which new technologies are developed and put into use; it directs the planning process regarding individuals, organizations, or groups involved; and it produces plans that facilitate innovation in development and use while avoiding roadblocks that might impede system integration. Stakeholders are appropriately involved in health innovation planning through the process's fairness and openness,

which is ensured by integrating stakeholder analysis into participatory planning procedures [[→74](#), [→75](#), [→76](#), [→77](#), [→78](#), [→79](#)].

The marketing approach known as customer relationship management (CRM) places a strong emphasis on the product's final user, even speculating that the user may be a pharmaceutical marketing patient. Nevertheless, salespeople in the industry are forming connections with doctors, stockists, and retailers rather than with patients. Physicians issue prescriptions for drugs manufactured by a particular business, but merchants and stockists are an essential part of the supply chain and may increase sales by telling local physicians about promotions or providing alternatives to certain prescriptions. Salespeople never have a conversation with their final customer, the patient, yet the patient still purchases and uses the product. Determining the true participants in the pharmaceutical industry is therefore essential. As [→Fig. 1.2](#) illustrates, several stakeholders are involved in the formulation as well as execution of a pharmaceutical policy framework. Regulatory agencies, patients, physicians, chemists, insurance companies, academic institutions, professional organizations, non-governmental organizations, nonprofit organizations, and consumer groups with no active participation belong to these stakeholders in addition to state and federal governments (all of which include all ministries like health, trade, commerce, and industry). National and multinational manufacturers of medicines are among them. Governments encompass state authorities, the federal government, and any associated ministries responsible for the manufacturing, importation, exportation, investment,

licensing, pricing, research and development, and quality control of pharmaceuticals.

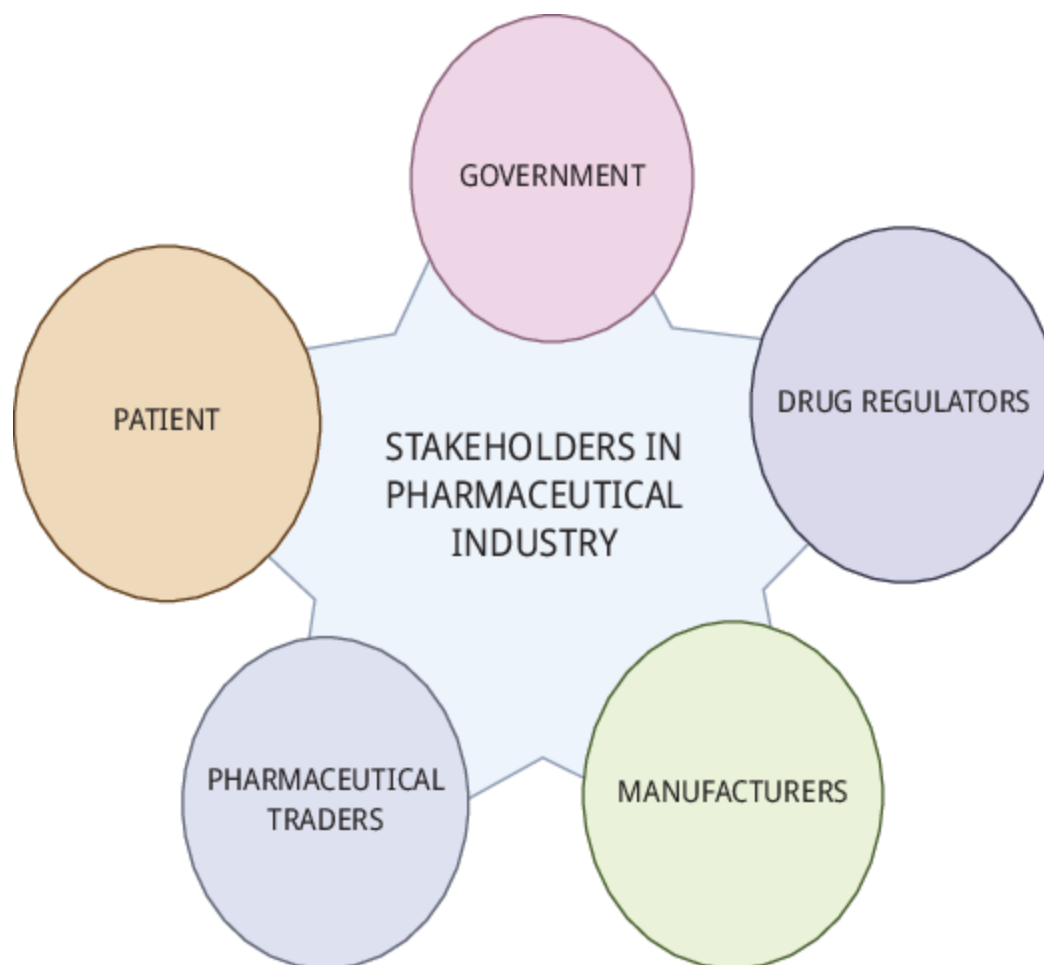


Fig. 1.2: Different stakeholders in the pharmaceutical industry.

When it comes to creating, planning, implementing, and supervising pharmaceutical policies, they are the key participants and, in a way, the pioneers. When creating new pharmaceutical policies, governments must take the initiative and confer with all pertinent stakeholders before issuing a policy framework. The political will of the government in each nation is a critical factor in the development of effective policies for the safe and

efficient utilization, procurement, distribution, and quality control of pharmaceuticals, as well as their equal availability, affordability, and financial risk mitigation [[→ 80](#), [→ 81](#)]. Drug regulators have a crucial role in the development and application of pharmaceutical laws in India, particularly in licensing, marketing, distribution, storage, clinical trials, manufacturing, import, export, and import-export. Drug regulators are not, however, at the forefront of the process in nations like India, where the ministries are directly responsible for creating pharmaceutical policy. Because they are knowledgeable about the difficulties and roadblocks that occur in putting these policies into action, they are crucial to the implementation of robust, thorough, and feasible policy creation. Drug regulators are the key players in India for the creation of comprehensive, practical, and robust regulations since they carry out these policies on the ground and have knowledge of the challenges and barriers that arise in doing so [[→ 82](#)].

Moreover, producers can make their goods available for consumption by physicians, pharmacists, and patients due to pharmaceutical supply networks and loosely regulated marketplaces. That puts manufacturers in a position to evaluate a drug's quality and, in turn, its suitability for treating common ailments. They also have the authority to determine drug costs, which lowers the cost of prescriptions for people from lower socioeconomic backgrounds. However, pharmaceutical companies are profitable enterprises whose main objective is to maximize shareholder wealth, as is widely known [[→ 83](#), [→ 84](#)]. A vital conduit between the pharmaceutical

industry and prescribers is provided by pharmaceutical traders, such as super-stockists, distributors, transportation and forwarding representatives, retailers, wholesalers, and medical representatives. Due to their commercial interests, they stand to gain a great deal from the advertising of branded medications and can significantly influence prescription practices [[→ 85](#)]. Patients are the last, but certainly not the least, significant stakeholders in the development of pharmaceutical policies since they are the focus of all other stakeholders. As the last users of medications made by licensed, regulated pharmaceutical businesses, they fall under government regulation. Pharmacists distribute these medications, while physicians prescribe them, and pharmaceutical merchants market, supply, and sell them. They serve as the pivot, providing support for all of the other chain links' tasks and endeavors. The efficacy of a policy framework is determined by its capacity to alleviate patients' distress and assist them in managing their sickness. As a result, throughout the formulation process, it is essential to carefully analyze the requirements, desires, and ambitions of patients to build a pharmaceutical strategy that is outcome-oriented and patient-driven [[→ 86](#)]. Other stakeholder groups include chemists, health care funds, health insurance plans, and pharmaceutical distributors. They should be considered clients in a wider sense, given their influence on prescription decisions. The three main clientele in the pharmaceutical sector are payers, physicians, and patients. Consequently, doctors, hospitals, wholesalers, pharmacies, end users, and patients are the three customer sectors that

pharmaceutical firms presently focus on with their sales efforts [[→ 87](#), [→ 88](#), [→ 89](#)].

1.5 Challenges and conquering strategies in smart pharma: foundation of AI and big data

Despite the prospective benefits of AI in drug discovery, there are also a variety of drawbacks and challenges that must be considered. One of the most significant obstacles is the acquisition of pertinent data. The training of AI-based techniques typically necessitates the use of large data collections. The reliability and accuracy of the results can frequently be influenced by the limited number, variable quality, or low quality of the supplied data [[→ 90](#), [→ 91](#), [→ 92](#)]. It offers a substantial amount of data, which is used by models that are trained gradually. Due to its higher cost, the company suffered greatly from the data acquisition model from multiple databases. Furthermore, a substantial amount of trustworthy data is required to guarantee the pharmacological analysis. Another obstacle keeping the biomedical and pharmaceutical industries from fully adopting AI is the shortage of skilled personnel to oversee the AI-based platforms [[→ 93](#)]. Job losses are the result of the use of AI and ML by large corporations, such as pharmaceutical companies, to supplant their human staff and the lack of financing for small organizations. In an ML-based process, decisions are generated by the input data; however, the paths leading from input to output are not transparent. The input data in CNNs is essentially altered by several

intricate layers, which eventually produce unpredictable behavior [[→ 94](#)]. Furthermore, human error in data interpretation influences machine learning predictions when there is no reasoning behind predictions specific to black-box machine learning models. But logical models and decision trees, like relational machine learning, don't show the fundamental problem. The dependence on model performance is impacted by human users' incapacity to understand anticipated outcomes. There are still concerns regarding prediction confidence despite the ML models' encouraging initial results. A pathological failure in the hypothesis generated by machine learning approaches is more likely when an algorithm tries to seek regimes outside of its application scope [[→ 95](#)]. When AI-based solutions are utilized to create innovative medications, ethical problems arise since prejudice and fairness issues may result. Inaccurate data can affect the reliability and accuracy of findings, thus human researchers need to make sure AI is used fairly and ethically. AI is limited to making predictions based on the facts at its disposal; human researchers are still required to confirm and explain the results. As a result, it is imperative to guarantee that AI is applied equitably and morally while creating novel pharmaceuticals [[→ 96](#), [→ 97](#), [→ 98](#), [→ 99](#), [→ 100](#)]. Millions of chemicals are available to pharmaceutical companies in their data sets for the development of new medications; it may be too much for conventional machine-learning technologies to handle. A computerized model that utilizes the quantitative structure-activity relationship (QSAR) can swiftly ascertain a significant number of compounds or critical physicochemical properties, such as log P or log D. Consequently, these

models are unable to accurately anticipate the efficacy and side effects of specific substances, which are complex biological characteristics. Additionally, QSAR-based models are limited by the absence of experimental validations, experimental data errors in training sets, and the brief length of their training sets [[→101](#)]. AI has demonstrated its exceptional utility in the improvement of medical imaging services by its capacity to manage vast quantities of radiology data. AI-based software has the potential to enhance the precision of patient diagnoses, thereby alleviating the burden on medical personnel and simplifying their responsibilities. AI-supported healthcare programs have the potential to improve intelligence as more data is collected and medical research is released. Patients who are not acquainted with AI may be convinced to participate in system-supported operations if they are apprised of the potential for more rapid, precise diagnosis, fewer errors, and reduced medical expenses. Healthcare systems should provide clients with information regarding the aids and disadvantages of AI-integrated medical systems [[→102](#)].

1.6 Conclusion

AI has transformed the pharmaceutical sector, leading to a significant enhancement in the efficiency of medication discovery and research. The application of AI methodologies, such as machine learning and natural language processing, facilitates the examination of large datasets, the

forecasting of medicinal compound effectiveness, and the creation of optimized drug delivery systems. The landscape of the pharmaceutical industry has also been impacted by big data, which has enabled healthcare systems to become more mobile and efficient. AI has the potential to enable personalized medicine, which would result in improved living standards, decreased resource waste, and reduced medical costs. However, there are still obstacles to overcome, such as the difficulty of locating relevant data, the lack of available trained personnel, and ethical concerns. Healthcare systems must provide patients with information regarding the advantages and disadvantages of AI-supported healthcare systems. AI can significantly improve medical imaging services and patient diagnostics, notwithstanding these challenges.

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Chapter 2 Emerging technology in pharma and the role of AI and blockchain

Abstract

The study, development, manufacture, and distribution of novel chemical entities (NCEs) and biological products intended to enhance human health constitute the core activities of the pharmaceutical industry, which follows a methodical and intricate business model. The chapter provides information about the innovative technologies introduced in pharma. In particular, drug discovery and regulatory approval are two areas where the pharmaceutical industry could undergo a radical transformation because of AI. Blockchain technology may significantly enhance the safeguarding of data confidentiality and the verification of intellectual property asset generation. Blockchain technology is also applicable to medical consultation frameworks, protecting users' privacy and security, an essential consideration from the perspective of the patient when it comes to psychological and sexual diseases. Nonetheless, there are several obstacles that blockchain technology must overcome, including data management, privacy issues, data redundancy, and integration with current healthcare systems. An overview of the latest developments in artificial intelligence and blockchain technology is provided in the chapter.

Keywords: blockchain, pharmaceuticals, ELN, LifeCrypter, GCoin, electronic health record data,

2.1 Introduction

The business model of the pharmaceutical sector is intricate and focused, with a primary emphasis on the production, marketing, research, and development of new chemical entities (NCEs) and biological products that are designed to improve human health [[→1](#)]. In the pharmaceutical industry, “advanced analytics” are utilized throughout the entire value chain, including development and research, safety, production, as well as regulation. These analytics are developed from various industries [[→2](#)]. The most important component of the healthcare system is the pharmaceutical sector, which is in charge of manufacturing and distributing drugs, biological substances, and medical equipment used in disease diagnosis and treatment. Additionally, the pharmaceutical industry engages in research to create new products that are beneficial to human welfare. Because numerous pharmaceutical goods are lifesaving, maintaining the final products’ quality is crucial to avoiding any possible health hazards [[→3](#)]. A straightforward prerequisite for the achievement of the international organization’s objectives for sustainable development is the establishment of a pharmaceutical manufacturing unit. Encouraging and improving the capabilities of the populace is vital to controlling their wellbeing and producing novel drugs that are both more affordable and effective. It will help achieve one of the main aims of sustainability, which is the reduction of poverty [[→4](#), [→5](#)].

“Traditional technology” refers to a three-step energy conversion process that was in use before the fourth industrial revolution. The needs of patients in both armed forces and civilian populations, along with healthcare providers, are not sufficiently met by the traditional manufacturing of pharmaceutical products. Traditional manufacturing is inadequately qualified to oversee and manage complex, computerized manufacturing processes for the proficient and profitable production of customized items. The capabilities of conventional technologies for the development of materials, including those used in medical applications, have reached their limits [[→6](#)]. It is possible to eliminate prescription errors that are caused by selecting the incorrect options from a selection box with the assistance of AI, which plays a significant role in the process as shown in [→Fig. 2.1](#). Artificial intelligence (AI) can recognize errors like these by analyzing the patient’s medical records. Lowering the chance of drug overdose along with additional health risks contributes to an improvement in the standard of care. Human error in the diagnosis of CT and MRI scans is reduced by AI, which improves the precision of treatment. It mitigates the danger to patient wellbeing, protects the hospital’s reputation, and decreases costs related to the breach of sensitive patient information [[→7](#)]. In recent decades, technology has significantly transformed every aspect of human life, including transportation, communication, manufacturing, business, and the pharmaceutical and medical sectors. The pharmaceutical industry has begun to shift away from traditional careers and toward more-advanced careers to facilitate the production of medicines for the next generation. The variety of

work available in the pharmaceutical research industry is expanding as career opportunities continue to improve. As a result of technological advancements, cutting-edge career opportunities are emerging. Through innovative research, these possibilities have been shown to greatly boost the pharmaceutical industry's output and introduce the next generation of medications onto the market [[→8](#), [→9](#)].

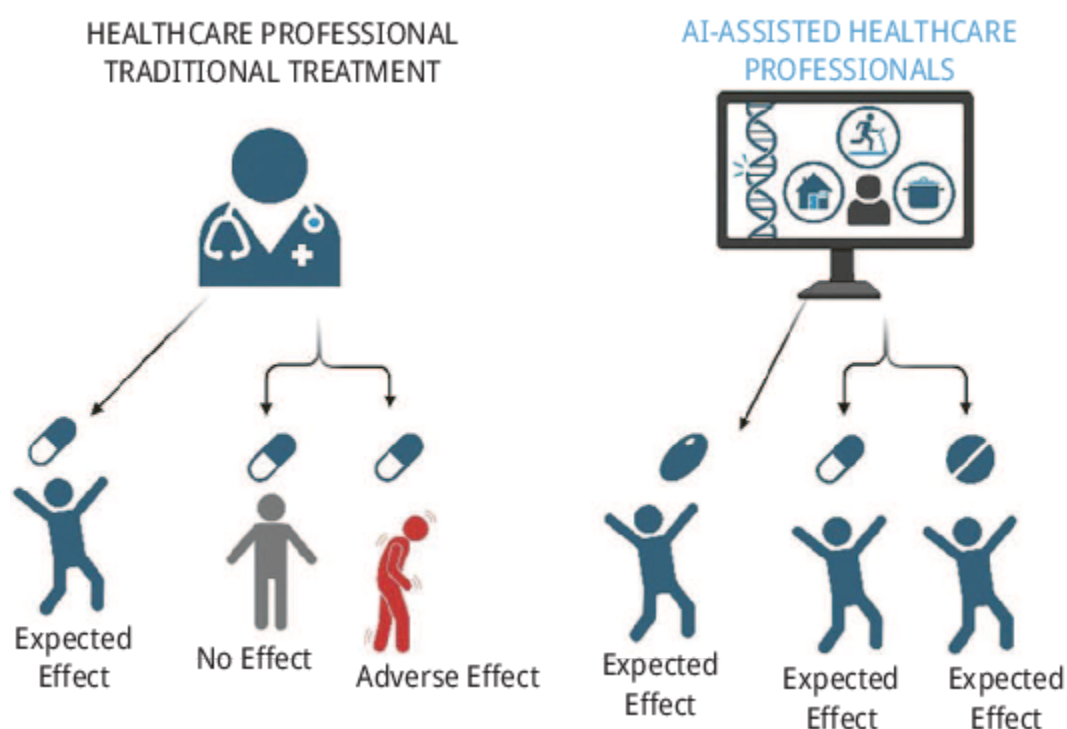


Fig. 2.1: Comparison of traditional treatment by healthcare professionals with AI-assisted healthcare professionals.

A growing number of institutions, companies, and scholars are becoming interested in blockchain technology, especially when it comes to using Bitcoin, a type of virtual currency. A diverse array of sectors and industries, such as aerospace, automotive, computing and electronics, engineering,

business and accounting, banking and finance, defense, and healthcare, have been transformed by the application of intelligent technologies other than blockchain. These technologies include AI, virtual reality, the Internet of Things (IoT), and machine learning [[→10](#), [→11](#), [→12](#), [→13](#), [→14](#)]. There is a novel, integrated, one-of-a-kind, and innovative computing technology that are a part of blockchain technology. The system includes transparent transactions, sophisticated consensus-building processes, dynamic public key algorithm encryption, programmable smart contracts, decentralized peer-to-peer networking, and decentralized retention [[→15](#)]. One of the numerous benefits of blockchain technology is that the contracts and transactions that are incorporated into the distributed database are impervious to modification. Decentralization, transparency, trust, privacy, and stability are among the numerous attributes of blockchain technology. A viable approach to preventing the sale of counterfeit medications has been identified through the implementation of blockchain technology in the pharmaceutical industry. Blockchain technology enables the monitoring of the procurement of raw materials, the distribution of medications, and the origin of pharmaceuticals. Furthermore, the implementation of blockchain technology leads to a decrease in the number of intermediaries, which in turn improves the safety ratings and reduces the costs of the pharmaceutical method [[→16](#), [→17](#)].

The supply chain system as a whole is equipped with verification processes that validate the product serial numbers within the parameters of the following technology. Additionally, the traceability of drugs has been

improved to prevent theft, and quality control has been accomplished through the utilization of digital signatures. In addition, to guarantee that the quality remains consistent, blockchain chain codes, medical records, and data mining have been instituted from the supplier to the pharmacy [[→18](#)]. Furthermore, delivery delays can hurt business operations as a whole, and in the pharmaceutical and medical sectors, they can result in fatalities or the exacerbation of maladies. The pharmaceutical supply chain has implemented blockchain technology. Drug monitoring and provenance are indispensable for regulatory compliance, patient health, and business operations. Sophisticated and secure monitoring and tracking technologies enable the timely delivery of products in transit to their intended destinations, thereby enabling the pharmaceutical business and patient management. To expedite the global shipment of pharmaceuticals, a secure global registry was established. Even though the technology offers substantial opportunities for large pharmaceutical companies, it is also beneficial for smaller businesses [[→19](#), [→20](#)].

2.2 Role of blockchain in the pharmaceutical industry

Drug discovery, drug progress, as well as regulatory certification, comprise the multi-year, intricate process of pharmaceutical development and research. Blockchain is a decentralized computer network that provides its members with access to a secure ledger of transactions and operates without

a central server. By time stamping each transaction made within the network, it eliminates the need for an outside organization. Every network node maintains its local ledger and maintains a local copy of the record. Modest changes made to the localized ledger are reflected across the network. Beyond just cryptocurrencies, blockchain technology enables the creation and monitoring of any sort of digital asset. Within the pharmaceutical supply chain, a product created by a manufacturer and registered on the network can be transferred to other participants like cryptocurrency, as it is considered a digital asset. Once again, blockchain ranks first for privacy protection. Each member of a blockchain network is given a key pair that serves as their identity [[→21](#), [→22](#)]. The blockchain approach maintains patient privacy and data security while offering a special benefit for patient recruitment. Patient recruitment times are shortened by using blockchain distributed ledgers because anonymously registered patients can be linked to various clinical trial sites. As a result, the researcher can select the necessary subjects from a group of patients by the study plan. Proposals have also been made for a blockchain structure that involves multiple contracts for clinical trials to manage the studies and involve patients [[→23](#)]. Also, the clinical trials are tracked using the blockchain technique, which encourages others to participate in clinical research while also instilling confidence in the outcomes. Because it can result in fewer clinical trials where positive results are covered selectively, transparency is an essential part of drug discovery research. Consequently, applying blockchain technology to clinical trials encourages improved

relationships between sponsors and different clinical research sites in addition to enhancing data integrity and transparency [[→24](#)].

Participants in clinical research can also access shared data due to blockchain technology. Even though big data increases CT storage capacity, its vast volumes, variety, and rapidity of records pose several challenges for decentralized storage frameworks [[→25](#)]. A compelling feature of blockchain is its ability to securely share real data with multiple stakeholders, which can simplify the administration of clinical trials. Clinical trials evaluate a novel drug's effectiveness after it is developed. Clinical trials are drawn-out procedures with intricate interactions between numerous regulatory review boards located in different places and different data sources. Clinical trial data must be collected and shared while maintaining confidentiality and proprietary information. Accurately and effectively managing clinical data from various information sources is difficult, though. By providing a unified platform for sharing information among all parties involved, blockchain technology can enhance the caliber of clinical trial results [[→26](#)]. First, all the data from different sources such as social media, laboratories, medical equipment, and many more is combined to create raw data. Later on, the following data grows into big data. The data that follows is indispensable to the blockchain-based healthcare system and functions as the primary building element for the initial layer of the stack. The fundamental framework for constructing a four-component, secure healthcare architecture is the unprocessed information layer, which is integrated with blockchain technology.

Consensus protocols and algorithms are among the features that vary amongst blockchain platforms. Numerous blockchain platforms have been developed to help users generate and oversee their transactions as shown in [→ Fig. 2.2](#) [[→ 27, 28](#)]. A blockchain-based solution called LifeCrypter has been introduced to improve security and credibility in the world's pharmaceutical supply chain. By employing clear identification tags, the system can maintain ownership data throughout its trade history, resulting in a reliable, transparent, and traceable drug supply chain network. Another reason the GCoin blockchain was introduced was to offer a transparent, consensus-driven, and unchangeable drug supply chain. To detect counterfeit and diverted medications, a Hyperledger-based framework is created to ensure patient privacy, security, and transparency [[→ 29](#)]. Additionally, researchers have used blockchain-based smart contracts to manage, secure, and analyze the sensors in real-time as well as to deal with PHI (protected health information) produced by such sensors. Researchers propose an entirely tier-based design for healthcare, where the ongoing medical monitoring system is managed by a patient-centric agent. To manage real-time data streams, some researchers have also introduced an architecture that involves a patient agent that coordinates blockchain implementation [[→ 30, → 31](#)]. The demonstration of the creation of intellectual property assets and the protection of data confidentiality can be significantly facilitated by blockchain technology. The blockchain-enabled Electronic Laboratory Note (ELN) is a digital notebook that facilitates the systematic, secure, and efficient sharing and archiving of laboratory data.

The system guarantees security and permits the delayed submission of a reduced number of novel medicinal products of legitimate interest. Before transferring the intellectual property documents to the blockchain and creating a record on the ledger, the researchers encrypt them to prevent unauthorized access. The researcher can obtain a blockchain certificate by submitting the encryption key produced during the document upload on the blockchain [[→32](#)]. Furthermore, before a medication is delivered to consumers, it may travel thousands of kilometers from its production facility. Logistics firms must keep an eye on the medication as it moves throughout the supply chain to make sure it is handled, transported, and stored in the best possible way. The effectiveness of the products may be adversely affected by improper maintenance of the appropriate conditions. Blockchain technology's decentralized structure, immutability, and transparency ensure that all participating authorities will abide by the rules. Smart contracts can be used to manage any deviations from complying with conditions by notifying the appropriate supply chain stakeholders. Researchers are conducting studies with the use of blockchain to monitor and trace products. Furthermore, researchers are incorporating Internet of Things sensors to monitor optimal temperature and humidity levels throughout the supply chain [[→33](#)].

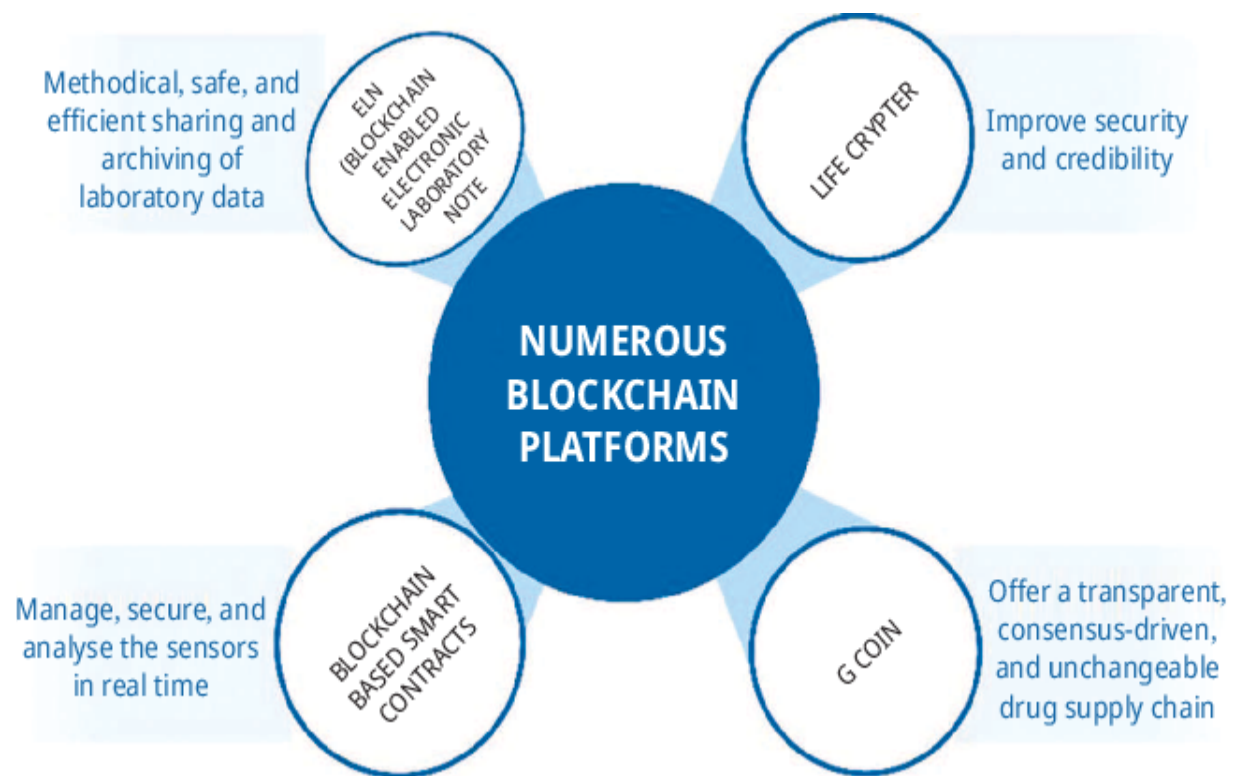


Fig. 2.2: Numerous blockchain platforms used in the pharma sector to help users.

The design, sharing, and preservation of electronic health record data within and between institutions have been the subject of extensive research in the field of blockchain technology. By creating secure online storage for patients' private medical records, researchers created a secure blockchain structure to facilitate medical data sharing. Within the structure, medical management of data is accomplished through the use of a digital archive with information owners' access control rights. By using cloud encryption underneath the chain, it is stored. Another researcher used blockchain technology and multiple authorities to create an attribute-based authentication scheme in a related study that guaranteed and validated EHRs. It makes broadcasting group messages easier and may be resistant to

collusion attempts. Additionally, some researchers have suggested a secure cloud-based electronic medical record system that is built on blockchain and attribute-based digital currencies. To encrypt medical data, digital signs were employed in conjunction with identity-based encryption and identity-based signatures. In addition to blockchain technology, other methods are implemented to ensure the authenticity and accountability of healthcare facilities [[→ 34](#), [→ 35](#)].

2.3 Convergence of AI and blockchain in the pharmaceutical industry

Blockchain is still considered a potentially transformative technology that has the potential to revolutionize how people communicate, conduct financial transactions, monitor markets, and record transactions. The potential of blockchain technology is to eradicate the need for a central authority to supervise and authenticate transactions and agreements between multiple parties. Another evident field that is rapidly gathering traction is AI. AI enables machines to acquire knowledge from data and make decisions based on their acquired knowledge. The idea of decentralized AI has advanced in recent years. Decentralized AI is constructed upon a combination of blockchains and AI. It facilitates the distributed, middleman-free processing and storage of meticulously classified, trustworthy, and shared data on the blockchain. It is now expected that the blockchain will serve as a reliable medium for storing the

data, and AI is said to be able to process enormous volumes of data. Smart contracts can be incorporated into blockchain programming, enabling dependable third parties to keep an eye on user data sharing and access [[→ 36](#), [→ 37](#), [→ 38](#), [→ 39](#)]. When combined with AI, blockchain technology has the potential to improve the supply chain in a variety of ways. The protection of local or regional supply systems and the provision of the requisite intelligence to improve operational efficiency may be achieved through the integration of blockchain technology and AI in a supply chain. It can facilitate decision-making and enhance data security and efficiency. Blockchain safely stores enormous volumes of data, while AI analyzes, deduces, and generates novel situations and patterns from data behavior [[→ 40](#), [→ 41](#), [→ 42](#)].

By integrating AI with the blockchain, the supply chain could become fully automated and undergo a fundamental transformation. The AI platform can gather data from a variety of sources, including sales data, point-of-sale information, and past purchases, through integration with the blockchain. It makes it possible to identify the characteristics of the data and apply predictive analysis, which is useful for tasks like route planning, demand forecasting, revenue projections, and network management. The researchers in their study recommended the implementation of blockchain technology and AI to ensure the secure distribution of medications throughout the entire supply chain. A triggered request-response mechanism allows for the transfer of any product in the chain between verified chain members. A product can be traced back to its source through the use of smart contracts,

which utilize the blockchain to record each exchange between entities. The experiment results show that the method is both feasible and considerably more secure than existing approaches [[→ 43](#), [→ 44](#)]. Researchers addressed the problem of minor medical evaluations by integrating AI and BC technologies. The use of BC in medical consultation frameworks ensures users' privacy and security, which is crucial when it comes to psychological and sexual diseases from the patient's perspective. Both AI and BC techniques have an extensive variety of applications in the medical insurance industry. AI has been used to create personalized offers for patients suffering from chronic illnesses, assist insurance (case) managers in managing insurance claims, and help them assess cases more accurately and decisively [[→ 45](#)].

Blockchain and AI are two technologies that have each shown promise for advancement in the healthcare industry. The use of these methods in the processing of EHR data is of interest to the scientific community. However, blockchain is starting to appear more frequently in the medical industry. It addresses the problems with current EHR frameworks' interoperability to prevent malicious misuse and record-altering of patient information. At present, several EHR-based system types, including Epic, employ live AI solutions for population health management and clinical decision-support platforms to predict patient deterioration, mortality rates, hospitalizations, and risk levels. In the same vein, MedRec, a blockchain-based platform, has enabled patients to access information [[→ 46](#), [→ 47](#)]. While blockchain technology can ensure security and integrity, adding AI capabilities could

be advantageous for the health sector. At least as effective as humans, AI is currently primarily used to identify anomalies in CT and X-ray scans, a task that ensures more individualized treatment. Due to the significant advantages that could result from reducing medical costs and raising the standard of care, experts predict that significant inventions will surface in the future [[→ 48](#)]. Therefore, the integration of state-of-the-art AI tools with blockchain technology will facilitate the decentralization and acceleration of biomedical studies and healthcare. It was demonstrated that deep learning (DL) is a potent AI technique that has been effectively employed in a variety of disciplines that have been the subject of research by certain researchers, such as drug classification, chemical synthesis, medical diagnostics, and cell image analysis. Additionally, it was demonstrated how DL has shown to be a useful tool for drug discovery and research [[→ 49](#)].

The goal of personalized cardiovascular medicine could be advanced by integrating blockchain technology with AI. Additionally, by sharing proprietary AI algorithms for generalization, decentralizing vendor or health system databases, providing incentives for solutions that perform better than those that do not, and greatly increasing the availability of data for AI development and training, other researchers studying the combination have discovered ways to accelerate it. These applications remain in their infancy, and there are some concerns about how they should be used. Further study is required in areas like technological trust, ethical considerations, compensation, and whether competing businesses would approve of the data flow in practice [[→ 50](#)].

Among other possibilities, blockchain-based smart contracts that integrate AI models may suggest the return of expired goods, the restructuring, payment, or buying of stock based on specified limits and events, the resolution of disagreements, and the selection of the most environmentally friendly shipping option. The impact of AI and blockchain technology on an automation service is examined by the researchers. The investigator opted for a secondary data collection approach to obtain pertinent data and details regarding the topic of the study [[→51](#)]. Blockchain also contributes to the advancement of AI by facilitating access to immense amounts of data from internal and external sources, enhancing control over data use and model sharing, generating more actionable insights, and promoting a more dependable and transparent information market. Using outside organizations, or oracles, to process data, researchers proposed a framework that leverages the use of blockchain as a ledger and smart contract platform for controlling robots. Researchers demonstrated how to use smart contracts to regulate robots, securely record events, and interface with external AI systems for the analysis of images [[→52](#)]. Data on blockchain is extremely safe because encryption is integrated right in. Storing sensitive and confidential data, such as health records or personal recommendations, on a blockchain seems logical. Massive, ongoing data sets are necessary for AI. Research and development efforts are currently concentrated on developing AI algorithms that can securely handle encrypted data.

2.4 Regulatory considerations of AI in pharma

Although its application in pharmaceutical regulatory affairs is still in its infancy, AI holds great promise for revolutionizing the sector. AI can help with regulatory affairs by providing information about the latest advancements in the field, particularly the potential benefits of AI for regulatory approval and drug discovery. AI can assist pharmaceutical researchers in expediting and enhancing decision-making by automating processes, identifying novel opportunities, and refining judgment. It facilitates the expedited introduction of novel pharmaceuticals to the market. Regulatory affairs refer to the divisions within pharmaceutical corporations that act as intermediaries with regulatory authorities, both national and worldwide, aiming to get product approval by prevailing rules and regulations. Regulatory bodies approve products based on the information and paperwork submitted during the drug discovery and product development process [[→53](#), [→54](#)]. Blockchain, internet-based computing, 3-D bioprinting, and AI are examples of consumer technology innovations with applications in the healthcare sector. These developments necessitate new regulations to keep up with the rapidly evolving field of technology [[→55](#)]. The Food and Drug Administration (FDA) of the United States is crucial in advancing the development of innovative technologies by establishing technical standards and guidelines, in addition to defining supplementary objectives to enhance the regulatory approval process. Recent developments in AI algorithms and their broad applicability point to significant consequences for medical technology and healthcare. Artificial general intelligence, or AGI, is defined as AI that demonstrates human-like

intelligence and can carry out any intellectual task that a human can. An FDA analysis indicates that artificial narrow intelligence (ANI), a category of AI commonly employed in medical technology, is designed specifically for a singular goal [[→ 56](#), [→ 57](#)].

AI must be regulated immediately because it is beginning to surpass human intelligence, and the widespread use of AI may have unfavorable consequences. The European Commission's "RoboLaw Project," which looked into the legal ramifications of robotics, produced the Guidelines on Regulating Robotics. Due to its emerging nature, AI cannot be confined to a single jurisdiction and thus requires global regulation. The risks that AI algorithm developers must deal with include dataset shifting, careless confounder fitting, and inadvertent discriminatory bias. The generalization to new groups and the potential unintentional adverse impacts of new algorithms on health outcomes are additional challenges that call for careful consideration and regulation [[→ 58](#)]. The hazards that AI algorithm developers have to contend with are inadvertent discriminatory bias, negligent confounder fitting, and dataset shifting. The generalization to new groups and the potential unanticipated negative effects of new algorithms on health outcomes are more concerns that require careful consideration and regulation. The FDA provides opportunities to extract validation information from real-world scenarios as well as release medical products based on it, with no need for setting up specialized clinical studies. Any information about a patient's health status that can be gathered from sources such as product and disease databases, electronic health records, or patient-

generated data gathered from settings used at home is referred to as real-world data. These initiatives demonstrate that the FDA views possibilities for software-based devices as well as patient health as being improved by data-driven strategies and dynamic development processes. AI systems must generally be viewed as dynamic systems. When more data is provided, it is meant to get better [[→ 59](#), [→ 60](#)].

As a result, AI-based devices are meant to be dynamic rather than to have a set status. Medical device regulations, on the other hand, demand that the release be in a fixed status. A thorough validation of the status is required before its release. Every time new data is received, the device would need to be released again. It prevents the database from changing continuously while the application is running. It would be required to validate and release the entire process. In concept, process-oriented validation extends beyond products to other software-based tools used in an organization's quality system. As a result, when medical devices are evaluated continuously using AI tools as mandated by the MDR – it is also relevant. The FDA's Pre-Cert program presently implements fundamental measures toward the approach [[→ 61](#)]. The regulation lays out the following: (i) standard operating procedures for the sale, use, and functioning of AI systems within the Union; (ii) limitations on the application of specific AI techniques; (iii) unique specifications for high-risk AI systems along with operator responsibilities; (iv) standard operating procedures for the application of AI and pharmacovigilance with COVID-19; rules governing transparency for AI systems used to generate or modify image, sound, or video content;

emotion recognition and biometric classification systems; and AI systems intended to interact with people; and (v) laws governing market surveillance and monitoring [[→ 62](#)]. The European Commission is prepared to create a legal framework that focuses on the risks associated with the specific applications of AI systems. Three categories of AI applications are distinguished by the risk-based approach: (1) unacceptable risk, (2) high risk, and (3) moderate to low risk [[→ 63](#)]. Certain AI systems that pose risks deemed “unacceptable” would be forbidden. Numerous “high-risk” AI systems could get approved, but only after meeting certain conditions and fulfilling certain duties to be allowed to enter the EU market. Very minimal transparency requirements would apply to AI systems that pose only “limited risk.”

The Health Ministry has been working to fulfill the regulatory role in quality management, consolidation, data collection, and consent standard-setting with the NDHB (2019) and the EHR Standards (2016). Enabling the ecosystem to share data and compile it in a way that is suitable for AI will advance it technologically. To facilitate data sharing, the government should do more than just establish plans and standards; instead, it should establish data marketplaces and cooperative initiatives. Since the healthcare industry is still in its infancy, the regulations governing AI are still changing. Additionally, there is a chance to have adaptable regulation and gain knowledge from past mistakes in finding the ideal balance between excessive regulation, which could impede the implementation of meaningful public health initiatives on a large scale, and insufficient

regulation, which could jeopardize user trust, safety, effectiveness, and adoption. As a result, the regulator must take into account two things: the clinical assessment of AI models and the post-market surveillance and tracking of AI models that are being used [[→ 65](#)].

2.5 Challenges of blockchain in pharma

The domain of blockchain technology research is expanding. Initially confined to the banking industry, blockchain technology has now been adopted by government agencies, the Internet of Things, reputational frameworks, and security services. The pharmaceutical business has lately implemented blockchain technology, mostly to address the issue of counterfeiting. Blockchain technology has two primary advantages due to its decentralization: information security and privacy. Nonetheless, it necessitates increased processing power. About fifty percent of the energy is allocated to thwarting hostile user attacks; without efficient tracking and tracing, users may suffer losses due to theft and misplaced items [[→ 66](#)].

Global secure banking transactions can be facilitated by blockchain technology, but interoperability concerns pose a serious challenge.

Distributed ledger technology can be used to standardize blockchain operations (DLT). Further research is necessary to ascertain the best approaches to implement DLT applications, even as the International Telecommunication Union (ITU) strives to identify and standardize DLT applications and their services. The utilization of edge computing, cloud

computing, and advanced Internet of Things applications has markedly enhanced data accessibility [[→67](#)]. The enormous growth presents many challenges for the implementation of blockchain, including data management, privacy issues, redundancy issues, and accessibility issues. Additionally, because of misbehavior and poorly run supply chains, it has tried to leverage blockchain technology to speed up medical transactions, safeguard patients, and track and trace [68, 69]. Ensuring interoperability between diverse healthcare equipment and systems is also crucial. For smooth exchange of information and interaction between the various hospitals and systems, robust interoperability solutions are required. The blockchain network must accommodate increased transaction rates without incurring performance degradation as healthcare data volume expands. To ensure blockchain implementations adhere to rigorous data protection requirements such as GDPR and HIPAA, meticulous design and enforcement protocols are essential [[→68](#), [→69](#), [→70](#), [→71](#), [→72](#)].

Researchers assert that the integration of blockchain technology with the existing healthcare system presents significant challenges. Since numerous hospitals still use legacy systems, careful planning is required to guarantee backward compatibility and a smooth transition. Furthermore, it is imperative to tackle the energy consumption issues associated with blockchain technology, specifically with proof-of-work consensus. Blockchain solutions that are energy-efficient and environmentally friendly should be taken into consideration by healthcare organizations [[→73](#), [→74](#), [→75](#), [→76](#), [→77](#)]. Also, managing storage capacity is another challenge.

Blockchain, engineered to document and manage transactional data, possesses a constrained scope, hence necessitating minimal storage capacity. As technology evolved and penetrated the healthcare sector, storage challenges emerged. Daily, substantial data from the healthcare sector need processing. In a blockchain context, all data, such as patient records, medical histories, test results, MRI scans, X-rays, and other images, will be available to every node in the network, requiring substantial storage capacity. As blockchain technology advances, it encounters not only the previously noted technological issues but also social obstacles, including cultural transformations. Embracing and implementing a technology that is entirely disconnected from conventional work methodologies is invariably challenging. The medical profession is progressively transitioning to digitalization; however, substantial advancements are required before it completely adopts technology, particularly blockchain, which now lacks clinical confirmation. To get doctors to rely on technology rather than paperwork, it will take time and work. Because the healthcare industry has been slow to adopt new technology and policies, the policies offered are not widely trusted [[→ 78](#), [→ 79](#), [→ 80](#), [→ 81](#)].

Blockchain technology remains in its early stages of development and refinement. The technology will initially cost institutions more and will need to be embraced and integrated over time. It's also necessary to address issues with non-standardization, scalability, information decay risk, and integration with current legacy systems. A related issue is blockchain

transaction verification time. Technical implementation challenges include unproven systems, hidden operating costs, and the need for a network of connected computers (nodes) to process transactions and create blocks [[→ 82](#), [→ 83](#)]. No incentives exist to use blockchain technology to build a vast network of interconnected information “blocks” to replace national record-keeping. Blockchain could improve the current system but not replace it. The majority of medical data would not be stored on the blockchain. To describe a particular patient or procedure, a small amount of information would be retained in each “block” [[→ 84](#)]. In some scenarios, the growing volume of daily transactions causes the blockchain to get heavy, which causes scalability problems. It makes developing blockchain-based real-time healthcare applications difficult because they cannot process millions of medical records. Furthermore, because of the limited block capacity at the moment, miners may choose to verify transactions with higher transaction rewards before completing smaller ones. On the other hand, large-size blocks decrease transaction speed. Blockchain networks necessitate security as users generate addresses rather than utilizing their actual identities. Due to the visibility of each private key’s balance and activities to all network participants, blockchain-based solutions are unable to provide transaction confidentiality. In the medical field, safeguarding sensitive patient data is essential, which makes the issue crucial [[→ 85](#), 86]. As a result, blockchain technology is still developing and addressing social issues like a culture shift. Technology that differs from traditional work methods is hard to accept. Blockchain technology is

gradually making its way into the pharmaceutical industry, but full adoption of the technology takes time. Cultural resistance may occur because most patients dislike having their data shared. To use blockchain in the framework, some people must change their behavior to accept distributed data sharing [87].

2.6 Conclusion

Healthcare relies on the pharmaceutical industry, which researches, develops, manufactures, and markets drugs, biological compounds, and medical devices. Traditional manufacturing methods have reached their limits, and AI can help eliminate prescription errors and improve care. Blockchain technology, particularly Bitcoin, has gained interest in the pharmaceutical sector due to its decentralization, transparency, trust, confidentiality, and stability. It has been utilized to avoid counterfeit drugs, track drug origins, improve drug transparency and quality control, and aid the pharmaceutical supply chain. Blockchain allows safe transactions as well as privacy protection, decreasing patient recruitment times and enhancing the precision of data and transparency. Platforms like LifeCrypter, GCoin, Hyperledger, and smart contracts have been developed to improve security and credibility in the pharmaceutical supply chain. Blockchain technology is utilized to create, exchange, and safeguard EHR data, ensuring the validity and accountability of healthcare institutions. AI has a significant potential to transform the pharmaceutical sector, especially

in drug discovery and regulatory approval processes. The US Food and Drug Administration (FDA) and the European Commission's "RoboLaw Project" have provided technical standards and guidelines for AI algorithms. However, blockchain technology faces challenges such as data management, accessibility, redundancy, privacy, and integration with existing healthcare systems.

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Chapter 3 Partnership and collaboration in the pharmaceutical industry

Abstract

A significant contributor to the creation of treatments and vaccines is the pharmaceutical industry. To develop vaccines or treatments (like COVID-19), relationships and open, cross-disciplinary collaboration among scientific and research communities, startups, and pharmaceutical firms have been critical success factors. The chapter looks at the evolving models of open innovation (OI) that pharmaceutical companies created, funded, and supported to interact with startups to accomplish sustainable corporate growth. The foundation of the open innovation model is the idea of an alliance between an established business and a startup. The public-private partnership (PPP) system is a startup-company asset-sharing arrangement that can be used to foster innovation in a variety of drug discovery and development contexts. The chapter primarily focuses on the various types of collaborations and their significance in the pharmaceutical industry. The conclusions of the chapter show that the pharmaceutical industry needs to work with startups, producers, and businesses from a variety of fields to collaborate and accelerate innovations when new pharmacological mechanisms, enabling techniques, and translational models are discovered.

Keywords: partnership, collaborations, pharmaceutical industries, public-private partnership, start-ups, challenges,

3.1 Introduction

It is unambiguous that the pharmaceutical industry needs to supplement its traditional approach to innovation with a quick approach that focuses on getting acquainted with startups and innovators quickly and repurposing concepts, expertise, and technologies that are easily accessible. The key success factors in developing vaccines and discovering treatments (such as COVID-19) have been the relationships and open, cross-disciplinary collaboration among the scientific and research community, startups, and pharmaceutical companies. The role of corporate venture funds, partnership initiatives, and internal and external systems for sourcing and scouting was never as crucial in generating ideas that pharmaceutical companies could adopt or in coming up with novel approaches to employing already-existing facilities, services, and products as coronavirus crisis solutions. Large companies' main strategic reasons to participate in open innovation are to forge new alliances, investigate emerging technologies, and spot new business prospects [[→1](#), [→2](#)]. Academic, commercial, and nonprofit partnerships can give projects with low financial returns enough momentum to advance; this is especially true for the drug discovery industry. Public-private partnerships (PPPs) offer strategic guidance; industry groups are dominated by strongly product-focused processes and wide expertise; academics typically collaborate on high-risk projects alongside little financial return; and each partner brings a unique set of skills and technical

know-how to the table that benefits the project as a whole [[→3](#)]. The need to sustain and grow shareholder value and successful manufacturing levels drives the pharmaceutical industry continuously. When it comes to advancing candidates through the development pipeline, small and medium-sized pharmaceutical and biotechnology businesses with stronger ties between research and development have outperformed their larger sisters. To address unmet market needs, they have become more adept at creating biological products, including monoclonal antibodies, vaccines, and peptides [[→4](#), [→5](#)].

PPPs are partnerships between private businesses and public organizations that are intended to carry out specific projects. These contracts are usually used to fund initiatives like building parks and public transport systems. Within the pharmaceutical sector, these agreements facilitate the funding of research and development (R&D) endeavors via public funds or charitable organizations to create novel drugs, particularly in therapeutic domains with unfulfilled medical needs and lower levels of pharmaceutical research activity [[→6](#)]. The drug industry's wide expertise and highly product-focused processes, combined with PPP strategic guidance and academics' propensity to work on high-risk, low-reward projects, can create a fertile and productive environment [[→7](#)]. Partnerships, alliances, and collaborations can lower overall costs and lengthen development times while significantly increasing clinical success. Through them, every business is limited to maintaining its fundamental skills and is dependent on outside parties to learn about its areas of non-competence. R&D executives

have realized in recent years that internal efforts alone are insufficient to sustain the entire scope and depth of research projects. Numerous pharmaceutical companies have moved their R&D centers close to some of the top universities [[→8](#)]. National Cancer Institute–Technology Transfer Center (NCI-TTC) facilitates partnerships and collaborations for licensing with businesses of all sizes, across a wide range of geographies and development stages, and with a variety of life sciences research and development interests. Leading healthcare corporations, possessing ample financial resources to conduct clinical trials, are essential for introducing innovative technologies to the market. Large pharmaceutical companies, along with small and early-stage companies, must thus collaborate and license partnerships because they have an increased need to have access to expert clinical and research expertise, hands-on experience, and new technologies [[→9](#), [→10](#)]. Researchers examined the efficacy of industry-university collaboration as a comprehensive endeavor with a variety of stakeholders and goals. Organizationally, one of the most important things from the company’s point of view is the firm’s capacity to take on new technology and incorporate it into the business value chain. Scholars bolster the idea by asserting that businesses prioritize the impact of research over project results. An important criterion for assessing the efficacy of knowledge transfer is how the new information will improve an organization’s performance. Strong leadership, senior management support, incentive structures, and researcher motivation are all significant university-level factors that impact collaboration [[→11](#), [→12](#), [→13](#)].

Research and development cooperation between universities and the pharmaceutical sector is not a new idea in and of itself. It was established some time ago to promote the development of new products and maintain pharmaceutical innovation [[→ 14](#), [→ 15](#)]. Technology and knowledge transfer from academia to business through licensing has increased in frequency. However, the industry and the university have different objectives. Conversely, researchers plan to publish their findings in specialized journals to disseminate their findings to the general public. Academic reputation is more significant to researchers than profit, but industrialists place a great deal of importance on financial performance [[→ 15](#), [→ 16](#), [→ 17](#)]. Moreover, although universities lead the way in scientific innovation, most fields see product- or process-oriented technical change taking place within businesses. To achieve sustainable corporate growth, the work investigates university-pharmaceutical industry cooperation in Portugal's Centro region. The discussion will center on whether the seemingly at-odds interests work to the advantage of pharmaceutical companies [[→ 18](#), [→ 19](#), [→ 20](#)]. It has developed into multi-stakeholder, more horizontal PPPs. One of the primary reasons is that the multi-stakeholder PPP model establishes the essential commonality that enables government agencies, educational institutions, patient organizations, health foundations, and business organizations to aggregate resources and expertise. PPPs enable these stakeholders to collaborate to resolve the primary concerns that are of mutual interest in the pharmaceutical innovation sector [[→ 21](#), [→ 22](#)]. In recent times and with

ever-increasing intensity, Big Pharma has been an important participant in transactions like collaborations and partnerships, mergers and acquisitions (M&A), outsourcing of products and procedures, and others that require a complete reorganization of the R&D. Pharmaceutical industry multinationals are starting to realize the full potential of open innovation, emulating other sectors that were early adopters of knowledge sharing. Furthermore, the FDA and EMA decided to open up their clinical trial database to anybody, enabling them to use the data to speed up the development of new medications and simplify the approval process [[→23](#)].

3.2 Types of partnerships involved in pharmaceutical industry

When more than one of these organizations collaborates, some of the best work toward the discovery of novel medications is produced. The drug industry's highly product-focused processes, broad expertise, inter-firm PPPs' strategic guidance, and academics' propensity to work on high-risk, low-reward projects can create a fertile and successful environment. These kinds of partnerships have surfaced as shown in [→Fig. 3.1](#).



Fig. 3.1: Various types of partnerships in the pharmaceutical industry.

3.2.1 Inter-firm collaborations

Although a lot of research has been done on venture capital contracts, less has been written about the difficulties that arise in inter-firm R&D partnerships, like those found in the pharmaceutical sector, where intellectual capital plays a major role. There are more reasons to research these inter-firm collaborations because it has been noted that patents and intellectual property rights are essential to the financial success of any pharmaceutical company [[→ 24](#)]. In the pharmaceutical sector, inter-firm

contracts are not uncommon; however, there is a notable variation in the terms of these contracts. Every contract partner may find different terms more or less appealing. Startup businesses typically prefer upfront payments, but large pharmaceutical companies may view them as risky if drug development fails. Large pharmaceutical companies usually prefer royalties as a profit-sharing mechanism after a drug is successfully commercialized because they lower the risk of initial payments [[→25](#)]. Although the pharmaceutical sector offers an abundant setting for examining inter-firm strategic alliances involving intellectual capital, these kinds of partnerships are not exclusive to the healthcare sector. Other industries, especially those involving cutting-edge technology, have inter-firm alliances where businesses collaborate to share intellectual property and other complementary skills and resources to support successful product development and commercialization [[→26](#)].

3.2.2 Public-private partnerships

Such initiatives, which vary from two-party agreements to large, both privately and publicly financed consortia, involve companies, colleges, universities, governments, and nonprofit and nongovernmental organizations within the life science sector. Open access and controlled collaborations can facilitate sharing, which can be highly beneficial for both presently under-researched and under-focused areas of study. It might create avenues for novel discoveries, which would then benefit public health. While the pharmaceutical industry offers numerous asset-sharing models,

the PPP model has garnered attention as a potent means of stimulating innovation across the spectrum of drug discovery and development. To further research in specific disease and therapeutic areas, or to achieve specific objectives, such as target identification, target confirmation, biomarker growth, and drug safety and efficacy evaluation, PPPs have also been introduced. PPPs that are specific to certain diseases or therapeutic areas include the Alzheimer's Disease Neuroimaging Initiative (ADNI), Dundee Kinase Consortium, Accelerating Medicines Partnership (AMP), National Cancer Institute (NCI) Community Cancer Centers Program (NCCP), and Coalition against Major Diseases (CAMD) [[→27](#), [→28](#), [→29](#), [→30](#), [→31](#)]. Furthermore, pharmaceutical companies have developed publicly accessible screening libraries, such as the Tres Cantos Antimalarial Set (TCAMS), which was assembled by GlaxoSmithKline (GSK). Numerous individuals within the malaria community have successfully implemented TCAMS [[→32](#), [→33](#), [→34](#), [→35](#), [→36](#), [→37](#)]. The Structural Genomics Consortium (SGC), a nonprofit PPP that was established to identify protein structures that may serve as targets for drug discovery initiatives, receives the backing of numerous pharmaceutical companies [[→38](#)].

3.2.3 Academic-industry partnership

Academic medical centers are investing resources to guarantee successful collaborations with pharmaceutical companies, as they have rekindled interest in doing so. These collaborations shouldn't be one-sided, with

information flowing both ways. Their pharmacological counterparts can provide academic colleagues with valuable insights. The majority of academic centers could benefit from the exchange of experiences regarding technologies, knowledge management, project management, and operational excellence. Pharma must establish a method to provide academicians with access to its extensive chemical libraries, confidential data sets, cutting-edge tools, and technological platforms. It is essential to participate in strategic, cooperative innovation in biomedicine to fully leverage the potential of novel biomarkers or disease targets and to stay up-to-date with the rapid tempo of discovery. Pharma's present business strategies are predicated on the identification of clinical candidates through acquisitions, partnerships, and mergers. It is essential to forge active partnerships with academia to address critical scientific inquiries and convert novel targets into therapeutic programs, thereby replenishing the future drug pipeline with innovative medications [[→ 39](#), [→ 40](#)].

New translational models, enabling technologies, and therapeutic mechanisms are being discovered at an ever-increasing rate. The practice of segregating pharmaceutical research within an organization was common in the past. Similarly, academic drug discovery has changed dramatically in recent years, and working with the private sector has become more valuable. The worth goes beyond money. These multidisciplinary collaborations bring together the most innovative concepts and methods for locating new therapeutic targets, solving shortcomings in translational

chemical biology, and creating new chemistry and technologies to increase the practical applications of these publicly funded discoveries [[→ 41](#), [→ 42](#)].

3.2.4 Collaboration with start-up

To remain competitive, expand into new markets, manage the mobility of qualified professionals, abbreviate product life cycles, and address the increasing complexity, specialization, and convergence of technologies, businesses have consistently sought new innovative approaches. The number of enterprises is increasing steadily. Startups are brimming with young talent who possess a strong sense of entrepreneurship, a willingness to take risks, a willingness to adapt to change, and the power of digitization, even though they lack resources, brand credibility, and easy access to the market. They are also awash in exciting ideas. To improve their chances of success, they typically need to collaborate with other businesses. Several organizations, including those in the pharmaceutical industry, are collaborating with startups to pursue innovation outside of their boundaries. These initiatives are well-designed and mutually beneficial, generating value for clients [[→ 43](#), [→ 44](#), [→ 45](#)].

Beginning with the significance of corporate relationships and open innovation in the pharmaceutical industry as mentioned in the preceding paragraph, researchers chose to examine the developing models of OI that pharmaceutical companies developed, funded, and supported to engage with startups. To examine the phenomenon, a database was created

containing the OI programs that a sample of startups had implemented [[→46](#), [→47](#)]. All businesses with billion-dollar annual sales, Italian businesses, and businesses engaged in the respiratory and central nervous system therapeutic fields were taken into account, and a rival analysis was initiated. Initiatives involving pharmaceutical companies and startups were found to be more successful than those involving non-partnership companies [[→48](#)].

3.2.5 Collaborations with R&D industries

There are several inherent risks associated with forming a successful strategic partnership for R&D activities, particularly when considering the context of multinational enterprises' (MNEs) knowledge misappropriation. According to resource-based theory, small and medium-sized enterprises (SMEs) nevertheless need to collaborate on R&D projects with large corporations despite the possibility of technology secrets being misappropriated. It requires strong intellectual property policies (IPPs) to protect the SMEs' vital proprietary technology. Additionally, R&D outsourcing and collaboration are brought on by the growth of large-scale cooperative multi-technological products [[→49](#), [→50](#)]. If SMEs don't have enough resources, big businesses won't be able to acquire the cutting-edge information and tools needed to add value. Trust in the ability to exchange sufficient information is one of the key success factors in the collaboration between large corporations and SMEs. Based on the types of new markets that are entered, knowledge management systems, and R&D systems,

researchers propose significant distinctions between open and closed innovation. Certain researchers conduct analyses of open innovation cases based on factors such as industry, size, and the types of open innovation, which include technology exploration and exploitation. According to the research, medium-sized businesses engage in more open innovation than small businesses do, and they are motivated to collaborate in the market by factors like new market development and customer needs realization. The open innovation strategy and how firm size affects the degree to which the tactic is chosen are discussed by other researchers [[→ 51](#), [→ 52](#)].

3.3 Motivation for alliances in the pharma sector

Small biotech companies provide the pharmaceutical industry with “external” sources of research and development. A deficiency in internal development is a primary reason for larger companies to seek new product technology from external sources. However, numerous minor factors frequently encourage the use of external procurement to facilitate the development of new technological innovations in the industry: (1) the licensee firm’s functional capabilities have a high absorptive capacity for new technologies, (2) the top management believes that external sources will complement internal sources, (3) the environment is perceived as too hostile and hazardous for R&D investment due to rapid product changes and regulatory uncertainties, (4) the company’s management is risk-averse, (5) the company has extensive experience with external sourcing, (6) the

management has a strong global orientation, and (7) there are substantial opportunities for external sourcing [[→ 53](#), [→ 54](#)]. Firms can contribute substantial resources through alliances that are greater than what each individual cooperating firm could provide. These alliances frequently unite businesses that possess complementary core competencies, which facilitate the expansion of the companies into new markets, the surmounting of trade obstacles, and the development of new products [[→ 55](#)]. The following strategic justifications for alliances are provided: (1) to resolve technological gaps; (2) to mitigate the risks associated with entering new markets, including international ones; (3) to accelerate the introduction of new products into the market; (4) to achieve scale economies; (5) to reduce trade and legal barriers; (6) to broaden the scope of current operations; and (7) to reduce divestment costs [[→ 56](#)].

The most well-known type of relationship between for-profit and not-for-profit organizations in the healthcare context is interactions among pharmaceutical companies, healthcare providers (which include professionals and healthcare organizations), and large-scale interactions [[→ 57](#), [→ 58](#), [→ 59](#), [→ 60](#)]. Over most of the doctors interact with pharmaceutical companies, according to research from multiple countries, and a few percentages of US biomedical researchers accept funding from the industry [[→ 61](#), [→ 62](#), [→ 63](#), [→ 64](#)]. Academics, professionals, the media, and politicians are scrutinizing pharmaceutical company-healthcare provider interactions more and more, but experts are divided on whether these interactions improve social value and have positive effects [[→ 65](#),

[→ 66](#), [→ 67](#), [→ 68](#)]. There are two categories of interactions: (a) interactions focused on education, and (b) interactions focused on research. Education-oriented interactions are those that aim to improve professionals' (such as doctors or nurses) awareness, knowledge, and mindset toward new goods and services through the dissemination of factual and instructive information [[→ 69](#), [→ 70](#), [→ 71](#), [→ 72](#)]. Research-oriented interactions are those that are specifically designed to sponsor, fund, or provide assistance with research initiatives. Research indicates that pharmaceutical companies and healthcare providers that prioritize education influence physicians' prescribing behaviors. In other words, physicians who participated in educational interactions prescribed a greater quantity of name-brand medications and administered a specific injection on a more frequent basis [[→ 73](#), [→ 74](#)]. Finally, the prescribing behavior of physicians is binary, influenced by education-oriented interactions in both investigations. In other words, the prescribing practices of physicians are not significantly affected by the increased monetary value of the education-oriented interaction [[→ 75](#)]. Further research on research-oriented interaction reveals that doctors who identify primarily as entrepreneurs and whose professional identities are closely aligned with pharmaceutical companies are more likely to allow pharmaceutical companies to exert greater control over different aspects of research projects [[→ 76](#)].

The corporate pharmaceutical sector is characterized by intense competition, with each organization striving to emerge as the winner in the sprint to introduce innovative blockbuster compounds. The industry knows

for a fact that companies that enter the market first profit more than those that introduce follower molecules. Moreover, early adopters have enough time to refine their offerings and determine the going rate. Pharma companies are always searching for and developing new compounds that target similar or very similar drug targets [[→77](#)]. These repeated research initiatives ultimately use resources that could be better allocated to the R&D of other diseases with unmet medical needs, given the enormous funding directed toward pharmaceutical R&D. Several cooperative agreements have been put forth and used to improve the performance of pharmaceutical R&D [[→78](#), [→79](#)]. Pharmaceutical companies, biotechnology firms, and academic drug discovery divisions share the goal of conducting fundamental research that will facilitate future drug discovery and innovation, even though they would otherwise compete. Consequently, they collaborate on precompetitive research. Precompetitive research, in its simplest form, determines the scientific feasibility of pursuing a specific therapeutic pathway before a full-scale research and development drug campaign is launched [[→80](#)]. Precompetitive research can be conducted in several areas, such as target validation and identification, compound library sharing, and assay and biomarker development. Precompetitive cooperation has many advantages, such as lower research expenses because businesses pool resources and knowledge, increased productivity because businesses concentrate on their core skills and improve their quality, and the exchange of ideas among scientists [[→81](#)]. Therefore, precompetitive studies, partnerships, and collaboration strategies have improved drug discovery

efficiency. Research on rare and neglected diseases should be expanded into underserved therapeutic areas. It would protect companies with blockbuster drugs that are on the verge of expiring.

3.4 Challenges in pharmaceutical collaboration

Even though industry-academia cooperation is on the rise, certain problems and obstacles still affect the mechanisms of collaboration, as shown in [→ Fig. 3.2](#). A distinct time horizon, confidentiality, and exclusivity; organizational variances may exist, and these can directly affect funding levels, university budgets, academic incentives, and research priorities. Industry is driven by profit-driven solutions to issues and outcomes, while university researchers are motivated by academic prestige and interest. There are certainly cultural differences between university and industrial R&D research, with the former concentrating on applied problem-solving research and the latter more exploratory. Due to divergent values and misunderstandings, it could lead to a serious conflict among academic and industrial partners [[→ 82](#), [→ 83](#)]. Collaborative and partnership research is impeded by the diversity of perspectives on knowledge production and use, as well as the connection between researcher and practitioner from top-down to bottom-up, sequential to interactive, and multidimensional [[→ 84](#), [→ 85](#)]. Based on research traditions and/or experiences, researchers and stakeholders may possess distinct fundamental assumptions regarding knowledge and learning [[→ 86](#)]. Researchers discovered that social

relationships could be weakened by role ambiguity, various positions, and role conflicts. They also identified barriers such as organizational change, cultural differences, and standards on research output, as well as positive effects on actual usage [[→87](#)]. The development of the collaborative knowledge platforms that serve as the foundation for many of these innovative working practices is also beset by numerous real-world obstacles. Creating a critical mass of contributors, standardizing and ensuring the quality of the data entered, curating the databases that are produced, utilizing the right tools for analysis and visualization, and protecting clinical sample data are a few examples [[→88](#)].

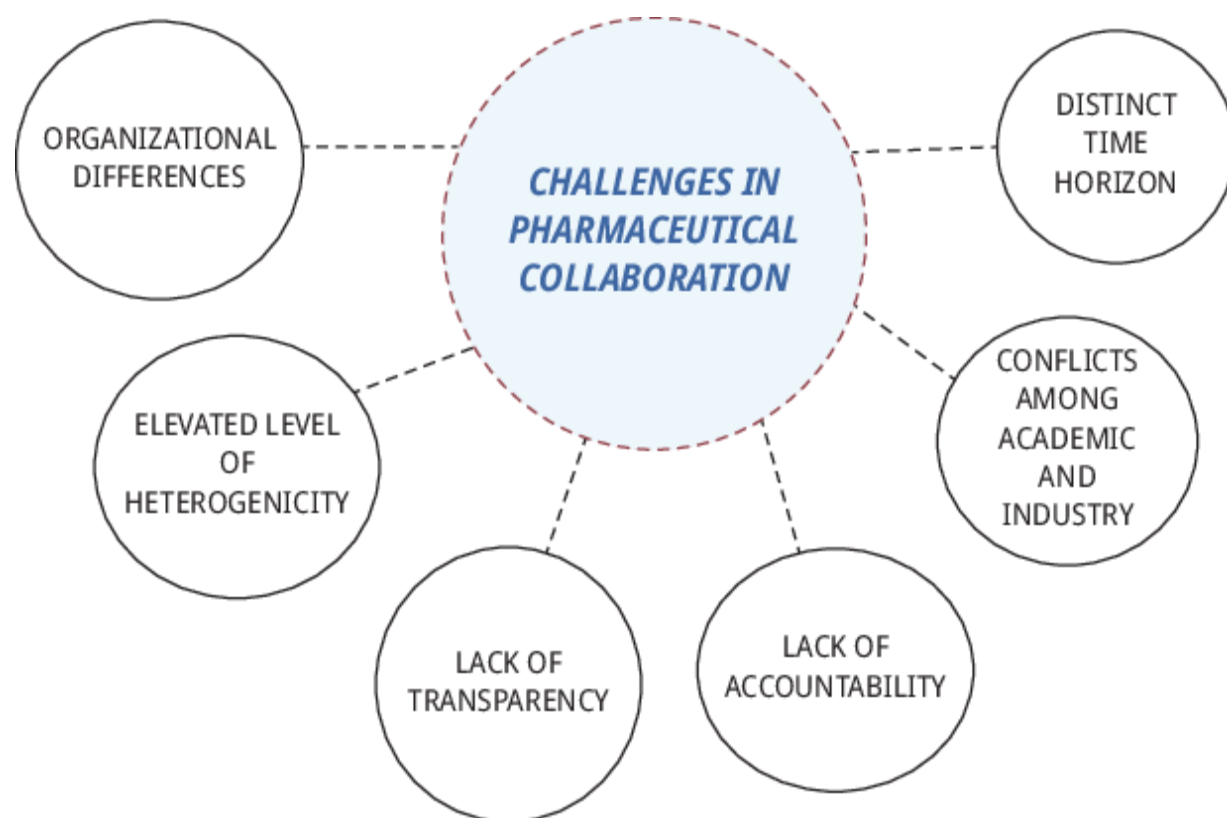


Fig. 3.2: Various challenges faced by collaborations in the pharmaceutical industry.

The absence of accountability and transparency in PPPs is an additional impediment to equitable health care on a global scale. Public inspection is frequently prohibited for partnership agreements with the private sector. The selection of private partners, the establishment of objectives to be achieved, and the development of management policies are all opaque processes. The apparent lack of transparency makes it difficult to ascertain the equity objectives that have been established and the individuals who should be held accountable for their fulfillment. Due to their intricate structures, differing processes of accountability between both the private and public sectors, and complicated governance, private companies can be challenging to hold accountable for failed PPPs [[→89](#)]. The achievement of the university-industry collaboration is impacted by multiple challenges. Organizational differences present a variety of challenges as well. However, the most frequent problems in university-industry (UI) partnerships are those brought on by organizational differences between universities and industries. Industries and academic institutions have distinct institutional settings and prioritize different things [[→90](#)]. The organizational structure of a university is more rigid and less adaptable than that of a business. Even though the challenges arise from these structural differences, we still don't fully understand how various collaborations address these challenges. There is a dearth of understanding about how these organizational differences impact collaboration, particularly when students are involved [[→91](#), [→92](#)]. Another issue that affects these collaborations is an elevated level of heterogeneities, usually during the execution stage [[→90](#), [→91](#)]. The

complexity of collaboration arises from the diverse characters, values, and traditions that come together to work together. Researchers frequently lament that the business isn't giving them the information they need to carry out their work. Prior studies indicate that there are fewer organizational differences the more closely industry and academia interact. Organizational differences have an impact on the collaboration process since the parties involved in it are accustomed to seeing things from the other partner's perspective [[→92](#), [→93](#), [→94](#), [→95](#)].

3.5 AI-assisted pharma collaborations

Partnerships between public/governmental, private/commercial, and civil society organizations can enhance the health of the impoverished by creatively integrating the diverse resources and talents of various organizations. Public agencies benefit from collaborating with the private sector in areas such as product development, manufacturing, marketing, distribution, and production process development, where the public sector is weak. Partnerships seem to be the most suitable option in the following scenarios: The impact of traditional independent methods of addressing a problem is limited. Potential collaborators can reach a consensus on the specific desired objectives. Both sectors possess relevant complementary expertise. The long-term interests of each sector are met, and the contributions of resources and expertise are fairly balanced. It is not advisable to assume that PPPs will fulfill obligations that are located

elsewhere. In general, PPPs should be considered social initiatives that aim to identify more effective strategies for addressing complex health issues [[→ 97](#)].

In response to the rapid adoption of AI in healthcare, numerous pharmaceutical companies have established joint ventures and invested in AI companies to develop more sophisticated medical instruments. These encompass the identification of pharmacological targets, the development of new medications, and the advancement of biomarkers or diagnostics. The ML/DL analyses of the underlying data are subsequently combined to develop algorithms that significantly contribute to the advancement of AI-infused modern healthcare [[→ 98](#), [→ 99](#)]. As a result, several international alliances between the pharmaceutical and artificial intelligence sectors have recently been formed. Benevolent AI is one of the AI technologies that are presently being utilized in the development of medications. It employs the text mining technique to analyze the extant patents, as well as additional genetic and biological data, to infer the connections between those entities. It leads to the development of diagrams that are highly illuminating and comprise dynamic maps with over one billion relationships [[→ 100](#)].

Exscientia is an AI venture that focuses on the discovery of phenotypic medications. AI significantly underperforms human analysis of exceedingly intricate datasets in the realm of high-content phenotypic drug discovery. Testing each newly developed compound and subsequently comparing its anticipated performance to other molecules facilitates the rapid evolution of drug designs. Numerate is also a notable AI enterprise. Numerate

underscores the potential of algorithms to revolutionize the search for new medications by addressing critical therapeutic gaps through the analysis of extensive drug development datasets, with a particular emphasis on the application of classical methods to combinatorial ML, ADMET, and ligand chemistry [[→101](#)].

The development of innovative and effective remedies for a variety of diseases is contingent upon the collaboration between pharmaceutical scientists and AI researchers. They can leverage their expertise and experience to create algorithms and models that are rooted in machine learning. This will allow them to predict the efficacy of potential drug candidates and accelerate the drug discovery process. Collaboration can improve the precision and efficacy of clinical trials by utilizing AI algorithms to analyze the data acquired during these trials to identify trends and potential side effects of the medications being tested [[→102](#)]. It can expedite the entire drug development process and assist pharmaceutical companies in making well-informed judgments regarding which potential drugs to pursue. Furthermore, the affordability and accessibility of healthcare can be improved through collaboration between pharmaceutical scientists and AI researchers. The efficacy of potential drug candidates for specific patient populations can be predicted by analyzing data from large populations and identifying patterns and trends. Artificial intelligence (AI) algorithms can be employed to achieve this. It enables the personalization of remedies to accommodate the unique needs of each patient [[→103](#)]. The pharmaceutical industry's advancement may be expedited by the

implementation of AI algorithms and big data, according to researchers. One of the most promising applications of AI has been designated by researchers as the pharmaceutical industry's pursuit of medications to treat uncommon ailments. Pharmaceutical companies are less inclined to allocate resources toward drug development when there is a lack of market demand [[→104](#), [→105](#)]. In contrast, AI can determine which patients would derive the most benefit from clinical trials. The risk of failure at the most expensive stage of the pharmaceutical industry, the clinical trials phase, can be significantly reduced by developing a profile of ideal candidates. Another area of intense research concentration is the role that AI algorithms will play in the displacement of humans [[→106](#), [→107](#)].

3.6 Conclusion

The pharmaceutical industry needs to work with startups, innovators, and businesses from different fields to collaborate and accelerate innovations. R&D is funded through private-public partnerships. Collaborations can save expenses, postpone development, and enhance clinical results. In the pharmaceutical sector, partnerships aid in the creation of new medications. Businesses, academic institutions, governments, and nonprofit organizations work together to share resources and intellectual property to develop and commercialize goods. Academic medical centers make partnerships with pharmaceutical companies' investments. Both parties should exchange information in these partnerships. These collaborations

provide value beyond financial gain by addressing significant scientific issues and creating new treatment targets. Working together with academic institutions is crucial to creating innovative medications when new therapeutic mechanisms, enabling technologies, and translational models are identified. Time horizons, confidentiality, exclusivity, and cultural differences between industry and university research are some of the challenges that face pharmaceutical collaboration. The lack of transparency and accountability in public-private partnerships can also impede global healthcare equity. Collaboration may be hampered by the disparities in organizational structures and heterogeneities between universities and businesses. Moreover, pharma AI, with assistance from organizations from the public, business, and civil society, works together to enhance health and tackle challenging health problems. Partnerships that have a shared objective, complementary specialties and equitable contributions are advantageous to public agencies. To create novel medications and biomarkers, pharmaceutical companies have teamed up with AI startups.

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Chapter 4 Education and training in smart pharma

Abstract

A “connected health” movement is being sparked by the digitalization of healthcare, which also lowers per capita costs, enhances care quality, and improves patient outcomes. The pharmaceutical industry is expanding at a rapid rate due to digitalization, which uses information shared via social media, mobile phones, and other technologies to comprehend consumer demand and optimize supply chain efficiency. Machine learning algorithms can retrieve drug-drug interactions and predict their effects. Big data analysis enables digital health care by eliminating the need for in-person visits and guaranteeing that clinicians have immediate access to all medical records. The physical networking of software, embedded systems, and electronic sensors that permit data sharing across a network from any location is known as the Internet of Things (IoT). The pharmaceutical industry is using blockchain technology to expedite regulatory approval, drug development, and discovery. The Internet of Things is also being used to monitor health problems, predict events about patients, alter data, reduce expenses, and save lives. The chapter offers state-of-the-art training approaches for the smart pharmaceutical industry.

Keywords: pharma, virtual reality, gamification, academic, collaboration, Internet of Things,

4.1 Introduction

The pharmaceutical sector is among the global economic domains exhibiting the swiftest growth rate. The efficiency of the supply chain is enhanced by digitalization, which involves the exchange of information through mobile devices, system integration, and numerous other channels to comprehend and anticipate consumer demand. In the aftermath of COVID-19, businesses in all sectors are more dependent on digital technology than ever to improve performance by increasing manufacturing productivity, enhancing competitiveness, improving long-term financial viability, and using more accurate forecasting and planning [[→1](#)]. The digital revolution in healthcare is fostering a “connected health” movement that has the potential to benefit all parties involved by enhancing the treatment experience, reducing per capita costs, and improving patient outcomes [[→2](#), [→3](#)]. For instance, the implementation of digitalization to create pharmaceuticals that are resistant to counterfeiting and have serial numbers that may be traced throughout the supply chain will guarantee quality and ensure adherence to forthcoming serialization regulations [[→4](#)]. Pharmaceutical companies may employ digitalization as an additional tool to meet the anticipated global market demand. They could utilize digitalization to more effectively communicate with distributors and suppliers, comply with regulations, and identify cost-saving production efficiencies through cloud-based information exchanges [[→5](#), [→6](#)].

Moreover, real-time manufacturing is made possible by the use of automation, advanced sensors, social media, and medical applications to forecast demand globally and track medication compliance [[→7](#), [→8](#)].

AI is capable of recognizing critical relationships in unprocessed documents that can be utilized for the diagnosis, mitigation, and treatment of disease.

Almost any discipline of medicine can benefit from the application of several of the more recent techniques employed in the emerging field of computational thinking. To resolve intricate clinical issues, it is necessary to confront the challenge of collecting, assessing, and implementing a plethora of knowledge [[→9](#)]. The development of numerous drug molecules from a chemical space is a difficult task due to the absence of suitable technologies. However, the drug development process can be enhanced by integrating artificial intelligence (AI) [[→10](#)]. The healthcare sector has experienced a substantial increase in the utilization of robotic technologies as a result of AI. Healthcare is undergoing a revolution due to robotics and AI, which are improving patient care, evaluation, and treatment. AI systems can examine sizable datasets, including metabolomic, proteomic, and genomic data, to pinpoint possible therapeutic targets and clarify the workings of diseases [[→11](#), [→12](#)]. Scientists can learn more about the molecular causes of different ailments and find possible targets for developing drugs through the process. Furthermore, promising drug candidates can be found in large compound libraries using AI-driven virtual screening techniques, which are more effective than conventional high-throughput screening techniques. AI algorithms can forecast the

pharmacokinetic, toxicity, and biological activity of possible drug candidates [[→13](#), [→14](#)]. As a result, less money and time are required for experimental validation. By employing deep learning (DL) techniques to forecast the properties of potential compounds, AI can facilitate the development of novel drug molecules from the ground up. Machine learning (ML) algorithms, known as generative adversarial networks (GANs), are capable of generating novel molecular structures that exhibit the desired biological function and drug-like properties [[→15](#), [→16](#)].

The effects of drug-drug interactions can be forecasted and mined using AI. To simplify the extraction of drug-drug interactions, researchers integrated kernel-based and feature-based methods with ML methodologies [[→17](#)]. Certain researchers implemented an automated drug-drug interaction detection method to enhance drug safety surveillance in a hospital setting [[→18](#)]. AI can be employed to predict the effects of drug-drug interactions. To extract drug-drug interactions efficiently, researchers integrated kernel-based and feature-based techniques with ML approaches [[→18](#)]. ML can considerably enhance pharmacovigilance by enabling its models to predict a significant number of potential drug-drug interactions by learning from a limited amount of drug-drug interaction combinations [[→19](#)]. Additionally, ML can be employed to identify populations that are susceptible to adverse drug reactions (ADRs) and to direct personalized care. Moreover, ML methodologies can be utilized to pinpoint specific patients, particularly those vulnerable to fluoropyrimidine toxicity. Researchers employed ML methods to analyze toxicity patterns, enabling them to predict the

prevalence of individuals with DPD-related toxicity. The model may yet prove advantageous in the future, notwithstanding the risk of considerable overfitting [[→20](#)].

4.2 Key technologies in smart pharma

Smart pharma involves a diverse array of stakeholders, such as hospitals, physicians, patients, and research facilities. It is a complex, organic entity that encompasses a variety of functions, including health decision-making, hospital administration, medical research, diagnosis and treatment, and illness prevention and monitoring. Information technologies, including blockchain, IoT, ML, DL, and big data, are the foundations of smart pharma, as shown in [→Fig. 4.1](#).

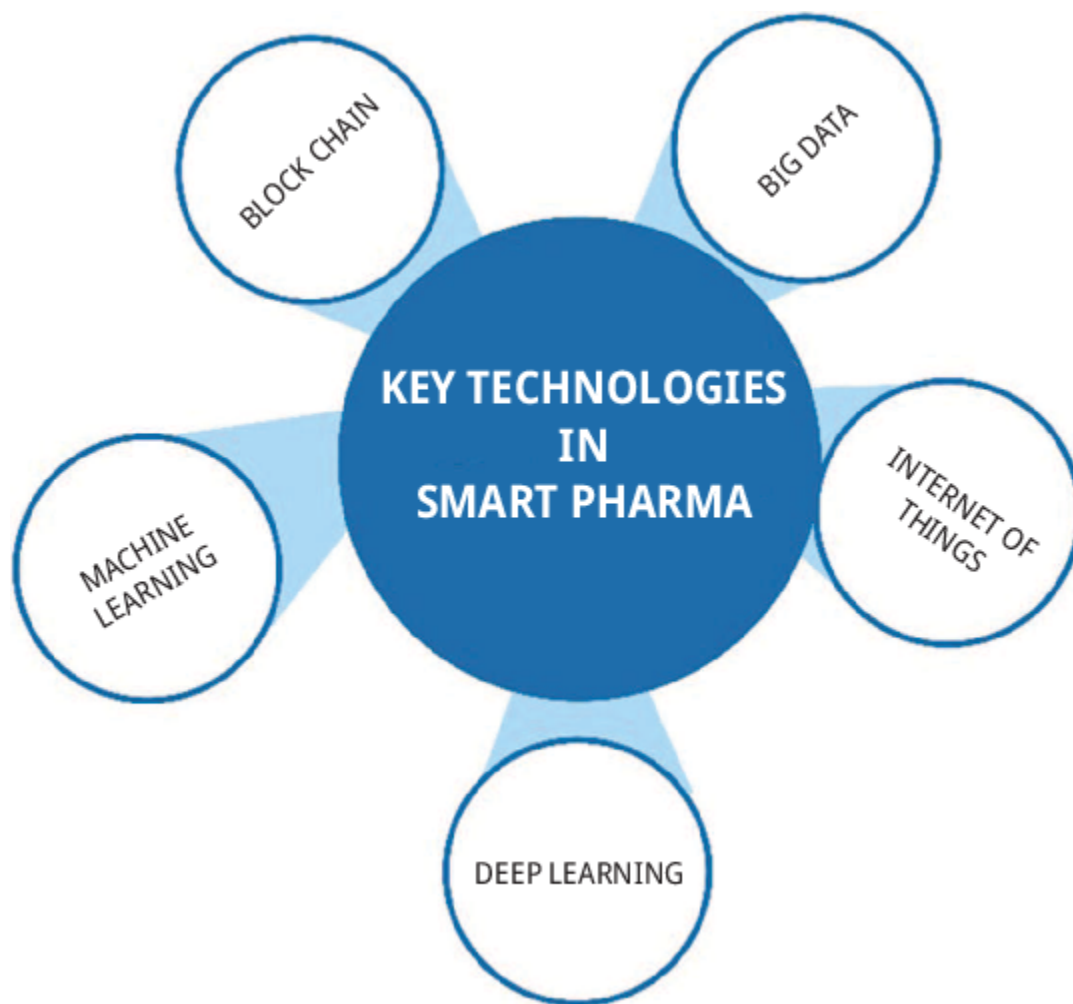


Fig. 4.1: Graphical representation of various key technologies involved in smart pharma.

4.2.1 Big data in smart pharma

Digitalization of healthcare data has aided in the expansion and advancement of big data. Due to the data's explosive growth in the last several years, a new field known as "big data" was established. Health care is becoming more mobile and intelligent thanks to big data analysis and other smart healthcare solutions that also reduce costs and increase efficiency. Chemists can use statistical analysis to predict pharmacologic

outcomes and recognize the patients who might benefit most from pharmaceutical treatments, which can aid them in better understanding the risks and side effects of pharmaceuticals [[→21](#), [→22](#), [→23](#)]. Big data has emerged as a popular research topic in recent years due to its possible uses in health and medicine. It has demonstrated promise in the treatment of conditions like Alzheimer's disease by utilizing big data-driven systems for clinical decision-making. Electronic medical record optimization has also been achieved through the application of big data techniques. Toxicology, pharmaceuticals, and pharmacology have produced data regarding chemical properties, pharmacokinetics, clinical behaviors, and drug toxicity. High-throughput screening (HTS) and next-generation sequencing (NGS) have led to a significant increase in the amount of data [[→24](#), [→25](#)].

By reducing the need for in-person visits, enhancing patient outcomes, and guaranteeing that clinicians have immediate access to all patient medical records, big data analysis will enable digital health care. Through digital health and aggregated real-time data, big data analysis, which offers more chances for proactive intervention and a more comprehensive picture of patients' circumstances, will become a crucial tool for healthcare providers [[→26](#), [→27](#)]. The present epoch is defined by the incessant proliferation of data streams, attributable to the extensive use of electronic medical records, the vast quantities of data relayed by various sensors, and the information patients disseminate on social media platforms. The healthcare sector consistently produces significant volumes of data, including genetic information, medical records, imaging data, and health-related behaviors.

Effective data utilization by healthcare organizations can aid in tracking illnesses, medical decision-making, and public health management [[→ 28](#)]. To solve the problem, electronic health records, or EHRs, are digital collections of electronic patient health data. The use of EHRs is growing in developing nations. These records can be shared between various healthcare settings via a robust network connection. Since digital health records can be shared, revised, and altered across various healthcare systems, there is less chance of missing any important paperwork, which contributes to the reduction of duplicate data. EHR initiatives provide instant benefits to hospital management, physicians, and patients [[→ 29](#)].

4.2.2 Blockchain in smart pharma

Pharmaceutical development and research is a multi-year, complex process that includes drug discovery, drug development, and regulatory approval. Blockchain is a decentralized computer network that provides its members with access to a secure ledger of transactions and operates without a central server. By timestamping each transaction made within the network, it eliminates the need for an outside organization. Every network node maintains its local ledger and maintains a local copy of the record. Modest changes made to the localized ledger are reflected across the network. Beyond just cryptocurrencies, blockchain technology enables the creation and monitoring of any sort of digital asset. Within the pharmaceutical supply chain, a product created by a manufacturer and registered on the network can be transferred to other participants like cryptocurrency, as it is

considered a digital asset. Once again, blockchain ranks first for privacy protection. Each member of a blockchain network is given a key pair that serves as their identity [[→ 30](#)]. The blockchain approach maintains patient privacy and data security while offering a special benefit for patient recruitment. Patient recruitment times are shortened by using blockchain distributed ledgers because anonymously registered patients can be linked to various clinical trial sites. As a result, the researcher can select the necessary subjects from a group of patients by the study plan. Proposals have also been made for a blockchain structure that involves multiple contracts for clinical trials to manage the studies and involve patients [[→ 31](#)]. Also, the clinical trials are tracked using the blockchain technique, which encourages others to participate in clinical research while also instilling confidence in the outcomes. Because it can result in fewer clinical trials where positive results are covered selectively, transparency is an essential part of drug discovery research. Consequently, applying blockchain technology to clinical trials encourages improved relationships between sponsors and different clinical research sites in addition to enhancing data integrity and transparency [[→ 32](#)].

4.2.3 ML in smart pharma

The initial phase of conventional drug development is the identification of a target protein that is related to a disease. The process is resource-intensive as well as time-consuming. The procedure entails the examination of extensive chemical libraries in search of compounds that interact with the

target. As a result, the drug-like properties are improved through iterative optimization procedures [[→33](#)]. The strategy is typically distinguished by a protracted implementation process and a high rate of attrition. Nonetheless, the paradigm is seeing a transition due to the application of ML techniques. The forecasting of biological activity, dissolution, stability, toxicity, absorption, distribution, metabolism, and elimination (ADMET) characteristics of prospective drug candidates exemplifies an effective application of ML. Using the analysis of extensive datasets that include physicochemical properties, activity profiles, and molecular structures, ML algorithms can detect connections and patterns that humans may have overlooked. Because of this, it is no longer necessary to conduct costly and time-consuming experimental studies to ascertain how a substance will operate in biological structures [[→34](#)]. Compound screening and selection are acknowledged applications of ML in drug discovery. Novel molecules can be anticipated to demonstrate activity against a given target by ML models trained on existing active and inactive compounds [[→35](#)].

4.2.4 Deep learning in smart pharma

Across a variety of domains, such as computer gaming, speech recognition, natural language processing, computer vision, and autonomous vehicles, DL has achieved remarkable success. It has reaped the rewards of the endlessly expanding availability of computational capacity and the expansion of the volume of data. DL is distinguished from the majority of other ML methods by the adaptability of its neural network architecture.

The designs consist of fully connected feed-forward networks, recurrent neural networks (RNNs), and convolutional neural networks (CNNs). The natural application of multilayered feed-forward neural networks for biological function predictions has resulted from the continued application of single-layer NNs in QSAR modeling, as data size as well as processing power have increased [[→36](#)]. It was not anticipated a few years ago that RNNs would be used in de novo design, which has been a somewhat unexpected development. CNNs have demonstrated amazing success in computer vision since high-throughput imaging technology has become widely available, and they are currently a popular choice for processing biological images. In recent years, DL applications have significantly accelerated pharmaceutical research and drug discovery. Their utility has exceeded expectations and demonstrated promise in addressing various drug discovery problems. Their numerous applications in scientific and technological fields, particularly in biomedicine and bioinformatics, are apparent. The automated synthesis of novel chemical compounds with defined characteristics, the prediction of molecular structure and function, and the sophisticated image analysis skills of DL render it an attractive contender for drug discovery. The application of DL can benefit every stage of the drug discovery process, including target identification, prognostic biomarker discovery, drug development, design, and prediction of synergistic antagonism [[→37](#)].

4.2.5 Internet of things in smart pharma

The pharmaceutical industry is one where Internet of Things (IoT) is important. Scholars worldwide are endeavoring to implement this fascinating technology within the pharmaceutical sector. Because of their mutual internet connectivity, most researchers predict how much work they are going to do in the field of the next-generation medicinal product revolution. The network uses dynamic data to connect to things, including appliances, cars, trucks, and industrial equipment, as well as other Internet-connected utilities [[→ 38](#)]. The IoT is the term used to describe the physical connection of software, embedded devices, and electronic sensors, which enables the sharing of data across a network from any location. IoT's primary contributions to the pharmaceutical industry are its ability to track health issues, forecast patient-related future events through the use of evolutionary technologies, transform data, save costs, and save lives [[→ 39](#)].

Radio Frequency Identification (RFID) technology has shown great promise for the pharmaceutical industry's manufacturing and supply chain processes. But unlike bar codes, Internet of Things (IoT) devices are never scanned because they communicate real-time data to other devices through the Internet. IoT is essential to the management of all services because it provides medical services in the field of pharmaceuticals [[→ 40](#)]. In addition, IoT could be helpful in monitoring and streamlining various departments for real-time control and monitoring to improve production efficiency during the pharmaceutical product's manufacturing. IoT can be used in pharmaceutical supply chain management and warehousing to

monitor the drug product's storage conditions in real-time and to improve visibility, which will increase operational effectiveness [[→41](#)].

4.3 Innovative training methods in smart pharma

The Indian pharmaceutical industry is undergoing significant transformation, and in the intricate and cutthroat business climate, an organization's ability to succeed is determined by a wide range of factors. Any company's capacity to foster a culture of continuous learning is one of the critical elements that can contribute to its success. To obtain a general competitive advantage, the learning must concentrate on all employee ranks. Pharma L&D has additionally demonstrated its capacity for creative problem-solving to guarantee that training is not only an activity but rather fits in with business and functions as a true business partner. Gamification and virtual training are two cutting-edge methods in smart pharma [[→42](#)]. Virtual reality (VR) has been shown in most research conducted over the last ten years to be safe, effective, and to increase user satisfaction to a high degree. With the rising affordability, adaptability, and portability of VR technology, it can now be used for medical treatment in inpatient as well as outpatient settings [[→43](#)]. Studies indicate that virtual reality is an efficacious approach for alleviating pain related to numerous medical and psychological problems [[→44](#), [→45](#)]. It has been identified as a potential treatment for a diverse array of conditions, such as migraines, pruritus,

diabetic neuropathy, cancer, and injuries that result in acute or chronic pain [[→46](#), [→47](#)]. VR has also demonstrated promise in the treatment of pain following medical or surgical procedures, such as dental work, IV-line implantation, and dressing changes for wounds. VR helps enhance the effectiveness of analgesics and supports conventional pharmacotherapies [[→48](#), [→49](#)]. Five of the seven studies that were reviewed showed that VR is a useful nonpharmacologic supplement to conventional opioid analgesic therapies [[→50](#), [→51](#), [→52](#), [→53](#)]. In those studies, both adult and pediatric patients undergoing uncomfortable medical procedures reported less pain and anxiety when using VR [[→54](#), [→55](#), [→56](#)]. Analgesia given either with or without a virtual reality game was the subject of a different randomized trial. The trial showed that the combination of virtual reality and analgesics proved significantly more efficient than analgesics alone in reducing pain in children receiving burn treatment [[→57](#)]. Additionally, research has been conducted to ascertain whether the advantages of virtual reality persevere beyond the initial therapy regimen [[→58](#)].

Virtual reality offers numerous benefits compared to alternative computer-based methodologies utilized in drug design and discovery, including enhanced three-dimensional visualization of compounds and targets. The FDA's approval of medications has not matched the rising costs of drug development in recent decades, necessitating a swift and effective approach to drug development. Moreover, numerous unknown pharmacological targets exist, fully immersed in virtual reality using molecular modeling to identify and create innovative drug candidates with significant potential.

Visualization techniques are essential for drug design and discovery when examining the interactions between ligands – potential pharmaceuticals – and macromolecules. Precise visualization of ligand binding to extensive complex structures, such as proteins, is essential. Virtual reality (VR) enables scientists to concentrate on ligand-binding sites and other pertinent regions while monitoring the dynamics of protein structural configurations [[→ 59](#), [→ 60](#)].

Gamification involves incorporating game elements or features into a product to enhance user engagement. While it may seem ridiculous to learn topics like physiology or pharmacokinetics by engaging in video games, it has been shown to enhance the confidence and skill sets of healthcare professionals. Conventional lectures concentrate on teaching a specific subject to learners, enabling them to apply the knowledge in a clinical context. There is much more to applying the knowledge that is taught in school than just memorizing it. It requires the application of critical thinking and decision-making abilities, as well as the cultivation of effective communication skills with patients and other healthcare providers. By utilizing class time to practice integrating their knowledge, the concept of gamification offers an additional opportunity to exercise these skills [[→ 61](#)]. Educational materials incorporate game design components, including game mechanics, as part of the methodology. Gamification is dedicated to providing consumers with motivational affordances, or “games,” to enhance their motivation. Research has revealed that the activities associated with the implementation of gamification exhibit

variations [[→ 62](#), [→ 63](#), [→ 64](#)]. With the following mode of delivery, learners can be targeted with motivational tools like progress trackers, leaderboards, points, and feedback [[→ 63](#)]. Studies reveal that depending on the individual user, gamification's effect on intrinsic motivation varies considerably. Gamification has the potential to both "facilitate and undermine intrinsic drive" by appealing to people's feelings of autonomy and competence [[→ 64](#)]. Furthermore, the literature currently in publication is ambiguous regarding the duration over which gamification affects intrinsic motivation. The challenge of comparing the efficacy of the strategy is made challenging by the imprecise description of gamification and the widespread use of various variations in the approach. Numerous studies back up the need to conduct additional comprehensive and repeatable research on the effects of gamification. One benefit of gamification that has been repeatedly demonstrated to be true in literature reports is its positive effect on user satisfaction [[→ 65](#), [→ 66](#), [→ 67](#)]. Commercial escape rooms cater to a wide range of people. Enigmas must be modified for educational escape rooms since they are made for an extremely limited target audience, which is typically students studying a particular subject [[→ 68](#)]. Therefore, a thorough understanding of the target students' abilities and attitudes is crucial for the Room of Escape to be successful [[→ 69](#)]. These factors suggest that the installation of an escape room in a school environment should happen after a significant portion of the curriculum has been taught to the students, allowing them to actively participate in solving the challenges [[→ 70](#)].

4.4 Industry-academic collaboration

Academic medical centers are spending money to ensure that their partnerships with pharmaceutical companies are successful because there is renewed interest in these kinds of partnerships. Information should be shared in both directions during these collaborations. Academic colleagues in the field of pharmacology can impart invaluable knowledge to one another. The majority of academic institutions stand to gain from exchanging experiences in the areas of technology, project management, knowledge management, and operational excellence. Pharma must find a way to grant academics access to its vast chemical libraries, private data sets, state-of-the-art instruments, and technological platforms. Strategic, cooperative innovation is the only way to keep up with the rate of biomedical discovery and to fully realize the potential of new targets or disease biomarkers. Pharma's current business tactics rely on collaborations, acquisitions, and integrations to find clinical candidates. Effective collaborations with academic institutions to tackle noteworthy scientific inquiries and integrate novel targets into treatment approaches are imperative to feed the forthcoming pharmaceutical pipeline with inventive pharmaceuticals [[→71](#), [→72](#)].

There is a constant search for new translational models, facilitating technologies, and therapeutic mechanisms. Pharmaceutical research used to be frequently divided into different departments within organizations, but this is no longer the case. Similarly, collaboration with the private sector

has become more valuable, and academic drug discovery has undergone significant change in recent years. It is worth more than cash. To increase the practical significance of these publicly funded discoveries, these multidisciplinary collaborations bring together the most creative concepts and methods for identifying new therapeutic targets, addressing gaps in translational chemical biology, and developing new chemistry and technologies [[→73](#), [→74](#)]. Training courses can be added to the curriculum as a way to help students develop their skills. Students can be required to attend courses in their areas of interest throughout their time as graduates. A foundational understanding of work skills will enable freshmen to achieve their objectives. Such a training program can be carried out as a joint venture between industry and institutions and should place greater emphasis on industrially relevant areas. As part of academic collaborations with industry, industries should step forward and align the educational content with their requirements for the development of training modules. Industry-academia interaction ought to be promoted and taken into account as a component of education. Numerous IT companies have begun collaborating with universities and engineering colleges [[→75](#)].

University-industry collaborative research and development (UIC R&D) is a term that encompasses pharmaceutical innovation processes that are more transparent. In the process, scientists and industry-based organizations collaborate to identify technological breakthroughs and knowledge, and subsequently develop new drugs based on these discoveries [[→76](#), [→77](#), [→78](#)]. Molecular, biological, and chemical pharmacology knowledge is

advanced through formal or informal collaborations between universities and industry groups. The goal of these collaborations is to translate the knowledge into new drug creation and commercialization initiatives that are beneficial to society and the economy. It is known as pharmaceutical UIC R&D [[→ 79](#), [→ 80](#)]. Using diverse UIC research and development strategies, such as establishing technology transfer offices (TTOs) and initiating academic start-up company incubation programs, universities endeavor to enhance and fortify the ties between academia and industry to advance an efficient and productive innovation ecosystem for novel drug discovery [[→ 81](#)]. However, as academic and industrial organizations have different goals, guiding principles, and schedules, there are still very few drug development processes that are fully and flawlessly driven by university-industry collaboration. These elements play a part in conflict resolution and negotiation issues. There is little, if any, comprehensive research on the composition and coordination of pharmaceutical R&D partnerships between academic institutions and private businesses. Due to the phenomenon, a large number of scholarly contributions have focused on the outcomes of pharmaceutical R&D in one or a small number of university-industry collaborations. Gaining insight into the causes and factors influencing university-industry R&D in pharmaceuticals can help increase productivity and effectiveness [[→ 82](#)]. The potential for productive specialization in a variety of institutional contexts may be presented by collaborating with others. Rather than incorporating industry practices into academia, industry collaboration could improve the academic framework by

decreasing patenting rates and increasing publication rates. Collaboration between academia and industry in research areas with both scientific and commercial potential may result in a more productive division of tasks and responsibilities, enabling each party to concentrate on the aspects of their institutional environments that are most rewarding: for academia, it would be the development of deeper scientific discoveries, while for business, it would be the commercialization of these discoveries. Previous research has suggested that academic institutions have a comparative advantage in the investigation of more fundamental, basic concerns, while companies have a relative advantage in development and commercialization. Depending on their interests, scientists tend to specialize in either industry or academia. Scientists who are more inclined to enter academia are those who have a stronger “taste for science” or who prefer non-financial rewards; on the other hand, scientists who are more inclined toward applied research and financial incentives tend to enter the industry. Furthermore, the intellectual and material exchange that occurs within academic institutions is especially conducive to curiosity-driven research, while the resources and experience of businesses facilitate effective commercialization [[→83](#), [→84](#)].

4.5 Challenges faced by smart pharma

[→ Figure 4.2](#) illustrates the hurdles and downsides that must be considered, notwithstanding the potential benefits of AI in drug discovery. The acquisition of pertinent data is a major challenge. Extensive data

aggregations are generally necessary for the training of AI methodologies. The reliability and precision of the results can sometimes be compromised by the limited number, inconsistent quality, or poor quality of the available data [[→ 85](#), [→ 86](#), [→ 87](#)]. It offers an important quantity of data, which is used by models that are trained gradually. The company suffered greatly from the data acquisition model via different databases because it was more expensive. Furthermore, a substantial amount of trustworthy data is required to guarantee the pharmaceutical analysis. Another obstacle keeping the pharmaceutical and biological sectors from fully adopting AI is the shortage of skilled personnel to oversee the AI-based platforms [[→ 88](#)]. The lack of funding for small organizations and the use of AI and ML by large corporations, such as pharmaceutical companies, to replace their human workforce, lead to job losses. In ML processes, decisions are generated by the input data; however, the processes from input to output are not transparent. The input data is transformed through several intricate layers in CNNs, leading to unpredictable behavior in the end [[→ 89](#)]. Furthermore, the absence of a rationale for predictions that are specific to black-box ML models has an impact on ML predictions as a result of human error in data interpretation. But logical models and decision trees, like relational ML, don't show the fundamental problem. The dependence on model performance is impacted by human users' incapacity to understand anticipated outcomes. There are still concerns regarding prediction confidence despite the ML models' encouraging initial results. When an algorithm attempts to search domains outside of its applicability domain, it

is more likely to result in a pathological failure in the hypothesis produced by ML techniques [[→90](#)].

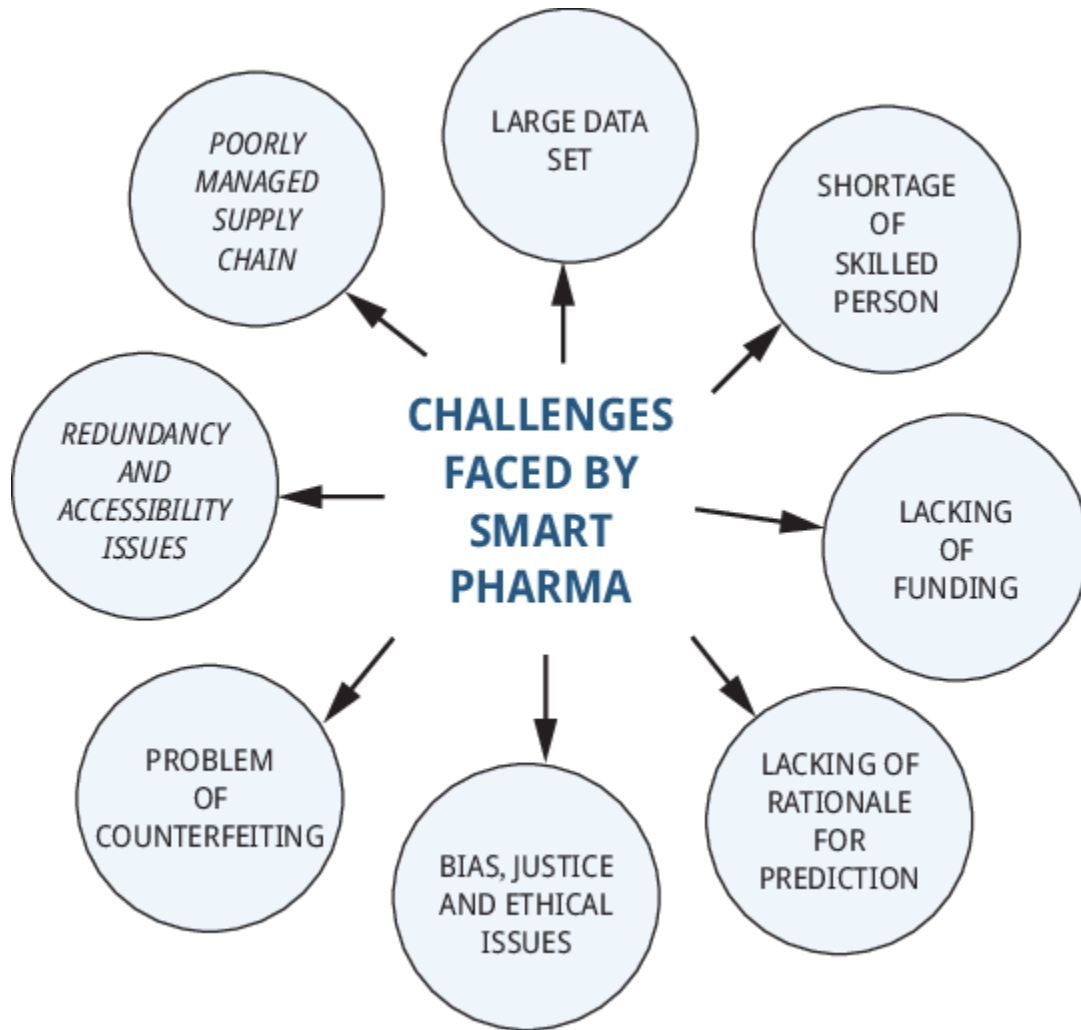


Fig. 4.2: Various challenges faced by smart pharma.

The reliability and accuracy of the results can sometimes be compromised by uneven quality, insufficient quantity, or substandard quality of the available data. Another ethical problem is the possibility that AI-based solutions may provoke inquiries over prejudice and equity. An ML system may produce inaccurate or unfair forecasts if the training set contains biased

or unrepresentative data. It is imperative to consider the ethical and equitable implementation of AI in the development of novel pharmaceutical products. The knowledge and experience that human researchers have acquired, as well as traditional experimental procedures, cannot be supplanted by current AI-based methods. As AI is limited to making predictions based on available data, human researchers are subsequently required to validate and interpret the results [[→ 91](#), [→ 92](#), [→ 93](#), [→ 94](#), [→ 95](#), [→ 96](#), [→ 97](#)]. Pharmaceutical companies have access to millions of compounds in their databases for drug development, which could prove too numerous for conventional machine-learning tools to manage. A computational model can employ the quantitative structure-activity relationship (QSAR) to rapidly forecast several chemicals or essential physicochemical characteristics, such as log P or log D. Nonetheless, these models exhibit little proficiency in forecasting intricate biological characteristics, like the effectiveness and side effects of certain drugs. The lack of experimental validations, the data derived from experimental errors in training sets, and the restricted size of the training sets hinder the advancement of QSAR-based models [[→ 98](#), [→ 99](#)].

The problem of counterfeiting is one of the main application areas for blockchain technology, which has been most recently adopted by the pharmaceutical industry. Information security and privacy are two benefits of blockchain technology, which is largely decentralized. More processing power is needed. Roughly half of the power is dedicated to thwarting malicious user attacks; without efficient tracking and tracing, users risk

losing goods and becoming victims of robbery [[→ 100](#)]. Blockchain technology can enable safe, international banking transactions, but interoperability issues present a significant obstacle. Blockchain operations can be standardized through the use of distributed ledger technology (DLT). Additional research is necessary to determine the most effective methods for implementing DLT applications, despite the International Telecommunication Union (ITU)'s efforts to identify and standardize DLT. The deployment of smart IoT applications, cloud services, and periphery computing has significantly improved the availability of data [[→ 101](#)]. The deployment of blockchain faces numerous challenges as a result of the massive growth, including issues with data management, privacy, redundancy, and accessibility. Furthermore, it has attempted to use blockchain technology to expedite medical transactions, protect patients, and enable tracking and monitoring due to misbehavior and poorly managed supply chains [[→ 102](#), [→ 103](#)]. It's also critical to guarantee interoperability amongst various healthcare systems and equipment. Strong interoperability solutions are required for effective communication and data exchange among the different hospitals and systems. The blockchain network must be capable of accommodating increased transaction rates without experiencing performance degradation as the volume of healthcare data increases. Careful design and enforcement procedures are required for blockchain deployments to abide by strict data protection laws like GDPR and HIPAA [[→ 104](#), [→ 105](#), [→ 106](#), [→ 107](#), [→ 108](#)].

The researchers claim that it is challenging to integrate blockchain technology with the present healthcare system. Since many hospitals continue to operate with legacy systems, careful planning is necessary to ensure a seamless transition and backward compatibility. In addition, it is critical to address blockchain's energy consumption problems, particularly those related to proof-of-work consensus. Healthcare organizations should consider blockchain solutions that are environmentally friendly and energy-efficient [[→ 109](#), [→ 110](#), [→ 111](#), [→ 112](#)]. Managing storage capacity presents another difficulty on the front. Because blockchain was developed to store and analyze transactional data, which is restricted in scope, it does not necessitate a significant amount of storage. Ethical concerns also pose a challenge, as AI-based solutions may raise concerns regarding impartiality and bias [[→ 113](#)]. For example, the forecasts generated by ML system may be erroneous or inequitable if the training data is biased or unrepresentative [[→ 114](#)]. Considering the ethical and fair implementation of AI in the creation of new medicinal molecules is crucial [[→ 115](#)]. Various strategies and techniques can be utilized to overcome the challenges that AI faces in the domain of chemical medicine. One approach is to employ data augmentation, which involves generating synthetic data to enhance current datasets. The accuracy and consistency of outcomes can be enhanced by increasing the quantity and diversity of data utilized to train ML algorithms [[→ 116](#)]. An alternative method that seeks to deliver succinct and intelligible elucidations for the predictions produced by ML algorithms is the application of explainable AI (XAI) approaches [[→ 117](#)]. It has the

potential to reduce concerns regarding the impartiality and bias of AI-based techniques [[→ 113](#)] and to provide a more thorough understanding of the fundamental assumptions and mechanisms that underlie the predictions.

DL approaches are not interpretable, in addition to the constraints on data. When it comes to DL methods, they are viewed as a mystery in contrast to traditional ML methods, which frequently undergo a thorough validation analysis based on mathematical reasoning. Even though DL outperforms ML on the majority of tasks, researchers frequently are unable to comprehend why ML produces such excellent results. If a DL model produces a novel outcome that deviates from prior research, its interpretability issues render the result unsatisfactory. Unlike other fields, drug discovery is rich in knowledge about the regulatory systems of biological pathways, the metabolic processes of molecules, and the mechanisms of action of compounds. A comprehensive understanding of pertinent biological processes is crucial to guarantee the safety and efficacy of medications. These processes include the physical features of medications, the proteins they bind to in the body, the binding mechanisms, the biological reactions they elicit, and their metabolic pathways. DL is restricted to the acceptance of input and the generation of predicted output; it is incapable of providing a comprehensive explanation of the process by which the output is generated [[→ 118](#)].

4.6 Conclusion

The pharmaceutical industry is experiencing rapid growth due to digitalization, which uses information shared via system integration, mobile phones, and other technologies to understand consumer demand and improve supply chain efficiency. The digital revolution in healthcare is bringing about a “connected health” movement, improving patient outcomes, reducing per capita costs, and enhancing the care experience. Robotic technologies improve patient care, diagnosis, and treatment, while AI can find significant relationships in unprocessed documents and aid in drug development. Drug-drug interactions can be retrieved, and their effects can be predicted by machine learning algorithms. By eliminating the need for in-person visits and guaranteeing that clinicians have immediate access to all medical records, big data analysis is enabling digital health care. The pharmaceutical industry is using blockchain technology to expedite regulatory approval, drug development, and discovery. The IoT is also being used to track health issues, forecast patient-related events, transform data, save costs, and save lives. The Indian pharmaceutical industry is undergoing significant transformation, and fostering a culture of continuous learning is crucial for success. Two cutting-edge methods in smart pharma include gamification and virtual training. Educational escape rooms in higher education should focus on student learning and industry-academic collaboration to advance knowledge related to molecular, biological, and chemical pharmacology.

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Chapter 5 Drug manufacturing and quality control with artificial intelligence

Abstract

Artificial intelligence (AI) uses advanced instruments and networks to mimic human intelligence. Artificial intelligence is increasingly used in the pharmaceutical industry to automate and manage large data sets. Quality by design (QbD) and AI can improve product quality, reduce waste, and speed up production in the pharmaceutical industry. The chapter provides information on drug manufacturing and quality control using AI in the pharma industry. The medical research sector must integrate technologies like blockchain and AI into its core competencies to thrive. The objective of the European Union's AI-based manufacturing compliance regulations is examined, aiming to align technology with the EU's core values, rights, and principles. The body of research on the Internet of Things privacy and security measures was reviewed, along with the myths surrounding widely used technologies that make extreme security promises. To raise the bar for pharmaceutical handling of data, sharing, processing, assessment, and classification based on outcomes, the chapter discusses cutting-edge blockchain and AI methods.

Keywords: drug, production, quality, management, security, regulatory compliance, quality-by-design,

5.1 Introduction

The application of artificial intelligence (AI) is growing across many domains in society, chief among them being the pharmaceutical industry. As processes become more digital, it's harder to acquire, evaluate, as well as apply knowledge to complicated clinical problems [[→1](#)]. It encourages AI use because it can automatically handle large amounts of data. AI simulates human intelligence using advanced tools and networks [[→2](#), [→3](#)].

However, it doesn't threaten to completely replace people in the physical world. AI makes use of hardware and software to perform independent analysis, learning, and decision-making to accomplish predefined objectives [[→4](#)]. After training, the ability to learn over time is frequently lost when an AI system is used to control procedures like clinical trials or manufacturing. Since the recent adoption of quality by design (QbD) techniques, the pharmaceutical industry has grown, and Industry 4.0 initiatives indicate rapid development [[→5](#)]. Successful treatment requires choosing the right target for drug molecule synthesis. Numerous proteins, a few of which are overexpressed, are involved in the disease's progression. Therefore, predicting the desired protein's molecular structure is crucial to designing a drug molecule specifically targeting that disease. AI can facilitate drug discovery through structural analysis by modeling the three-dimensional configuration of proteins that corresponds to the chemical milieu of the target protein site. It aids in forecasting a compound's target impact and ensuring safety before synthesis or manufacture [[→6](#)].

AI has the potential to supplant trial-and-error methodologies. QSPR enables many computational techniques to address formulation design challenges, including stability, dissolution, porosity, and more [[→7](#), [→8](#)]. A feedback mechanism monitors the entire process and makes occasional adjustments to power decision-support tools. Based on the medication's physicochemical properties, they choose the excipient type, amount, and type using rule-based systems [[→9](#)]. AI offers a plethora of options for streamlining development and production manufacturing processes. AI can result in improvements in quality. Additionally, by using AI in several ways, including accelerating processes, it can reduce material waste and increase production efficiency [[→10](#)]. To ensure quality and produce the pharmaceutical composition within the intended timeframe, a well-planned, scientific approach utilizing AI is required for quality assurance and control. Researchers examine the characteristics of pharmacological substances as well as how they interact with equipment, unit activities, and other substances. Applications include correlations, the first-principle model, heuristics, and decision trees. Among other things, decisions about machine sizes, excipients, and manufacturing processes are made using the information. To increase product quality and reduce waste, AI is used in conjunction with related networks, methods, and tools. Pharmaceutical companies will benefit as a result. To guarantee product quality, quality by design, or QbD, is applied as shown in [→Fig. 5.1](#).

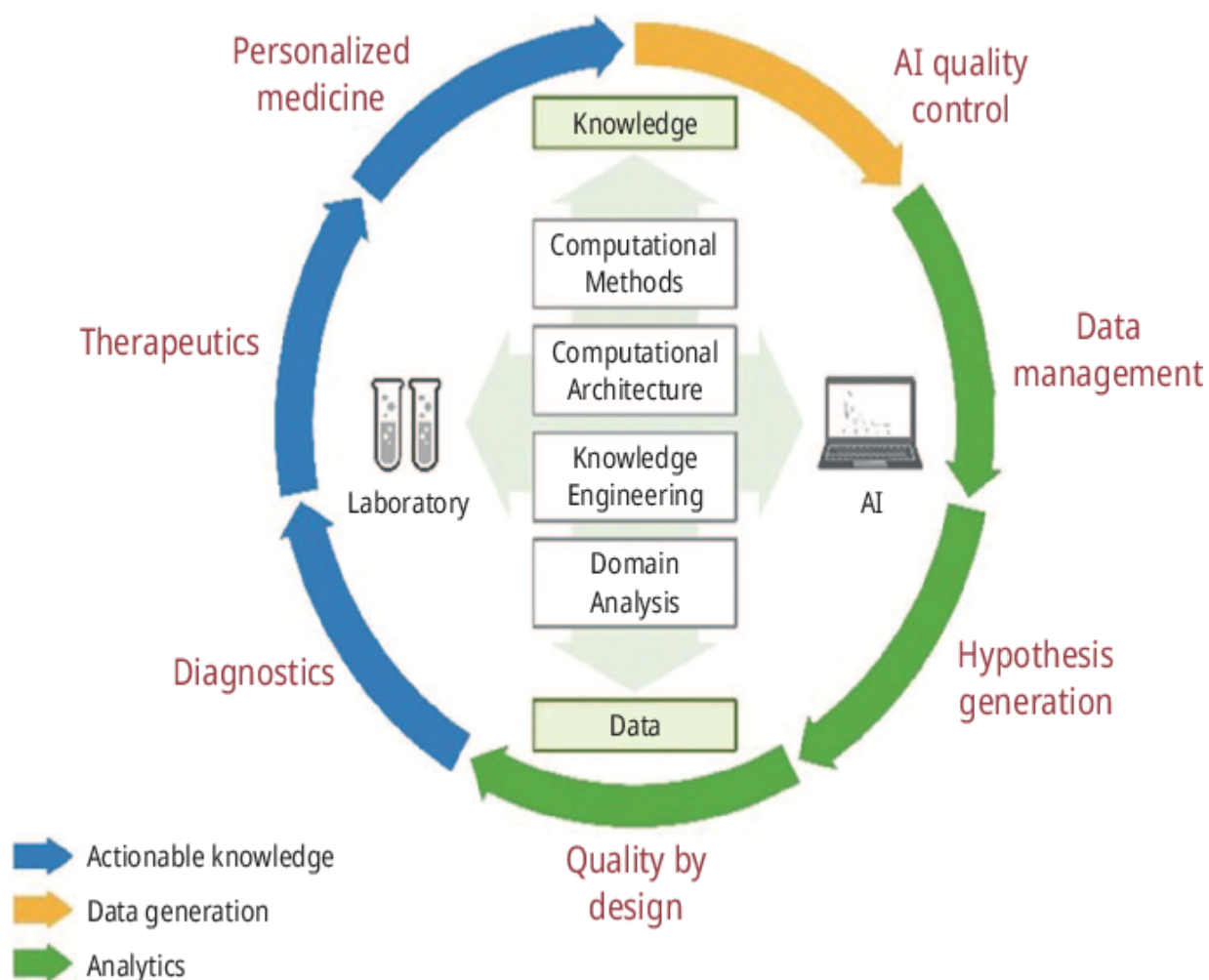


Fig. 5.1: Graphical representation of the utilization of AI and laboratories in the pharmaceutical industry.

The complexity of manufacturing has grown compared to previous years due to notable advancements and recent advancements in pharmaceutical technology. The quality and efficacy of a tested product are also essential for the intricate process. Modern pharmaceutical manufacturing systems incorporate human expertise into machines to improve processes. AI in pharmaceutical production may boost the industry [[→ 11](#)]. AI tools will help mechanize many pharmaceutical activities. Not only can it help with manufacturing process issues, but it can also support numerical simulation.

By utilizing other state-of-the-art manufacturing technologies, AI can be incorporated into the medication manufacturing process to achieve the desired level of outcomes. Potential applications of AI in pharma production processes were listed in a discussion paper released by the U.S. FDA. AI-based pharmaceutical quality control and assurance systems are anticipated to yield benefits [[→12](#)]. Pharmaceutical companies use bands to ensure product quality, even though they may be inaccurate. AI applications in the pharmaceutical sector can forecast minute changes that may affect the efficacy and security of the products. AI-based pharmaceutical quality assurance and control can enhance process accuracy and product quality [[→13](#)]. Moreover, preclinical lead molecule failure rates can be decreased by using computational methods with multi-objective refinement. AI in drug design uses computer software to evaluate, explore, and learn from enormous quantities of pharmaceutical data to find new therapeutic compounds. Combining ML advances in a unified, completely automated way achieves it [[→14](#)]. Machine learning (ML)-based methodologies, a derivative of AI, emphasizes the transformation of extensive biomedical large data into novel insights and practical knowledge rather than the theoretical advancement of intricate physicochemical principles [[→15](#)]. These days, deep learning (DL) paradigms and other AI technologies have incredible feature extraction and generalization capabilities that make them highly intriguing for drug design. While DL approaches can automatically extract characteristics from the input data and reorganize fundamental traits

into complex characteristics via multi-layer feature extraction, conventional ML approaches use manually generated attributes [[→16](#), [→17](#)].

5.2 AI application in pharmaceutical manufacturing

The pharmaceutical sector is urged to innovate via process analytical technology (PAT) and QbD methodologies. The QbD approach underscores the importance of understanding the product and process in determining the key material attributes (KMAs) and critical process parameters (CPPs) that substantially affect the critical quality attributes (CQAs) of both the product and process. It results in the creation of a design area with acceptable quality standards. Operating inside a space that has been approved by regulations for design does not constitute a change; therefore, production flexibility can be increased. AI primarily refers to computational techniques that carry out tasks that are commonly associated with cognitive processes similar to those of humans, like pattern recognition as well as decision-making. These tasks are carried out by ML techniques in AI, which learn from a dataset and generate a response without explicit programming. As the training dataset expands, ML can adapt the model behavior to continually enhance its performance, which is particularly useful for data-driven manufacturing applications [[→18](#), [→19](#), [→20](#), [→21](#), [→22](#)].

Additionally, the medical regulatory bodies are becoming more receptive to AI/ML techniques. However, more investigation is needed to prepare for

smart manufacturing concerning the ML process's real-time applicability in the pharmaceutical manufacturing process. ML can use past process data to estimate the crystallization result. For example, researchers used data from various industrial batch crystallizations to create ANN and GA models. Nine descriptors were used to model the crystal density, which, after optimization, led to a notable improvement in the final product and assisted in identifying the critical parameters. Based on a few experimental runs as well as four input variables, the design area of a co-crystallization process could be investigated in a different study. ANN inputs for the operating variables (moisture, supersaturation, agitation speed, along with seeding properties) can also yield a more accurate crystal growth rate estimate than numerous non-linear regressions [[→ 23](#), [→ 24](#)]. Furthermore, the behavior of the compacted powdered mixture within the tablet press must be considered initially when constructing a tableting process. For each filling, the blend's flowability must be sufficiently high to ensure that almost identical quantities of powder enter the die. Certain researchers employed a feed-forward and back propagation artificial neural network (FF-BP ANN) to forecast the rate at which specific powders would traverse a circular aperture. The input parameters for powders typically include bulk density, tapped density, particle size, aspect ratio, roundness, convexity, and true density. The flow equation proposed by other researchers was less precise than the flow rate values obtained as shown in [→ Fig. 5.2](#) [[→ 25](#), [→ 26](#)].

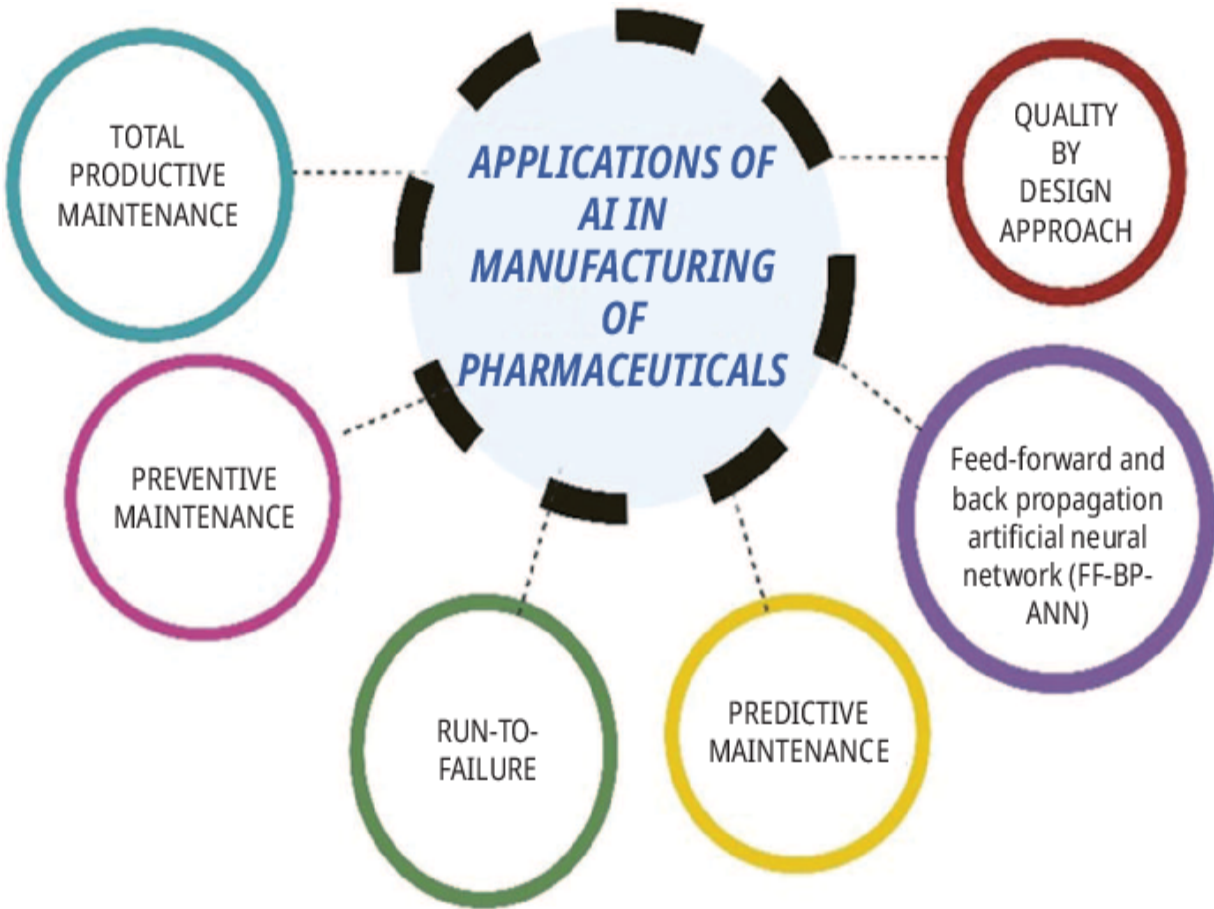


Fig. 5.2: Various applications of AI in the manufacturing of pharmaceuticals.

With downtime and financial losses becoming more and more important, predictive maintenance, or PdM, has become a popular anomaly prediction technique in the manufacturing system. PdM can forecast the health state using ML algorithms and continuously monitor manufacturing machinery with smart electrical sensors. Preventive, predictive, and run-to-failure (R2F) are the three categories into which maintenance strategies employed by manufacturing companies can be divided. The simplest is R2F since it is only used in the event of a system failure. The direct costs of manufacturing processes will increase, nevertheless, if production lines are stopped,

industrial equipment is repaired, and production facilities are adjusted. In contrast, PdM works better when implemented before a system fails because it examines, observes, and prevents early faults. The idea of total productive maintenance (TPM) has been established based on the productivity principle to increase employee satisfaction at all organizational levels, decrease manufacturing costs, and improve the overall performance of production equipment [[→27](#)]. Using ML algorithms to identify when preservation chores are required can save operational expenses. Second, by utilizing statistical techniques, engineering approaches, and historical data, it can produce more accurate predictions. Thirdly, because it can detect faults early on, it can extend the life of industrial equipment and enhance working conditions [[→28](#)]. There are two challenges even though PdM is the most efficient for fault detection. On the one hand, PdM uses ML algorithms, but the quality of the data representation frequently limits the performance. Specifically, poor ML algorithm performance is produced by data that have redundant attributes and low attribute importance [[→29](#)]. Supply chain design and configuration strategic decisions can also be informed by ML techniques. A Bayesian Network-based risk propagation model has been proposed by researchers. The suggested model considers the peculiarities of a supply-chain network structure in addition to the interdependency of different risks to produce specific risk measures. Lastly, a simulation study shows how output risk measures can be used in a supply chain environment [[→30](#)]. By automating risky tasks, they were able to decrease inventory costs through improved supply and demand planning,

shorten development cycles, increase engineering productivity, decrease errors, improve safety, increase revenue through improved sales lead identification and cost optimization, and more. As a result, they experienced rapid growth. The novel idea of “intelligent manufacturing” is a clever method of production in which humans and machines collaborate side by side under minimal supervision. Intelligent manufacturing is best demonstrated by the Siemens manufacturing sector. The worker creates programmable logic circuits and uses a virtual factory that mimics a real factory floor to supervise and manage their production. Barcodes let items talk to the machinery that makes them, and the machinery may talk to each other to exchange parts and troubleshoot problems [[→ 31](#), [→ 32](#), [→ 33](#)].

Manufacturers are optimizing important indicators of performance and reviewing them in real-time with the help of AI. For instance, customizing a model with virtual programs can improve energy consumption and better anticipate, detect, and prevent staffing and material bottlenecks. More significantly, it notifies the engineers in advance of issues and suggests fixes. AI-enabled manufacturing speeds time to market, reduces waste, cuts costs, and efficiently executes assembly line procedures. It has been demonstrated that utilizing collaborative robots, self-driving cars, and machine-learning algorithms can lower inventory levels and improve warehouse expenses [[→ 34](#)]. DL is a subclass of ML that utilizes extensive experimental data to learn through advanced approaches such as artificial neural networks. Neurons in multi-layered neural networks generally alter data non-linearly in DL models. Recently, convolutional and recurrent

neural networks have been effectively employed in pharmaceutical sciences for various applications in developing accurate dosage formulations. These applications include the detection of tablet defects, prediction of particle flowability, prediction of storage stability, and prediction of drug dissolution profiles [[→ 35](#), [→ 36](#), [→ 37](#), [→ 38](#), [→ 39](#)]. The predictive accuracy of the models must be assessed once the ML modeling process is finished. Researchers looked into how deep neural networks could be used to forecast how long tablets would dissolve. In the following work, the improved maximum dissimilarity algorithm (MD-FIS) was used to separate training and testing subsets of data on various drug formulations that had been gathered through literature mining. Notably, validation and testing of data sets are made possible by MD-FIS, an advanced data selection technique [[→ 40](#)].

5.3 Application of AI in quality control of pharmaceuticals

Based on the “quality philosophy,” which places a high priority on management by objectives, full engagement, sharing of technology, solving issues, adaptability to change, heritage preservation, and people-oriented practices; additionally, the “core values,” which emphasize the significance of system integration as a means, sustainable development for a healthy environment, and pragmatic benefits as an inducement, are the foundation of the organization [[→ 41](#)]. Participants will need to think through the

matter thoroughly and work very hard during the brewing process. Since the term “Industry 4.0” has become highly contested in the industry, the writers have been able to concentrate solely on the development of people in the high-quality expert field, and they have not developed any countermeasures [[→42](#)]. Any AI-based application used in a clinical environment needs to have a quality assurance workflow in place to regularly and individually verify the effectiveness of the AI output.

All validation procedures that confirm the outcomes of AI-based applications created specifically for each patient or machine are included in case-specific quality assurance (QA). The output can be used in subsequent stages of the RT workflow once the QA results are deemed satisfactory. If the quality check’s case-specific assurance is unsuccessful, the output will go through verification. The effectiveness of certain applications may be assessed using a separate secondary algorithm [[→43](#)]. Additionally, recent research has been done on auto-segmentation quality assurance using ML. Using distinct features for every ROI, decision trees can be created to classify the morphology of a particular auto-segmentation input [[→44](#)]. Research and development in the fields of DL and ML have significantly increased in recent years. Researchers have developed increasingly complex algorithms and models owing to the enhanced availability of data and computational capacity, which has markedly progressed in various areas. One area of research that has advanced significantly is natural language processing (NLP) [[→45](#), [→46](#), [→47](#)]. Researchers have used DL models such as transformers and recurrent neural networks (RNNs) to construct

more accurate and efficient language models. Improved human-machine connection, simpler process automation for customer service, and more sophisticated search engines and automated assistants could all result from these models. Another area that has made significant strides is computer vision [[→ 48](#), [→ 49](#), [→ 50](#), [→ 51](#)].

More accurate image classification and object recognition have been made possible by convolutional neural networks (CNNs). These models have the potential to improve autonomous driving, manufacturing quality control, and medical imaging analysis [[→ 52](#), [→ 53](#), [→ 54](#), [→ 55](#), [→ 56](#), [→ 57](#), [→ 58](#)].

DL approaches may autonomously identify and generate abstract features, while conventional ML methods generally necessitate domain expertise for feature construction or engineering [[→ 59](#)]. In a variety of inspection applications, DL techniques that utilize recurrent neural networks (RNNs), autoencoders, and CNNs demonstrate exceptional outcomes [[→ 60](#)].

Researchers use a deep ensemble learning model to examine surface flaws on automobile bodies. When the identical task is finished, their method performs better than human inspectors [[→ 61](#)]. A CNN is used by researchers in [[→ 62](#)] to locate flaws in textured surfaces. The results demonstrate high defect detection accuracy on a multi-class dataset.

Researchers [[→ 63](#)] identify manufacturing defects in semiconductors using a CNN-based transfer learning technique; however, other researchers [[→ 64](#)] suggest using a CNN model that can handle time series data. Researchers [[→ 65](#)] recommend using deep CNNs for sewer system inspection. They claim that CNN-based techniques perform better than manual feature

extraction methods using highly variable image data. An optimized VGG model is used by researchers [[→ 66](#)] for laser welding inspection. A transfer learning methodology is utilized when a diverse array of images is employed to pre-train the VGG model. Researchers [[→ 67](#)] describe a system for defect identification that integrates Random Forest (RF) and Support Vector Machines (SVM) with a pre-trained AlexNet for feature extraction. The smart quality inspection (SQI) method seeks to improve the model's efficacy and address many elements affecting the visual inspection process. Outsourcing segments of the inspection process enables the regulation of various task aspects, external influences, and other variables. Research gaps are found through the literature review process, which is closely linked to the development of the assessment algorithm. The strategy for creating SQI proposes a methodology for utilizing AI-driven visual inspection on-site. The inspection task was allocated to CNN, an AI model selected for its unique architecture. The application enables users to upload images of the product as it exits the assembly line. The CNN model will subsequently analyze the images to ascertain whether the product is adequate or defective and whether it has passed or failed the required inspection. Furthermore, the operator has the option to add the results to the inspection record by utilizing the SQI application [[→ 68](#), [→ 69](#), [→ 70](#)].

5.4 Data management and security

Interoperability is essential for enabling intelligent healthcare data management systems to receive requisite data input, as various data sources often store and process information in many forms. The challenge of obtaining relevant information is apparent in the immense volume of data that is continuously produced. The Office of the National Coordinator for Health Information Technology in the United States recently issued a draft rule regarding information blocking and interoperability, emphasizing the need to grant patients unfettered access to their electronic health records [[→71](#), [→72](#)]. Securing privacy and interoperability at the same time seems incompatible, which makes it difficult to achieve. Recently, the potential uses of cutting-edge blockchain technology for the administration of medical data have generated a lot of conversation in both academia and business [[→73](#), [→74](#), [→75](#), [→76](#)]. Blockchain technology can enhance and expedite the sharing of private health information by providing users with complete control over their data and privacy, eliminating the necessity for a centralized authority. Data sharing is essential in the realm of chronic diseases due to patient mobility, the involvement of multiple hospitals for diagnosis and treatment owing to specialized centers, the requirement for a “second opinion,” and the use of multiple medications by patients [[→73](#), [→77](#), [→78](#)]. Optimizing the pharmaceutical supply chain with blockchain technology can benefit clinical trials and medical research overall [[→79](#), [→80](#), [→81](#)].

By storing only the metadata, which may include patient consent, on the blockchain instead of medical records, blockchain technology can guarantee

the traceability and inviolability of patient health data. Individual network nodes can store vast and sensitive healthcare data, and their intelligent representation will be kept off-chain [[→73](#)]. An alternate method involves utilizing compliant cloud-based services for temporary storage (i.e., within a timeframe specified by the patient) and data sharing [[→78](#)]. The FHIR chain is a blockchain-based approach for data sharing that encapsulates clinical data by the Fast Healthcare Interoperability Resources (FHIR) standard. The sharing of primary care data will be expedited and simplified while safeguarding privacy through the effective management of on-chain consent and the establishment of access-control mechanisms as specified in the consent. Applications utilizing blockchain technology throughout the supply chain are starting to materialize. These include improving the public health supply chain, protecting medical equipment, improving the usability of healthcare Internet of Things (IoT) devices, and using the immutability and transparency of the blockchain to combat counterfeit medicines [[→79](#)]. Additionally, these applications ensure that the pharmaceutical company retains control over returned drugs. Researchers show how blockchain technology can offer features that improve pharmaceutical traceability in particular and supply chain management in general [[→80](#)]. Ensuring compliance in the pharmaceutical supply chain is crucial, as is confirming the conditions of transportation and storage. Pharmaceutical companies are also moving closer to their objective of obtaining patient record data directly from the source and putting it into an electronic system so that it can be sent gradually and steadily to sponsors or customers. The switch

from paper-based data collection to an electronic system is referred to as “electronic data capture.” Unprocessed patient or volunteer data from clinical trials is kept in a software system called an electronic data system, or EDC [[→ 81](#), [→ 82](#), [→ 83](#), [→ 84](#)]. Before being entered into the system, the information is first written down and then saved as an electronic case report form (eCRF). Clinical trial software is beneficial to all kinds of organizations, including pharmaceutical, biotech, medical, and research. These programs assist with all facets of conducting a clinical trial, encompassing study submission, archiving, and recruitment. These programs are relevant for both small-scale preliminary research and extensive investigations involving thousands of participants. The phrase “clinical data management systems” or “software” refers to the collection of applications available for data management. These programs are essential because they can manage massive volumes of data in multicentric systems or trials. Many software programs, including ORACLE CLINICAL, MACRO, RAVE, CLINTRIAL, eClinicalSuite, EZentry, and others, are utilized as CDM software in pharmacies [[→ 85](#), [→ 86](#), [→ 87](#)]. The functionality of these programs is essentially similar, so they cannot outperform one another. These software programs have the drawback of being costly, and for an organization to operate, sophisticated information technology communications must be acquired. Several benefits are provided by the medical records management software, including the ability to expedite study timelines and control costs, provide precise clinical trial predictions, enable robust program modeling, ease recruiting, scheduling,

and screening processes, support physicians, and effortlessly adjust for foreign exchange rates [[→ 88](#), [→ 89](#), [→ 90](#), [→ 91](#)].

The secret to success is ingraining readily available technologies like AI, blockchain, and other technologies into a company's core competencies [[→ 92](#)]. To enhance clinical research and attain patient-centricity, the industry must employ technology to create user- and customer-centric interfaces, make decisions based on data for innovative data processing methods, and enhance outcomes [[→ 93](#), [→ 94](#)]. AI could help, for instance, with patient identification and prioritization for medication development and monitoring, which is necessary for standardized drug production and expedited schedules [[→ 92](#)]. AI and numerical drug design techniques were used to analyze clinical trial data to determine dosage, evaluate the efficacy of medication formulations, and repurpose pharmaceuticals that were already on the market. By incorporating secure and automated data analysis, blockchain facilitates the creation and maintenance of ledgers, or blocks of content, in systems design. Secure recording and analysis of all health-related data allow for prompt updates to be provided to payers, medical professionals, and doctors. Nevertheless, the blockchain would be exceedingly inefficient for storing a substantial volume of records, such as comprehensive electronic medical records or genetic data records, due to the substantial computational resources that would be necessary. It is a substantial defect in blockchain technology, as it complicates the process of querying data within a blockchain [[→ 95](#), [96](#)].

However, there are ways to work around the issue with the blockchain and AI algorithms integrated. Risk assessment must be the initial phase in protecting sensitive personal information and PHI within the life sciences and healthcare sectors. Implementing the Cyber Security Framework (CSF) from the National Institute of Standards and Technology (NIST) is advised for establishing a cybersecurity program. The NIST Cyber Security Framework encompasses standards, recommendations, and best practices for managing cyber security in critical infrastructure. But cloud- and AI-based products, services, and technologies are employed to construct business processes utilizing frameworks to develop security initiatives that will enable those employed in the sector of the economy to be ready to defend the sensitive data and able to respond promptly to any incidents or security threats [[→97](#), [→98](#), [→99](#)]. To create a comprehensive security program from process and policy viewpoints, researchers began with the NIST Cyber Security Framework (NIST CSF). They then used technology from SAP Cloud, specifically SAP S4/HANA and its AI-based cloud services, to make up for it. A source of veracity and a record system would be the SAP S4/HANA. Digitization and digital transformation of the life science industries are more important than ever. SAP S4/Cloud offerings provide all required products and services along with control features like private cloud services, enabling life science companies to access state-of-the-art technologies and solutions that leverage cloud and AI services. They are still in charge of their private information, and it helps them comply with global privacy and data laws. With the public, private, and hybrid

models that SAP S4/HANA cloud offering offers, users can select the one that best suits their needs and yields the highest return on investment (ROI) while maintaining all the controls required to protect their company's objectives, safeguard sensitive data, and carry on innovating and producing new medications through clinical research and R&D [[→100](#), [→101](#), [→102](#)]. Blockchain technology and AI powered by the IoT will improve biomedical and healthcare data ledgers, accelerating biological and clinical research. These assessments are predicated on fundamental aspects of blockchain and AI, including improved security, resilience, decentralized administration, unchangeable audit trails, and above all the reinstatement of characters' rights. Innovative and state-of-the-art techniques that will improve medical data handling, collaboration, processing, assessment, and result-based classification are being developed using blockchain and AI. The healthcare industry, which is driven by the IoT, is being redefined by blockchain technology. It is achieved through enhanced productivity, access control, advances in technology, safeguarding privacy, and operational data process safety [[→103](#)].

IoT solutions' security and privacy are rapidly approaching but largely ignored. The research on privacy and security measures in the IoT is reviewed by the study's authors, who also look at common misconceptions about technologies that claim to be incredibly secure. Researchers discuss user authentication methods and safe key management for fog computing-based IoT environments. The SAKA-FC security system comprises a dual-component framework featuring three-factor key management and a user

authentication mechanism [[→ 104](#)]. In another study, researchers [[→ 105](#)] examined the safety of implanted medical devices (IMDs), including pacemakers and insulin pumps. The study's authors propose a three-factor lightweight remote user authorization approach for implantable medical devices, utilizing elliptic curve cryptography. The security scheme illustrates how the recommended scheme should be implemented using the Dolev-Yao threat structure. It is a system model that is linked to the user (U), the Controller Node (CN), and the IMD. The security of remote monitoring is improved by employing a three-factor verification strategy. To guarantee secure communication, it necessitates the establishment of bilateral keys between CN and IMD, as well as mutual authentication between U and CN through the use of a session key. Other researchers have suggested a lightweight data-sharing plan with traceability, or LiST, for mobile health solutions. The system in the model is comprised of healthcare professionals (the data user), the public cloud, WBSN (the data owner), and the Key Generation Center (KGC), which is considered a completely dependable organization that generates public characteristics for the entire system and provides secret keys to data users. The authors assert that LiST provides lightweight user revocation, lightweight test algorithms, lightweight encryption, lightweight backdoor generation using keywords, and lightweight traitor tracing [[→ 106](#)]. The scheme implements a lightweight revocation mechanism to eradicate unauthorized users and their decryption privileges to safeguard the privacy of the EHR.

5.5 Regulatory compliance in AI-based manufacturing

Today's legislators and regulators have considerably higher standards for corporate governance and compliance. There is a noticeable shift in favor of obligatory and complete corporate trust and transparency. Legislators and regulators in the European Union are considering creating compliance regarding the use of digital technologies, specifically AI, for the first time. A few years ago, the EU Commission released an AI regulation to guarantee that technology is applied in a way that upholds EU ideals, principles, and fundamental rights [[→ 107](#), [→ 108](#)]. An illustration of a measure implemented to ensure the security, transparency, ethics, neutrality, and human oversight of AI is the EU Commission's AI Act. The EU now categorizes AI applications according to their risk levels, and corresponding norms and regulations have been established based on the classification. For example, social scoring is too dangerous to use, and transparent rules are all that low-risk chatbots need to abide by. Despite the significance of compliance in the framework of digital technologies, neither legal nor digital technology experts are currently providing thorough scientific approaches or focused attention to the issue. Consequently, AI continues to be the sole notable instance in which the European legislature has looked into these matters. The EU Medical Device Regulation imposes stringent regulatory requirements before medical devices, including AI software devices, may be used in clinical practice. The EU MDR [[→ 109](#)], which

replaced the EU Medical Device Directive, must be fulfilled. For the first time, many of the terms and specifications of the EU MDR also apply to medical device software created by healthcare organizations for internal use.

Software meant to offer data for any of the objectives quantified in the explanation of a medical device [[→ 109](#)], including monitoring, treatment, prevention, diagnosis, and alleviation of disease, is subject to EU MDR regulation. It covers both software that offers a diagnosis or treatment on its own and software that only offers data to help a medical practitioner arrive at the best diagnosis or course of treatment. The EU MDR remains applicable regardless of whether the software is an executable program, dynamic website, web service, script, or even a simple spreadsheet macro. Additionally, the software is considered a medical device if it is intended for medical purposes and is intended to be used with humans or their data, regardless of the complexity of the processing, the risk it poses to the patient or user, whether or not a healthcare professional uses it, or the computing platform on which it is running [[→ 110](#), [→ 111](#), [→ 112](#)]. AI could potentially make predictions regarding the complex relationships between drug efficacy and immunological responses to substantially improve drug discovery [[→ 113](#), [→ 114](#)]. AI could also be beneficial for more intricate tasks, such as the analysis of multiple genomic issues, or for simpler ones, such as the best practices for electronic documentation [[→ 115](#), [→ 116](#)]. AI has been deployed to enhance medical reporting related to oncology and radiology [[→ 117](#), [→ 118](#)]. For analytical labs, where protocols and

procedures can occasionally total hundreds or even thousands of pages, and where the quantity of pages grows with every new treatment, it is encouraging. AI can quickly create useful methods and predict text based on input data [[→ 119](#)]. AI language translation tools can also be applied to worldwide shared methods to improve interlaboratory reproducibility. Furthermore, data produced during stability and new batch cGMP testing needs to be carefully examined, and any inconsistencies need to be looked into through quality investigations. AI tools are capable of conducting investigations and proofreading analytical data to find errors [[→ 120](#)].

In delicate areas, AI systems should be employed with extreme caution, as judgments made using skewed data or incorrect criteria may have far-reaching effects [[→ 121](#)]. Consequently, these systems necessitate extensive validation that effectively addresses issues that are distinctive to AI, such as the black box problem. Numerous writers speculate that XAI might be useful in the validation, but particular strategies have not yet been created and assessed. Four phases make up the validation concept, which can be carried out repeatedly or step-by-step. The typical design tasks involved in creating and implementing a new IS are handled in the first phase, known as the setup phase. The technical setup and performance phase that follows covers the steps specifically related to AI development to create a dependable AI system. Thus far, the process bears similarities to validation methodologies or black-box AI system performance assessments [[→ 122](#)]. Predefined performance measures may suffice for AI systems operating in insensitive environments; however, further effort is necessary to verify that

the AI system has been developed and is functioning as intended under the proposed validation concept. The third phase of the XAI Assessment offers the recommended XAI approach, which gives an example of how to use decision rules and/or spaces appropriately. The final stage, the stress test, is designed to ascertain the limits of the AI system's use, building upon the evaluation. A validation approach was developed in a research and development project with the assistance of specialists in technology, academia, industry, and GMP auditing. This strategy was applied with success in the pharmaceutical manufacturing sector [[→123](#)]. The business putting the AI systems into place must still evaluate a few elements of the created validation strategy, though. It was therefore recommended that future research focus on creating and establishing guidelines and standards for sensitive use cases, such as the pharmaceutical industry. It might result in less validation work being needed, which would then make comprehensive validation techniques more feasible for use cases involving less sensitive data [[→124](#)].

5.6 Conclusion

The pharmaceutical industry is using artificial intelligence more and more to manage massive data volumes and enhance automation. Artificial intelligence can simulate human intelligence through sophisticated tools and networks, but it cannot take the place of humans in the real world. The pharmaceutical industry has been enhanced by quality by design (QbD)

techniques, and AI has the potential to enhance product quality, reduce waste, and expedite manufacturing processes. AI can also be used in drug design to assess, learn from, and uncover big data related to pharmaceuticals, integrating machine learning (ML) advancements in a highly automated and unified manner. Intelligent manufacturing, where humans and machines work together under minimal supervision, has been used to optimize performance indicators, improve energy consumption, and predict staffing and material bottlenecks. AI-based applications in clinical settings must have quality assurance tools to confirm the efficacy of the AI output. Scientists have been able to develop object recognition and classification models for autonomous driving, manufacturing quality control, and medical imaging analysis using convolutional neural networks (CNNs). Recurrent neural networks (RNNs), autoencoders, and CNN-based deep learning techniques have produced outstanding results in a variety of inspection applications. Blockchain, artificial intelligence (AI), and other technologies must be incorporated into core competencies for the medical research industry to succeed. The European Union is considering implementing compliance regulations for AI-based manufacturing, aiming to align technology with EU principles, values, and fundamental rights.

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Chapter 6 Drug development and clinical trial via artificial intelligence

Abstract

Clinical trials and medication development have been profoundly impacted by artificial intelligence (AI), with deep learning techniques outperforming traditional machine learning tactics. Though its potential and accessibility are limited, high-throughput virtual screening (HTVS) has been proposed as an alternative to expensive and time-consuming methods. The chapter overviews computational drug development techniques that identify drug targets, validate drugs, and conduct clinical trials using artificial intelligence (AI). Additionally, by simply acquiring the governing equations from data, a novel deep learning technique has been described that is designed to build PK/PD models that can forecast patient response times and simulate the impact of unknown dosage regimens. The technique keeps the dose-concentration effect pharmacologically sound while enabling the model to predict treatment outcomes and replicate the patient's dosage schedule. Drug target interactions (DTIs) are essential for locating possible drugs and offering information on side effects and drug mechanisms. Additionally, ethical issues about social gaps, security of data, informed consent, medical consultation, empathy, as well as sympathy, are raised in the last section and affect AI-enabled clinical trials in the healthcare industry.

Keywords: drug target interactions, PK/PD models, high-throughput virtual screening, validation, ethical issues,

6.1 Introduction

The application of machine learning (ML) methodologies in computer-aided drug development has increased markedly in recent years. The ability of deep learning methods, such as artificial neural networks with numerous hidden layers, to autonomously extract features from input data and model nonlinear input-output relationships has recently garnered heightened interest [[→1](#), [→2](#), [→3](#), [→4](#), [→5](#), [→6](#)]. The characteristics of deep learning methods supplement conventional machine-learning methods that use molecular descriptors created by humans. Deep learning has made a somewhat tardy comeback to the drug discovery space, but in the interim, it has yielded an unprecedented wave of novel modeling techniques and applications. The continued advances in deep learning have proven beneficial to many domains within the chemical field [[→7](#), [→8](#), [→9](#), [→10](#)].

Contemporary medicinal chemistry techniques primarily rely on stringent testing protocols and a cyclical methodology [[→11](#)]. These methods evaluate several potential pharmacological substances to identify those possessing the requisite qualities. However, these procedures may be expensive, labor-intensive, and produce inaccurate results [[→12](#)]. Their limitations may stem from the lack of appropriate test substances and the challenges associated with precisely forecasting their behavior within the body [[→13](#)]. These problems can be addressed by various artificial

intelligence (AI) algorithms, including reinforcement learning, evolutionary algorithms, rule-based systems, and both supervised and unsupervised learning methods. These strategies generally rely on the analysis of extensive datasets that provide many applications [[→ 13](#), [→ 14](#), [→ 15](#)]. These algorithms can more precisely and effectively predict the safety and efficacy of new therapeutic molecules compared to existing methods [[→ 16](#), [→ 17](#)]. Furthermore, AI-driven algorithms can pinpoint novel targets for pharmaceutical development, encompassing particular proteins or genetic pathways linked to illnesses [[→ 18](#)]. Enhancing drug discovery beyond conventional approaches may ultimately result in the creation of novel and more efficacious therapies. These constraints can be overcome with the implementation of ML and other AI techniques. ML algorithms can discern patterns and trends in extensive datasets that may escape human researchers. It facilitates the expedited proposal of contemporary bioactive substances with fewer adverse effects compared to traditional techniques. Recently, a dataset comprising recognized medicinal substances and their associated biological processes was utilized to train a deep learning algorithm [[→ 19](#)]. Subsequently, the program showed remarkable precision in forecasting the action of new chemicals. Substantial contributions to mitigating the toxicity of potential therapeutic molecules have been made by utilizing extensive datasets of recognized hazardous and non-toxic substances to train ML algorithms [[→ 20](#)]. Furthermore, the use of AI in drug discovery requires a high-quality dataset. Technology and high-throughput sequencing advances have accelerated the creation of many free,

open-access drug discovery databases. As seen in [→ Fig. 6.1](#), the drug discovery procedure is accelerated and assisted in entering the big data era by these databases. Presently, the manually selected databases like ChEMBL contain more than two million compounds with characteristics resembling pharmaceuticals [[→ 21](#)].

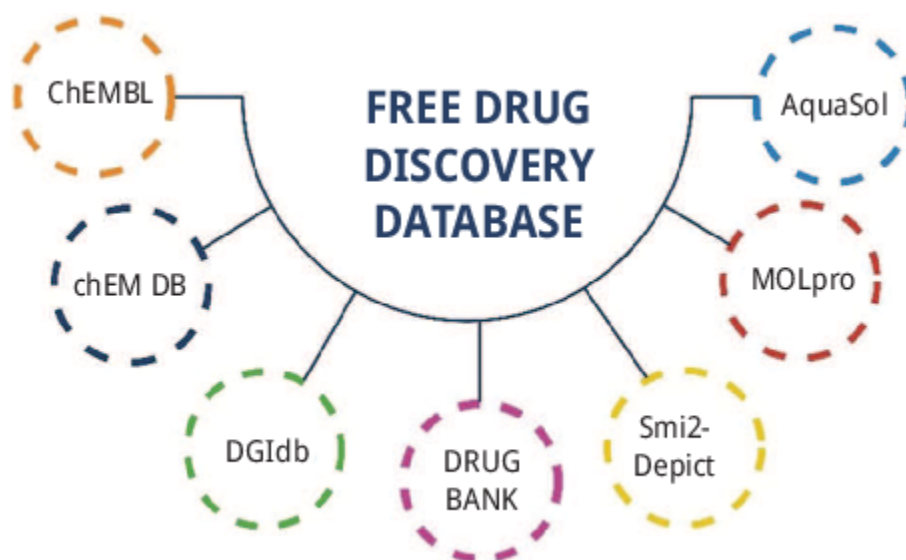


Fig. 6.1: Various databases for drug discovery.

ChEMBL collects data on the deposited compounds' molecular characteristics, toxicity, target interactions, metabolism, elimination, absorption, distribution, and excretion mechanisms. ChemDB, a publicly available database, provides physicochemical properties (such as rotational bonds, molecular mass, and dissolution rate) and information on almost 5 million readily accessible small molecules [[→ 22](#)]. Moreover, ChemDB includes several cheminformatics tools, including Reaction Predictor, MOLpro, AquaSol, and Smi2Depict that help in drug discovery.

Information on genes and gene products that may interact with medications is available in the Drug-Gene Interaction Database (DGIdb) [[→23](#)]. To find drug-gene interactions, users can use the search field module to enter a list of drugs or genes, or they can browse the genes in each group. Additionally, DGIdb can be automatically fetched via the web-based interface's API.

DrugBank is one easily accessible reference drug database. It currently includes many medications along with a plethora of knowledge about drug classes, interactions, target-drug relationships, and reactions. Utilizing the integrated tools, users can search, investigate, and retrieve text, images, and structural data from DrugBank. At present, it is the most extensively utilized resource globally for drug screening, layout, and metabolic prediction [[→24](#)]. The two main issues with these platforms' integration are the strict laws governing AI usage and the projected loss of jobs. However, these technologies are meant to make the task easier rather than completely replace people [[→25](#)]. In addition to making hit compound identification quick and error-free, AI can also identify relevant SAR, recommend the best chemical structure, understand drug-target relationships, and suggest synthesis routes for these agents. After a basic medication is added, AI can help with quick decision-making and offer vital information for improving the drug in the right dosage form. Higher-quality products can be produced batch-to-batch more quickly and consistently as a result. Through in-depth market analysis as well as forecasting, AI can ensure the best alignment and pricing in the industry and potentially assist in determining the safety and efficacy of the agents in clinical trials.

Through the analysis of patient-relevant genome-exposome attributes, AI can assist in identifying specific patient cohorts for participation in Phase II and III clinical trials. It could enable the monitoring of the existing therapy targets of the selected participants by more advanced surveillance methods [[→26](#), [→27](#)]. The utilization of adjunct AI capabilities, such as predictive ML and alternative predictive algorithms, facilitates the preclinical evaluation of molecules and the identification of lead compounds, thereby enhancing the forecasting of lead molecules suitable for clinical trials within the designated patient population [[→27](#)]. Patient dropout accounts for about one-third of clinical study failures, resulting in additional enrollment requirements that must be met for the trial to be completed, which adds to the trial's financial and time constraints. Such issues can be avoided with close patient supervision and support in following the correct clinical trial protocol [[→28](#)]. In a Phase II trial, “AiCure” has released smartphone software that tracks patients with schizophrenia who take their medications regularly. Consequently, there was a minor percentage increase in the patients' compliance frequency, which guaranteed the clinical trial's success.

6.2 Computational drug discovery techniques

Computational methods are vital and practical tools in the drug development process. Finding a chemical compound that can both geometrically and chemically fit into a particular cavity of a target protein is

the aim of drug development. Computational methods have made many tasks easier, as shown in [→ Fig. 6.2](#), including lead optimization, design, search, comparison, modeling, binding energy estimation, and pharmacokinetic predictions. Many advancements in the field of developing and discovering drugs have been made possible by advances in computational power and analytical algorithms and approaches [[→ 29](#)]. High-throughput virtual screening, also known as in-silico virtual screening (HTVS), has produced a valuable alternative to the costly and time-consuming high-throughput screening experimental methods. The capability to virtually screen catalogs of compounds to enhance ligand enrichment and move forward with experimental validation has resulted in the production of many lead compounds. Large databases of virtual compounds that are either complementary with the solved receptor structure or similar to a known inhibitor (ligand-based) are screened computationally by HTVS. It enables researchers to search through big databases or compound libraries to find a narrowly targeted subset that can be used for experimental confirmation of activities [[→ 30](#)]. But instead of requiring an exhaustive list, traditional HTS, de-novo drug design relies on methods of search to efficiently scan the chemical space. HTS can handle thousands of library members, verify binders, reproduce findings via biophysical experiments with high precision, and remove false positive hits, irrespective of the target 3D structural information. The subsequent, time- and money-efficient phase of potentially enhancing the physicochemical characteristics of hits to a lead medication depends heavily on HTVS [[→ 31](#)].

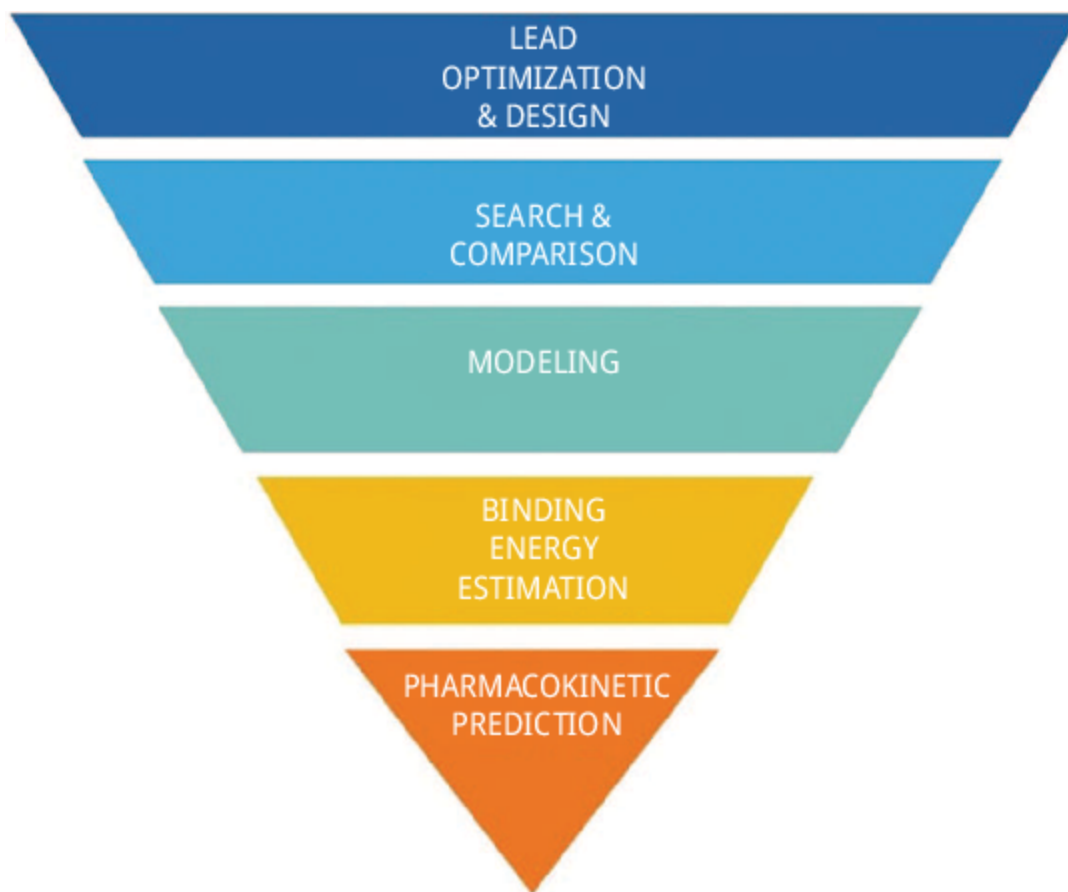


Fig. 6.2: Pictorial representation of computational methods of drug discovery.

In HTS, various biological effectors and modulators are evaluated and tested against distinct, particular targets. Consequently, HTS is generally favored when target knowledge is insufficient and structure-based drug design is impractical. HTS may also be utilized in conjunction with alternative tactics, including computational methods and fragment-based drug creation [[→ 32](#), [→ 33](#)]. Target discovery, molecule management, chemical preparation, assay optimization, and screening are the elements of high-throughput screening (HTS). High-throughput screening, when combined with multi-well cell-based platforms, enables the identification of small molecule modulators of associated signaling and metabolic pathways.

The development of new medications for central nervous system disorders, especially neurodegenerative diseases, is fraught with risk due to the elevated failure rate of candidates after long and expensive clinical and non-clinical research. Consequently, the establishment of accurate assays for quantifying neurodegeneration is an essential initial step in the formulation of therapies for neurodegenerative diseases. The ability to pinpoint the related processes that initiate and result in cell death is essential to the development of effective assays. To this aim, assays have been developed to identify biochemical occurrences, such as energy metabolism, metabolic fluctuation, and DNA fragmentation that lead to neuronal death [[→ 34](#), [→ 35](#)]. Historically, cytoprotective studies utilizing fluorescent markers or dyes have been crucial for classifying therapies that induce neuronal death [[→ 35](#), [→ 36](#), [→ 37](#)]. Pharmaceutical companies regularly employ several tests, which may be readily adapted for high-throughput screening methods, to assess the neurotoxicity of medications and their possible adverse effects. Researchers devised a high-throughput screening (HTS) method for primary neurons suitable for extensive investigations, encompassing compound libraries and whole-genome RNA interference (RNAi) [[→ 38](#), [39](#)]. The method may be utilized for the screening of mammalian cell lines, including CHO cells, for minute particles [[→ 40](#), [→ 41](#)]. The selection of the readout for a cell-based experiment targeting a signaling pathway can be flexible. Target-oriented and cell-oriented screening procedures have been utilized to explore various innovative drug delivery systems for Alzheimer's disease, Parkinson's disease, bipolar disorder, autism, and

schizophrenia. Cell-based screening concurrently assesses many targets, with the results indicating the effects on a cellular pathway or network. The models' failure to precisely represent the pathophysiological environment of neurodegenerative diseases is a primary challenge faced by high-throughput screening in its quest for prospective therapeutic candidates. High-throughput screening is crucial to the drug discovery process for neurodegenerative diseases [[→ 42](#), [→ 43](#), [→ 44](#)]. The efficacy of high-throughput screening (HTS) is constrained by both the quality and amount of the chemical library evaluated, as well as the absence of human-relevant neurodegenerative disease (NDD) models. Thus, increasing the number of tiny molecules accessible through high-throughput screening improves the likelihood of identifying effective disease-modifying therapies for neurodegenerative conditions. The advancement of cheminformatics in the pharmaceutical industry, the rise of AI in drug discovery, and the development of dynamic combinatorial chemistry (DCC) are anticipated to produce numerous compounds that could result in more targeted pharmaceuticals exhibiting efficacy in preliminary high-throughput screening (HTS) assays [[→ 46](#), [→ 47](#), [→ 48](#)].

The substantial increase in the utilization of in-silico methodologies, including chemoinformatics, molecular simulation, and AI, over recent decades is rational, given the advancements in computing technology and the exponential proliferation of structural, chemical, and biological data related to a widening spectrum of therapeutic targets. Molecular docking is a preferred and effective structure-based in-silico technique for forecasting

interactions between molecules and biological targets [[→ 49](#), [→ 50](#)]. The technique typically begins with predicting the molecular orientation of the ligand within the receptor, followed by assessing their complementarity by a scoring function. A ligand is positioned within a protein's active region with matching algorithms that use chemical and shape information. Pharmacophores are used to illustrate the ligand and the protein. The distance matrix that links the pharmacophore, as well as the corresponding ligand atoms, governs new ligand conformations. For each match, the pharmacophore's distance within the ligand as well as the protein is computed. Programs like DOCK [[→ 54](#)], FLOG [[→ 55](#)], LibDock [[→ 56](#)], and SANDOCK [[→ 57](#)] offer algorithms for matching ligand docking. A more recent approach that integrates multiple scores to evaluate the docking conformation is consensus scoring [[→ 58](#)]. A ligand or potential binder may be approved if it exhibits good performance across multiple scoring systems. Consensus scoring usually produces notable gains in bound conformation, pose prediction, and virtual screening enrichments [[→ 59](#)]. Nonetheless, the prediction of binding energies may still be off. Receptor flexibility continues to be a significant barrier in docking studies, especially when it comes to the movement of several crucial extra components of the receptor that are involved in ligand and catalyst binding in addition to the flexibility of the core structure. Certain techniques to address side chain flexibility are suitable and successful in specific circumstances. A popular solution for global flexibility, which makes sense from the standpoint of conformer selection, is a collection of proteins. Protein structures capable of

accommodating the ligand should be included in the ensembles, as docking requires an efficient approach for acquiring and selecting reliable protein structures. Furthermore, the method's computing expense is an additional disadvantage. The scoring function is a critical component for enhancing those merits of docking. Effective examples demonstrate the utilization of computational methods to engineer novel small compounds and evaluate results from an extensive database. However, experimental technology is still needed to achieve realistic interactions between small molecules and receptors. Docking applications may be enhanced by precise and inexpensively computed scoring functions [[→ 60](#)].

6.3 AI-powered target determination and validation for clinical trials and drug development

Creating new medications is a costly and time-consuming process. The pharmaceutical industry faces challenges in maintaining a satisfactory return on investment due to exponential cost growth. Precise forecasting of clinical trial results can streamline pharmaceutical companies' pipelines by cutting down on failure costs. It can also help representatives of investment banks and hedge funds make better decisions when managing investment portfolios [[→ 61](#), [→ 62](#), [→ 63](#), [→ 64](#)]. Early clinical trial outcome prediction may enhance the efficacy of pharmaceutical R&D, facilitate the creation of effective funding sources, and lead to the creation of innovative financial instruments for biotechnology research financing. Deep learning methods

are quickly spreading into the field of biomedicine since they have begun to outperform humans in various tasks, such as image recognition. DL is widely applied in drug discovery and repurposing [[→ 65](#), [→ 66](#), [→ 67](#)]. Pharmaceutical companies employ ML extensively in clinical trial operations; however, no validated reports have been published regarding the utilization of ML/AI predicted chances for technical as well as regulatory success methods throughout pharma R&D processes [[→ 68](#)]. Although there are numerous methods and resources available for predicting clinical trial outcomes, the main techniques used in these methods were readouts from previous phases of clinical trials or trial design [[→ 69](#), [→ 70](#), [→ 71](#)]. The goal was to integrate all scoring strategies, biological backgrounds, and multimodal data sources with the most recent deep-learning models.

Furthermore, employing a pipeline consisting of biological analytical instruments and deep learning algorithms, the in-silico team accurately forecasted the outcomes of phase I and phase II clinical trials for the first time. Interatomic, transcriptome, and pathway activity scores were employed to compute the target network scores. The scores were derived using the unique biomedical knowledge graph and the Panda Omics platform employed by in-silico Medicine. Pathway perturbation analysis and dimensionality reduction were performed using the in-silico pathway activation network decomposition analysis (iPANDA) tool. The algorithm provides a dependable approach for target identification and assessment of the synergistic effects of dual-target inhibitors and combinations, as demonstrated in [→ Fig. 6.3](#) [[→ 72](#), [→ 73](#)]. Furthermore, ML is a potent

subset of AI that has enhanced translational research and possesses the ability to expedite drug discovery through sophisticated data-driven decision-making solutions [[→74](#), [→75](#), [→76](#), [→77](#), [→78](#)]. The utilization of ML in medication development has demonstrated significant benefits in modern times. Target identification, patient categorization, lead generation, de novo chemical synthesis, biological activity forecasting, medication repurposing, and precision medicine are included in these facets. In-silico methodologies utilizing AI and ML can enhance every stage of the drug discovery process [[→79](#), [→80](#), [→81](#)]. The development of software and analytics platforms that integrate ML algorithms and in-silico statistical tools has made preclinical research and screening analyses much easier over the past few years. If phenotypic screening is designed and assessed based on preconceived notions about biology, it is more biased than target-based screening, but it still has some inherent bias [[→82](#), [→83](#), [→84](#), [→85](#)]. Virtual screenings have been a successful way to find novel therapeutic targets and identify leads for a while now, saving time and money [[→86](#)]. Ultra-screens, which use high-throughput docking (HT docking) and structure-based algorithms to identify high-quality hit molecules and sample a significant portion of the available chemical space, are now a reality due to DL algorithms and in-silico advancements in hardware and computing power. As a result, there are fewer false-positive results and more high-affinity binding agents are produced [[→87](#), [→88](#)].

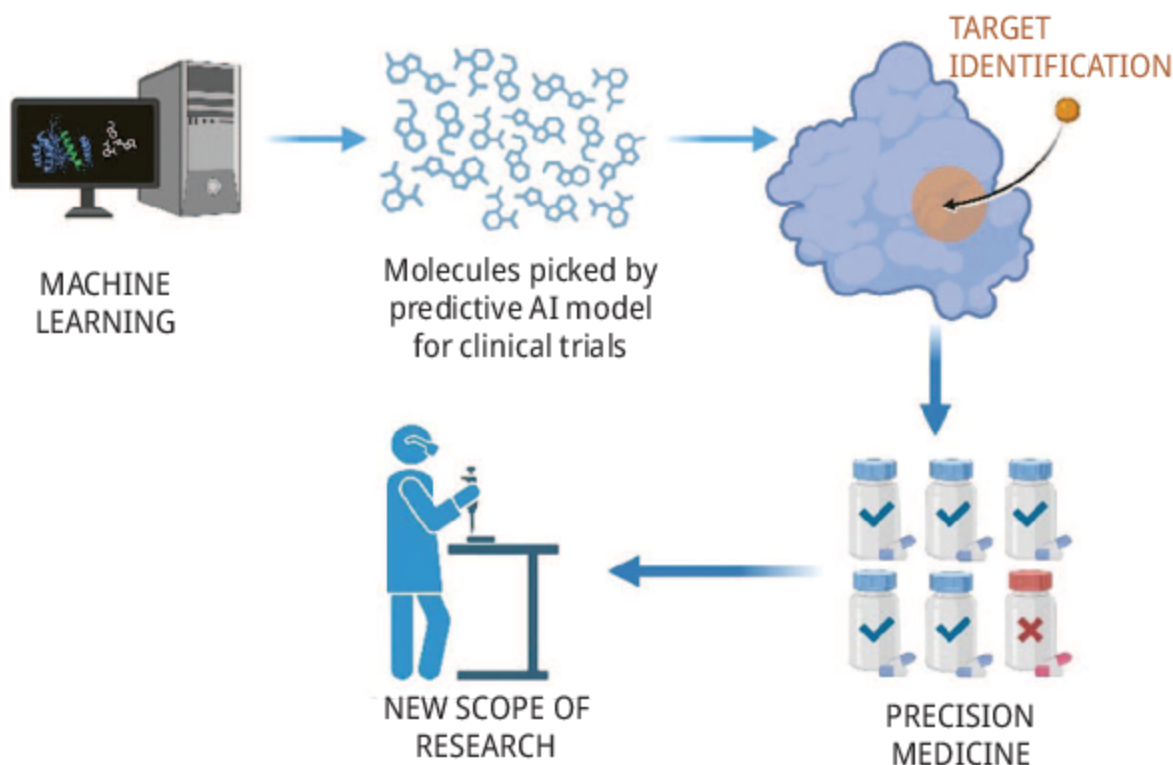


Fig. 6.3: Pictorial representation of drug target identification via machine learning.

Target deconvolution and analysis of chemical abundance (MoA) can be conducted utilizing various methodologies. These technologies encompass protein microarrays, chemical proteomics, affinity chromatography, and functional genomics techniques, including CRISPR and RNA interference. A team of researchers involved in the project created a platform that can forecast drug targets [[→ 89](#)]. It uses a Bayesian ML method that combines various forms of data collection to predict drug-binding targets. Furthermore, it used hierarchical clustering as a case study to find de novo tubulin possibilities. It also predicted targets for small compounds with unknown targets but promising clinical activity. Additionally, drug target interactions (DTIs) are helpful in the identification of potential medications,

which further restricts the scope of research that can be conducted in the field of biochemical experimentation. Moreover, they may provide substantial new insights into the drug's mechanism of action and adverse effects [[→ 90](#)]. For this reason, DTI prediction is a necessary prerequisite to drug discovery, utilizing ML that integrates chemical and genomic space [[→ 91](#)].

Validation of the pharmaceutical process is one of the most important and widely accepted aspects of current good manufacturing practices (cGMPs). It would appear that the quality system (QS) regulation necessitates the confirmation of the process [[→ 92](#)]. It is the purpose of a quality system to ensure that the products being manufactured are always suitable for the functions they are designed to fulfill. The training and validation divide from the target selection and trial design approaches was applied to the meta-model to train and validate it. Retrospective validation of the human trial transition prediction [[→ 93](#)] revealed impressive performance. Conversely, the retrospective validation method may be prone to various biases and may not correctly represent the precision of predictions produced in real-world scenarios. The majority of companies providing estimates are publicly traded small and mid-cap pharmaceutical enterprises. The study design was shaped by the proactive distribution of press releases by publicly traded small and mid-cap pharmaceutical companies concerning the results of their phase II clinical trials. As part of pipeline updates for the former, the big-cap and privately held companies, on the other hand, might either not disclose any clinical study results at all for the latter or disclose

the annual discontinuation of phase II clinical programs [[→ 94](#)]. Both of these scenarios are possible. On the other hand, when compared to large-cap pharmaceutical companies, the securities and derivatives that are available for small- and mid-cap pharmaceutical companies are significantly less liquid. Because of this, it is becoming increasingly difficult to short small and mid-cap pharmaceutical stocks. As a result, it may be necessary to make use of derivatives such as put option contracts to construct investment portfolios that are of high quality.

6.4 Machine learning algorithm in pharmacokinetics and pharmacodynamics

Pharmacokinetic and pharmacokinetic/pharmacodynamic (PK/PD) modelings are essential methodologies in drug discovery, development, regulatory assessment, and approval. Pharmacokinetic/pharmacodynamic modeling is crucial for the design of clinical trials, data analysis, and dosage determination. Non-compartmental analysis, lacking structural model assumptions, produces macroscopic pharmacokinetic parameters (area, mean residence time, and variation of residence time) that define the temporal variations of concentration levels in the circulation [[→ 95](#)]. Nonetheless, the methodology has constraints in extrapolating for applications such as including temporal pharmacodynamic data and fails to delineate the temporal evolution of pharmacokinetic data. Furthermore, blood circulation and the anatomical composition of the organs are used in

physiologically based pharmacokinetic modeling (PBPK) to more precisely represent actual physiology. Nevertheless, the methodology continues to rely on structural models and requires rather extensive data for modeling. Furthermore, it is based on numerous difficult-to-validate assumptions [[→96](#)]. A multitude of structure-based PK/PD models exist, each dependent on the drug's mechanisms of action, the feasibility of doing PK/PD sampling, and additional factors. Deep learning is presently widely utilized in numerous fields, including pharmaceutical research, due to the development of increasingly robust deep learning neural networks. There has been an increasing interest in employing neural networks for modeling pharmacokinetics (PK) and pharmacokinetics/pharmacodynamics (PK/PD) [[→97](#)]. The researchers demonstrated that the network proficiently captured the temporal correlations in the pharmacokinetic/pharmacodynamic (PK/PD) relationship by utilizing a long short-term memory (LSTM) network to assess the PK/PD data of a hypothetical drug based on an indirect response model. The LSTM network was applied to examine the data used for the evaluation of PK/PD data [[→98](#)].

A novel deep learning technique creates PK/PD models that can simulate the effects of unknown dosage regimens and forecast patient response time courses by directly learning the governing equations from data. The method is unique in two respects. First, the deep learning model's architecture guarantees the preservation of the pharmacological principle of the dose-concentration effect. This means that the model presupposes a causal relationship in which drug concentration is driven by dosage, and effect

dynamics are subsequently driven by dosage. Second, the network architecture is set up so that the dosing data enters the model through two different ports, guaranteeing that the model can simulate the patient's dosing regimen in addition to predicting the treatment data that is currently available [[→ 99](#)]. Quantitative structure-activity relationship (QSAR) models, used to predict ADME properties and analyze complex pharmacokinetic profiles of drugs and nanoparticles, have been developed using ML techniques. Furthermore, researchers initially used support vector machines, a notable ML method, as a prospective tool in quantitative structure-activity relationship (QSAR) studies to forecast the tissue/blood partition coefficients (K_p) of organic compounds in various tissues. Subsequent research utilized ML techniques to predict many pharmacological parameters, including intestinal absorption in individuals. Researchers conducted multiple studies evaluating tissue and plasma concentrations, as well as various pharmacokinetic parameters (including AUC, C_{max} , and $t_{1/2}$) in both people and animals, by combining a ML-based QSAR model with a simplified PBPK model [[→ 100](#), [→ 101](#)]. Likewise, additional researchers computed the pharmacokinetic parameters, such as AUC, C_{max} , V_d , and mean residence time, employing the ML/AI-based pharmacokinetic/pharmacodynamic modeling program BIOiSIM. BIOiSIM, an AI biosimulation program, utilizes the PBPK model, distinguished by numerous separate compartments that represent essential organs. The model was incorporated with existing in vivo datasets to

forecast the first-order absorption rate constant (h_1), K_p , and the blood-to-plasma ratio. The PBPK model was subsequently executed [[→ 101](#)].

Researchers computed the interdose AUC of everolimus in recipients of renal transplants using XGBoost ML algorithms. Because the training dataset containing real patient data was smaller than that of other immunosuppressants, ML algorithms were taught on patient, simulated, and hybrid data. The estimates were subsequently compared between MAP-BE AUC estimates, derived from trapezoidal AUCs obtained from an independent database of transplant recipients with kidneys, which had been compiled through extensive sampling and a three-point limited sampling strategy utilized by the ISBA expert system. Consequently, simulated training datasets of varying sizes were evaluated using many measures, with imprecision (i.e., RMSE) as the primary parameter [[→ 102](#)]. The best outcomes were obtained by ML algorithms that were trained on numerous simulations without patient data. However, RMSE does not represent the exact error in a specific patient; rather, it represents imprecision within the dataset. Researchers trained a PK model on a limited set of PK profiles, and during that process, they generated a MAP-BE. The model was not, however, independently validated. In a similar line, PK in whole blood and mononuclear cells in peripheral blood was explained by a different team of researchers using a PK model that did just a little bit better than the algorithm that was trained on patient data [[→ 103](#), [→ 104](#)]. Despite training on a large amount of data, the XGBoost algorithm's performance was only slightly better than the previous MAP-BE. Because of the extremely high

peak concentration at an hour, these unusual flat profiles either led to an underestimate of the atypical cases or an overestimation of the cases. The modeling approach, which intentionally anticipated a wide range of scenarios, presumably addressed these scenarios better. To enhance the ML algorithms and attain an equivalent degree of precision and accuracy for a pair of drugs, it would take a long time to gather the same amount of real patient data [[→ 105](#), [→ 106](#)]. Researchers chose to enhance ML as a test that could be used for other drugs by combining patient as well as simulated profiles in the training dataset or by relying exclusively on a high number of simulations. The last strategy turned out to be the most successful since it yielded superior outcomes over the whole AUC value range of the external validation dataset [[→ 107](#)].

The primary difference between pharmacometrics (PM) and ML methodologies in drug development, as noted by various researchers regarding physiological plausibility and clinical relevance, is that PKPD models are grounded in both statistical analysis and fundamental biological principles; these models can vary from compartmental algorithms to comprehensive mechanistic models, such as physiology-based pharmacokinetic models. In contrast, ML is frequently, though not always, a purely data-driven approach. The biological interpretation of PM parameters can facilitate the identification of underlying mechanisms. On the other hand, ML parameters can be understood mathematically, though less so from a biological standpoint. However, some approaches to AI, like causal AI, offer more interpretability [[→ 108](#)]. In PM, bias can also be

introduced into the analysis if previous mechanistic knowledge is imputed incorrectly; however, it is not the case in ML. In PM, a model is manually constructed iteratively, with the modeler determining the next step by continuously assessing the model's performance at each stage. It leads to a laborious and time-consuming process that may result in varying models from the same dataset by different modelers. Although ML is a more rapid and efficient approach compared to PM, it fails to adequately consider it [[→ 109](#)]. ML algorithms encounter challenges with sparse and tiny data sets, prompting the development of PKPD modeling tools to elucidate temporal variations in biomarkers. To address the deficiency in a minimally documented RWD dataset, PKPD predictions are incorporated into a rudimentary ML model [[→ 110](#)]. Future research may evaluate the efficacy of the hybrid method against more sophisticated ML approaches under varying degrees of data sparsity. The hybrid PKPD/ML model must be integrated into MIPD software to facilitate clinical decision-making [[→ 111](#), [→ 112](#), [→ 113](#), [→ 114](#)]. The hybrid PKPD/ML model, equipped with numerous predictive characteristics, must be connected with the EHR to eliminate cumbersome data entry at the point of treatment. Despite utilizing publicly available data, certain values may remain unknown in specific therapeutic settings. Utilizing imputation with standard values may be suitable for specific missing data, such as laboratory results that minimally influence model predictions. The user's predicting uncertainty may then be amplified by the resultant uncertainty. Augmenting the training data led to diminished prediction bias and enhanced recall in the PKPD model

simulations; however, it concurrently decreased accuracy on other measures. Larger synthetic data sets resulted in significantly lower effectiveness as the hybrid ML model became more like the data-generating PKPD model, requiring extensive manual tuning [[→ 115](#)].

Nonetheless, forecasts must be continually assessed against empirical data, as ML methodologies can easily reassess the pharmacokinetic-pharmacodynamic model and its intrinsic model inaccuracies. Therefore, exercise caution when executing the plan. The PKPD model utilized for enrichment was constructed using the same data as the ML models, which is a limitation of the methodology. It suggests that specific data obtained from the PKPD parameters are redundant with other ML features. For instance, regimen-related attributes are integrated into the advanced ML models with the PKPD components, which encompass the chemotherapy regimen. The disparities in regimen feature significance between the enriched model and the base ML model exemplify the link. If the data communicated between the PKPD and the ML models were diminished, the efficacy of both the enriched and augmented models may differ [[→ 116](#)].

6.5 Ethical challenges AI-enabled for clinical trials

Studies on the ethics of AI in healthcare are blatantly unbalanced, emphasizing topics other than global health, and more especially disease surveillance. There are few explanations of the ethical, legal, or practical

issues needed for widespread adoption or large-scale implementation of AI-driven treatment in global health, according to a large portion of research in this area. Also, there is the interdependence of sources and data types needed for reporting and decision-support, as well as the surveillance of diseases at the individual and population levels [[→117](#), [→118](#), [→119](#)]. Applications of AI in imaging, electronic medical records (EMR), laboratory assessment and treatment, novel drug discovery, preventive strategies, precision medicine, biological data analysis, procedural optimization, and information management for healthcare organizations have profoundly transformed the medical sector. Nevertheless, the scientific domain must confront several ethical and legal issues. It is essential to consider several parameters, as depicted in [→Fig. 6.4](#). The elements encompass social disparities, data protection and privacy, informed permission, medical consultation, empathy, and sympathy [[→120](#), [→121](#), [→122](#), [→123](#), [→124](#)]. Union-based data processors or controllers manage personal information in compliance with these requirements, alongside the operations of international organizations and corporations to guarantee sufficient protection of personal data [[→125](#)]. The Genetic Information Non-Discrimination Act (GINA) prohibits employers in the United States from making discriminatory decisions based on an individual's genetic health information. AI can assess consumer health data and medical device imagery, thereby expediting health research and improving diagnosis and outcomes. Moreover, the procedure of engaging with a healthcare professional regarding ethical disclosure, decision-making ability, and

documentation is referred to as informed consent [[→ 127](#)]. Ethical duty dictates that patients are entitled to information regarding their diagnosis, health status, planned care, therapy outcomes, test results, charges, insurance plans, and other pertinent medical details. Patient consent must be unequivocal, explicit, and suitable for the intended purpose. Concerns about the issue have intensified when AI is implemented in healthcare applications.

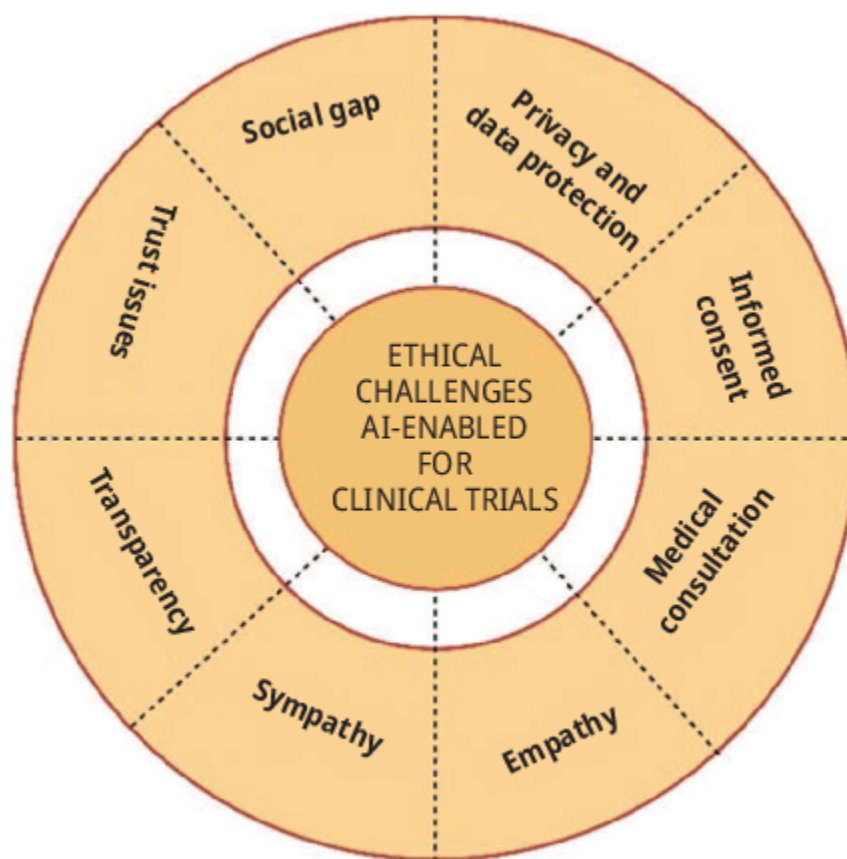


Fig. 6.4: Pictorial representation of ethical challenges in AI-enabled clinical trials.

Another issue that threatens societies as a result of AI development is the problem of social gaps. People in every nation on Earth face increasing

levels of social inequality and a dwindling sense of social justice with every new advancement, discovery, and invention. Even though AI makes it easier to access more information about science, technology, global politics, environmental concerns, and international occurrences, it enhances socioeconomic disparity [[→128](#)]. Due to the uniqueness of human emotions, it is plausible that the evolution of medical robots and humans will not occur concurrently at an accelerated pace. Medical practitioners and other healthcare professionals ought to consult with or provide counsel to their peers, in contrast to autonomous (robotic) systems. In autonomous systems, it is impractical. Conversely, it is highly improbable that patients will acquiesce to “machine-human” medical contacts instead of “human-human” interactions. The idea that medical professionals, including doctors and nurses, would provide treatment in a compassionate and nurturing setting can significantly influence patients’ healing capacity [[→129](#)]. Robotic doctors and nurses are not capable of doing it. When patients interact with robotic nurses and doctors, they will no longer feel empathy, kindness, or appropriate behavior because these machines are not human. Patients also won’t be capable of behaving appropriately [[→130](#)].

Similar to the way data privacy concerns the scope of such data, confidentiality deals with the appropriate handling, distribution, and storage of sensitive information [[→131](#)]. Consulted sources indicate that security threats significantly affect confidentiality in the deployment of AI inside healthcare. Transferring and disseminating patient data to external parties may render it susceptible to security breaches or privacy infringements

[[→132](#), [→133](#), [→134](#)]. Furthermore, mobile devices are used in the majority of research projects on mobility disorder detection systems to gather relevant signals, transfer them to other computers for evaluation, visualize the findings, and notify users of the findings. Transparency is the capacity to understand how an AI system decides. To be more precise, the black box's basic analytical steps and decision-making process are opaque. Algorithms that are programmed on biased data or that disregard particular demographic traits make these problems worse [[→134](#), [→135](#)]. For example, an AI system used in the United States to predict recidivism for defendants charged with crimes showed that African Americans with less serious offenses had higher risk scores than White Americans with more serious offenses [[→136](#)]. Trust from the public, patients, and patients themselves will ultimately depend on transparency and the widespread application of AI in healthcare. Explainability is the next sub-issue associated with the main ethical problem, following transparency. Because ML algorithms are opaque, it can be challenging to comprehend and articulate the relationships between input data and outputs. It is related to explainability and transparency [[→137](#), [→138](#)]. Users' trust in AI systems can be fostered or strengthened by explainability [[→139](#)]. The level of confidence in the application of AI in medicine is enhanced by the accuracy of the explanations offered by the AI system. Moreover, given that AI is an essential element of healthcare and is employed in the medical sector, patient safety remains a paramount concern. The sub-issue addresses the potential harm or risks posed by AI tools or hazardous AI in the medical

domain. A few benefits of incorporating AI into healthcare include improving patient safety and care quality, increasing access to healthcare, providing patients and clinicians with local real-time advice, and identifying medical emergencies like sepsis. AI-enabled support services, however, are not error-proof, and their algorithms might provide patients with risky advice or choices that could jeopardize their health [[→ 140](#)].

6.6 Conclusion

AI has profoundly influenced medication research and clinical trials, with deep learning methodologies enhancing traditional machine learning techniques. AI can forecast the toxicity and efficacy of novel therapeutic molecules, pinpoint new targets, and formulate more potent pharmaceuticals. Premium datasets, including the ChEMBL database, ChemDB, and DGIdb, enhance drug development in the era of big data. AI can propose optimal chemical structures, drug-target interactions, synthesis pathways, and structure-activity relationships (SAR). AI can facilitate the identification of specific illness cohorts for Phase II and III clinical trials, enabling more accurate predictions of therapeutic targets. High-throughput virtual screening (HTVS) presents a cost-effective and efficient alternative to traditional approaches, although its potential and accessibility remain constrained. Methods for target deconvolution and molecular abundance analysis encompass protein microarrays, chemical proteomics, affinity chromatography, and functional genomics techniques. Deep learning

methodologies have been utilized to create pharmacokinetic/pharmacodynamic models that anticipate patient response time profiles and simulate the impacts of experimental dosage regimens. Furthermore, ethical issues related to AI-driven clinical trials in healthcare include societal inequities, data privacy and security, informed consent, medical oversight, empathy, and compassion. Thus, AI can greatly enhance the optimization and integration of the formulated pharmaceutical in the suitable dosage form in the future. AI can assist in assessing the safety and efficacy of products during clinical trials prior to market introduction. However, several challenges remain before widespread adoption of the technology. Nevertheless, it is highly probable that the pharmaceutical industry will soon recognize AI as a valuable asset.

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Chapter 7 AI-driven clinical decision support systems for pharma executives

Abstract

Artificial intelligence (AI) is a powerful tool that makes data processing, analysis, and interpretation possible. The following chapter offers information on the AI algorithms employed by pharmaceutical corporations in their clinical decision support systems. There are difficulties in implementing AI-CDSS, and future possibilities are also covered. AI-based algorithms such as logistic regression, support vector computations, naive Bayes, random forests, decision tree models, and ANN are expected to be critical in optimizing medication compositions and enhancing treatment outcomes, while also anticipating potential toxicity along with side effects in patient-specific dose formulation and safety evaluations. AI-powered CDSS helps doctors, nurses, and patients make informed decisions by evaluating patient data. NLP improves patient communication and electronic health records. Through meticulous analysis of the patient's signs, symptoms, medical records, and contextual information, an NLP-driven Clinical Decision Support System can propose potential diagnoses, treatment strategies, and pharmacological interventions. These difficulties include organizational and social factors, human factors, and a lack of confidence among medical professionals.

Keywords: clinical decision support, pharma, support vector machine, artificial neural network, decision tree, ethical considerations,

7.1 Introduction

The rising significance of artificial intelligence (AI) and machine learning (ML), along with the expanding accessibility of clinical data, has intensified interest in AI applications, namely computerized clinical decision support systems (CDSS). Initially, CDSSs were regarded as capable of supplanting doctors in decision-making processes. By analyzing significant data, a more sophisticated and contemporary understanding of the role of CDSSs is to aid physicians in making better decisions than the healthcare provider or the CDSS could accomplish separately. Contemporary Clinical Decision Support Systems (CDSSs) often offer recommendations to doctors, who are anticipated to make independent decisions and dismiss any suggestions they consider unsuitable [[→1](#)]. Large multinational pharmaceutical corporations have led the industry's vigorous use of AI, ML, and data mining technology to minimize the duration and expenses associated with medication research and development. During the business and management planning stage of drug development, artificial intelligence technologies must be strategically employed in addition to the exploratory drug discovery procedure [[→2](#)]. Pharmaceutical companies wonder what kind of medications to target for their next research project, taking into account the managerial and market conditions of their business [[→3](#)]. The majority of the time, choices about drugs under development may have been made qualitatively based on the

opinions of the board of directors of the company or the technologies that the companies own. In other words, human judgment has historically played a larger role in drug development decision-making than formal analytic or data-based methodologies. An efficient model or approach that can raise the success rate of drug development is therefore desperately needed [[→4](#)].

Artificial neural networks, decision trees, random forests, naïve Bayes, logistic regression, support vector machines, and numerous other AI methodologies have been successfully utilized in drug development. The integration of various models enhances and optimizes the drug discovery pipeline more efficiently than conventional machine learning methods. This is where collective education proves advantageous. Random forest analysis is an ensemble method for decision trees that produces individual trees by picking random samples using the bagging technique [[→5](#)]. The support vector machine (SVM) is an additional tool that improves predictive accuracy by utilizing a unique approach to minimize structural risk throughout the learning process while increasing the complexity of the mathematical functions necessary for categorization. The multilayer perceptron, characterized by a complex arrangement of concealed layers, constitutes the many hidden layer architecture in deep learning neural networks. It enhances model generalization by reducing overfitting and optimizing hyper-parameter selection, resulting in superior performance [[→6](#)]. Artificial neural networks (ANNs) are widely employed in drug release studies, pharmacokinetics/pharmacodynamics (PK/PD) modeling,

and the simulation of biological processes. The ANN is employed to develop controlled release formulations by determining the appropriate dissolution properties, bioavailability, pharmacokinetic profile, and in vitro–in vivo correlation (IVIVC) [[→7](#)]. Implantable microchips are effectively used in continuous/pulsed drug delivery methods as well as drug reservoirs for regulated drug delivery. According to reports, AI technology can provide integrated implantable robots for gene and drug delivery applications, allowing for the optimization of target-specific drug delivery. Ongoing research on magnetized microrobots for ocular drug delivery may prove beneficial for ocular targeting therapy. Nanorobot path optimization is supported by quorum sensing and bacterial foraging strategies. Bio-nanorobots are used to count the number of biomarkers associated with cancer as shown in [→Fig. 7.1](#) [[→8](#), [→9](#)]. Theranostics agent delivery along with disease monitoring is made easier with medical capsule robots (MCR), which use artificial intelligence systems in place of wireless capsule endoscopy (WCE) [[→10](#), [→11](#), [→12](#)].

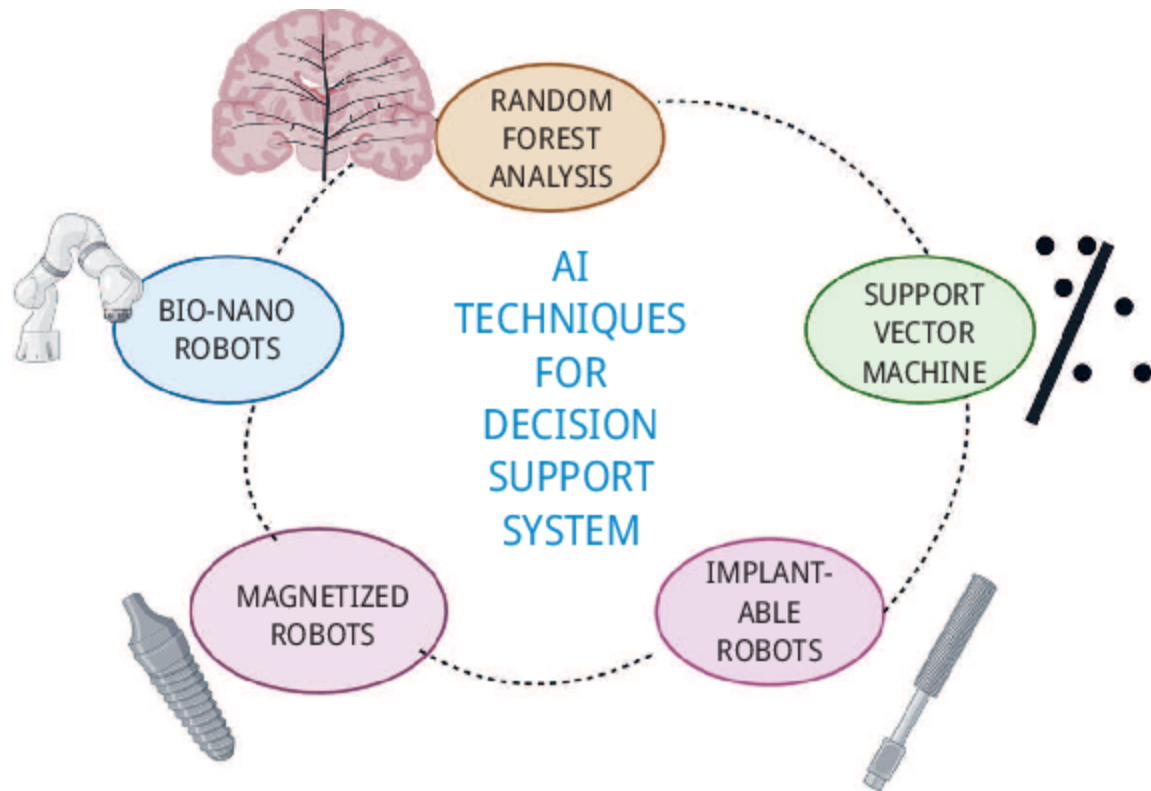


Fig. 7.1: Various AI techniques for decision support Systems in Healthcare.

AI and ML algorithms are becoming more and more important in biomedical research, and they are influencing clinical decision-making processes. Consequently, they directly affect medical practice and the exchange of information between doctors and patients [[→ 13](#)]. The foundation of the traditional doctor-patient relationship is open communication about medical advice and treatment choices, as well as personal trust. Decisions become opaque if the doctor is no longer the only one making the diagnosis or choosing the best course of treatment, but is instead heavily influenced by a machine using AI and learning algorithms. CDSS come in a variety of forms and classifications, with the ability to be delivered actively or passively and the timing of interventions among their

many subcategories. It is frequently separated into categories based on knowledge and those based on non-knowledge [[→ 14](#), [→ 15](#)]. Knowledge-based systems generate rules based on patient, practice, or literature-based evidence. The system subsequently collects data to evaluate the rule and produces an output or action. Non-knowledge-based CDSS require a data source; however, rather than utilizing specialist medical knowledge for decision-making, they depend on statistical pattern recognition, ML, or AI. Despite being a quickly expanding use case for AI in medicine, CDSS is fraught with difficulties, such as those relating to data collection and comprehending the logic that the AI employs to generate recommendations (black boxes) [[→ 16](#)].

Pharmaceutical formulations are extremely complex structures with a wide range of variables that can occasionally have poorly defined effects on how they behave. Pharmaceutical formulations are extremely difficult to characterize using conventional analytical methods due to their complex preparation technology. Hundreds of well-defined physicochemical parameters are being reduced to theoretically useful descriptions that will not influence decision support in the future. The characterization of specific difficulties in pharmaceutical technology continues to depend significantly on empirical data. Because of this, empirical modeling is the method of choice in the field to create the right model when numerical analysis of the data is used [[→ 17](#)]. It makes it possible to build the model solely from the data, negating the need for prior knowledge and a priori assumptions. The data, which encapsulates the current understanding of the issue, is utilized

to construct the model. The primary advantage of a data-driven model is its impartiality, as it is free from established theories. In the absence of well-founded theories, partially validated theories or hypotheses from other domains may lead to misconceptions [[→18](#)]. ANNs exemplify an empirical modeling tool that has become highly effective for empirical modeling applications. Additionally, CDSS are employed for triage, detecting alterations in health-related symptoms, extracting patient data from medical records, providing in-patient assistance, evaluating treatments, and monitoring. As researchers and developers prepare to implement CDSS, they must consider the five “rights” of CDSS: the right information, the right person, the right intervention format, the right channel, and the right workflow time, as shown in [→Fig. 7.2](#) [[→19](#), [→20](#)].

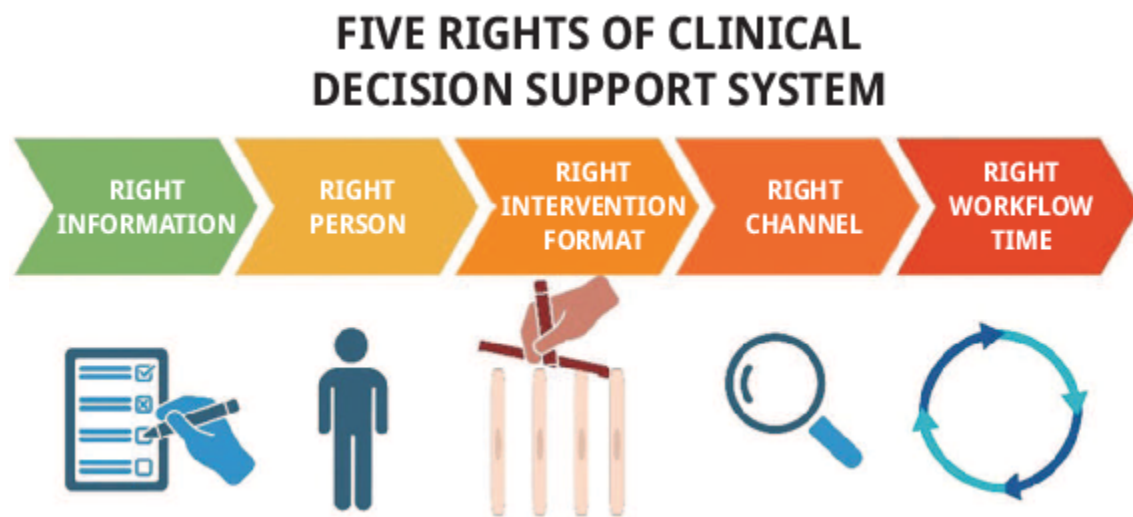


Fig. 7.2: Pictorial representation of the five rights of a clinical decision support system.

7.2 Key components of AI-driven decision support system

AI-driven CDDS can aid physicians, nurses, patients, and other carers in making more informed choices. It analyzes patient data from the past, present, and future to identify faults or deficiencies in the treatment route and propose enhancements or safety issues to the user. The two types of decision support systems are ML systems and rule-based expert systems (ES) [[→21](#)]. Many hospitals' electronic health records (EHR) have been incorporated into decision support systems, which employ pre-established rules to guide decision-making. Because these systems only require binary choices, they are simpler to configure. ML systems, on the other hand, rely on statistics to make decisions. It necessitates a thorough comprehension of the standard process by which medical professionals evaluate patient data and reach important conclusions [[→22](#)]. An AI system designed to diagnose *Helicobacter pylori* infection during endoscopy was shown to detect the infection with good accuracy and exploitability in a recent case-control study. These tools can assist pathologists in identifying *H. pylori* infections or take their place. When it comes to gastric cancer, AI's diagnostic precision and efficacy are on par with or even better than those of general pathologists. Through convolutional neural networks, deep learning models can identify and classify esophageal squamous cell carcinoma and its depth of invasion in a matter of minutes or seconds. It works well for detecting adenocarcinomas as well [[→23](#)]. AI models have been effectively developed to assess prognosis in patients with colorectal cancer (CRC), biliary tract cancers, and hepatocellular carcinoma (HCC) [[→24](#)]. The prognosis for gastric cancer is presently uncertain. Whole slide

images (WSIs) were utilized to develop a deep learning model in cohort research that produced the Tumor Risk Score (TRS) as a prognostic marker. Clinical staging approaches failed to equal TRS's capacity to uniformly categorize patients into up to five groups with markedly different prognoses, attributable to its predictive efficacy. Though they have a lot of potential, AI-driven decision support systems still need to be worked out before being applied in clinical settings. These challenges include data uniformity, effectiveness, morality, government approval, as well as medical acceptability. For gastroenterology to begin to utilize AI technology, these issues must be resolved. AI-powered decision support systems can enhance gastroenterology by offering quick and accurate diagnosis, individualized treatment regimens, and enhanced patient care [[→ 25](#)].

The healthcare sector has experienced a significant rise in the use of AI applications, driven by the swift advancement of AI and other emerging technologies, which possess the capacity to generate new opportunities and eliminate current obstacles through effective data processing and analysis [[→ 26](#)]. Today's healthcare industry is better off avoiding many mistakes and unfortunate incidents due to clinical decision-making technologies and electronic health records (EHR). Examples of these technologies include those for entering pharmaceutical orders, predicting several critical diseases, and more [[→ 27](#), [→ 28](#)]. Healthcare institutions in wealthy nations are adopting AI with enthusiasm, as evidenced by a review of several real-world applications in the medical field. These facilities are utilizing AI-

enabled systems to help medical staff diagnose and treat patients for a range of illnesses, as well as to maximize the effectiveness of nursing and administrative duties related to hospital services. When choosing a CDSS, there are two primary factors to take into account: the CDSS's quality and its suitability for bridging the clinical quality gap. A CDSS's quality must be evaluated at least on two levels: the technological platform level and the data or knowledge level utilized in its construction. Generally accepted quality assurance techniques for developing software used in the field of medicine should be used in the design, implementation, testing, and documentation of CDSSs, as they are software that may also be medical devices [[→29](#)]. In the instance of CDSS built around models learned through machine learning or statistical analysis, an evaluation of the source data's quality is required [[→30](#), [→31](#)]. Further hazards linked to AI-based CDSSs encompass bias and overfitting to the training data, among other factors. It may lead to undetected poor performance at the local location, as the CDSS cannot extrapolate from the training data. Therefore, considerable attention must be paid to alleviate any possible negative effects of CDSSs. It is crucial to carefully choose the CDSS that aligns with the site's requirements. CDSSs necessitate rigorous acceptance testing, commissioning, and quality assurance at the local facility, similar to any other medical technology [[→32](#), [→33](#)]. ML is a discipline that examines the creation and advancement of algorithms capable of learning from and predicting outcomes based on data. These algorithms generate a model from sample inputs, enabling data-driven predictions or judgments rather than

relying solely on static program instructions. A technique known as SVM constructs a hyperplane to identify outliers and analyze data for regression and classification. The technique utilizes a set of labeled training samples to create the maximum separation between two linearly distinguishable categories. The instances are depicted as points in a spatial framework, and new instances are categorized according to the side of the gap they occupy when projected into the same space. The kernel trick employs the dot product of vectors in a designated space to convert non-linear data into a linear representation. SVM is a popular substitute for AI in clinical management decision models. Nonetheless, the limitations of SVM encompass its incapacity to provide direct probability estimations and suboptimal performance when the feature count surpasses the sample size [[→ 34](#), [→ 35](#), [→ 36](#), [→ 37](#), [→ 38](#)]. Decision tree learning is a predictive modeling technique that employs statistics, data mining, and machine learning to create a decision tree from item observations, resulting in inferences about the item's target value. Decision trees are classified into two main types: (1) discrete decision tree models, in which the goal variable assumes only discrete values, and (2) continuous decision trees. Decision-making and execution can be clearly and visually depicted through a decision tree. Random forests, also known as random decision forests, are ensemble learning methodologies employed for classification, regression, and various other applications. They function by generating many decision trees during the training phase and producing a class that represents the average prediction (for regression) or the most frequent class (for

classification) from the individual trees. Random decision forests reduce the tendency of decision trees to overfit their training datasets [[→ 39](#), [→ 40](#)]. A machine learning technique called Deep Learning Neural Network (DLNN) uses representation learning to enable systems to automatically find the necessary representations. Deep artificial neural networks and other methods are used for identifying and categorizing features from unprocessed data. DLNN may acquire intricate features from data using a hierarchical feature representation by employing several hidden layers. In contrast to networks with fewer layers, DLNNs execute input data (features) via a higher volume of mathematical operations, rendering their training more computationally intensive [[→ 41](#), [→ 42](#)]. The ANN, CNN, and DLNN models have been thoroughly analyzed as AI-based CDSS engines. Radiology is frequently employed for the diagnosis of medical conditions (e.g., breast cancer through mammography), stratification/classification (e.g., lung diseases), imaging-based prediction (e.g., CT imaging in smokers), imaging outcome prediction (e.g., pulmonary edema on V/Q scans), and multi-omics genotype-phenotype evaluation of disease behavior (e.g., multiple sclerosis). In medical imaging applications of AI, DLNN have been especially favored. These applications encompass picture segmentation, expedited image capture, lesion detection, and imaging-based clinical decision support [[→ 43](#), [→ 44](#), [→ 45](#)].

Better data-driven decision-making, more efficient communication among healthcare professionals, improved patient care are all impacted by AI-based CDSS enhancement in documentation quality [[→ 46](#), [→ 47](#), [→ 48](#)].

Furthermore, treating glaucoma patients presents a daily challenge for clinicians. Treatment of glaucoma requires the synthesis of data from multiple visits over an extended period from various sources (tonometry, pachymetry, perimetry, optical coherence tomography disc pictures, ocular exams, patient history, and so on) [[→49](#), [→50](#)]. These data have a high test-retest variability and are frequently complex. The patient's unique circumstances must be taken into consideration when interpreting the data. An additional challenge is that, in the middle of a busy glaucoma clinic, decisions must be made swiftly based on the evaluation and integration of the data. Certain challenges may be addressed by recent developments in the area of predictive modeling [[→51](#), [→52](#)]. Predictive models for glaucoma must deliver clinically relevant information to decision-makers at the optimal time within the clinical workflow to enable informed judgments that substantially improve clinical practice. Clinical decision support (CDS) technology can aid in mitigating these challenges [[→53](#), [→54](#)].

Contemporary Clinical Decision Support (CDS) systems, which are computer-based instruments, are intended to aid clinicians in enhancing decision-making for specific patients [[→55](#)]. These devices have successfully addressed various non-ocular conditions, including diabetes, cancer, sepsis, acute respiratory distress syndrome, hyperglycemia, and hyperbilirubinemia in neonates [[→56](#)]. The use of diagnostic tests and treatment choices can both be enhanced by CDS systems [[→57](#), [→58](#), [→59](#)]. An example of an effective CDS system is an add-on app for an electronic health record (EHR) that manages neonatal bilirubin levels.

7.3 Integrating AI into pharma executive decision-making

The increasing usage of AI-based applications has given rise to worries regarding data security and privacy. Health information is often the target of data breaches due to its sensitivity. Thus, patient data security is crucial [[→ 60](#)]. Legal actions have been initiated due to data sharing between major healthcare institutions and AI companies, raising concerns among patients regarding potential breaches of their privacy throughout the data collection process. After data acquisition, the ensuing challenge is the advancement of AI technology [[→ 61](#)]. Overfitting may arise when a system identifies spurious associations between patient attributes and outcomes. It results from having an excessive number of variable parameters compared to outcomes; consequently, the algorithm predicts using unsuitable features [[→ 62](#)]. Recent years have seen a significant increase in interest in the application of AI in the pharmaceutical industry and clinical healthcare, which has resulted in a growing corpus of relevant literature and research. It offers information on how AI can improve clinical results and patient care, which is consistent with healthcare management and personalized medicine [[→ 63](#)]. Big data practices consider upcoming talent requirements and fresh applicants in the healthcare industry. Big data presents a significant opportunity for medical practitioners, pharmacy personnel, and health policy specialists to make data-informed decisions that will eventually improve patient care [[→ 64](#), [→ 65](#)]. It has the power to alter how healthcare

HR practices are ingrained and equipped to boost productivity across the board, gain a deeper understanding of data repositories, and reach a stated opinion or choice. Big data has been essential to any human resources department in the following situation, not only for keeping employees but also for their ability to better understand their surroundings and make the best decisions [[→ 66](#), [→ 67](#)]. Three strategies can be used in a big data-driven decision-making process as shown in [→ Fig. 7.3](#) [[→ 68](#), [→ 69](#)].

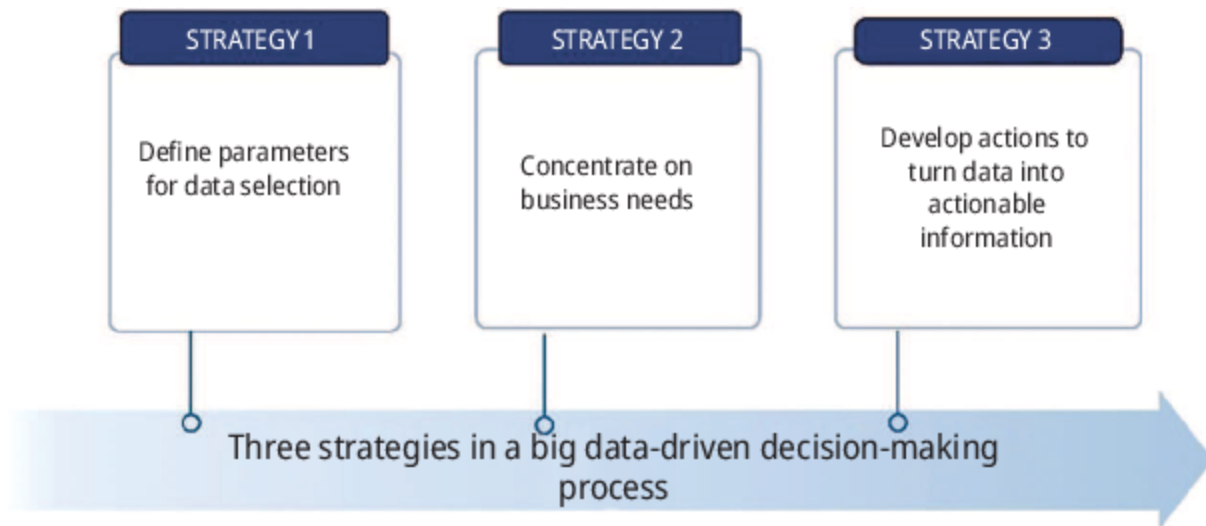


Fig. 7.3: Three strategies used in a big data-driven decision-making process.

Researchers have proposed utilizing ANNs' ability to reveal concealed information and values to enhance care efficiency and support the healthcare sector in adopting "lean thinking" or value-based decision-making [[→ 70](#)]. An example of how ANNs aid in the incorporation of Lean thinking in healthcare is through the modeling of "information flow" among cancer patients, demonstrating the relationship between quality of life and clinical outcomes [[→ 71](#), [→ 72](#)]. Support vector machines are considered

advanced methodologies for addressing issues that conventional statistical analysis cannot answer. They are utilized for modeling high-dimensional data. Despite their analytical capabilities, broad implementation remains challenging, mostly due to methodological complexities and scalability concerns [[→73](#)]. A systematic examination of deep learning models employing data from electronic health records identified issues associated with information inconsistency and timeliness, including concealed correlations among clinical features arising from both short-term and long-term events. The models' performance may decline if these difficulties are not sufficiently addressed [[→74](#)]. Research demonstrates that ANNs are extensively utilized in business applications, especially in decision assistance, stock price prediction, and financial distress and bankruptcy matters. Furthermore, hybrid networks have been developed to improve the efficacy of traditional models in commercial environments. The integration of ANNs with secondary AI and meta-heuristic approaches, including fuzzy logic, genetic algorithms, bee colony optimization, and artificial immune systems, has been suggested to address challenges related to ANNs, such as network structure selection, initial weight assignment, and control parameter optimization. Hybrid intelligent systems are employed in various fields, including natural language processing, robotics, medical evaluation, and manufacturing process oversight [[→75](#)].

The efficacy of AI-CDSS deployment is influenced not only by technological progress but also by human issues; including explainability, privacy, and equity, together with the social and organizational dimensions

of medical practice [[→76](#), [→77](#), [→78](#), [→79](#), [→80](#)]. For example, because end-user clinicians in the following case do not trust or comprehend AI-CDSS's capabilities, they may be reluctant to adopt or use it. One reason could be that many AI-CDSS operate like "black boxes," meaning that clinicians are unable to understand how they come up with the recommendations. These problems might result in low system uptake and user acceptance [[→81](#), [→82](#)]. Researchers involved in the project assert that the success of a cooperative alliance between human decision-makers and AI-integrated systems heavily relies on the trust that users have in these systems. It indicates several variables influencing confidence in CDS systems [[→83](#)]. For instance, the two sources from which the participants sought validation of medication recommendations were the guidelines published by professional societies and the articles published in reputable clinical journals. Although guidelines are regarded as more reliable recommendations, physicians are aware that guidelines change more slowly than findings from publications. Fundamentally, faith is placed in the underlying processes of clinical research, particularly randomized clinical trials [[→84](#)]. The randomized clinical trial method has long been subject to criticism due to its purportedly tenuous connection to the actual medical experience and is not without controversy [[→85](#)]. It has been demonstrated that participants in clinical trials tend to be younger, sicker, and more male than those who would use the experimental treatment or medication in question.

Validating and putting AI to use in medicine are still major tasks that need to be completed. AI technology currently requires significant innovation in governance and regulatory oversight before it can be widely used in a safe and morally responsible manner. It is imperative that AI be validated in clinical settings and that algorithms be published in peer-reviewed journals [[→86](#)]. Perhaps in the future, AI should collaborate with scientists and medical professionals to give patients the best care possible rather than trying to take the place of specialists. AI is unlikely to handle all aspects of patient admission, diagnosis, and treatment selection. Therefore, it is imperative to attend to a patient's spiritual and social needs as well. It is doubtful that machines will be able to provide these kinds of care [[→87](#)]. For any algorithm to be validated in a clinical setting, some training is necessary. Patient data must therefore be exchanged between healthcare facilities and occasionally across international borders [[→88](#)].

7.4 Regulatory compliance and ethical consideration

The pharmaceutical regulatory affairs sector may experience a comprehensive revolution with the application of AI, notwithstanding its present nascent adoption in the domain. AI can assist in regulatory matters by providing current knowledge on the newest developments in the sector. It is particularly accurate, considering its potential benefits for drug discovery and regulatory approval. AI can expedite and enhance decision-

making for professionals in drug discovery by identifying novel prospects, automating tasks, and augmenting decision-making capabilities. Their ability to expedite the introduction of new medications is enhanced as a consequence. The term “regulatory affairs” refers to the divisions or units inside pharmaceutical companies that act as a point of contact for national and international regulatory bodies whose job is to grant product approval by current rules and regulations. Regulatory bodies, on their part, grant product approval based on data and records submitted during drug discovery and product development [[→ 89](#), [→ 90](#)]. Examples of consumer technology advancements with applications in the healthcare industry include blockchain, internet-based computing, 3-D bioprinting, and AI. In light of these advancements, new regulations are necessary to keep up with the quickly evolving field of technology [[→ 91](#)]. The US Food and Drug Administration (FDA) is a primary catalyst for the advancement of these breakthrough technologies, establishing new objectives to enhance the regulatory approval process and offering technical standards and recommendations. The extensive uses and recent progress in AI algorithms indicate that medical technology and healthcare will be profoundly affected. Artificial general intelligence (AGI) denotes artificial intelligence that can do any cognitive activity achievable by a human, demonstrating intelligence akin to that of a human being. Artificial narrow intelligence (ANI), a type of AI commonly utilized in medical devices, is defined as AI engineered for a specified purpose, as per an FDA assessment [[→ 92](#), [→ 93](#)].

The GDPR establishes a new legal framework for scientific research that profoundly affects the rights of data subjects. Union or Member State legislation may permit exceptions to the rights of access (Article 15), rectification (Article 16), erasure (Article 17), restriction of processing (Article 18), and objection (Article 21) when personal data is used for research and development purposes. These exceptions are granted only if the fulfillment of the research objectives would be impossible or would be materially hampered by the rights; otherwise, no exceptions will be granted [[→ 94](#)]. The GDPR does not offer complete protection against AI systems since it solely addresses two aspects of AI system regulation: the processing of private information and shielding data subjects from ADM. In the domains of healthcare services and medical research, the regulation of these entities requires a more complex legal framework accompanied by rigorous enforcement [[→ 95](#)]. The European Commission (EC) underscored the necessity for a regulatory framework to facilitate the new technology while managing its associated dangers, notwithstanding the absence of a distinct legal structure for AI within the EU to date (European Commission, 2020). The ramifications of AI technology for the economy, legal framework, and society have rendered it a significant policy issue for the European Union in recent years [[→ 96](#)].

All relevant players must comply with and enforce the same ethical standards and principles governing the development and application of AI technology in the healthcare sector. Artificial intelligence technology utilizes many data sets and techniques, including supervised, semi-

supervised, and unsupervised learning. Health practitioners and academics exercise caution regarding these “machine-driven” analytical techniques, despite their potential, due to their intricacy. AI for Health, unlike other AI applications, directly influences human life and can have significant repercussions for patients in various aspects. The implementation of AI technologies in healthcare increases the likelihood that the system may become autonomous, hence jeopardizing human autonomy. Artificial intelligence in healthcare possesses the capacity to enable machines to make decisions. Medical decision-making and the AI-based healthcare system should be fully under human control. Under no circumstances should AI technology interfere with patient autonomy. The user or patient may not know which to believe when it comes to clinical decisions made by the AI system and the doctor because of differences in judgment [[→ 97](#)]. The patient must be given both options in such circumstances. All research initiatives and evaluation programs must adhere to a consent protocol before the deployment of any AI technology within the healthcare sector. The advantages of employing AI technologies, along with the related social, psychological, and physical concerns, must be adequately communicated to patients. Patients must be completely empowered to accept or reject AI technologies. Moral considerations and human values must be diligently and transparently monitored at every phase of AI development and implementation.

The implementation of AI technologies in healthcare increases the likelihood of the system becoming autonomous, hence jeopardizing human

autonomy. Artificial intelligence in healthcare possesses the capacity to enable machines to make decisions. Medical decision-making and AI-driven healthcare systems must remain entirely under human oversight. AI technology must not, under any circumstances, compromise patient autonomy. The patient may be uncertain about which authority to trust regarding clinical judgments made by the AI system and the physician due to these discrepancies. The patient must be presented with both alternatives in such situations. All research initiatives and evaluation programs must adhere to a consent protocol before the deployment of any AI technology within the healthcare sector. Patients must be fully acquainted with the advantages of using AI technologies as well as the risks – social, psychological, and physical – that come with using them. The ability to accept or reject AI technologies must be fully granted to patients. At every stage of the development and application of AI, moral considerations and human values should be effectively and transparently monitored [[→98](#)].

The most ideal feature for any prognostic or diagnostic tool used in AI healthcare is trustworthiness. Similar to how they must develop confidence in other tools they use, clinicians must also develop confidence in AI technologies. Clinicians and healthcare providers want a straightforward and reliable method to evaluate the validity and reliability of AI technologies for optimal use. A reliable AI-driven system must deliver precise health data analysis while also being legitimate, ethical, dependable, and authentic. AI-driven technology must ensure data security and privacy during the complete development and implementation phases. Securing the

confidence of all stakeholders, particularly medical receivers, in the safe and secure use of their data is essential for the effective and extensive deployment of AI [[→ 99](#)]. The primary aim of data privacy is to avert unauthorized access, alteration, and destruction of personal information. The use of personal data by artificial intelligence must not excessively limit individuals' perceived or actual freedom. These procedures are essential in the healthcare business, as medical information contains sensitive data that could be exploited against patients or result in prejudice, even inadvertently. Data about specific patients should ideally be anonymized unless clinical or research needs require that it be kept in an identifiable format. The duty of an individual or organization to take responsibility for their actions, account for their activities, and transparently disclose the outcomes is known as accountability. AI systems that are meant to be used in the healthcare industry need to be prepared to be examined at any time by relevant authorities. For AI technologies to operate at their best, frequent internal and external audits are required. These audit reports have to be released to the general public. The data used to train and test the system is what essentially determines the outcome of artificial intelligence, as it is a data-driven technology. It is particularly important when it comes to AI for health, as biased and skewed datasets can cause issues with bias and discrimination, among other issues. Artificial intelligence for health and other data-driven technologies are thought to be most vulnerable to data bias [[→ 100](#)]. It is imperative to implement measures to guarantee that the

“training data” is free from established biases and accurately represents a substantial segment of the intended audience.

7.5 Future trends of AI in the pharmaceutical industry

Artificial intelligence will usher in a transformational era for the pharmaceutical industry. Given the potential side effects and toxicity, AI algorithms are anticipated to be crucial in tailored dosage formulation and safety assessments, refining pharmaceutical compositions, and improving treatment outcomes. Wearable technologies, sensors, and artificial intelligence-enabled monitoring devices will enhance remote patient care and optimize medication adherence via tailored treatment recommendations. Artificial intelligence impacts patient selection, recruitment, and the design of clinical studies. Genetic profiles and electronic health records are employed to find qualifying patients, reduce trial costs, and accelerate approval procedures. Furthermore, it is anticipated that AI’s continuous surveillance and oversight capabilities will optimize pharmaceutical manufacturing processes, guaranteeing efficiency and uniformity [[→ 101](#)]. As stated in certain current conceptualizations, strategic decisions are not limited to plant design or global supply chain management but rather affect all functional areas and levels of the organization. In a similar vein, decisions made in the short and medium term now span far larger domains and are not limited to production

scheduling and planning tasks. Furthermore, to better meet the demands along with the dynamics of the real world, it is now imperative to integrate the various decision levels [[→102](#)].

Therefore, a company's current decision structure is far more intricate and wide-ranging, encompassing various time scales as well as enterprise functions and activities, impact and scope variations, physical network variations, and finally, multiple technological settings. In the future, the patient will be the primary influencer in determining the pertinent operational performance metrics as well as the specifications for the product or service [[→103](#)]. To improve the client experience from initial discovery to therapy delivery, companies must integrate the patient's perspective into their decision-making processes. Decision-support models will proficiently tackle challenges such as a broader range of products and volumes, patient-focused distribution networks, improved service level precision, and real-time evaluation of the whole customer experience via closed-loop information systems [[→104](#), [→105](#)]. All of the company's technological assets are housed in the technological layer. The organization must make use of all of those capabilities to optimize the value of all assistances that can be provided with decision models that have been developed. Therefore, whenever possible, economical and/or environmentally friendly alternatives should be considered when making decisions about the most pertinent future features and particular technology selections. At the manufacturing level, production expenses, lead times, and inventory levels will inevitably diminish as more economical technologies,

such as continuous manufacturing, are adopted and advanced decision-support tools, based on intricate optimization and simulation methods, are developed. Enhancements in resource and energy efficiency, along with increased flexibility and responsiveness to market variations, are anticipated to result in elevated capacity utilization [[→106](#), [→107](#)].

A multidisciplinary process should take into account the requirements of all stakeholders when developing an ML-based CDSS. This is particularly valid when these systems are integrated with XAI. Decision-makers who utilize the systems, people impacted by the decisions' outcomes, and system designers should all be taken into account [[→108](#), [→109](#)]. It is recommended that experts in the domains of psychological and social behavioral psychology, mental health, and cognitive science collaborate when developing models [[→110](#)]. Unstructured medical data continues to be a difficulty for the advancement of practical AI-based systems, despite the ability of XAI to aid in identifying data-related difficulties. It is necessary to consider domain-specific requirements, such as having a clear grasp of the system's goal, evaluating the effectiveness and interpretability of the systems already in place, and determining the kind and volume of necessary explanations. Moreover, scientists [[→108](#)] recommend against using those black-box simulations unless necessary. They recommend against using complicated algorithms that call for the post-mortem application of XAI methods whenever feasible. Rather, they suggest using transparent or interpretable by-design algorithms. The audience's cognitive abilities and limitations, as well as implications for ethics, justice, and

safety, must all be taken into account when determining the kind of explanations that should be given. More thorough user studies are required [[→ 111](#)]. Some authors [[→ 112](#)] claim that while giving doctors a more detailed explanation of the data used to generate the system's recommended diagnosis boosted trust, it also led to issues with over-reliance. Less thorough explanations, on the other hand, had a reverse impact, leading participants to doubt the system's dependability and develop problems with self-reliance [[→ 113](#)]. Some researchers discovered through an investigation that inaccurate ML recommendations can have an impact on clinicians and reduce decision accuracy, and that explanations alone are insufficient to address an over-reliance on an algorithm that proposes incorrect decisions. It was discovered that the choice of explanation strategies should be based on the clinicians' experience with ML and that the recommendations from ML were more useful to those who had prior experience.

However, generative chemistry, ML, and methods like Pareto optimization are increasingly being used in AI approaches for hit and lead discovery. Community-based platforms that provide resources and information for AI4DD technologies are strong supporters of the trend. Predictive toxicology is making significant strides, and regulatory bodies are positioning themselves to profit from AI [[→ 114](#)]. AI technology can lower healthcare expenses by increasing productivity and decreasing mistakes. It is essential to acknowledge that there remain certain constraints and challenges associated with the utilization of AI in pharmacy that require resolution. For example, using AI-powered systems to track information

about patients may raise security and privacy issues. Pharmacists might also require additional training to properly apply and incorporate artificial intelligence into their work. Future research may also focus on the ethical implications associated with the utilization of AI in pharmacy. As technology advances, it is essential to address concerns such as algorithmic bias, data security and confidentiality, and the possibility of AI systems supplanting human labor. By addressing these concerns judiciously and proactively, pharmacists can integrate AI in a manner that is both ethical and efficacious [[→ 115](#)].

7.6 Conclusion

AI, ML, and data mining are being used by pharmaceutical companies more and more to improve model generalization, drug discovery, and decision support systems (DSS). There are two categories for these technologies: knowledge-based and non-knowledge-based. Empirical modeling is the preferred method for creating the right model in pharmaceutical formulations, eliminating the need for prior knowledge and assumptions. AI-powered CDSS can help doctors, nurses, and patients make informed decisions by examining patient data from the past, present, and future. Two types of decision support systems are ML systems and rule-based ES. AI systems have shown good accuracy in diagnosing *H. pylori* infections and gastric cancer, while deep learning models identify and classify esophageal squamous cell carcinoma and adenocarcinomas. Natural language

processing, or NLP, is being used increasingly in the medical field to give clinicians a way to communicate with patients and to enhance the use of unstructured electronic health records, or EHRs. AI-CDSS implementation in medicine faces challenges due to human factors, social and organizational aspects, and a lack of trust among clinicians. AI technology in healthcare has the potential to become autonomous and threaten human autonomy, but it must be fully under human control and informed about its advantages and risks. Trustworthiness, data security, privacy, and frequent audits are crucial for AI-based diagnostic tools. In the future, patients will be the primary influencers in determining operational performance metrics and product specifications. Cost-effective technologies and advanced decision-support tools can lead to decreased production costs, lead times, and inventories. Collaboration between experts in social and behavioral science, mental health, and cognitive science is recommended for developing ML-based decision-support systems.

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Chapter 8 AI-based pharmacovigilance and drug safety

Abstract

Pharmacovigilance (PV), a scientific and applied field, aims to identify, evaluate, understand, and minimize adverse drug-related effects. The chapter provides information about artificial intelligence-based tools for pharmacovigilance, drug safety, and its regulatory aspects. It encompasses the scientific and applied endeavors made to identify, assess, comprehend, and mitigate the adverse consequences of drugs or possible drug-related issues. ADR detection and reporting, technological AE coding, safety individual reporting, severity assessment, and connection to suspected drugs are among the most significant duties in the PV sector. To gather data regarding adverse drug reactions (ADRs) along with adverse drug events (ADEs), process reports according to international clinical standards, obtain drug-drug interactions, estimate drug side effects, indicate clinical trials, and incorporate estimation unpredictability toward diagnostic classification algorithms, artificial intelligence (AI) is being utilized for pharmacovigilance and patient safety. The lack of cross-specialty opportunities for training, privacy issues, data models for health care organizations, and insufficient education and training in nations with middle and low incomes are some of the challenges that have been addressed in the context of pharmacology as well as drug safety.

Keywords: pharmacovigilance, drug safety, adverse drug effects, natural language processing, XAI, regulatory aspects,

8.1 Introduction

The Greek word “pharmakon” (meaning drug), as well as the Latin word “vigilare” (meaning to keep watch), are the sources of the term “pharmacovigilance.” Detecting, evaluating, comprehending, and minimizing drug side effects or potential drug-related issues are the scientific and practical endeavors known as “pharmacovigilance” (PV) [[→1](#)]. On a global scale, pharmacovigilance systems are gathering a substantial volume of data daily. Processing the extensive dataset poses a significant challenge. Recently, the program’s scope was extended to include herbals, blood-related products, medical devices, traditional and complementary medicines, herb vigilance, hemovigilance, and materio vigilance [[→2](#)]. Two examples of the unique characteristics of artificial intelligence (AI) are machine learning (ML) and natural language processing (NLP). ML techniques examine structured data, including genetic and imaging data. Unstructured and free-text forms can be distinguished by NLP, an interpretive system that can understand and process human language. AI is also utilized in search engines, clinical settings for treatment, detection, and outcome prediction, and medical diagnosis. With appropriate monitoring, it is also utilized in the healthcare industry to prevent health issues in people [[→3](#)]. Prioritizing patient safety was the primary objective of PV’s design since patients’ exposure to treatment medications throughout clinical trials as well as studies is limited.

It allows for the long-term observation and use of drug profiles. It also covers specific populations, including the elderly, racialized, pediatric, and pregnant populations. PV studies are necessary due to the lack of complete data from long-term drug exposure [[→4](#)]. Lifesaving medications like antiretroviral, antitubercular, and anticancer drugs are approved under a fast-track system, making them readily available to patients. PV is in charge of weighing the advantages and disadvantages of these medications.

After a new drug is introduced to the market, drug safety is currently facing significant risks. The main cause of erosion during a clinical trial or post-marketing is the drug's unpredictable toxicities, which can result in morbidity and mortality even at normal dosages. ADR data must be gathered from pharmaceutical companies by the license holder and submitted to the local drug regulatory body. ADR detection and reporting, technical AE coding, safety individual report preparation, severity assessment, and linkage to suspected drugs are the most crucial tasks in the PV industry. As a result, a new technology is required for the detection of ADRs, as current methods rely on time-consuming human intervention. An AI and ML system was created by a multinational pharmaceutical company and a consulting firm to make it easier to process and maintain quality and compliance standards [[→5](#)–7]. AI and ML are being investigated for their potential to influence PV throughout its whole lifecycle [[→6](#), [→7](#)].

Furthermore, ML has been used to find complex patterns, or syndromes, in safety data and to perform a range of predictive modeling tasks.

Furthermore, pharmacoepidemiology, the study of drug interactions with

human populations, is using machine learning (ML) extensively to examine the advantages, drawbacks, and practical applications of drug use [[→8](#), [→9](#)].

Pharmacovigilance also uses machine learning to assess how well NLP skills classify unstructured text in patient safety reports of incidents. NLP can serve as a layer of protection by identifying cases that result in severe harm or death, based on researchers' testing of the system's capacity to classify free text within patient security incidents autonomously and decide the severity of harm outcomes. Several studies looked at the possibility of using machine learning to filter reports on patient safety from sources such as electronic health records [[→10](#)]. Researchers have discovered that near-miss reports gathered by passive surveillance monitoring systems, as well as large, semi-structured, or disorganized collections of adverse events, can be screened and evaluated effectively using text mining and machine learning algorithms [[→11](#)]. A more focused strategy was employed by the researchers, who created a deep-learning model and tested it on various data sets to identify allergic reactions in the free-text narration of hospital safety reports and assess the model's generalizability [[→12](#)]. According to the study, the model may be applied to enhance allergy treatment, possibly allowing for system enhancement and real-time monitoring events for medical errors. The absence of risk to a patient that can be avoided and the decrease in the possibility of harm coming from receiving medical care are known as patient safety. The healthcare system is integrating AI, which is

changing current dynamics (like the role of medical providers) and creating new opportunities to improve patient safety [[→13](#), [→14](#)].

Due to machine learning, computer systems can now identify latent information and make predictions about the data using both labeled (supervised learning) and unlabeled (unsupervised learning) data, all without the need for explicit programming. Among AI's many applications, ML, along with natural language processing, stands out for its social impact and frequent use in the medical field [[→15](#)]. Reinforcement learning is the third type of machine learning, where an algorithm learns from its successes and failures as it works to complete a task ([→Fig. 8.1](#)). Deep learning and artificial neural networks are also included in machine learning. The goal of natural language processing is to enable computers to comprehend human language and convert text into machine-readable, organized information that can be further studied using machine-learning techniques [[→16](#), [→17](#), [→18](#)]. Electronic health records (EHRs), which may be gathered from mobile devices and regular medical attention, also contain real-time, practical health data. It might result in a potentially proactive monitoring and safety signal identification strategy. The advent of ML and NLP techniques has created new opportunities for identifying drug safety signals in the free-text EHR data, which is primarily unstructured [[→19](#)].

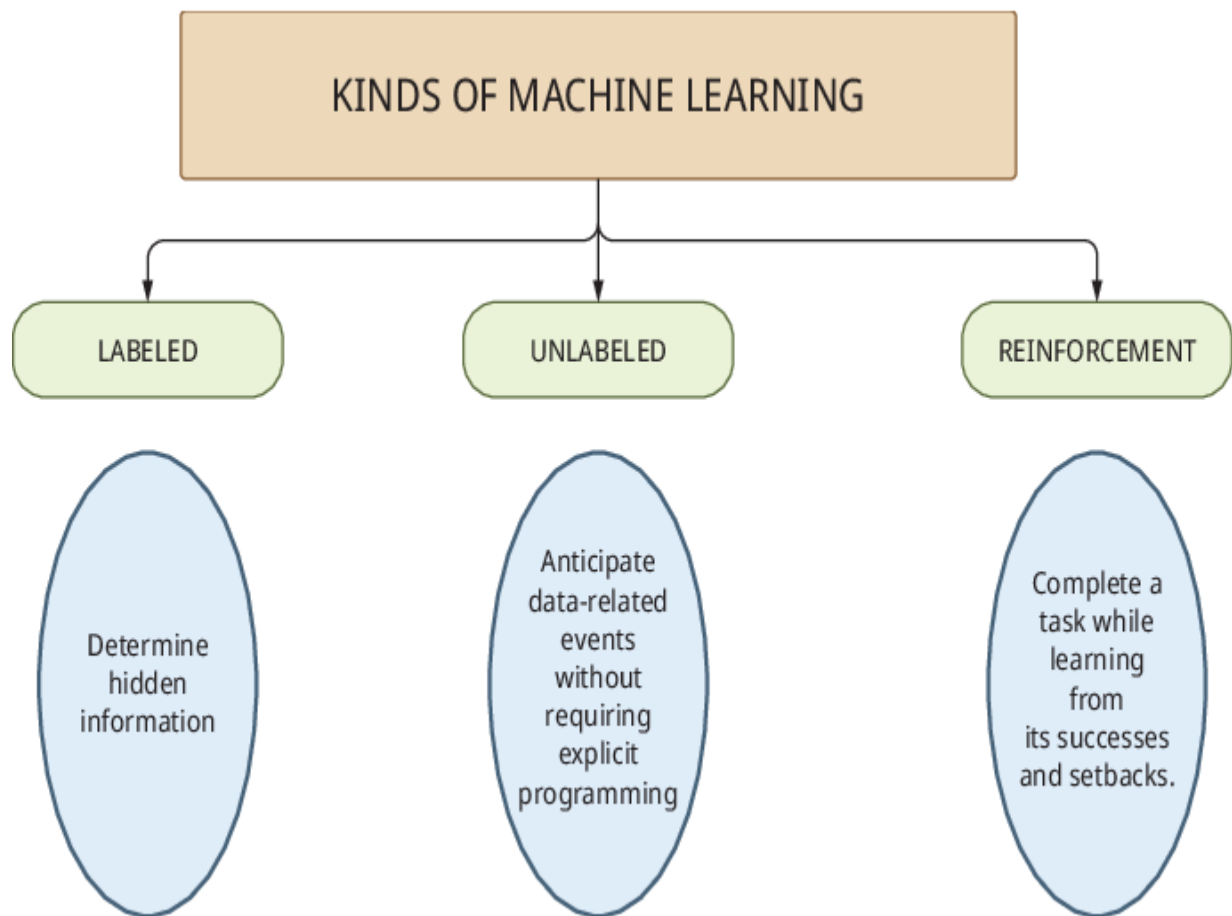


Fig. 8.1: Different kinds of ML for the study of pharmacovigilance.

8.2 Foundation of AI in pharmacovigilance

To improve patient safety and pharmacovigilance, AI is being actively used to collect data on adverse drug reactions (ADRs) and adverse drug events (ADEs), process international clinical standards reports (ICSRs), extract drug-drug interactions and predict their effects, identify high-risk populations for ADRs and inform personalized care, predict drug side effects, portray clinical trials, and incorporate prediction unpredictability into diagnostic classifiers [[→ 20](#)]. Researchers identified patient diversity

and polypharmacy as two potential applications of machine learning for ADR detection [[→21](#)]. Pre-marketing to aftermarket safety evaluations are two stages of the drug development process where these opportunities may arise [[→22](#)].

Automation using machine learning techniques is growing in value due to the variety of disease states, medications, as well as ADRs that patients encounter. A reference standard is compared by certain organizations using machine learning to speed up the utilization of adverse event reports, which is semi-supervised and rule-based [[→23](#)]. Researchers examined and contrasted information posted on social media platforms regarding patient-reported medication outcomes for type 2 diabetes, rheumatoid arthritis, cystic fibrosis, and asthma. While they find that social media platforms are appropriate and complementary sources for analyzing drug outcomes, they also urge greater attention to be paid to developing more dependable techniques for text mining and natural language processing. Furthermore, researchers [[→24](#)] evaluated patient-written reviews of pharmaceutical oncological drugs that were posted to user forums, and they made a comparison between official drug labels and the reported issues. Support vector machines were used to efficiently categorize attitudes toward ADRs. Additionally, other scholars [[→25](#)] looked into the potential of modeling the data exchange among users regarding drug-related therapies for depression using content created by users from a depression-related forum. They discovered that NLP can be used to more precisely identify the ADRs of these medications, and they validated the ADRs with the help of relevant

medical literature. Another use of machine learning in pharmacovigilance is evaluating how well NLP can classify unstructured text in patient safety incident reports. Researchers discovered that by identifying cases that cause serious injury or death, NLP is a kind of safety net. To determine the degree of harm outcomes, they assessed how well free text in patient safety incident reports could be automatically classified. However, it is not a perfect method, and it cannot yet take the place of manual review entirely [[→ 26](#)].

Moreover, XAI can guarantee the interpretability and explainability of AI models by offering a more user-friendly support system to physicians and patients. Along with improving the model's transparency and traceability, it can also streamline the evaluation process. There is a lot of promise in methodically managing and tracking XAI-based models and working closely with clinicians to improve the system through ongoing fine-tuning that will enable more accurate and efficient clinical decision support delivery. By managing rules, risks, and other requirements, PV XAI's transparency can also lower the costs associated with PV model governance, cut down on the need for expensive mistakes made during manual inspection, and lower the possibility of unintentional bias [[→ 27](#)]. Supervised ML learns methods to reduce the gap between its predictions and the (known) reality represented in the labels by using labeled data. Conversely, because patterns appear in high-dimensional spaces and are typically invisible to humans, unsupervised machine learning techniques try to identify them in unlabeled data. Consequently, the labels in supervised

learning can reveal whether or not a patient has had an adverse drug reaction in a pharmacovigilance setting, and the resulting model is trained to differentiate individuals [[→28](#), [→29](#)]. Large volumes of textual data are produced, among other places, on social media, blogs, and forums. Parallel to it, tremendous advancements in artificial intelligence have produced effective NLP techniques and algorithms that make it possible to process and comprehend text produced by humans. It created new avenues for text analysis and social media mining [[→30](#)]. These fields have seen notable innovations in recent years. Text mining is the process of analyzing textual data, such as unstructured or semi-structured text, to find patterns and information hidden within. It thus blends NLP and data mining. The necessity to examine vast volumes of human-language text to extract insights that support data-driven decision-making gave rise to text mining [[→31](#), [→32](#)].

To improve pharmacovigilance, statistical analysis and machine learning techniques have gained increasing traction in recent years. In a study, two machine learning techniques were used: decision trees as well as inductive logic programming, as shown in [→Fig. 8.2](#), to look into common side effect profiles depending on combined drug and target descriptors. Of these, the inductive-logic programming approach is the most effective in relational learning, outperforming decision trees and leveraging not only drug properties, but also past knowledge [[→33](#)]. In a different study, single nucleotide mutations were analyzed using artificial neural networks; however, no significant findings were obtained. Several commonly used

algorithms for drug toxicity predictions, including cancer risk, mutagenicity, hepatotoxicity, acute oral toxicities, as well as cardiovascular toxicity, include naive Bayesian, support vector machines, random forests, and back propagation neural networks. These techniques have a well-established theory and are simple to apply [[→34](#)]. Moreover, Naive Bayesian (NB) is a simple Bayesian probabilistic classifier. The feature vectors in the analysis are largely unaffected by the decision variables. Every piece of data needs to be categorized as input, and the output serves as the training sample and characteristic attribute. Each feature attribute for each class of conditional probability estimations is categorized during the classifier training stage, after which the frequency of occurrence of every group in the training sample is calculated. A later classifier is used to categorize the classification items. Using sources of drug information, the following approach has been studied for side-effect prediction [[→35](#)]. It has also been researched for the preclinical safety evaluation of novel medications, as well as for the forecasting of the carcinogenicity, mutagenicity, and hepatotoxicity of various substances. ANNs are inspired by biological neural networks, which use example-based learning to solve problems in the real world [[→36](#)]. It is a paradigm for using highly networked neurons to process information. It is positioned as a scientific and engineering study specialist due to its remarkable capacity to handle imprecise datasets and discover intricate patterns from complex data sets. It has been researched to predict side effects from medication side information [[→37](#)]. By extracting word subsequences that characterize ADRs, the

technique has been used to classify them [[→ 38](#)]. Potential applications have included the analysis of single nucleotide variation changes for drug safety, preclinical safety evaluations of new drugs, and modeling and simulation-based prediction of the carcinogenicity and mutagenicity of chemicals [[→ 39](#), [→ 40](#), [→ 41](#), [→ 42](#)].

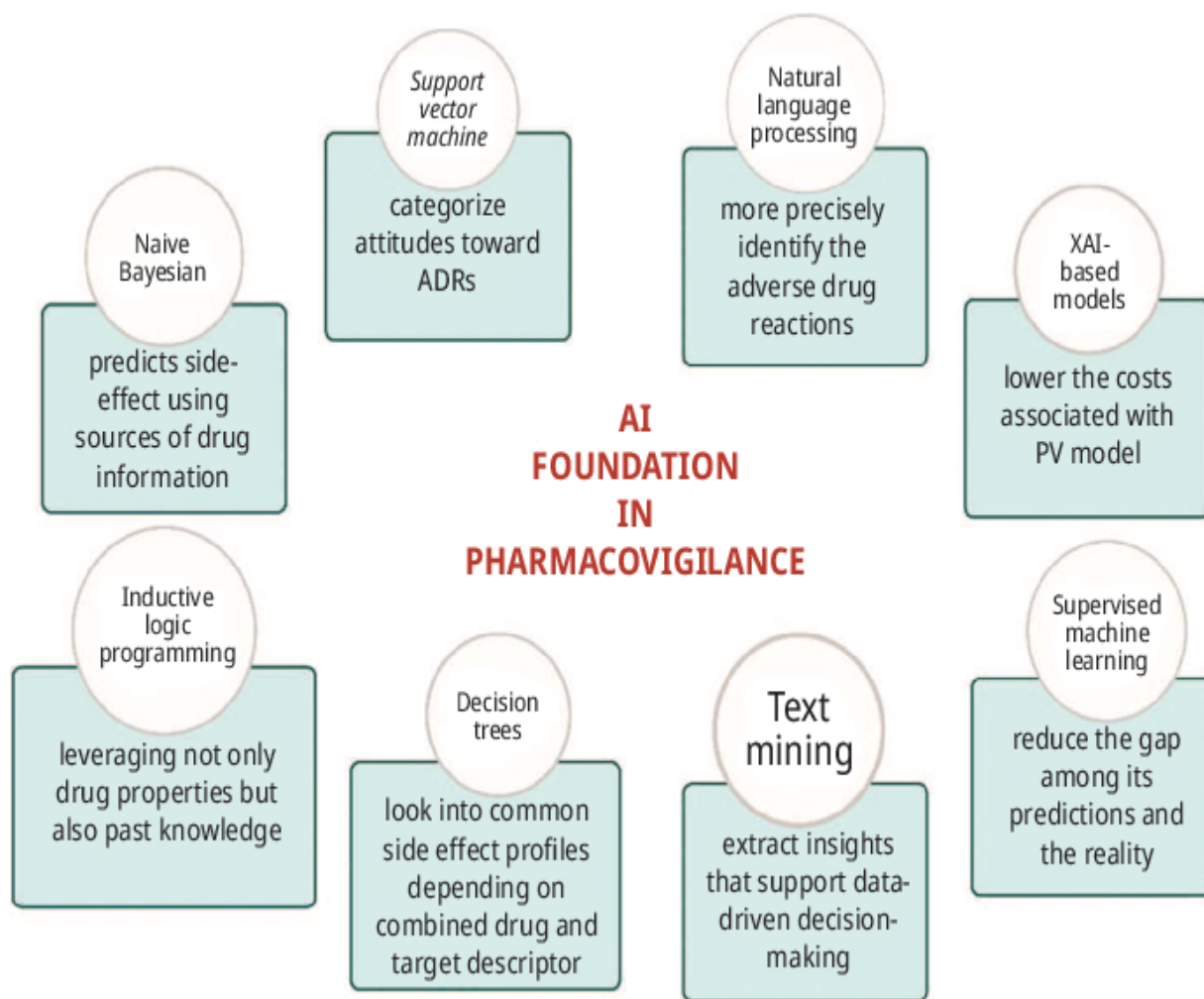


Fig. 8.2: Various AI-based tools for pharmacovigilance and drug safety.

Representational learning, which generates predictive features from data, is a basic concept in machine learning. The process lessens the need for the

laborious, sometimes necessary, human feature engineering found in non-ML techniques. One method for representational learning that can be enhanced is regression. Non-ML AI expressly uses symbolic representations of human reasoning and coded domain knowledge. These analyses can be regarded as falling outside the purview of artificial intelligence in pharmacovigilance (AIPV) because they initially relied on static rules that were unable to self-correct or self-learn [[→43](#)]. However, AI in general, and AIPV in particular, are still evolving, if unfortunately, sometimes in lockstep with hype cycles. The twenty-first century is seeing a resurgence of sorts for non-ML symbolic AI, so it's important not to write it off entirely. There are two illustrative examples. First of all, description logic is a web-based system for knowledge representation that enables data to be analyzed and read by machines [[→44](#)]. Better adverse event cases and queries, in addition to data set integration and inferencing, are made easier by the language. These elements matter in many different PV domains. Second, hybrid methods combine symbolic AI and deep learning to enhance each other. One example of this is neurosymbolic AI. These methods are currently being used in PV-related fields [[→45](#)].

8.3 Challenges and opportunities of AI in pharmacovigilance and drug safety

In LMICs, there is a dearth of PV healthcare professionals (HCPs) as a result of inadequate training and educational opportunities. Professionals in

a traditional PV system operate in four domains: qualified persons for pharmacovigilance (QPPV), systems, operations, and surveillance. Traditional PV requires QPPVs, data/system security administrators, drug safety physicians, and supervisors of drug safety. But for AI-based PV, specialists in a variety of AI domains are also required. For example, machine learning, as well as natural language processing (NLP) algorithm development, requires engineers. It is yet another significant barrier for LMICs. Very few LMICs are among the very few nations that provide cross-specialty, even cross-language learning for AI-based PV. Although AI technologies have already been used in pharmacovigilance systems, there are currently very few dedicated training programs for AI-based pharmacovigilance systems available globally. The lack of funding for healthcare professionals (HCPs) from low- and middle-income countries may also restrict the opportunities for HCPs to attend these courses. Despite this, HCPs in developed nations could receive training in AI technologies like data mining, deep learning, and natural language processing. However, there aren't many options for AI training in LMICs [[→ 46](#)].

Every medical facility supports a different data model than the others, and every kind of data model has the potential to be distinct with its own set of terminologies and value representation. Across several institutions, data from disparate sources must be combined and transformed into a format that is compatible with the common data model (CDM). To do this, it will take time and effort for engineers and HCPs to transform the data. Because algorithms and training models in AI-based PV rely on data with

annotations to make predictions on unannotated data, data annotation is crucial. The high cost of annotation and the dearth of excellent quality data for model training inhibit research on AI-based PV. Many ICSRs can be found in structured databases, but these databases are unable to recognize possible PV signals [[→ 47](#)]. The availability of structured as well as curated data for software training to detect possible drug safety issues is one of the primary obstacles to AI adoption. Privacy concerns are also raised by using AI in pharmacovigilance because participant consent may not be obtained before data is used for other purposes, as shown in [→ Fig. 8.3](#) [[→ 48](#)].

Everyone is entitled to privacy and the safeguarding of their private medical data. Furthermore, laws are in place to safeguard the privacy of sensitive information that can be used to identify a specific person on a regional and national level. It is necessary for people working in the PV system to have specific knowledge and abilities [[→ 49](#), [→ 50](#)]. These structures are currently constructed using tried-and-true methods, but conventional PV and IT interfaces become disrupted when machine learning is applied, for example. Existing evidence suggests that artificial intelligence has the potential to match or even surpass human performance in a variety of medical science domains. Change management involves individuals and functions that are affected by the change [[→ 51](#), [→ 52](#)]. The PV system spans many domains beyond the safety division of the MAH, and the introduction of new IT will have an impact on numerous stakeholders both upstream and downstream [[→ 53](#), [→ 54](#)]. The implementation of such a system will probably bring about unanticipated difficulties, but it will also

bring about unanticipated opportunities to enhance the PV system throughout its entirety. Intelligent automation's effects on business continuity planning are significant. As automation increases, it might be wise to concentrate on starting up automated processes again and stop using manual workarounds. However, to follow PV regulations and maintain vigilance during a disruption to safeguard patient safety, it will be crucial to remain adaptable [[→54](#), [→55](#), [→56](#)]. Furthermore, distinguishing between causal and associative relationships using observational data in the presence of confounding variables (i.e., variables related to both exposure and outcome) and colliders (i.e., factors that are affected by both exposure and outcome) presents a particularly significant challenge when incorporating machine learning. Although a multivariable regression approach is frequently utilized to account for the possibility of confounding factors, it is not possible to directly estimate the effects of causality. In addition, before causal connections can be established, observational studies must be conducted to capture and evaluate temporal relationships [[→57](#), [→58](#)].

Consequently, it is believed to present a chance to apply causal inference paradigms to the main machine-learning issues that the pharmacovigilance industry is currently dealing with. Examples of causal study designs, including Granger causality, hypothetical reasoning, as well as do-calculus, are less commonly used in pharmacovigilance tasks. It is especially important to investigate these types of study designs. In addition, there is a dearth of research that investigates the possibility of addressing the fairness problem in machine learning by incorporating causal paradigms. It could be

a new path for pharmacovigilance studies to pursue [[→59](#)]. Even though organizations in the life sciences are interested in AI and ML, they are concerned that ML and AI would not produce the same results as traditional case processing. Because they are unable to fully comprehend the potential that automated PV reporting systems could bring to their operations, many businesses in the health sciences industry struggle to define demanding requirements for their developers and to carefully evaluate vendors and platforms that are available on the market. Because they are unsure of the full potential of the platform, they may choose to limit their forecast and instead choose or develop solutions that adhere to established procedures rather than seeking room to reconsider [[→60](#)]. Even though it may only have a moderately favorable output, it will result in a significant reduction in the amount of time and money spent on pharmacovigilance operations, as well as an improvement in the quality of the results. Pharmaceutical companies can hire technology experts rather than industry experts, only to find out that these partners are not knowledgeable about the complex rules and regulations that govern international pharmacovigilance practices, even though they have almost adopted an existing platform for these kinds of applications. Furthermore, the extent to which developing and impoverished nations engage may be hindered by challenges such as limited accessibility, reliability, insufficient expertise in handling, and acceptance of technological use”. [[→61](#)].

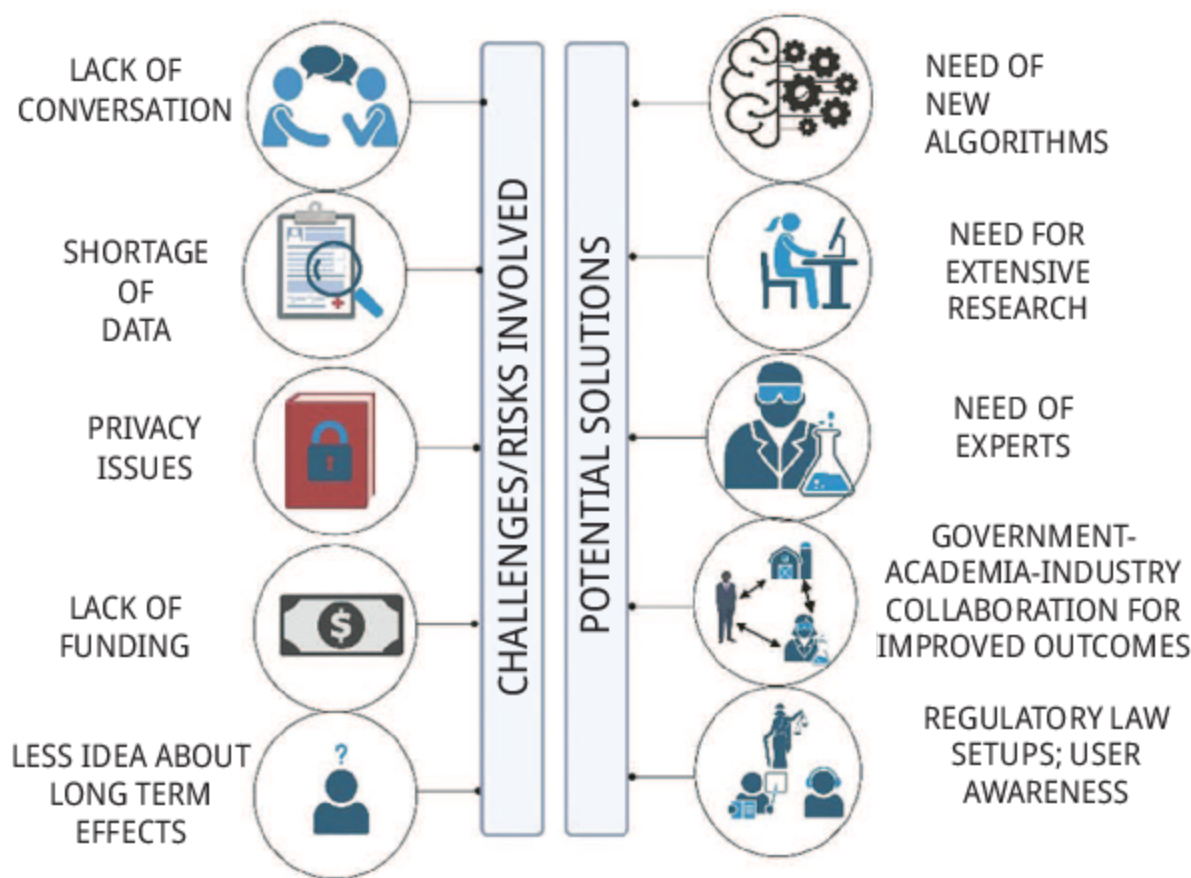


Fig. 8.3: Various challenges and solutions of AI in pharmacovigilance and drug safety.

8.4 Collaboration and partnership for drug safety

An agreement between the public and private sectors that is cooperative is known as a “public-private partnership,” or PPP. In this arrangement, the two sectors work together to accomplish a goal that is compatible or mutually beneficial by sharing resources, risks, and duties. The premise of public-private partnerships (PPPs) is founded on the conviction that, from the bench to the bedside, neither the public nor the private sectors can successfully monitor and sustain innovation across an extended period on

their own [[→62](#)]. A research and development procedure created exclusively for the public sector could be excessively sluggish and ineffectual. The ability of the public and private sectors to work together and share resources, including financial resources, knowledge, data, samples, medical knowledge, patient base, facilities, risks, and rewards, is one of the main benefits of a public-private partnership (PPP). More importantly, they create wider networks of experts with diverse viewpoints. In addition to making the innovation process more efficient, cost-effective, and time-bound, a public-private partnership (PPP) can also help to facilitate and improve the overall fluidity and holistic nature of the research and growth process [[→63](#)]. The private sector routinely uses medications at a substantially higher frequency than the public sector does. Adverse drug reactions, or ADRs, are one of the leading causes of death and are predicted to become a major public health concern in the not-too-distant future [[→64](#), [→65](#)]. In both urban and rural areas, private physicians are the only providers of healthcare, according to the findings of the National Sample Survey Office (NSSO). Furthermore, due to the private pharmaceutical industry's explosive growth, the private sector urgently needs handholding support to overcome obstacles like the underreporting of ADRs, the creation of a national patient safety database, as well as the pursuit of mutual benefits like recognition and avoiding regulatory action [[→66](#), [→67](#), [→68](#)]. Pharmacovigilance requires the involvement of healthcare professionals to be successful. Consequently, a strong system and tools for the tracking, reporting, and assessment of ADRs have been established as a

result of the collaboration between the PvPI and academic medical institutions, public and private hospitals [[→69](#), [→70](#)]. Consequently, the PvPI tackles the problems by informing and arguing with the hospital administration and the medical staff that reporting ADRs is entirely voluntary and does not result in disciplinary action. Reports obtained from the public sector even have higher quality completion scores than those from the private sector. The private sector's reports have a low overall rating because they withhold information about alternative dispute resolutions out of fear of lawsuits and a desire to maintain confidentiality [[→71](#)].

Historically, the primary goal of NRAs has been to collaborate with healthcare providers and the industry to report passive ADRs through hard copy and online reporting channels. NRAs are essential regulators in the field of pharmacovigilance. Pharmacovigilance has looked into the possibilities of using artificial intelligence and keeping electronic health records over the last few years [[→72](#)]. More people are coming to understand and accept the idea that patients and caregivers should be involved in the process of creating and regulating health products. Since they will ultimately be using the products in the healthcare industry, patients should be educated, engaged, and treated as participants in their care. They also possess the greatest depth of knowledge regarding their perspectives, experiences, and insights. Therefore, the viewpoints of patients and their life experiences can help to improve a health product at different stages of its life cycle [[→73](#)]. Regulatory approvals, post-approval

monitoring, clinical trial designs, research priorities, and technology evaluations are a few of these stages. The involvement of patients in pharmaceutical surveillance is crucial because it provides more context than the reports that healthcare providers present. One illustration of this would be the simple fact that patient reports usually include a more thorough description, especially when it comes to the impact on daily life, which may not be fully understood by medical professionals. It enhances patient participation in their care, adds to the reports provided by healthcare providers, and raises consciousness about ADRs [[→74](#), [→75](#)].

In the digital age, patients can now more easily report ADRs due to online reporting via the Internet and smartphones [[→76](#)]. In the aforementioned French study, which examined the variables influencing patient participation in pharmacovigilance observations and the development of patient involvement, it was found that the availability of digital solutions affected the rise in patient reporting of adverse drug reactions [[→77](#)]. In the field of drug safety communication, which is a vital part of pharmacovigilance, a further significant collaborative partnership among patients is essential. A study that examined the pharmacovigilance environments in eight ASEAN nations discovered that a common problem is communication [[→78](#)]. This is even though open and honest communication about the advantages, drawbacks, and efficacy of medications increases the likelihood of safe medication use among patients. For the sake of the general public's health, information regarding drug

safety must be made accessible, easily accessed, as well as successfully communicated.

8.5 Regulatory aspects of AI in pharmacovigilance and drug safety

AI holds a great deal of promise for revolutionizing the pharmaceutical industry, even though its application in pharmaceutical regulatory affairs continues to be in its infancy. Through the dissemination of information regarding the most recent developments in the industry, artificial intelligence can be of assistance in the management of regulatory affairs. This is especially true regarding the potential advantages of AI for regulatory approval as well as drug discovery. The term “regulatory affairs” refers to the groups or departments within pharmaceutical companies that act as a conduit to regulatory bodies, both nationally and globally, to obtain product approval in accordance with the laws and regulations that are currently in place. In turn, regulatory agencies give their approval to products based on the information and documentation that is provided throughout the entire process of drug discovery along with product development [[→79](#)]. The Central Drugs Standard Control Organization (CDSCO) mandates that Marketing Authorization Holders (MAHs) report the Individual Case Safety Reports (ICSR) of any drug marketed in India to both the National Coordination Center for Pharmacovigilance Program of India (NCC-PvPI) and the MAHs themselves. The management and

handling of these documents are becoming more costly, time-consuming, and laborious due to the increasing volume of ICSR reports. Therefore, to cut down on the time, resources, and labor needed for processing ICSR cases, PvPI may also use machine intelligence techniques like artificial intelligence [[→ 80](#)]. Regarding the acquisition of precise data, the use of AI technologies in pharmacovigilance (PV) is very advantageous. Almost all aspects of PV in handling cases and risk tracking can be automated or made easier with the help of AI tools, which contributes to a reduction in the overall processing time. One idea that has been proposed is that of Beneficial AI, which is artificial intelligence developed in a secure manner and for the benefit of society. To construct AI that is beneficial, two different types of measures have been proposed. Because of extrinsic measures, artificial intelligence designers are required to adopt beneficial designs. Including certain requirements, incentives, and constraints is one way that these designs could be implemented. Measures that are deemed intrinsic include the formation of social norms and the organization of communications. Extrinsic measures may be impacted by the influence of intrinsic measures [[→ 81](#)].

Furthermore, artificial intelligence technologies in PV greatly aid in the accurate extraction of information processes. Almost all PV case processing and risk tracking tasks can be automated or simplified with the use of AI tools, which helps to shorten processing times overall [[→ 82](#)]. The PV database known as VigiBase stores the information in a format that is both structured and ordered, making it possible to easily analyze the data that has

been recorded. The WHO International Classification of Diseases, the WHO Drug Classification, and the WHO-Adverse Reaction Terminology/Medical Dictionary for Regulatory Activities (MedDRA) are some of the medical and drug classifications linked to the system. Vigibase is a web application that the general public can use to browse and easily access information on the negative effects of drugs [[→ 83](#)]. In addition, Vigilyze is an online resource that offers a clear and concise overview of Vigibase, which can be investigated online for more information [[→ 84](#)]. Vigigrade, an online ICSR management system, is employed for worldwide drug surveillance. It does so by gathering, preparing, and disseminating data to enable efficient data analysis. MedDRA and WHO Drug support the system [[→ 85](#)]. As a result, Vigimatch is an algorithm that finds individual case reports that are similar to one another by using probabilistic matching of patterns. It is significant to remember that the main objective of Vigigrade is to evaluate the degree of completeness of clinically relevant data in an organized manner on an individual's case report. It is carried out as part of the process of informing nations about the caliber of the data [[→ 86](#)].

The various regulatory bodies are transitioning from paper-based submissions to electronic submissions of the CTD. CTD (paper-based) submissions have been stopped from being accepted by several different authorities. It has been determined to take action to expedite the submission process and centralize the procedure. Regulations worldwide have been greatly impacted by the switch from paper to electronic submission of

Common Technical Documents (CTD) [[→ 87](#), [→ 88](#)]. The acceptance of submissions in the electronic submission of the CTD format has resulted in the optimization of resources, the increase of efficiency, and the improvement of application tracking. Regulations worldwide have been greatly impacted by the switch from paper to electronic submission of Common Technical Documents (CTD). The Food and Drug Administration (FDA) in the United States has implemented requirements for the submission of electronic clinical trial data (eCTD) for a variety of applications, such as biologics license applications, new drug applications, as well as abbreviated new drug applications. There has been a positive impact on regulatory procedures as a result of the implementation of electronic submission of CTDs [[→ 89](#)]. Individuals who have experience with electronic document submission (eCTD) can show financial savings over paper submissions and shorten the time it takes for their submissions to be approved. This phenomenon has been noted in a broad range of organizations, regardless of the magnitude or volume of submitted work. In addition, the utilization of electronic submission and review (eCTD) has made the process of reviewing and managing the lifecycle of electronic submissions easier, which has resulted in the harmonization of electronic submission and review initiatives [[→ 90](#), [→ 91](#)].

Due to their effects on drug research, medical practice, and regulatory professionals' responsibilities, each of these trends has a variety of related issues that need to be addressed. Ethical considerations encompass a wide range of topics, such as privacy, wellbeing, and the preservation of human

autonomy [[→91](#)]. It is crucial to ensure transparency and allow capability explanation when it comes to deep learning models that are used in machine learning and artificial intelligence. Furthermore, these technologies must be utilized in a responsible and accountable manner. The emergence of digital disruption presents notable obstacles for enterprises seeking to improve workforce competencies and comprehension via employee upskilling and reskilling programs. Implementing a mentorship program, in which experienced employees offer advice and support to less experienced colleagues during their transition period, is one possible way to address these issues [[→92](#), [→93](#), [→94](#)]. Moreover, India's National Pharmacovigilance Program was launched in November 2004 by the Central Drugs Standard Control Organization to improve drug safety through the collection and assessment of adverse drug reaction data. Using a tripartite structure and multiple hubs expedites the implementation of regulations to guarantee public health. Its main objective is to promote a notification culture among doctors and other medical staff members, like nurses and pharmacists. The program permits reporting of common or surface-level side effects of even established drugs, even though drug regulators would be more intrigued to learn about adverse drug reactions of recently marketed drugs [[→95](#)]. The purpose of it is to incentivize all healthcare providers to begin disclosing adverse events. The reporting forms protect the privacy of the patient. Nonetheless, to facilitate information verification and deter the submission of false information, notification identification is required. Peripheral Pharmacovigilance Centers

can now be established by licensed medical professionals, including pharmacists [[→96](#)]. Pharmacovigilance (PV) systems of today handle safety data using algorithmic techniques based on binary logic. These user-managed algorithms use standardized, objective expert knowledge to ensure that users are familiar with and understand the results of data processing. The reporting forms protect the privacy of the patient. Nonetheless, to facilitate information verification and deter the submission of false information, notification identification is required. Peripheral Pharmacovigilance Centers can now be established by licensed medical professionals, including pharmacists [[→97](#)].

8.6 Conclusion

Pharmacovigilance (PV) is a scientific and applied endeavor that attempts to identify, evaluate, understand, and minimize the negative effects of drugs or potential drug-related issues. Natural language processing and machine learning are two distinct features of AI used in web searches, clinical settings, treatment, surveillance, outcome prediction, and medical diagnosis. Data intake, anomaly detection, and duplicate detection have all benefited from AI/ML. To gather information on ADEs and ADRs, process reports according to international clinical standards, obtain drug-drug interactions, predict adverse reactions to drugs, represent clinical trials, and incorporate estimation unpredictability into diagnostic classification algorithms, AI is being utilized for pharmacovigilance and patient safety.

Machine learning techniques are becoming more and more useful for predicting medication side effects, identifying high-risk populations for adverse drug reactions, and incorporating prediction unpredictability into diagnostic classifiers. A developing area in pharmacovigilance is text mining, which combines NLP and data mining. The classification of ADRs, side-effect prediction, and drug toxicity prediction are all done with artificial neural networks (ANNs). Nevertheless, there are obstacles in the way of pharmacovigilance and drug safety, including the scarcity of opportunities for cross-specialty training, privacy concerns, and data models for healthcare facilities, as well as limited training and education in LMICs. PPPs, or public-private partnerships, can aid in resolving these problems and enhancing the creative process.

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Chapter 9 Intellectual property and data privacy in the pharmaceutical sector

Abstract

In the pharmaceutical sector, data privacy is essential to providing quality care. However, protecting large volumes of data has proven to be a difficult mandate, and the healthcare sector is still vulnerable to data breaches that make headlines. Intellectual property rights (IPR) are crucial to the pharmaceutical industry because they allow inventors to legally maintain the confidentiality of their inventions for a predetermined amount of time. The exclusive rights that inventors and creators have to make money off their inventions are known as intellectual property rights or IPRs. These rights are protected under various laws, including patents, copyrights, and trademarks. The chapter also discusses how blockchain technology can be used to manage IPRs and improve patient privacy. It highlights the challenges of data privacy in the pharmaceutical industry, including cybersecurity risks and compliance with data protection laws. In the intricate pharmaceutical supply chain, cooperation between distributors, manufacturers, and other stakeholders is essential for safeguarding data privacy. Overall, the chapter emphasizes the necessity of a balanced approach to data privacy and intellectual property, as well as the intricate and regulated nature of the pharmaceutical industry. It suggests that

addressing unfair practices and promoting innovation can contribute to the welfare of consumers and the advancement of society.

Keywords: data privacy, copyright, patent, trademark, authentication, encryption, cybersecurity,

9.1 Introduction

In most economies, the pharmaceutical industry is highly regulated and complicated. The use of “learned intermediaries” such as physicians and pharmacists, restrictions on advertising and promotion, public and private insurance programs, and government purchasing and price controls all have a major impact on the demand for pharmaceuticals. Competition is shaped equally by strict product safety reviews, manufacturing regulatory oversight, and legal frameworks governing the transfer of technology between commercial entities and publicly funded biomedical research institutions on the supply side [[→1](#)]. All original works of human intellect, whether they are in the arts, sciences, technology, or literature, are considered intellectual property (IP). The legal entitlement of a creator or inventor to maintain the confidentiality of their idea or innovation for a specified duration is referred to as innovation for a specified duration is referred (IPRs) [[→2](#)]. For a specified duration, these legal rights confer onto the inventor or their assignee the exclusive authority to exploit their invention. The importance of intellectual property in the modern economic environment has been acknowledged for an extended period. Furthermore, it has been clearly established that the intellectual effort behind the

discovery deserves recognition for its contributions to the public welfare. Research and development (R&D) expenditures have escalated, leading to a corresponding increase in the investments necessary to introduce innovative technology to the market [[→3](#)].

Europe is where the rules and regulations about intellectual property originated. The fourteenth century marked the inception of the patent-granting tendency. England exhibited superior scientific advancement in specific domains compared to other European nations, hence enticing immigrant workers under advantageous conditions. Italy is the origin of recognized copyrights. The majority of legal thought that resulted in the first laws and legal systems in history, which eventually spread to other countries, occurred in Venice, which is why the city is recognized as the birthplace of the intellectual property system [[→4](#)]. In general, multinational corporations controlled the Indian medicine market in the post-independence era. India was among the most expensive countries in the world for drug prices, and the cost of the drugs was higher when they were imported. The committee correctly noted that a clear recommendation for guaranteeing the food, medicine, and medical device needs of the Indian population should be included in the patent act. These ought to be made as inexpensively available to the general public as feasible, with the patent holder receiving just compensation. In 1957, the Indian government took two significant actions: first, it formed Hindustan Antibiotics Limited through a UNICEF agreement; second, it designated the Justice Rajagopala

Ayyangar Committee to amend the patent law and establish a market that could be sustained locally [[→5](#), [→6](#), [→7](#)].

IPRs are typically regarded as having two principal areas of impact within the pharmaceutical sector. The primary concerns are the pricing and availability issues, namely the interaction between IPRs, particularly patent rights, and the cost and accessibility of innovative medications. The function of IPRs in providing incentives for the development, commercialization, and discovery of innovative pharmaceuticals is the second issue, namely research and development (R&D) incentives. Furthermore, IPRs influence research and development expenditures and their allocation among diseases, countries, and organizations. The two concerns are intricately connected, and their interplay presents numerous formidable policy and economic challenges [[→8](#)]. Utilizing IPRs to effectively reconcile “static” customer advantages from reduced prices and a competitive medicine supply with “dynamic” benefits from breakthrough products presents significant challenges, even within the confines of a single nation. On the one hand, declining effective patent life and growing R&D costs are severely pressuring industry finances. However, despite the enormous positive effects that pharmaceutical innovation has on the economy and public health, even in wealthy nations, the high cost of over-the-counter medications often prompts contentious political debates about issues of access and equity for underprivileged or low-income populations, as well as how best to allocate public health care funds [[→9](#)].

The present condition of the pharmaceutical sector indicates the inequitable promotion and exploitation of IPRs to the detriment of competition and customer welfare. The inequity affecting societal welfare is underscored by the drug industry's deficiency in innovation and reluctance to embrace risk. It is an injustice that cannot be remedied by mere legislative reform.

Although legislative initiatives aimed at addressing deficiencies in existing statutes and enacting new regulations to further curtail the pharmaceutical industry's inequitable business practices may offer some respite, antitrust law must effectively intervene [[→ 10](#)]. Antitrust laws have rigorously examined certain corporate activities in the pharmaceutical sector, including non-competition agreements and mergers and acquisitions; however, numerous additional practices require attention. Antitrust law can contribute to maintaining equilibrium between incentivizing innovation and safeguarding competition in various domains, such as the issuance of patents for minor elements of obsolete pharmaceuticals, reformulating drugs to secure new patents, and employing advertising and brand expansion to elevate barriers for generic rivals [[→ 11](#)].

Consequently, in the wake of globalization, it is essential to stay innovative and creatively ahead of the curve to compete in the competitive trade and technology markets. India is renowned for its intellectual prowess in a variety of technological domains, including software engineering, missile technology, missions to the Moon or Jupiter, and other areas [[→ 12](#), [→ 13](#), [→ 14](#)]. IPRs and the framework of its policies have a direct impact on any society's development. IPRs ignorance leads to the demise of inventions,

increased infringement risk, financial loss, and the end of the nation's intellectual era. Therefore, to support domestic inventions and advancements in the field of research and technology, there is an urgent need to disseminate information about IPRs [[→15](#), [→16](#), [→17](#)].

9.2 Intellectual property rights landscape

IPRs refer to thoughts, creations, and artistic works that the public recognizes as proprietary. IPRs confer exclusive rights to owners, enabling them to monetize their inventions or reputations. A variety of laws, such as those about patents, copyrights, trademarks, and others, protect intellectual property as represented in [→Fig. 9.1](#).

9.2.1 The patent

A product or technique that presents a novel solution to an old problem or a new technical approach is called an invention. An invention is entitled to an exclusive patent, which gives the patent holder protection for their invention. This protection is implemented for a limited duration. A patented innovation cannot be manufactured, utilized, sold, or disseminated for profit without the consent of the patent holder. The patent holder is responsible for determining who may utilize the innovation and under what conditions while it is protected by the patent. Patent owners may license third parties to utilize the invention under mutually acceptable conditions. The owner may transfer the rights to the invention to a third party, who will then

assume ownership of the patent. Upon the expiration of a patent, an invention forfeits its exclusive rights and becomes available for economic exploitation by others, thus concluding its protection [[→18](#)]. The phrase “invention entering the public domain” denotes this concept. The majority of nations have enacted domestic laws to safeguard their IPRs. The protection granted to inventors and creators in specific countries, such as India, or regions, such as the European Union, is localized and cannot be transferred to other countries or regions, with the exception of copyrights [[→19](#)]. A patent granted in India is valid just within that jurisdiction and cannot be enforced in the USA. The primary motive for patenting an idea is to generate profit from its exclusivity. The resultant patent would confer exclusive market rights to the patent owner if the inventor or their assignee: (a) developed a significant invention after considering customer modifications; and (b) the invention was precisely articulated and claimed in the patent specification prepared by the patent agent. The patent holder may exercise their exclusive rights by selling the patented invention directly to consumers or by licensing it to another entity [[→20](#)].

Furthermore, a multinational accord referred to as the Patent Cooperation Treaty (PCT) was enacted in 1978. By specifying the countries of interest in the PCT application, an inventor from a PCT contracting state may concurrently get priority for their invention in any or all member nations. It obviates the necessity for the inventor to file a distinct application in each member nation [[→21](#)]. To protect an innovation internationally, a distinct patent application must be filed in each pertinent country; in some

circumstances, it must occur within a designated timeframe to get priority in those jurisdictions. A substantial and prompt financial investment is necessary to address filing fees, translation costs, legal charges, and other related expenditures. Moreover, the limited timeframe may render conclusions regarding the necessity of filing a patent application in a country misguided [[→ 22](#)].

9.2.2 The copyright

The term “copyright” refers to the legal protection afforded to published works, which prohibits anyone other than the author from publishing or selling them. A set of exclusive rights known as copyright governs how an idea or piece of information is used in a specific way. In its broadest sense, it refers to “the right to copy” an original work. Literary works such as plays, novels, poems, and reference books, as well as technical drawings, maps, and advertisements, as well as computer programs, databases, movies, musical compositions, photographs, sculptures, and architecture, are a few examples of works that are covered by copyright. Works that are the product of human intelligence and creativity are protected by copyright. Both published and unpublished works are included. Expressions of ideas are protected by copyright, not the ideas themselves. For a predetermined amount of time, it permits the copyright holder to prevent others from reproducing the work [[→ 23](#)]. As it gives creators of content exclusive rights over their original works, copyright law is crucial. Among other things, it offers protection for works of art, music, literature, and film.

Authors can manage how their works are reproduced, distributed, performed in public, and altered due to copyright laws. Content creators are better equipped to protect their works when they are aware of the extent of copyright protection, including the duration of rights, the requirements for copyright registration, and the notion of fair use [[→ 24](#)].

A multitude of research initiatives have been conducted across different countries to examine copyright legislation from diverse perspectives. The study's author focused on copyright and public domain legislation. The study indicates that the public domain is a fundamental component of copyright law, and its incorporation meets both historical and logical criteria. Without citing "public domain copyright," one cannot justify their actions. The study goes on to say that copyright and public domain are similar in that they have similar evaluative values. According to a related study, the general public is a stakeholder in copyrights. According to the author, the US copyright law grants creators several exclusive rights, many of whom either knowingly or unknowingly give up these rights to business entities [[→ 25](#)]. Citizens should support a copyright system that considers their interests since the general public is now considered to be a stakeholder in copyright. Traditionally vested interests must come first. The ethical and financial aspects of copyright were demonstrated by three additional studies on copyright policy. The term of protection was the subject of one study. It concluded that extending the protection would be detrimental and needless. According to the second study, there are two different schools of thought on copyright law: the first emphasizes the creators' inherent rights to the

money made from their works, while the second highlights the high cost of creating a framework for copyright that maximizes social welfare. The study's findings ultimately demonstrated that the "natural rights view" had a stronger impact on Canada's copyright laws than the economic perspective [[→26](#)].

9.2.3 The trademark

Article 15 of the TRIPS Agreement stipulates that "any sign, or any combination of signs, capable of distinguishing the goods or services of one enterprise from those of others, is necessary to constitute a trademark." Due to industrialization, the trademark has gained prominence in contemporary international trade [[→27](#)]. A trademark is a unique symbol or logo that signifies the manufacture or provision of a specific product by an individual, organization, or economic sector. Service marks, akin to trademarks, serve to differentiate service providers from their competitors. A business may employ various types of trademarks for distinct items, while trade names serve to differentiate it from competing enterprises. Trade names and trademarks assist companies in establishing their brand, enhancing awareness, and fostering trust with consumers. Consumers typically depend on trademarks due to the challenges associated with swiftly evaluating a product or service to determine its quality [[→28](#)]. Any distinguishable mark, or one that can be graphically represented and used to differentiate the goods and services of one company from another, is eligible for trademark registration under the Indian Trademarks Act.

Trademarks don't grant exclusive rights that could be abused, so their validity shouldn't be restricted. However, if there were no time limit on trademark validity, there would be an excessive number of registered trademarks that have no use [[→29](#)].

9.2.4 The trade secret

A trade secret is any knowledge that is kept confidential, not widely known to the public, and gives its owner a financial or competitive advantage. Trade secrets have a high value – tens or even hundreds of millions of dollars – and large damage awards have been given in trade secret lawsuits. A trade secret is misappropriated when it is obtained through “improper means.” Nonetheless, if the claimed trade secret was created independently, became known to the public, or was not kept a secret, then those defenses may be able to successfully contest a claim of misappropriation of trade secrets. Trade secrets are not as well known as other IPRs, even though they are an intrinsically valuable intangible asset for any business [[→30](#)]. Its unpopularity is attributed to numerous factors: First off, registration is not necessary for trade secrets. Trade secrets are generally credited with developing and being preserved due to standard business practices and business ethics. Even though trade secret laws are based on evolving legal theories, there is currently a weak enforcement system. Third, disputes involving such secrecy are seldom open to public discourse, implying that access to information regarding such matters is usually limited, rendering trade secrets comparatively unpopular. Fourth, trade secrets do not grant

their owner the sole right to utilize the information, in contrast to other types of intellectual property. The right may belong to others when knowledge is produced independently or obtained through the process of reverse engineering. Trade secrets provide the business with several financial advantages [[→ 31](#)]. Ensuring appropriate rewards for any invention that is supported by appropriate trade secret protection gives the company financial stability. To prevent breaches, no expensive system needs to be set up. Moreover, it always worries about knowledge spillovers leading to the misuse of company information. Trade secrets strike a balance between appropriate disclosures and secrecy, even though they rely on reasonable methods to maintain confidentiality. As a result, trade secrets may lead to independent invention or knowledge diffusion. The non-exclusivity restriction strikes the perfect balance between maintaining the essential distinction from absolute secrecy and ensuring confidentiality.

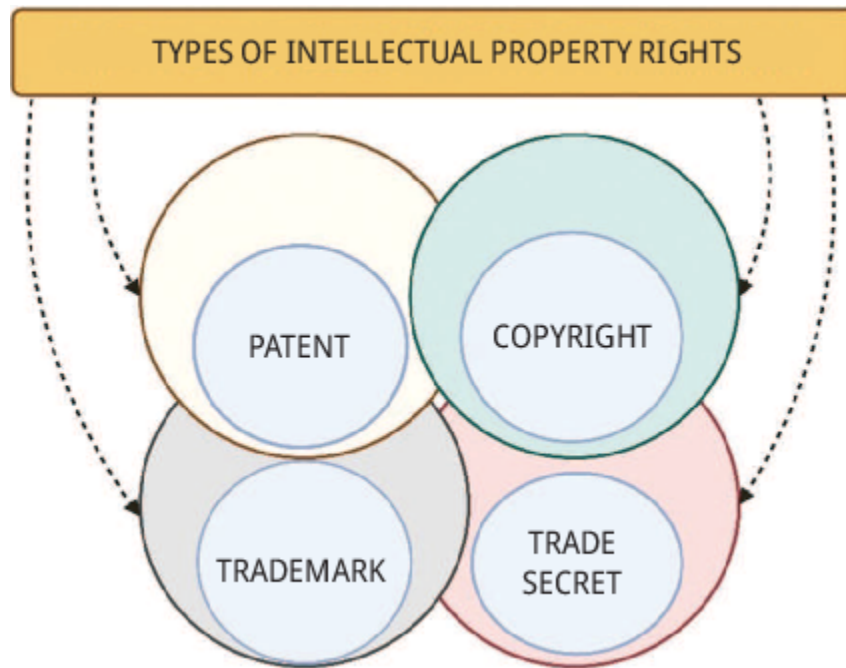


Fig. 9.1: Schematic representation of types of intellectual property rights.

9.3 Data privacy in pharmaceutical industry

Healthcare organizations send, store, and manage substantial volumes of data to ensure the delivery of effective and appropriate care. For many years, safeguarding the data has been a challenging imperative. To make matters worse, the healthcare sector is still among the most vulnerable to data breaches that are made public. Indeed, data breaches occur when hackers discover sensitive information through data mining techniques and procedures and make it publicly available. The complexity of putting security measures in place doesn't change, but the stakes keep getting higher as sophisticated methods of getting around security controls are developed. Consequently, organizations must adopt healthcare data security measures that comply with regulatory standards and safeguard critical

assets. In recent years, the incidence of advanced persistent threats has risen, characterized by focused assaults on information systems aimed primarily at exfiltrating recoverable data by the attacker. Consequently, breaches of patient privacy are regarded as a significant concern in big data analytics, complicating organizations' efforts to address these varied, interconnected, and pressing challenges. Data privacy delineates data access in accordance with privacy rules and regulations, specifying who is permitted to access financial, medical, personal, or confidential information. Conversely, data security regulates access to data across its entire lifecycle [[→ 32](#)]. A multitude of technologies ensures the confidentiality and privacy of healthcare data. The most popular technologies are ([→ Fig. 9.2](#)):

9.3.1 Authentication

Authentication is the process of verifying the veracity of assertions made by or regarding the topic. Numerous cryptographic methods incorporate endpoint authentication to mitigate man-in-the-middle (MITM) attacks. Authenticity generally relates to the truthfulness of one's origins, claims, commitments, and motives. Authentication is the process of verifying the veracity of assertions made by or regarding the topic. Information authentication often occurs with identity verification; however, it might provide distinct issues, especially regarding MITM attacks. Most cryptographic methods use endpoint authentication to mitigate MITM attacks in various forms. For example, cryptographic protocols such as

Transport Layer Security (TLS) and Secure Sockets Layer (SSL) facilitate safe communications over networks like the Internet. At the Transport Layer, network connection segments are secured end-to-end through TLS and SSL encryption. Multiple iterations of the protocols are widely employed in diverse applications, including voice-over-IP (VoIP), instant messaging, internet faxing, email, and web browsing [[→ 33](#)].

9.3.2 Encryption

Effectively encrypting data prevents unauthorized access to confidential information. The ownership of data is safeguarded across the whole data lifecycle, encompassing the data center, endpoints (including mobile devices of administrators, clinicians, and physicians), and the cloud. Encryption safeguards against security breaches such as packet sniffing and theft of storage devices [[→ 34](#)]. Efficient methods with minimal time and cost constraints should be employed, as the encryption of medical images is an expanding application of cryptographic systems. To convert an input image into a cipher image, an encryption method, either symmetric or asymmetric, must be employed. A variety of techniques and environments can be employed to encrypt medical images. Medical images can be encrypted by several techniques, including bitwise XOR diffusion, chaotic approaches, edge mapping, high-speed scrambling, and others [[→ 35](#)]. Furthermore, most of the time, encryption methods stop encrypted data from being processed, so data is processed in its original format. On the other hand, after computing encrypted data, homomorphic encryption

allows the user to obtain encrypted results. Homomorphic encryption not only makes it possible to process encrypted data but also safeguards privacy throughout [[→ 36](#)].

To provide more effective healthcare services and care, it is now feasible to quickly access data due to the digital age of healthcare. Electronic health record (EHR) systems are currently used by a large number of medical professionals. Most medical practices have shifted to using third-party clouds for the storage of patient data in centralized data warehouses, which are insecure and unreliable. Concerns regarding information preservation, access security, and privacy emerge when EHR data is stored on a potentially unreliable third-party cloud. Given that healthcare organizations routinely manage sensitive data, encryption is a key source of security and reliability. Numerous apprehensions have been raised regarding cloud storage's ability to provide scalable, regulated, and adaptable access [[→ 37](#)]. Researchers authored a study article addressing security and privacy concerns associated with cloud-based personal health record systems. The research analyzed various attribute-based encryption (ABE) methods applicable to cloud systems for the provision of fine-grained, scalable, and flexible access control. Contemporary cryptographic techniques, such as ABE, are included in the PHR system to mitigate security and privacy concerns related to cloud-based PHRs [[→ 38](#)].

9.3.3 Data masking

The original value cannot be obtained from the masked value, as masking does not constitute a genuine encryption method. It substitutes sensitive data items with indistinguishable values. It utilizes a method for de-identifying or obscuring data sets, suppressing or generalizing quasi-identifiers such as zip codes and dates of birth, and concealing personal identifiers including names and social security numbers. Data masking is hence one of the most prevalent techniques for real-time data anonymization. Initially, certain researchers introduced the k-anonymity approach, which safeguards against attribute disclosure but fails to protect identity revelation [[→ 39](#), [→ 40](#)]. The target application is deemed indifferent to the masked data in numerous prevailing data masking strategies. It may pertain to identifying information such as social security numbers (SSNs), but depending on the intended application, especially in the healthcare context, it may also encompass other sensitive information such as age. In such instances, the data is often encrypted prior to transmission to prevent unauthorized access. Nonetheless, encryption methods impair performance and often lead to considerable cost concerning reaction time and storage capacity. Moreover, they do not prevent the exploitation of information post-decryption by reputable external AI/ML researchers; hence, when such material is disseminated, legal agreements are often incorporated to prohibit the disclosure or inference of private data [[→ 41](#), [→ 42](#), [→ 43](#)].

9.3.4 Access control

An access control policy, typically founded on the privileges and rights conferred to each practitioner authorized by the patient or a reliable third party, will continue to regulate users' access to the information system. Consequently, the policy evolves into an efficient and flexible instrument for bestowing permissions upon users. It offers advanced authorization controls to guarantee that users can only execute tasks for which they are permitted, including data access, job submission, cluster administration, and other responsibilities. Security and access control concerns have been mitigated through several methods, with Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) being the most used models for EHRs [[→ 44](#), [→ 45](#), [→ 46](#)]. Furthermore, blockchain technology is a viable solution for enhancing healthcare privacy due to its inherent advantages. Data is dispersed across several network nodes owing to the ledger's distributed and decentralized characteristics. As a result, the probability of a singular point of failure or assault target diminishes. Secondly, blockchain technology can facilitate the creation of smart contracts that autonomously disseminate data while complying with privacy regulations. Blockchain technology has the potential to enhance patient privacy; nevertheless, some issues remain to be addressed [[→ 47](#)]. A primary issue is the incompatibility among various blockchain platforms, potentially resulting in healthcare providers utilizing fragmented and non-integrated systems. Blockchain technology integrates many privacy-enhancing technologies to bolster the security of sensitive patient information. Encryption enables the secure transfer and storage of

healthcare data, restricting access to authorized individuals who possess the necessary decryption keys. Patients can determine who accesses their data using consent management in smart contracts, thereby ensuring that healthcare data governance honors patient preferences [[→ 48](#), [→ 49](#)].

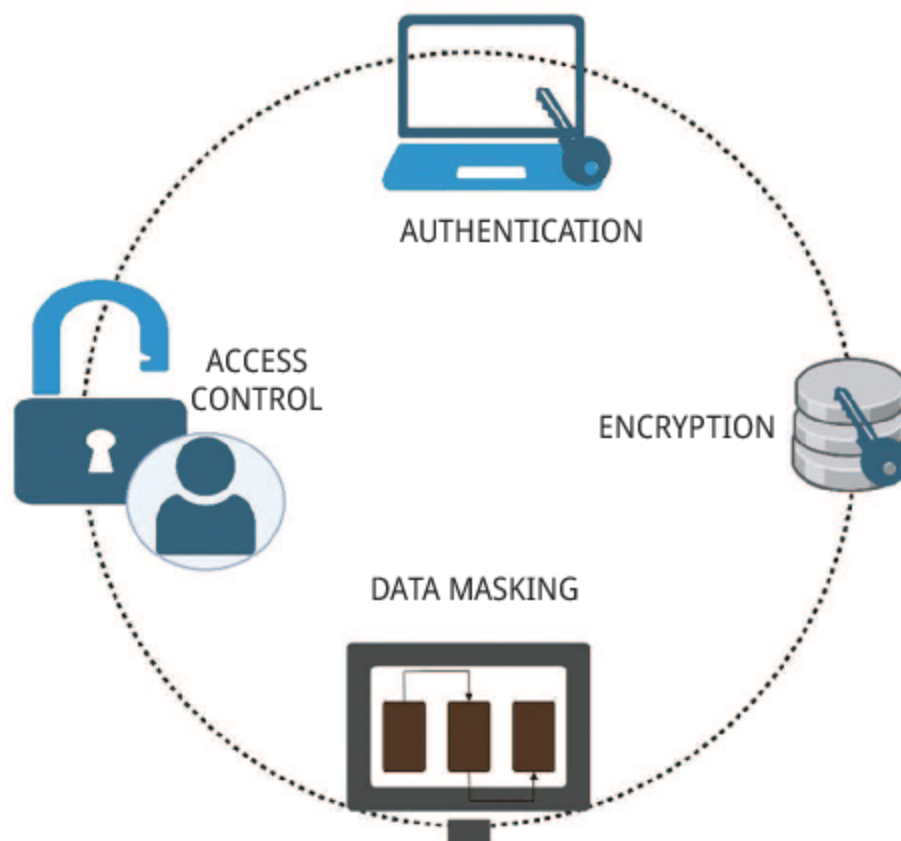


Fig. 9.2: Various techniques for data privacy in the pharmaceutical industry.

9.4 Emerging technology and intellectual property

The application of blockchain technology has grown in several sectors, including medical [[→ 50](#), [→ 51](#), [→ 52](#)]. Blockchain technology offers a

transparent, irreversible, decentralized shared database that may be utilized to construct a dependable value chain. Medical information systems are being created due to the digitization of the healthcare sector [[→53](#)]. An individual's existence relies on their healthcare, as do the accompanying statistics that assist in illness diagnosis and inform treatment options. Historically, information was inscribed and documented on media that were susceptible to destruction or modification [[→54](#), [→55](#)]. These systems must convey data efficiently and securely [[→56](#)]. They must also provide each user with enhanced access control, privacy, and anonymity. A deficiency in security, privacy, or trust may deter individuals from seeking therapy or disclosing sensitive information. The reliance on blockchain technology may evolve due to its distributed nature. It possesses the ability to steadfastly and collectively endure challenges and assaults. It also offers documentation of the ownership and veracity of the data [[→57](#)]. Consequently, blockchain is emerging as a versatile technology with applications across multiple sectors and use cases, including contract administration, identity management, management of supply chains, and healthcare [[→58](#)].

Data regarding various IP objects is more accessible owing to blockchain-based IPRs systems. The significant “trusted timestamping” functionality of the BCT can furnish the information. Consequently, the BCT enhances the competitiveness and administrative ability of all stakeholders engaged [[→59](#)]. To reconcile the interests of intellectual property holders and end users, it is imperative to create technical and legal standards prior to the

integration of blockchain technology into the IPRs system. BCT represents a groundbreaking approach to managing the patent system specifically and the IPRs system broadly. The secure attributes of blockchain facilitate the efficient storing and use of secret information, enabling patent applicants to accelerate the patent acquisition process for their intellectual property while safeguarding their rights against potential infringement [[→60](#)].

Additionally, researchers suggested using blockchain technology to improve the functioning of IP offices, strengthen customs processes for identifying fake goods, and boost the effectiveness of IP rights holders' management of their rights. Additionally, they offered some recommendations on how to make blockchain technology progress and reach a wider audience. They also discussed how to successfully integrate blockchain technology into the different IP management services and registration/transaction channels [[→61](#)]. Obtaining IPRs confers benefits beyond mere entitlements. Additionally, it is a resource that the owner can use to their advantage. As a result, once the author or inventor owns the work, they are free to dispose of its rights as they see fit and to prohibit others from using, duplicating, or selling it without permission. Since IPRs safeguard authors and inventors throughout the entire lifecycle of their work, managing these rights should consist of three main tasks: monitoring each IP right; ensuring that authors and inventors receive just compensation for using their work; and ensuring that no third party infringes upon their rights. Blockchain technology has the potential to eliminate the need for an intermediary and lower the costs associated with rights exploitation,

particularly in the context of copyright laws. Blockchain technology can also be utilized to increase an owner's rights during management and exploitation. As we've already shown, blockchain technology can be utilized to store information about a work's ownership or associated details. Rightsholders can use a smart contract to track the use of work once all relevant information – such as the work's attributes, author or inventor, ownership, type of granted right, and localization – has been recorded in the blockchain. An AI program, such as an Oracle or an Internet of Things device, can notify its owner each time it recognizes how its work is being used [[→ 62](#)].

A different study suggested managing data and processes related to product development using private blockchain technology, with an emphasis on additive manufacturing. The study used a multi-agent system built on blockchain technology to distribute and verify manufacturing intellectual property. The process entails the storage of intellectual property (IP) or production phase data on a blockchain among managers, designers, engineers, and stakeholders, both inside and external to the organization. The transaction is authenticated on the blockchain with the sender's private key to inhibit any alterations by participating stakeholders. All product-related information, including CAD files and other technical documentation, is housed in a cloud-based repository [[→ 63](#)]. To mitigate the risk of fraud, a verifiable cryptographic hash is generated for each block by amalgamating the public key checksums of the sender and recipient within the CAD file. The authors established secure virtual Network

Innovation Rooms (NIR) to enhance communication and collaboration among small and medium-sized enterprises (SMEs). Prior to gaining access to the secure collaborative environment, the SMEs opting to collaborate must electronically sign a non-disclosure agreement (NDA) specifying the intellectual property (IP) or information that will be preserved with a time stamp in a blockchain as part of their contribution to the NIR. The Ethereum platform is used to implement the NDA as a smart contract. It provides a detailed explanation of the co-creations and intellectual property that must be protected, as well as how they can only be used within the NIR [[→ 64](#)].

9.5 Challenges of data privacy in the pharmaceutical industry

The delicate and valuable nature of the information handled by the pharmaceutical industry presents several challenges for strong data privacy. The most significant of these challenges is the inherent difficulty of maintaining patient data, which includes genetic information, medical records, and treatment histories. Every data protection law, including the Health Insurance Portability and Accountability Act (HIPAA), the General Data Protection Regulation (GDPR), and others, has specific requirements that the industry must follow. Because of this, there is a complicated web of global regulatory compliance involved. In addition, breaches in cybersecurity are a persistent risk because they put the privacy of patients at

risk, and the intellectual property worth of the pharmaceutical industry is enormous [[→ 65](#)]. As well as the financial risks, the reputations of pharmaceutical companies and the trust of their patients are put in jeopardy when cyberattacks, such as ransomware and data breaches, occur. In addition, the vulnerability extends beyond the boundaries of the corporation in the complex pharmaceutical supply chain, where the protection of data privacy necessitates collaboration between distributors, manufacturers, and other stakeholders. Such a collaborative effort brings with it a delicate equilibrium that necessitates the exchange of information to advance research while simultaneously protecting proprietary knowledge. The fact that human error continues to be a significant factor in the occurrence of data breaches highlights the importance of ongoing employee education and awareness campaigns [[→ 66](#)]. Data privacy now faces additional difficulties as a result of emerging technologies like machine learning, artificial intelligence (AI), and the Internet of Things. These new challenges require a re-evaluation of the appropriate security measures. In addition, as pharmaceutical companies expand their operations across international borders, they face increased challenges in maintaining the confidentiality of data transfers across international borders and ensuring compliance with a variety of data residency requirements. Equally important is the careful management of third-party risks associated with partners and vendors, as well as the efficient administration of the data lifecycle, which includes everything from data collection to data disposal [[→ 67](#)].

The volume of data that is stored, which includes everything from the results of clinical trials to personal health profiles, is another factor that makes the protection of privacy even more crucial. It is made even more difficult by the fact that pharmaceutical operations are conducted on a global scale, which requires a thorough understanding of and adherence to a large number of data protection laws. Cybersecurity threats add a layer of complexity to the situation. Because malicious actors are constantly evolving their strategies, whether they take the form of social engineering or advanced hacking attempts, security precautions must be reviewed regularly. At every stage of the pharmaceutical supply chain, which is essential to the operation of the industry from manufacturing to distribution, vulnerabilities are introduced [[→ 68](#)]. It is because the supply chain is essential to the industry. Attempting to find a middle ground between the requirements of efficient teamwork and stringent data security procedures is an ongoing challenge that never seems to end. The industry is embracing emerging technologies at an increasing rate at the same time that new privacy concerns are emerging alongside the potential benefits of these technologies. Applications of AI, for instance, raise concerns regarding the ethical use of patient data and the transparency of algorithmic processes. As research and development activities are spread out across multiple locations, the complexities of data residency and the legalities that surround the transfer of data across international borders are brought back into focus. The management of the entire data lifecycle, including the implementation of safe disposal procedures, is an essential component in an industry where

data that is either obsolete or underutilized may still contain extremely sensitive information [[→69](#)]. Additionally, by relying on third parties from the outside, an additional element of external risk is introduced, which must be thoroughly investigated. The acquisition of patents for new treatments and medications is an essential step for businesses that are interested in innovating; however, it also places them in a never-ending competition to find a balance between the need for exclusivity and the desire of the general public to have drugs that are both affordable and easily accessible. Not only are trademarks and branding essential for market positioning and product recognition, but they also present challenges when it comes to the protection of brands and the prevention of counterfeiting. In a dynamic field where inventive processes and research techniques are frequently proprietary, protecting trade secrets becomes a delicate task. It is especially true in environments where researchers collaborate on their projects. The very foundation of innovation in the pharmaceutical industry is changing as a result of its investigation of various cutting-edge technologies, including machine learning and blockchain. AI algorithms that analyze large datasets for drug discovery raise ethical questions regarding the use of the data and how it is protected. It is in addition to the enormous potential that these algorithms possess. The use of blockchain technology to secure supply chains raises concerns about standardization and accessibility in an industry that is primarily dependent on diverse global networks. It is in addition to the fact that the technology introduces a new paradigm for transparency [[→70](#)].

The pharmaceutical industry's reaction to these challenges is further complicated by the fact that it is frequently involved in legal battles and disputes, which often involve intricate assessments of intellectual property laws. Not only do well-known cases teach us important legal strategies, but they also teach us how to strike a careful balance between protecting proprietary information and instilling a healthy level of competition in the market. In contrast, success stories are examples of how businesses are successfully integrating data privacy as well as intellectual property into their primary business plans. These stories showcase creative solutions and best practices in the industry. One of the most challenging tasks that the industry must complete is to keep up with the latest emerging trends. The landscape is constantly shifting as a result of the development of innovations such as digital twins as well as the growing convergence of fields that rely heavily on data. Given the rapid pace at which these technologies are being developed and implemented, ethical concerns are becoming increasingly important. As a result, a proactive strategy is required to anticipate and resolve potential hazards [[→71](#)]. Users have reaped enormous benefits from the Internet of Things; however, it has also brought about several challenges. According to the researchers and security specialists who were cited, the primary concerns that they have are related to cybersecurity and privacy risks. Due to these two factors, many public and commercial organizations have encountered a significant challenge. The susceptibility of Internet of Things technologies has been evidenced by the frequency of prominent cybersecurity breaches. As technology becomes

more passive and integrated into daily life, users must trust that Internet of Things devices and services are adequately safeguarded from vulnerabilities. The absence of adequate security protocols for Internet of Things (IoT) devices and services renders them prime targets for cyberattacks and data breaches. It is because unprotected data streams are left behind [[→72](#)].

Owing to the interconnectedness of IoT devices, a device's security and connectivity could potentially affect the Internet's resilience and security globally if it is not sufficiently secured. This behavior is merely the result of the challenges associated with widely utilizing identical devices within the Internet of Things [[→73](#)]. It suggests that consumers and creators of the Internet of Things must make sure they are not endangering other users and the Internet itself by putting them in situations where they could potentially harm one another, in addition to the fact that some devices can mechanically bond with other devices. Currently, a collaborative approach is needed to create a suitable and successful response to the problems facing the IoT. Owing to the intricacy of these matters, a thorough plan that strengthens the pharmaceutical sector's safeguards against data privacy violations is desperately needed. The plan ought to take into account every pertinent aspect. The strategy should include a proactive approach toward emerging threats in addition to technological innovation, legal knowledge, staff education, and employee training [[→74](#), [→75](#)].

9.6 Conclusion

In the pharmaceutical industry, IPRs are essential because they enable inventors to maintain the confidentiality of their creations for a set amount of time. The public grants rights to concepts, innovations, and creative expressions known as intellectual property rights, or IPRs. These rights are protected under various laws, including patents, copyrights, and trademarks. Moreover, data privacy in the pharmaceutical industry is crucial for effective care provision. However, safeguarding vast amounts of data has proven to be challenging, and the healthcare industry is susceptible to data breaches. Businesses need to implement data security solutions that meet compliance requirements and safeguard valuable assets to protect sensitive information. Popular technologies for safeguarding healthcare data security and privacy include authentication, encryption, and endpoint authentication. These technologies help protect sensitive information and ensure the protection of sensitive information in the pharmaceutical industry. Blockchain technology is being utilized across multiple sectors, including medical information systems, identity management, supply chain management, healthcare, insurance, and contract administration. It offers a decentralized, transparent, and immutable distributed database suitable for constructing a dependable value chain. Blockchain-based IPRs systems increase the availability of information about various IP objects, allowing for efficient storage and use of sensitive data. Researchers suggest using blockchain to manage patent systems, strengthen customs processes, and

improve IP management services. Hence, there is an immediate need for a comprehensive strategy that strengthens the pharmaceutical industry's defenses against breaches in data privacy. This is because the complexity of these issues makes it imperative that such a strategy be developed. The strategy must take into account all of the pertinent factors. At the same time, the strategy should incorporate technological innovation, legal understanding, employee training, and staff education, and it should also incorporate a proactive attitude toward new threats.

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Chapter 10 Regulatory affairs and compliance in the pharmaceutical sector

Abstract

The pharmaceutical sector is accountable for guaranteeing the safety, efficacy, and quality of therapeutic products. Attaining worldwide competitiveness in pharmaceutical manufacturing requires rigorous adherence to ongoing quality, environmental protection, occupational health, and safety standards. The study analyzes the recommendations and effective implementation strategies of the United States Food and Drug Administration. The objective of the chapter is to deliver an exhaustive overview of US regulatory compliance, emphasizing US FDA rules. Artificial intelligence (AI) is an advanced technical instrument that is significantly transforming several GMP sectors. AI systems can understand complex regulatory documents, therefore aiding pharmaceutical companies in staying abreast of the continually evolving compliance demands for medication research and approval. The application of AI extends to crucial areas such as compliance with packaging and labeling, significantly improving the efficiency and accuracy of regulatory affairs.

Keywords: regulatory affairs, US-FDA, DCGI, CDSCO, GCP, GLP, GMP,

10.1 Introduction

Regulatory affairs (RA), or government affairs, is a profession within regulated industries such as banking, energy, pharmaceuticals, and medical devices. “Regulatory affairs” holds significant relevance in the healthcare industries, encompassing medicines, medical devices, biological products, as well as functional foods [[→1](#), [→2](#), [→3](#), [→4](#)]. Most companies have dedicated divisions of RA personnel, regardless of whether they are small, innovative biotechnology firms or giant, multinational pharmaceutical enterprises. AI is swiftly revolutionizing the pharmaceutical sector, including regulatory issues. The application of AI in pharmaceutical regulatory concerns is developing, although it has the capacity to transform the business. The incorporation of AI into RA will facilitate the monitoring of current industry trends and enhance understanding of AI’s potential advantages in medication creation and its regulatory approval processes [[→5](#)]. The pharmaceutical division is responsible for being aware of the legal specifications needed to have new products authorized. They are aware of the promises the business has made to the authorities responsible for approving the product. They also provide the authorities with annual reports and additions. RA generally interacts with one of the centers, including the Center for Drug Evaluation and Research, at FDA headquarters instead of the local district offices. Although not directly linked to regulatory issues, companies must comprehend and assess alterations in drug manufacture and testing protocols to ascertain the proper timing and necessity of alerting the FDA [[→6](#)].

In the present competitive environment, the success of a product, and consequently the business, hinges on its speed to market. Consequently, the company's proficiency in managing its RA is vital to its financial performance. Insufficient data reporting may impede a prompt and favorable assessment of a marketing application. The creation of a novel pharmaceutical can incur costs amounting to millions of euros, dollars, or pounds, and a three-month postponement in its market launch might lead to substantial financial repercussions. A product recall may be required if all pertinent data was not comprehensively disclosed or if a product with erroneous labeling was distributed [[→7](#), [→8](#), [→9](#)]. The Department of Regulatory Affairs often acts as the company's primary liaison with the government. The RA team collaborates with marketing, R&D, and other departments to expedite time to market by creating innovative products that use recent technological and regulatory breakthroughs. Anticipated new goods are predicted to provide considerable benefits to the company's core operations, outweighing a slight reduction in time to market, hence promoting a major improvement in outcomes and productivity. Promoting adaptable clinical trial volunteers, relinquishing faster regulatory approval, and circumventing procedural dangers can all facilitate the swift development of novel drugs while minimizing costly errors and delays [[→10](#)]. The RA team collaborates with marketing and R&D, utilizing recent regulatory as well as technology breakthroughs to create innovative products that expedite market entry. The anticipated introduction of new products is predicted to significantly enhance the company's profitability, as

even minor savings in time to market result in considerable revenue and profit increases. Expedited regulatory approval, adaptable clinical trial designs, and process mitigation can facilitate the swift development of novel products while minimizing expensive errors and delays [[→ 11](#)].

The Indian economy has evolved over the last 20 years, moving from a quota-based, government-controlled system to a more liberalized business environment. Nonetheless, there is a general perception that the changes in legal and regulatory compliances are onerous and unfavorable to businesses and have not kept up with the times [[→ 12](#)]. The respondents identified the main problems with company structure, employee, ecological, and factory-related compliances as being outdated requirements, unfriendly procedures, and unclear laws and regulations. The main suggestions made by the respondents to lessen the burden associated with compliance are to simplify laws and procedures, boost self-certification and self-regulation, increase process automation, and manage government officials' discretionary powers more skillfully [[→ 13](#)]. Achieving global competitiveness in the manufacturing of pharmaceutical drugs and drug products requires a high degree of compliance, encompassing ongoing quality, environmental protection, and occupational health and safety topics. Competitor ability will be adversely affected by both excessive compliance measures and poor compliance. As a result, it is becoming more and more crucial to assess compliance levels, look into compliance expenses, and establish a link between compliance levels and operational measurements inside businesses. Furthermore, there was a greater emphasis on getting foreign

money to invest in the manufacturing sector [[→14](#)]. Compliance errors can have a financial cost to your company, but compliance officers who take on risks run the real risk of personal liability because they are responsible for compliance within the organization. Compliance heavily relies on ethical behavior, and it is the individual's responsibility to abide by the regulations that apply to the particular business. Too frequently, compliance is restricted to a few departments or even individuals, but other functions may need to provide input to comply with certain regulations. The company may incur negative expenses if you choose not to use a team of experts. Everyone ought to be aware of what compliance entails, how it may impact their role, and how it all fits into the greater scheme of things. Time is another expense of noncompliance. While many countries require the forms to be submitted in person, some allow businesses to submit filings online. As a result, businesses require the assistance of a specialist to complete the forms in the local tongue and submit all necessary documentation to the local authorities [[→15](#)].

10.2 Regulatory bodies and framework

The objectives of every national pharmaceutical legislations worldwide are to assess and address the global convergence and harmonization of pharmaceutical laws. The predominant basis for national pharmaceutical evaluations is the uniformity of efficacy, safety, and quality requirements. Nonetheless, national legislation and technical specifications have differed,

at times considerably, and the execution of procedures and systems to attain these identical aims has been markedly dissimilar. Globally, numerous frameworks exist for the regulation of drugs [[→ 16](#)]. These are determined by public health requirements, the magnitude of the pharmaceutical market, and the resources at hand. Various institutional pharmaceutical systems have been built as a result of the diverse histories, cultures, political experiences, and economic profiles of nations. Furthermore, there are still significant gaps in the regulatory capacities of industrialized and developing nations, even with some advancement in recent years. The availability of medications and the state of pharmaceutical systems and regulations vary greatly among global regions. The disparity in financial and human resources between regions with abundant resources and those with limited resources is widening rather than narrowing the regulatory system gap, as depicted in [→ Fig. 10.1](#).

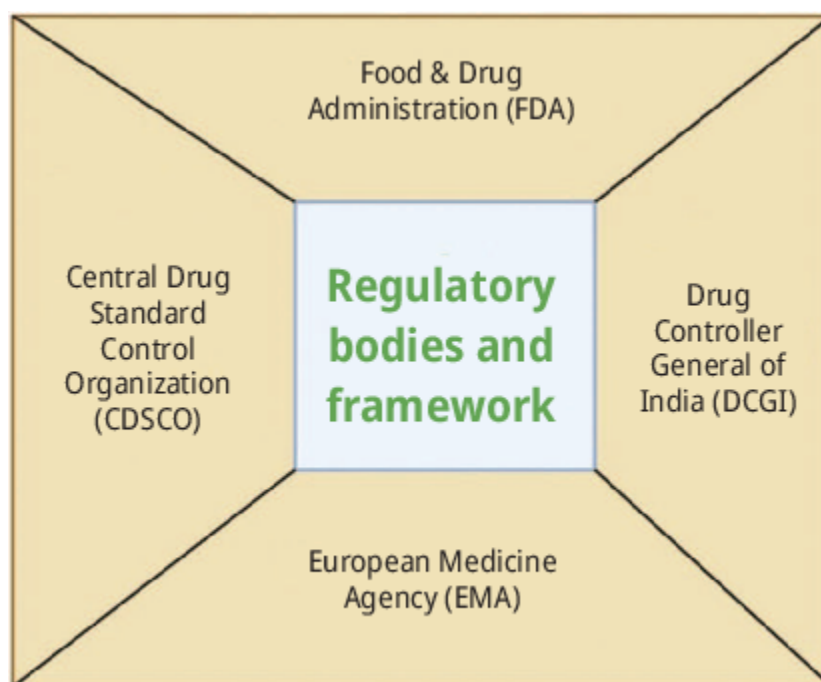


Fig. 10.1: Schematic representation of various regulatory bodies and frameworks in the pharmaceutical field.

10.2.1 Food and Drug Administration (FDA)

The safety of prescription medications is a vital public health issue for millions of Americans dependent on these pharmaceuticals for disease management and treatment. The Federal Food, Drug, and Cosmetic Act (FDC Act) designates the U.S. Food and Drug Administration (FDA) with the responsibility of ensuring the safety and efficacy of all prescription medications for patient use. The entity that developed the drug must submit an application to the FDA for assessment and approval, accompanied by sufficient data and information to demonstrate the drug's efficacy and safety, before its authorized sale to American consumers. The FDA assesses the advantages and disadvantages when deciding on the approval or

possible withdrawal of prescription medication by the FDC Act and its related regulations. The statute and regulations do not establish a quantitative threshold for drug safety findings or a specific method for drug safety analysis [[→17](#)]. The FDA authorizes the application of a drug for specific indications within the regulatory framework based on the supporting data that is presented. The drug sponsor conducted numerous clinical trials throughout three phases to assess the medication's efficacy and safety for its intended purpose. Typically, six comprehensive Phase III trials are randomized clinical trials that include treatment and control groups for comparative analysis. The FDA also authorizes the medication's labeling, which includes exhaustive information about the drug's chemistry, toxic effects, clinical data, evidence, and warnings, as well as usage characteristics. These clarifications are intended for healthcare professionals, including physicians and chemists. The labeling is present during the drug's transfer from the sponsor to the dispenser [[→18](#)].

10.2.2 European Medicine Agency (EMA)

Two novel procedures for pharmaceutical approval were instituted with the formation of the EMEA. Biotechnology and other high-tech items are required to undergo a centralized approval process, while other products are exempt from the following requirements. A single national entity is responsible for conducting the scientific evaluation, and the Committee on Proprietary Medicinal Products is responsible for overseeing the assessment process. At the national level, a secondary, decentralized procedure is

implemented. Applications are submitted to the “Reference Member State,” which is the market in which the business intends to introduce its product [[→ 19](#), [→ 20](#), [→ 21](#)]. The agency thereafter assists other “Concerned Member States” in acknowledging the marketing authorization. The current national process is still in place for applicants who are only interested in one country. The new regulations have not altered the fundamental components of these procedures, but they have brought about certain modifications. Furthermore, the EMEA has a history of receiving criticism for its lack of transparency [[→ 22](#), [→ 23](#), [→ 24](#)]. To address this, multiple provisions are introduced in the new rules. National authorities are now mandated to disseminate meeting minutes and details in conjunction with their procedural regulations. Upon approval, the authorities will promptly issue the marketing authorization, assessment report, justification for the opinion, and Summary of Product Characteristics. Historically, national agencies were not mandated to reveal the results of their evaluations. The new rule pertains to European Public Assessment Reports (EPARs), which provide appraisals of new applications sanctioned by the EMEA. While not to the extent desired by many critics, they are now required to compose their information “in a manner that is intelligible to the public” and provide a clear delineation of the terms and conditions associated with the product’s use [[→ 25](#), [→ 26](#)].

10.2.3 Other regulatory bodies

The National Regulatory Authority of India is known as the Central Drugs Standard Control Organisation (CDSCO). The Directorate General of Health Services (DGHS), a division of the Ministry of Health and Family Welfare, operates under the Government of India. The Drugs and Cosmetics Act appoints the CDSCO as the principal regulatory authority for pharmaceuticals, tasked with executing the responsibilities assigned to the national government. The headquarters are located in New Delhi [[→27](#), [→28](#)]. The authority to issue licenses for intravenous fluids, vaccinations, sera, and blood and blood products is shared between the CDSCO and state regulatory bodies. The Drug Controller General of India (DCGI) regulates medical devices and pharmaceutical items under the CDSCO. The DCGI solicits counsel from the Drug Consultative Committee (DCC) and the Drug Technical Advisory Board (DTAB). The Central Licensing Approval Authority (CLAA) oversees the categorization and licensing of medical equipment. It is responsible for formulating and enforcing safety standards, conducting post-market surveillance, alerting consumers, and recalling pharmaceutical products in instances of adverse effects [[→29](#)]. The Therapeutic Goods Administration (TGA) is the Australian regulatory authority responsible for overseeing all operations associated with therapeutic products, including product development, production, and distribution. The TGA regulates prescription medications, over-the-counter drugs, complementary and alternative therapies, sunscreens, medical equipment, biological products, blood and its components, and other therapeutic items [[→30](#), [→31](#), [→32](#), [→33](#), [→34](#)].

The UK Department of Health and Social Care's Medicines and Healthcare Products Regulatory Agency (MHRA) is another regulatory body that likely has a significant global impact. This might be due to a few major pharmaceutical companies that distribute their products worldwide and have their headquarters and manufacturing facilities in the UK, such as GlaxoSmithKline and AstraZeneca. Both regulatory bodies may have regulations regarding Good Manufacturing Practices (GMP) that the market will implement [[→35](#), [→36](#)].

10.3 AI-based compliance strategies in the pharmaceutical industry

Regulatory compliance is crucial for ensuring the safety, efficacy, and quality of products in the food and healthcare industries. The US FDA is responsible for formulating policies and regulations to safeguard public health and supervising the regulatory structure in the United States. Entities operating in regulated sectors, including medicines, medical devices, and food and beverage, are required to comply with US FDA regulations [[→37](#)]. Regulatory compliance is crucial since it ensures product integrity, protects consumer interests, mitigates health hazards, and maintains public faith in several industries. The USFDA, as the primary regulatory authority, formulates and enforces laws via various legislation and directives that govern the production, marketing, labeling, and distribution of regulated products [[→38](#)]. Good Manufacturing Practices (GMP), Good Clinical

Practices (GCP), and Good Laboratory Practices (GLP) are legal compliance standards employed in the pharmaceutical sector, as shown in [→ Fig. 10.2.](#)

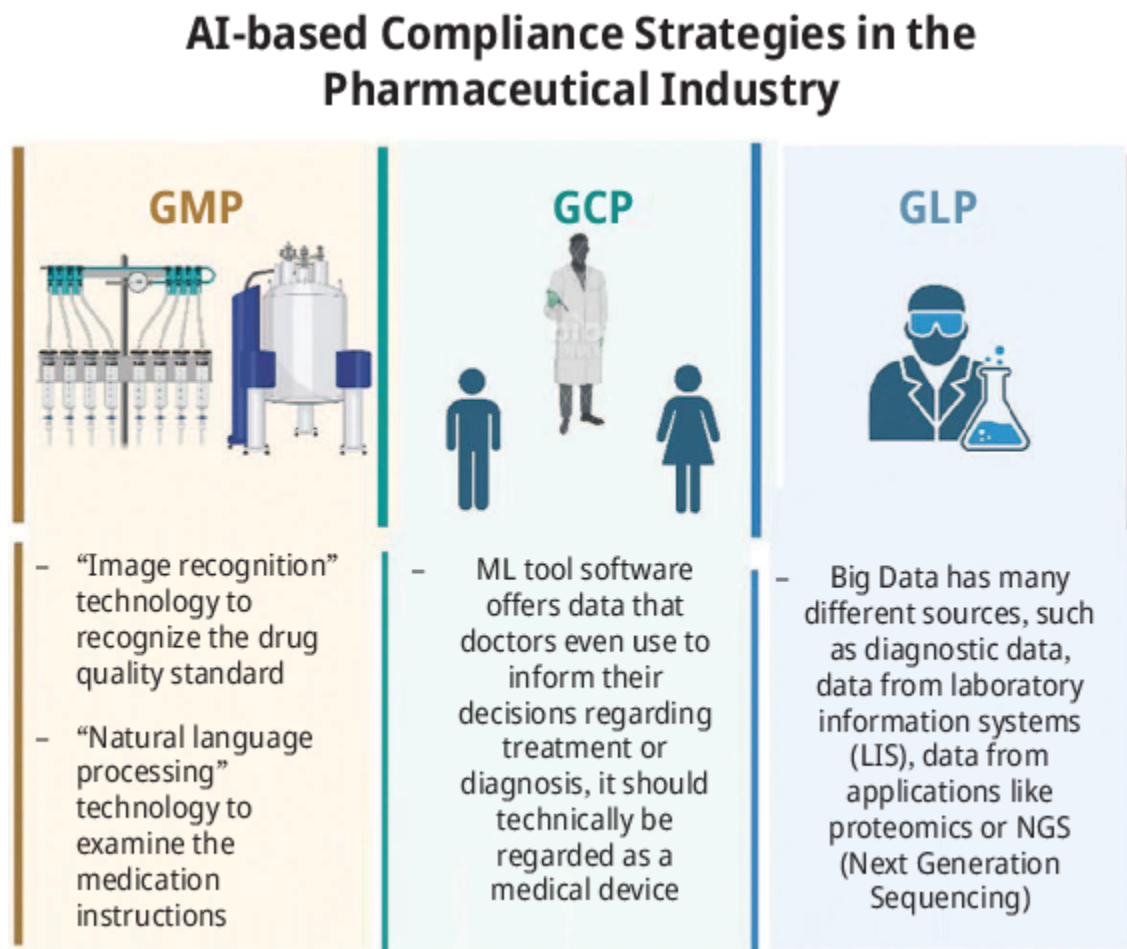


Fig. 10.2: Various AI-based compliance strategies applied in the pharmaceutical industry.

10.3.1 Good Manufacturing Practices (GMP)

A mandatory set of guidelines for the pharmaceutical industry is known as Good Manufacturing Practice (GMP) for Pharmaceutical Production, or GMP. Pharmaceutical manufacturers must comply with GMP to guarantee

the safe manufacturing of medications, uphold a reliable pharmaceutical equipment process, and rigorously maintain the quality management and verification system by relevant national requirements. The research and development of new pharmaceuticals increasingly rely on modern science and technology. Artificial intelligence is a state-of-the-art technology instrument that is significantly advancing and transforming various GMP domains. A variety of quality assessments, including drug composition analysis, microbiological quantitative evaluation, and pharmacological and toxicological testing, are essential for pharmaceuticals manufactured under GMP quality control. Traditional detection methods may exhibit low accuracy and require substantial labor and material resources. Utilizing deep learning and product data analysis, AI technology can enhance the efficiency and precision of detection. To improve detection effectiveness, “image recognition” technology to evaluate drug quality standards [[→39](#)] and “natural language processing” technology can be employed to scrutinize pharmaceutical instructions. AI technology is mostly utilized for data collection in GMP quality control. It can collect and compile comprehensive production data, encompassing production status, drug inspection data, and instrument data, to create an accurate and full database. AI technology can assess and improve all types of data produced throughout the pharmaceutical production process, including data screening, categorization, and analysis, employing machine learning (ML) and other technical methods. It allows for automated production link monitoring. AI technology can expeditiously perform extensive inspection

tasks, as well as effectively analyze and compare the inspection outcomes, thereby expediting the elimination of potential quality issues in the production process [[→ 40](#)].

Artificial intelligence can completely automate the pharmaceutical production lifecycle, thereby diminishing the necessity for human involvement. Conversely, the conventional medication manufacturing method requires a substantial workforce to manage the production operations. In an optimal scenario, adherence to GMP management specifications entails the maximal integration of automation and information within defined parameters and constraints, aiming to supplant human actions and operations, thereby achieving complete automation of batch control and management processes [[→ 41](#)]. It is frequently observed in domestic pharmaceutical production that variations in drug quality arise from errors in each link in the drug production process. AI can also meticulously oversee and manage each step of the production process to reduce the need for human involvement. Artificial intelligence (AI) technology keeps an eye on the entire pharmaceutical production process in real-time to identify possible issues early on and improve the safety, efficacy, and quality of the finished product [[→ 42](#)].

10.3.2 Good Clinical Practices (GCP)

Massive data availability and exceptional computing power are encouraging the widespread application of AI, but they are also posing ethical and

societal issues that need to be resolved to maximize the advantages and minimize the risks. While there are many different ways to apply ML in clinical trials (CTs), including patient recruitment, study design, endpoint definition, and more precise diagnosis, the evaluation of these technologies is influencing regulatory authorities' (RAs) work in authorizing clinical studies [[→43](#)]. From a regulatory standpoint, the lack of explicit guidelines and standardized protocols engenders uncertainty for applicants and regulatory authorities [[→44](#)], complicating the definition of these instruments, which, based on their declared intended use, may sometimes be employed in clinical trials, such as for inpatient enrollment selection, to reduce costs associated with labor-intensive procedures, and consequently may not be categorized as medical devices. Nonetheless, since ML tool software offers data that doctors even use to inform their decisions regarding treatment or diagnosis, it should technically be regarded as a medical device [[→45](#)]. An ML predictive model was developed by researchers to determine the most important issues that should be taken into consideration when using AI or ML systems in a CT setting. The effectiveness of the training model depends on the quality of the data, particularly its representativeness, which is essential for the learning process of any ML algorithm. Three data sets are typically required for training, model optimization, and evaluation: the training set, validation set, and testing set. The initial sample dataset must be rigorously created and suitable for its intended purpose, encompassing the prognostic characteristics of the specific patient cohort having the CT scan to aid in the

training and validation of the ML predictive model [[→46](#)]. The most impacted issue combination areas demonstrate the importance of data in informing the adaptive algorithms. Data are crucial to the assessment process and are closely linked to technical safety and reliability. The software's accuracy relates to its ability to make precise predictions; an incorrect output may lead to unintended consequences, such as suggesting a medication that is clinically less effective and has more adverse effects. The quality of the data utilized to shape the final CT study conclusions ultimately dictates the evidential strength generated by the clinical trials that underpinned the instrument's development phase, which also impacts the technical integrity and safety of the tool. Performance metrics assess the predicted efficacy of the ML model by analyzing risks and accuracy [[→47](#)].

The representativeness of datasets that may exhibit inequalities constitutes another potential source of bias. Utilizing these datasets to train the algorithm may lead to overestimation or underestimation of findings [[→48](#)], producing biased systems with diminished predictive accuracy that either establish or intensify prejudice among population groupings, contravening fairness standards. Transparency can enhance the scientific rigor of clinical trials, which may be undermined by publication bias, a significant concern where only favorable clinical study outcomes are disclosed, potentially generating data suitable for algorithm training. Algorithmic technologies possess an intrinsic opacity that complicates the ability of the scientists who created the algorithms to thoroughly understand the rationale behind the intricate outputs they generate and to elucidate the

technical procedures involved [[→49](#), [→50](#)]. This trait, known as the “black box,” makes AI less predictable and increases concerns about the possibility of losing human oversight, which in turn makes AI less trustworthy.

10.3.3 Good Laboratory Practices (GLP)

Laboratory medicine consistently ranks among the medical specialties with the highest degree of digitalization. Since their beginnings, automation, electronic data transfer, and electronic reporting have become increasingly prevalent [[→51](#)]. Medical laboratories maintain extensive databases comprising quality control reports and test results. Moreover, they typically have complex quality management systems established. Thus, it is not unexpected that laboratory medicine exemplifies a model discipline for the digitization of healthcare. Conversely, recent advancements in data science, including ML and artificial intelligence, have not yet completely permeated the field of laboratory medicine. The experience-based opinion paper does not primarily concentrate on the latest and forthcoming breakthroughs in AI and ML within laboratory medicine, since several recently published publications offer a comprehensive assessment. Big Data encompasses various sources, including diagnostic data, laboratory information systems (LIS), proteomics, and Next Generation Sequencing (NGS) [[→52](#), [→53](#), [→54](#), [→55](#)]. Data that can be easily found must be stored to facilitate its retrieval. To assign individual results to the respective patients and collection times for “standard” examinations is typically accomplished through the use of a patient identifier (PID) and date. It is easier said than

done, depending on how the laboratory is set up. Some potential pitfalls include the possibility that different patients at various branch laboratories may have the same PIDs assigned to them, or that analyses done on emergency patients who are not identified may not be able to be linked to the correct individual once their identity is known. Furthermore, for research purposes, results from various patients may be consolidated under a “collective” PID [[→56](#)]. Moreover, data may get convoluted when samples are recorded based on the scheduled collection date instead of the actual collection date, resulting in analysis timestamps preceding the collection. Specialized examination equipment is frequently not integrated with the LIS, complicating its retrieval. Semantic interoperability represents a significant and potentially important facet of Big Data in laboratory medicine. Each data item must be assigned distinctly and semantically; the optimal method for it is the utilization of standardized coding, such as LOINC (Logical Observation Identifiers Names and Codes) [[→57](#)]. The specifications of research efforts employing laboratory analysis require precise granularity; it is insufficient to identify them merely by their superficial title. Thus, any type of creatinine measurement may be enough for a “safety lab measurement,” but it would be inadequate for technique comparison studies or for determining reference intervals [[→58](#)].

10.4 Applications of AI in regulatory affairs

AI can aid RA by supplying knowledge on the latest advancements in the industry, particularly about the prospective benefits of AI for drug research and regulatory approval. By finding new opportunities, automating tasks, and enhancing decision-making, artificial intelligence (AI) can assist professionals in drug discovery in making decisions more quickly and effectively. It enables them to launch new medications faster.

Pharmaceutical companies' RA departments or groups act as a channel to national and international regulatory bodies whose purpose is to approve products following current laws and regulations. Consequently, products are approved by regulatory agencies based on data and records submitted during the drug discovery and product development process [[→ 59](#), [→ 60](#)].

Case-specific quality assurance (QA) includes all validation processes that validate the results of AI-based applications developed specifically for each patient or machine. The output can be utilized in later phases of the RT workflow if the quality assurance (QA) outcomes are considered acceptable. The output will undergo a second verification if the case-specific assurance of the quality check proves to be unsuccessful. A different secondary algorithm may be used to evaluate the efficacy of some applications [[→ 61](#)]. Furthermore, ML-based auto-segmentation quality assurance has been the subject of recent research. Decision trees can be developed to classify the morphology of a particular auto-segmentation input by employing distinct features for each ROI. Researchers have developed increasingly sophisticated algorithms and models owing to the enhanced availability of data and processing capacity, which has markedly

advanced various sectors. A notable domain of study that has progressed markedly is natural language processing (NLP) [[→62](#), [→63](#), [→64](#)]. Researchers have employed deep learning architectures, including transformers and recurrent neural networks (RNNs), to develop more precise and efficient language models. These models could enhance human-machine connection, streamline process automation for customer care, and develop more advanced search engines and automated assistants. This domain that has experienced substantial advancement in computer vision [[→65](#), [→66](#), [→67](#), [→68](#), [→69](#), [→70](#), [→71](#)]. Convolutional neural networks (CNNs) have facilitated enhanced object recognition and image classification. These models have the potential to improve medical imaging analysis, manufacturing quality assurance, and autonomous driving. Although deep learning approaches can autonomously identify and generate abstract features, conventional ML techniques generally necessitate domain expertise for feature engineering or creation [[→72](#), [→73](#), [→74](#), [→75](#)]. Deep learning methodologies utilizing RNN, autoencoders, and CNN yield favorable outcomes across many inspection applications [[→76](#), [→77](#), [→78](#)].

Artificial intelligence is actively employed to collect data on adverse drug reactions (ADRs) and adverse drug events, process international clinical standards reports, extract drug-drug interactions and assess their impacts, identify populations susceptible to ADRs and facilitate personalized care, indicate drug side effects, represent clinical trials, and integrate prediction uncertainty into diagnostic classifiers to enhance patient safety and

pharmacovigilance [[→79](#)]. The increasing variety of disease states, medications, and ADRs that patients experience has led to a growing value in automation through ML techniques. Some organizations use ML to compare a reference standard to accelerate the use of semi-supervised, rule-based adverse event reports [[→80](#)]. Furthermore, using user-generated content from a depression-related forum, researchers [[→81](#)] investigated the possibility of modeling the data exchange among users concerning drug-related therapies for depression. They discovered that the adverse drug responses (ADRs) of these pharmaceuticals can be more precisely recognized by natural language processing (NLP), and they corroborated the ADRs with relevant medical literature. ML is also employed in pharmacovigilance to evaluate the efficacy of natural language processing (NLP) in classifying unstructured text within safety incident reports. Researchers discovered that NLP serves as a protective mechanism by detecting instances that lead to severe harm or fatality. They assessed the efficacy of automatically classifying free text in safety event reports to quantify the severity of injury outcomes [[→82](#)].

Intelligent packaging serves as an effective instrument for product monitoring and condition tracking, allowing for fast information access and interchange, thus supporting prompt responses and timely decision-making. It is particularly accurate when clever packaging is combined with scientifically grounded ideas. These attributes are essential for any approach concerning drug safety or biosecurity [[→83](#)]. In the pharmaceutical business, the utilization of AI encompasses essential

domains such as adherence to packaging and labeling regulations, thereby enhancing the efficiency and precision of RA. Artificial intelligence technologies, particularly ML algorithms, may analyze extensive databases of regulatory requirements [[→ 84](#)]. This allows regulatory bodies to ensure that drug packaging and labeling follow the stringent guidelines that they have established. Through the utilization of Natural Language Processing (NLP), artificial intelligence systems can comprehend intricate regulatory texts, thereby assisting pharmaceutical companies in remaining current with the ever-changing compliance standards. Furthermore, automation driven by AI makes it easier to identify potential errors or discrepancies in packaging and labeling, which in turn reduces the likelihood of companies failing to comply with environmental regulations. Not only does the incorporation of AI make the process of compliance more efficient, but it also reduces the likelihood of errors that could put the safety of patients at risk. AI applications in packaging and labeling compliance are becoming increasingly important as the pharmaceutical industry continues to embrace digital transformation. These applications are becoming indispensable tools for ensuring that the highest standards of regulatory compliance and product integrity are preserved [[→ 85](#)]. Moreover, AI has a significant influence on packaging and labeling compliance that extends beyond simple regulatory compliance; it is essential to the optimization of these procedures. AI-powered solutions can improve labeling design efficiency and ensure that consumers and healthcare professionals receive important information in an understandable and thorough manner. Predictive analytics

allows AI to foresee possible problems with labeling and packaging, allowing for proactive corrections and avoiding expensive delays in regulatory approval. Continuous improvement is also facilitated by real-time monitoring capabilities since AI systems can quickly make the required adjustments and adapt to changing regulatory environments. In addition to guaranteeing compliance, the flexible strategy puts pharmaceutical companies in a position to easily handle regulatory changes. Essentially, the use of AI in packaging and labeling compliance represents a paradigm shift in which technology is viewed as a tactical ally in maintaining regulatory standards and encouraging creativity and adaptability in the pharmaceutical sector [[→ 86](#)].

10.5 Conclusion

In conclusion, Regulatory Affairs (RA) is essential in the pharmaceutical sector, guaranteeing adherence to legal and regulatory standards for drug development and approval. The regulatory framework differs among nations, with organizations like the FDA in the United States, EMA in Europe, and NRAs in India supervising the clearance procedures. Adherence to rules such as GMP, GCP, and GLP is crucial for guaranteeing the safety, efficacy, and quality of pharmaceutical goods. AI is increasingly being utilized in regulatory affairs to enhance efficiency and accuracy. AI can automate processes, analyze large amounts of data, and assist in decision-making, particularly in drug discovery and regulatory approval. It

can also assist in adverse drug reaction monitoring, pharmacovigilance, and intelligent packaging compliance. However, the implementation of AI in regulatory affairs also presents challenges. Clear guidelines and standardized procedures are needed for the application of ML in clinical trials, and ethical and societal issues must be considered. Additionally, the digitalization of laboratory medicine and the use of AI in quality assurance and auto-segmentation are areas that require further development. To improve the regulatory environment, suggestions include simplifying laws and procedures, increasing self-certification and self-regulation, and managing government officials' discretionary powers more effectively. Harmonization of pharmaceutical laws across different countries is also a goal for regulatory bodies. The incorporation of AI in regulatory activities can enhance efficiency, refine decision-making, and guarantee adherence to regulations. The field is under continuous development and can transform the pharmaceutical business.

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Chapter 11 AI and big data in post-marketing surveillance

Abstract

Post-marketing surveillance, or PMS, involves monitoring a drug's safety after clinical trials and successful market introduction. The “timely identification” of “novel” adverse drug reactions (ADRs) that differ from one another in terms of their “clinical nature, degree, and frequency” is the main goal of PMS. Examples of new technologies covered in the chapter include machine learning, artificial intelligence, and big data analytics. Many obstacles exist when using AI in post-marketing surveillance, and these technologies have made data privacy protection more difficult. Drug safety monitoring benefits greatly from big data analytics since it makes it easier to identify issues early on and to keep an eye out for potential signals in real time. The creation of adaptable, fact-based quality assurance frameworks that encourage innovation and more informed decision-making can be aided by AI solutions. They can also assist in meeting EU MDR/IVDR regulations for PMS practitioners and manufacturers of medical devices. Moreover, many obstacles are discussed regarding privacy protection when using AI for post-marketing surveillance. Therefore, complicated algorithms are used in healthcare AI applications, which give private AI firms access to patient medical data even when “anonymization” is employed, thus posing a risk to privacy.

Keywords: surveillance, data analytics, drug safety, machine learning, adverse drug reaction, stakeholders,

11.1 Introduction

Within the domain of pharmaceutical sciences, the process of drug innovation is an ongoing endeavor that frequently results in the identification of novel therapeutic agents. A medicine's development, from the idea to the final product, is thought to be an extremely intricate process that examines every facet of the medication to ensure sufficient safety at the time of approval. Post-marketing surveillance (PMS) can be defined as the process of monitoring a drug's safety following successful market release from clinical trials [[→1](#)]. The primary objective of PMS research is to identify novel beneficial and detrimental effects. Other important factors include off-label use of drugs, orphan drug concerns, and difficulties in conducting international research studies in the pediatric population. It is a crucial tool that helps paint a clearer picture of the benefits and drawbacks that a drug could start to pose over an extended period of administration by tying the strength of the drug-exposed to the cause of the adverse events [[→2](#)]. Phase IV studies are those related to PMS because most medications cannot be commercialized without first completing clinical trials in phase III. Rare adverse drug reactions (ADRs) will only be partially disclosed by clinical trials conducting safety and efficacy evaluations of novel pharmaceuticals. Results are more realistic because they take place in more natural environments and provide proof to support or improve the safety of authorized medications [[→3](#), [→4](#)]. Pharmacovigilance and

pharmacoepidemiology are the two areas of pharmacology that are covered by post-marketing drug monitoring programs. Pharmacovigilance, also known as medication safety surveillance, has as its main objective the “prompt detection” of “novel” adverse drug reactions (ADRs) that differ from one another in terms of the “clinical nature, severity, as well as frequency.” “Population-based investigation into the use of drugs and the risks associated with these uses” is the definition of pharmacoepidemiology. It is important to stress the importance of pharmacovigilance as well as the idea that a drug’s practical application begins the moment it is marketed [[→5](#), [→6](#), [→7](#)].

The World Health Organization suggests that companies look for information in literature and other sources to find out about any new risks associated with using or operating a device. The new Medical Device Regulation (MDR) and the In Vitro Diagnostics Medical Device Regulation (IVDR) from the European Union (EU) demonstrate the substantial increase in PMS demands placed on in vitro diagnostic and medical device companies. Gathering and assessing pertinent specialized, scientific, and technical sources is one requirement [[→8](#)]. The regulations do not specify the use of literature records or the format in which the documentation of the search process should be done; they only call for a thorough description of the chosen data sources, gathering strategies, and evaluation techniques. New EU regulations mandate that companies that produce in vitro diagnostics and medical devices actively search for information regarding the safety and/or efficacy of their products. It will ensure that gadgets

remain current and improve early problem detection. The industry will undoubtedly benefit from the shift to proactive surveillance [[→9](#), [→10](#)]. To comply with the new regulations, businesses will need to invest a significant amount of money in new procedures, staff, and rules. It is noteworthy that many small and medium-sized businesses, particularly those with outdated or fragmented data systems, find it difficult to adequately review and analyze the massive volumes of data offered from various sources. New EU regulations mandate that companies that produce in vitro diagnostics and medical devices actively search for information regarding the safety and/or efficacy of their products [[→11](#), [→12](#)]. To keep up with the new regulations, businesses will need to invest a significant amount of money in new procedures, staff, and rules. It is noteworthy that many small and medium-sized businesses, particularly those with outdated or fragmented data systems, find it difficult to adequately review and analyze the massive volumes of data available from multiple sources. It is valid across a wide range of sectors, including healthcare [[→13](#)]. As shown in [→Fig. 11.1](#), artificial intelligence (AI) tools that improve the speed and accuracy of signal detection and enable a wider range of data coverage include machine learning (ML), deep learning (DL), as well as natural language processing (NLP) [[→14](#), [→15](#), [→16](#), [→17](#)]. Numerous aspects of healthcare, such as risk management, medication discovery, diagnosis as well as clinical decision-making, and discovery, are areas where these systems exhibit great promise [[→18](#), [→19](#)].

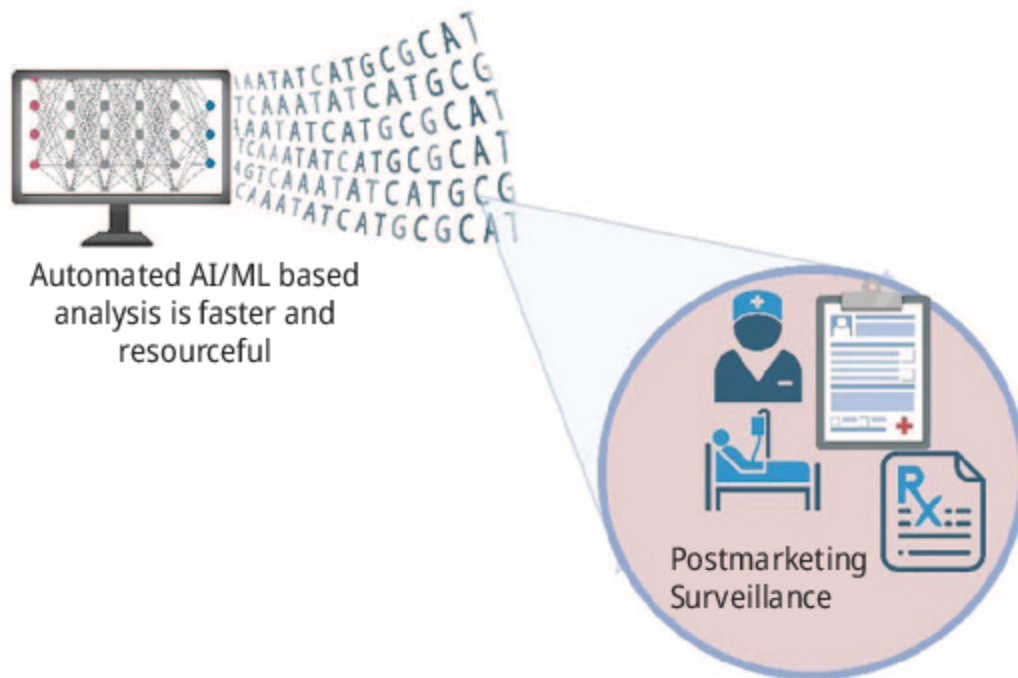


Fig. 11.1: Schematic representation of the analysis of post-marketing surveillance via AI.

In a study published by Celgene and IBM, deep-learning techniques to evaluate ICSR processing exceeded many accuracy rate thresholds, suggesting the potential of AI to improve data consistency and quality along with reporting efficiency [[→20](#)]. AI in pharmacovigilance is favored by drug safety professionals, per a survey conducted by Celgene’s Global Drug Safety and Risk Management division. This is because it would free them up to concentrate on information evaluation and analysis as opposed to labor-intensive tasks [[→21](#), [→22](#), [→23](#), [→24](#)]. Similarly, artificial intelligence (AI) solutions have a lot to offer PMS practitioners as well as medical device manufacturers. They can accomplish this by lowering the time and cost associated with adhering to EU MDR/IVDR regulations and by establishing flexible, fact-based quality management systems that promote innovation and more deliberate decision-making. Understanding

that AI can enhance and simplify adherence to QIAGEN, Huma, and EU MDR/IVDR, AI developed a state-of-the-art post-market intelligence platform to enhance signal detection and expedite PMS workflows linked to risk and performance evaluation through literature surveillance [[→25](#)]. Keeping an eye on published research is essential to PMS because it can offer important insights into pathogen genetics, patient demographics, emerging technologies, the safety and efficacy of devices, and possible off-label use. Nonetheless, more than 35 million citations and abstracts are stored in biomedical databases like PubMed [[→26](#)], and approximately 1.4 million new citations are added annually [[→27](#), [→28](#)].

11.2 Role of machine learning in post-marketing surveillance

There has always been a need for trustworthy, effective, and cost-effective techniques to evaluate students' written and spoken abilities. Project Essay Grader (PEG), created in 1966 by Ellis Page, is an application that generates computer-generated scores according to writing characteristics like grammar, essay length, and structure. With it, automatic essay scoring (AES) technological advances were introduced [[→29](#), [→30](#)]. AES systems, like PEG, are designed to make essay scoring faster and easier, with the goal of either supporting or replacing human scorers [[→31](#)]. These kinds of systems use predictive models that analyze the text and, depending on how well the results correlate with scores generated by humans, generate

standardized scores at varying degrees of accuracy in forecasting. Since the release of PEG, numerous other AES systems have been invented to improve predictive model accuracy and provide extra features like critical feedback and plagiarism checks. AES systems are currently divided into two main categories: those that use raw-text techniques and those that use hand-engineered essay features [[→32](#)]. Hand-engineered features, which employ domain expert knowledge to produce numerical illustrations of the text that allow basic statistical modeling techniques to find relevant patterns for scoring, have long been the industry standard. In raw text approaches, the space of possible words is mapped to a higher multidimensional integrating space using statistical models, usually neural networks, encoding a deeper comprehension of the word [[→33](#)].

Moreover, BERT turned out to be the most efficient technique for all the macrostructure components. The reason for BERT's success could be traced back to a method known as transfer learning, which involves pre-training the system on a sizable corpus to give it a deep understanding of the English language. To help explain words with unclear semantics, the technique may be employed as well to train the model to understand each word's embedding in the wider context of its sequence. Lastly, word embedding creates a semantically connected lower-dimensional space in which unknown phrases can still be portrayed despite their association with known words, which allows BERT and GVEL to handle unseen words in training [[→34](#)]. Alternatively, without any prior training, the CMRF and TIRF methods were compelled to model the narrative structures to score

these different elements from the bottom up. For instance, CMRF and TIRF had no structures to deal with the sequential nature of the text, so to score “Initiating Event,” they had to figure out the way an initiating incident could be stated from a small sample of multiple narratives. The inability of these models to handle words that were not included in the initial training set makes them even more limited in practice, and it may have additionally contributed to their subpar results on the tests that were not shown.

However, GVEL uses some transfer learning principles and manages sequences naturally through the LSTM. Given the literature’s reports of LSTM performance, it’s plausible that an LSTM with sufficient training data could beat BERT. However, the experiments showed that training an LSTM from unknown weights is very difficult and ultimately not feasible, requiring many narratives to acquire these kinds of dependencies and patterns [[→ 35](#), [→ 36](#)].

While AES systems have many obvious applications, their main use in development has been to score high-stakes written assessments like the SAT and GRE. AES technology is very useful but not always readily available, although well-known systems like PEG, E-Rater, and IntelliMetric are exclusive technologies that are not accessible to the general public. Thus, if clinical practitioners in a variety of disciplines, including psychology, pathology, and education are to benefit from AES technology open-access technology must be developed. The fact that many AES systems rely on holistic scoring techniques, which aim to assess writing quality overall using a point system, is another disadvantage of these systems. Although it

is the preferred method for grading standardized exams such as the GRE, scoring systems based on a rubric can provide more detailed feedback by highlighting specific areas of a writer's work that are strong or weak. Educators, psychologists, and clinicians frequently use tools with a rubric to identify language deficit areas in children as well as track changes in the children's language abilities over time [[→37](#)]. It is especially recommended that medical professionals, like speech-language pathologists (SLPs), take language samples from their patients to identify their strengths and weaknesses and track improvement during the intervention. During the school-age years, narrative examples of language are the ideal elicitation context; however, due to time constraints, many clinicians do not use them [[→38](#)].

In the medical field, clinical accounts are the primary means of communication. Compared to the generically coded portions of electronic health records (EHRs), narrative notes offer a more comprehensive and distinct account of the patient's history and assessments, which enhances the context for clinical decision-making. The study of natural language processing, or NLP, is a branch of artificial intelligence that examines the analysis and synthesis of data expressed in natural language. It has repeatedly demonstrated that evidence can be taken out of clinical narratives and made available for large-scale analysis at a later time. The primary justification for machine learning's popularity as a panacea for the knowledge-gathering information obstacles is that manually annotating data is a simpler task than actually eliciting knowledge [[→39](#)]. However, the

amount of data needed to train a machine-learning model might take just as long to annotate as the process of eliciting knowledge [[→ 40](#)]. Similar to the principle of energy conservation, it appears that the information needed to guide the development of a precise computational model is merely converted from one format to another. Machine learning relies on implicit knowledge in the form of annotations and their distribution, with acquiring moments of these annotations being almost constant, as opposed to explicit understanding in the form of rules. The accessibility of clinical narratives poses a challenge to the machine learning approach due to the sensitive nature of medical data and security concerns [[→ 41](#)]. Due to these issues (i.e., lack of manually annotated data), the training data may not be representative enough, which could lead to poor performance from the associated ML models [[→ 42](#)].

As a component of governmental control over the distribution of medical devices (MDs), safety monitoring is crucial both globally [[→ 43](#), [→ 44](#), [→ 45](#)] and in Russia [[→ 46](#), [→ 47](#)]. To guarantee the effectiveness and safety of medical devices used daily, the following kind of monitoring is carried out. Greater oversight is necessary when using high-risk (class III) MDs. Therefore, even in the case that there are no risks or adverse events associated with the product, the Board of the Eurasian Economic Commission's Decision [[→ 48](#)] mandated that risk class III medical devices be inspected annually for three years following the issuance of a registration certificate. Software as a medical device (SaMD-AI), which is based on machine learning technologies, uses technological solutions to mimic

human cognition and produce outcomes that are at least equivalent to human performance [[→49](#), [→50](#)]. SaMD-AI cannot interpret data, so it needs extra supervision while in use. When applied to a population that differs from that used in machine learning, etc., it could lead to bias [[→51](#), [→52](#)]. Furthermore, through ongoing (self-)learning, it may integrate deep neural networks, in contrast to SaMD without AI technologies. The expeditious updating of SaMD-AI as an invisible medical device and the ensuing extension of the SaMD-AI re-registration period have raised concerns among researchers. The FDA (Food and Drug Administration) states that a unique registration system covering the entirety of the product life cycle is necessary for the following kind of medical device. The plan needs to be flexible enough to accommodate changes during SaMD-AI adaptation and use, all the while preserving the medical device's efficacy and safety [[→53](#)]. Other authors go into detail about the procedures that must be followed to guarantee that SaMD-AI is promptly and securely registered and that these products are integrated into actual clinical settings. The authors emphasize that the only way to rule out any potential scatter of SaMD-AI features is to assess SaMD-AI performance using actual clinical data. SaMD-AI's developers should be aware of any potential risks, such as difficulties when interacting with novel patient populations and inadvertently harmful health outcomes. It is imperative to examine both the predictive indicators, such as positive or negative predictive values, and the SaMD-AI main performance indicators (metrics), including the area under the feature curve [[→52](#)].

11.3 Role of big data analytics in post-marketing surveillance

The enormous amount of structured and unstructured data that is produced daily by machines, groups, and people is referred to as “big data.” It includes enormous volumes of data that are difficult to handle or evaluate with conventional data processing methods. Big data is growing exponentially and has several advantages ([→ Fig. 11.2](#)). One major factor contributing to the growth is the introduction of social media, the web, and connected devices. Big data’s velocity is one of its distinguishing features. To gain actionable insights into the unprecedented rate at which data is being generated, organizations must be able to gather, process, and evaluate data in real time. Social media platforms, for example, produce enormous amounts of data from interactions between users, posts, likes, and shares [[→ 53](#)]. Businesses can make better business decisions and increase customer engagement by utilizing the data to better understand trends, sentiment analysis, and customer preferences. The diversity of big data is another important feature. It includes an extensive range of data types, for example, transaction records, text, photos, videos, and sensor data. Such diversity is beyond the capabilities of conventional databases and information processing methods. However, due to technological developments, such as distributed computing frameworks such as Hadoop and Spark, businesses are now able to store, handle, and analyze sizable and diverse datasets. Organizations can improve decision-making and

innovation by integrating and analyzing multiple data sources to find previously hidden correlations, patterns, and insights [[→ 54](#)].



Fig. 11.2: Various advantages of big data analytics for post-marketing surveillance.

A crucial component of the healthcare system, drug safety surveillance is essential to guaranteeing patients' safety and wellbeing. It entails the methodical gathering, examining, and assessment of information about the risks and side effects connected to the use of drugs. Drug safety

surveillance monitors and assesses the safety profiles of pharmaceuticals to identify potential safety concerns, guide regulatory actions, and facilitate the development of risk-reduction and patient safety-improving plans. Data for drug safety surveillance is gathered from a range of sources [[→55](#)]. These sources include unofficial reporting systems, such as the Adverse Event Reporting Systems (AERS) and the Yellow Card Scheme, where patients and healthcare providers can report suspected adverse events. Scientific literature, clinical trials, post-marketing research, and EHRs are additional significant sources of safety information [[→56](#)]. By combining data from multiple sources, drug safety monitoring systems can offer a comprehensive perspective of drug safety, enabling more accurate assessments and timely interventions. The healthcare industry has undergone a revolution due to big data analytics' capacity to compile vast volumes of health data from numerous sources. Social media platforms, databases of insurance claims, EHRs, and patient forums are some examples of these sources. Combining data from various sources can lead to a thorough understanding of drug usage developments and potential safety issues. The vast amount of medical data that big data analytics can gather presents a unique opportunity to identify and analyze drug use patterns.

Drug safety monitoring benefits greatly from big data analytics since it makes it easier to identify issues early on and to keep an eye out for potential signals in real time. Early warning indicators of adverse events can be recognized and their progression tracked by continuously analyzing

data streams gathered from social media posts, online health forums, and spontaneous reporting systems. The capacity of big data analytics to access data from immediate sources is one of its primary advantages [[→57](#)]. User-generated content about drug use on social media and in health forums is abundant and shares actual experiences and viewpoints. By keeping an eye on these platforms, big data analytics techniques can quickly identify conversations, mentions, and sentiments associated with potentially negative events. Real-time monitoring makes it possible to swiftly identify new issues, which enables medical professionals and regulatory agencies to take the necessary action. Using big data analytics, these systems can be continuously monitored to identify trends and patterns in reported adverse events. By sifting through the deluge of reports, algorithms can swiftly spot potential safety signals and provide early warnings of drug-related risks [[→58](#), [→59](#)]. It is simpler to carry out timely investigations and take the required actions to safeguard patient safety when the capability is present. The ability of big data analytics to capture the evolving nature of drug safety signals is a benefit when it comes to real-time monitoring. Since adverse events can change over time, continuous surveillance is required to effectively monitor them along with any related patterns. Big data analytics methods can continuously analyze data streams to track the development of unfavorable events. They can identify any shifts in the frequency, intensity, or patient demographics of these occurrences due to it. The comprehensive viewpoint makes it possible to approach drug safety proactively, enabling potential issues to be swiftly identified and fixed [[→60](#)].

Another advantage of big data analytics is its ability to integrate multiple data sources, which can aid in validation and prioritization. Integrating information from various sources, including EHRs, clinical trials, and assertion databases, can lead to a more comprehensive and reliable comprehension of the safety indicators. The integration adds to the reliability and importance of the identified signals by bolstering the evidence and validating the findings [[→61](#)]. Integrating information from various sources, including EHRs, clinical trials, and assertion databases, can lead to a more comprehensive and reliable knowledge of safety indicators. This is because these networks most closely resemble big data's features in everyday pharmacovigilance. Most database networks consist of EMRs and claims databases, but they can also be linked to social media, biological samples kept in biobanks, or genetic information [[→62](#), [→63](#), [→64](#)]. Due to record linking, healthcare databases can combine information from several sources to present a complete picture of the medical condition of an individual and facilitate international patient research. This can be done while maintaining patient privacy by using anonymized patient identifiers. Due to the increasing use of healthcare database networks, a growing number of analytical techniques have been employed to leverage the enormous quantities of heterogeneous data [[→65](#), [→66](#), [→67](#)]. Not only are dispersed data networks being constructed for analysis, but networks in general are changing. Research collaborations like Observational Health Data Sciences and Informatics (OHDSI) create techniques and resources to make it possible to build the infrastructure of database networks, and data

collaborations like the Innovative Medicines Initiative's PROTECT have demonstrated how multiple database collaborations can be conducted via shared protocols rather than analyzing summary data centrally [[→ 68](#)].

Big data is becoming more widely used in pharmacovigilance, but it shouldn't diminish the importance of more conventional methods for observational research. In actuality, most middle-class and low-income nations lack accurate and systematically gathered electronic medical data [[→ 69](#)]. Until there is a more consistent process of computerizing medical information, observational studies using primary data collection and other methods will continue to be crucial in those countries, along with the use of spontaneous reporting. However, as manipulation and evaluation of data methods proliferate and become more widely available, it is conceivable that the use of big data will only grow in those situations where they are available [[→ 70](#), [→ 71](#)].

11.4 Challenges of AI in post-marketing surveillance

When institutions are incentivized to work together to guarantee data protection by design, robust privacy protection can be achieved [[→ 72](#)]. Privacy can be managed for business healthcare AI applications, but there are new goals at play. It is evident that businesses may not be sufficiently motivated to continuously uphold privacy protection if they can profit from the data or monetize it in some other way and if penalties imposed by law

are insufficient to discourage this kind of behavior. There is an additional risk associated with using commercial AI for big data: privacy violations could result from the incredibly intricate algorithmic systems themselves [[→73](#), [→74](#), [→75](#)]. The US, Canada, and Europe are just a few of the regions where there has been a rise in healthcare data breaches globally. Furthermore, criminal hackers are making it more difficult to protect medical data, even though they may not be heavily utilizing AI in conjunction with other algorithms just yet. It increases the privacy risks of granting control over patient health data to private AI companies, even when “anonymization” is employed. It is distinct from situations in which state institutions directly manage patient data in that liability, insurance, and other pragmatic considerations are also brought up [[→76](#), [→77](#)]. Carefully drafted contracts outlining the rights and obligations of the parties involved, as well as responsibility for any potential negative outcomes, will need to be made due to the diverse and complex nature of the legal risks that private AI builders and maintainers may be taking on when interacting with massive amounts of patient data [[→78](#), [→79](#), [→80](#)].

The industry is required to adhere to specific requirements set forth by every data protection law, such as the General Data Protection Regulation (GDPR), the Health Insurance Portability and Accountability Act (HIPAA), and others. This results in a complex web of international regulatory compliance. Furthermore, because patient privacy is compromised and the pharmaceutical industry has significant intellectual property worth, cybersecurity breaches pose a continuing risk [[→81](#)]. Cyberattacks, like

ransomware and information breaches, jeopardize pharmaceutical companies' reputations and the faith of their patients in addition to their financial risks. Furthermore, the vulnerability is not limited to the corporation; it also exists in the intricate pharmaceutical supply chain, where distributors, manufacturers, and various other stakeholders must work together to protect data privacy. Such a cooperative endeavor entails a fine balance that requires information sharing to progress research while also safeguarding proprietary knowledge. The significance of continuous employee awareness and education efforts is underscored by the fact that human error remains a major contributing factor to data breaches [[→ 82](#)]. Big data, AI, and machine learning are a few examples of emerging technologies that have increased the difficulty of protecting personal information. The right security measures need to be reevaluated in light of these new challenges. Another factor that increases the importance of privacy protection is the volume of data that is stored, which can range from personal health profiles to the outcomes of clinical trials. The fact that pharmaceutical activities are carried out on a worldwide scale, necessitating strict adherence to numerous data protection laws, makes it more challenging. The situation becomes more complex due to cybersecurity threats. Regular reviews of security precautions are necessary because malevolent actors are always refining their tactics, whether they take the shape of sophisticated hacking attempts or social engineering. Vulnerabilities are introduced at every stage of the drug supply chain,

which is crucial to the industry's operation from manufacturing to distribution [[→ 83](#), [→ 84](#)].

The fact that the pharmaceutical industry is regularly embroiled in court cases and disputes, many of which entail complex analyses of intellectual property laws, makes responding to these challenges even more difficult. Not only can well-known cases impart valuable legal tactics, but they also show us how to carefully balance safeguarding confidential information with fostering a healthy degree of rivalry in the market. Success stories, on the other hand, show how companies are effectively incorporating intellectual property and data privacy into their main business strategies [[→ 85](#)]. These narratives highlight innovative fixes and industry best practices. Keeping up with the newest emerging trends is one of the hardest things the industry has to do. The emergence of innovations like digital twins and the increasing convergence of data-intensive fields means that the landscape is always changing. Ethical considerations are becoming more and more crucial given how quickly these technologies have been developed and used. Because of this, a proactive approach is needed to foresee and address possible risks. Each type of medical facility supports a distinct data model with a unique set of terminologies and monetary representation, and each type of data model differs from the others. Data from various sources must be merged and formatted to be compatible with the common data model (CDM) across multiple institutions. To do this, engineers and HCPs will have to devote time and effort to data transformation. Because AI-based training models and drug safety

algorithms rely on data with annotations, data annotation is crucial for unannotated data. The lack of excellent information for model training and the high annotation cost are obstacles to studies on AI-based drug safety [[→86](#)]. The lack of organized and vetted data available for software creation to detect possible drug safety issues is one of the primary obstacles to AI adoption. Privacy concerns are also raised by the use of AI in pharmacovigilance since participant consent might not be obtained ahead of data being used for other purposes [[→87](#)].

The problem of social gaps is another issue that societies face as a result of the development of artificial intelligence. With every new development, discovery, and invention, people in every country on Earth deal with rising levels of social inequality as well as a dwindling conviction in social justice. Artificial intelligence exacerbates social inequality, despite making it easier to access more information regarding science and technology, international politics, environmental issues, and world events [[→88](#)]. It's also probable that medical machines and people won't evolve simultaneously because human feelings are different. Unlike autonomous (robotic) systems, physicians, as well as other health care providers, should consult with or refer colleagues for advice. It is not possible in autonomous systems [[→89](#)]. Robotic physicians and nurses cannot perform the following kind of work. Because robotic nurses and doctors lack human qualities, patients will no longer interact with them with compassion, kindness, or appropriate behavior. Patients won't be able to behave appropriately either. Additionally, mobile devices are used in the majority

of studies on mobility problem detection techniques to gather relevant signals, send them to other devices for analysis, visualize the outcomes, and notify users of the results [[→ 90](#), [→ 91](#)]. Transparency is the capacity to understand how an artificial intelligence system makes decisions. More precisely, the black box's basic analytical procedures and decision-making process are opaque. Algorithms that are programmed on biased data or that disregard particular demographic traits make these problems worse [[→ 92](#)].

The sub-issue focuses primarily on the unnecessary or possible harm that AI tools or risky AI in the healthcare industry can cause. Enhancing the quality of care and patient safety, expanding healthcare access, giving patients and doctors local real-time advice, and spotting medical emergencies such as sepsis are just a few advantages of integrating AI into healthcare. However, AI-enabled support services are not error-proof, and their algorithms may suggest dangerous options or advice to patients that could endanger their health [[→ 93](#)].

11.5 Conclusion

A crucial procedure in the pharmaceutical sciences, post-marketing surveillance (PMS) keeps an eye on a drug's safety after it has been put on sale following clinical trials. It assists in identifying adverse and advantageous effects, off-label usage of drugs, orphan drug issues, and difficulties encountered when carrying out global clinical trials on pediatric patients. The Medical Device Regulation (MDR) of the European Union

(EU) and the In Vitro Diagnostic Medical Device Regulation (IVDR) have increased PMS requirements for in vitro diagnostic and medical device companies. Compliance with these regulations requires a significant financial investment in new policies, procedures, and skilled staff. Natural language processing, machine learning, and deep learning are examples of artificial intelligence (AI) technologies that can cover a wider range of data and improve signal detection speed and accuracy. Deep-learning methods outperformed the accuracy rate threshold in a Celgene and IBM study, indicating that artificial intelligence (AI) may be able to increase data efficacy, consistency, and reporting efficiency. AI solutions can assist in the development of flexible, evidence-based quality control frameworks that promote creativity and better-informed decision-making. They can also assist in meeting EU MDR/IVDR regulations for PMS practitioners and manufacturers of medical devices. In 1966, automatic essay scoring (AES) technology was first introduced to simplify and expedite essay scoring. It can either supplement or replace human scorers. However, there are a lot of obstacles to privacy protection when using AI for post-marketing surveillance. If businesses can profit through or monetize the data, they might not be inspired to constantly uphold privacy protection, and the existing legal penalties are not enough to deter such behavior. Since healthcare AI applications involve complex algorithms, giving private AI companies access to patient health data presents privacy risks, even in situations where “anonymization” is used. Because of the speed at which

these technologies are developing and being used, ethical issues are becoming more important.

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Chapter 12 Challenges and ethical considerations

Abstract

In the healthcare industry, artificial intelligence (AI) is crucial because it helps with disease diagnosis, treatment planning, and operational efficiency. However, there are issues with biases, data breaches, and ethical consequences. The chapter aims to provide some knowledge on the challenges faced by the researchers of AI in healthcare along with some ethical considerations. Collaboration between AI clinicians can succeed if appropriate governance frameworks are in place. An overview of the challenges and ethical quandaries surrounding AI in healthcare is given in the chapter. Explainable AI techniques address issues with bias and transparency. Data privacy, algorithmic transparency, and ethical issues are among the difficulties. Up skilling medical staff, addressing biases, and obtaining patient consent are all necessary before implementing AI in healthcare. Furthermore, healthcare professionals need to believe in AI for it to be adopted. Research suggests that AI holds promise for enhancing patient outcomes and healthcare delivery. More research is necessary to fully grasp how AI will affect healthcare professionals' roles and responsibilities.

Keywords: challenges, ethics, XAI, data privacy, safety, algorithms, patient perspective, patient compliance,

12.1 Introduction

Artificial intelligence (AI) is applied as a system that consists of hardware and software. In terms of software, algorithms are the primary focus of AI. Limited resources, a rise in patient demand, and chronic illness put pressure on healthcare systems. As digital health technologies advance, so does the amount of data in every healthcare facility [[→1](#)]. When appropriately applied, medical professionals could monitor the effectiveness of interventions and preventative measures while concentrating on the causes of illness. Therefore, decision-makers, legislators, and others must be aware of it. Data scientists, computer scientists, along clinical entrepreneurs all agree that AI, particularly machine learning (ML), is important to the success of healthcare reform [[→2](#), [→3](#)]. In computing, the term “artificial intelligence” describes a computer program’s capacity to carry out operations akin to those of a human intellect, such as learning and reasoning. When it comes to its application, new technology raises concerns as it can result in data breaches and errors. Errors in the high-risk medical domain can yield deadly consequences for the affected patient. It is crucial to recognize that patients engage with doctors during moments of heightened vulnerability in their daily life [[→4](#), [→5](#)]. Collaborations between AI and clinicians, where AI assists in guiding medical decisions and delivering evidence-based management, can be effective if appropriately utilized (AI-Health). It can provide healthcare services in the

domains of diagnosis, medication discovery, epidemiology, personalized care, and operational efficiency. Researchers indicate that a robust governance system is essential to protect individuals from harm, particularly that arising from unethical conduct, when integrating AI solutions into medical practice. The Hippocratic Oath was established due to Hippocrates, the physician, who formulated the ethical foundation of medicine [[→6](#)].

In general, AI in medicine can be divided into three categories: virtual, physical, along hybrid (which combines robots and virtual reality). In both clinical and research contexts, each of these fields has proven to have extraordinary advantages and/or promise [[→7](#)]. A subset of AI known as virtual AI, as well as ML-based AI, consists of a collection of techniques for automatically identifying patterns in data to forecast trends or make decisions in unpredictable circumstances. Machine learning is still being used to help with research in genetics and molecular medicine [[→8](#), [→9](#), [→10](#), [→11](#)]. Furthermore, by combining genotype data from “omics” with phenotypic data from electronic health records (EHRs), AI can support precision medicine. The advent of high-throughput sequencing in recent times has enabled the acquisition of extensive data on proteins, DNA, and molecular interactions. The majority of therapeutic targets and diagnostic biomarkers identified today can be found directly or through interaction with algorithms. These techniques can also be used to optimize research on the creation of novel drugs and treatments, thereby lowering the complexity and expense of these studies. These investigations may encompass data-

driven, accurate treatment planning, clinical outcome prediction, and process management simplification (e.g., patient recruitment and retention) [[→ 12](#), [→ 13](#), [→ 14](#), [→ 15](#), [→ 16](#)]. Furthermore, AI has proven to be very beneficial in the digital healthcare era with the extensive use of EHRs and high-throughput sequencing. AI may be able to actively advise doctors when they make clinical judgments. Pathological diagnoses and treatments, imaging, and prediction all show promise and success stories. Robots are the best example of the physical AI branch in healthcare; they help senior patients and support surgeons. Senior care facilities are increasingly using care robots. Sensible Surgeons, a US corporation, manufactures surgical systems that can be operated from a console using a method that is minimally invasive to perform complex surgeries [[→ 16](#), [→ 17](#), [→ 18](#)].

AI methods and their applications can be used to diagnose diseases more accurately, care for and monitor the elderly with robots, detect cancer earlier and more quickly than in the past, etc. Massive volumes of data can be processed by ML techniques, which also enable increasingly precise evaluations and forecasts [[→ 19](#), [→ 20](#), [→ 21](#)]. While ML has aided in the advancement of AI, it also raises ethical concerns, particularly in the field of healthcare. Prioritizing ethics in the creation and use of AI across industries. The analytical process of evaluating and selecting alternatives in accordance with social, ethical, and legal standards is referred to as ethical decision-making by AI systems [[→ 22](#)]. The subsequent ethical considerations affect the ongoing development and implementation of AI, especially in the healthcare sector, where technology must adhere to

regulations, laws, and privacy norms to safeguard the public interest. Advanced ML algorithms employing deep learning and other complex techniques can generate opaque and challenging-to-interpret models referred to as “black boxes”. The makers of black box models encounter difficulties in elucidating the decision-making processes of AI systems [[→23](#), [→24](#)]. Users are compelled to make decisions devoid of justification. The issues confronting AI in the domain of chemical medicine can be addressed through multiple strategies. Data augmentation is a technique that entails generating synthetic data to improve existing datasets. As a result, a broader and more varied dataset may be utilized to train ML algorithms, resulting in the generation of more accurate and dependable outcomes. An alternate tactic that seeks to provide clear reasons for the predictions made by machine learning algorithms is the implementation of explainable AI (XAI) tools. It enhances comprehension of the core processes and assumptions that support the predictions, while also alleviating concerns regarding the fairness and bias of AI-based methodologies [[→25](#), [→26](#), [→27](#), [→28](#)].

12.2 Challenges in implementing AI in healthcare

To reliably define or predict a variety of tasks, ML as well as deep learning models requires large datasets. Machine learning has advanced most in industries where large datasets are available for backing more complex,

precise algorithms. However, data accessibility in the healthcare industry is a complicated problem. Health data is costly for organizations, and sharing it amongst hospitals is viewed with reluctance because each hospital views it as its property, used to manage its patients [[→29](#)]. The data's continuous availability after the algorithm that analyzes it was introduced is another problem. Ideally, training with ever-larger datasets would necessitate constant improvement for ML-based systems. However, it is frequently challenging to accomplish because of organizational resistance. To achieve it, it has been suggested that for information technology along with AI to make advancements in the healthcare sector, a drastic shift in emphasis from treating specific patients to all-around patient outcomes is required. It will promote data sharing to improve clinical outcomes [[→30](#), [→31](#)]. Applications using AI create privacy and data security concerns. Security breaches often target sensitive health data. Protecting patient data is therefore essential. People may begin to mistake AI systems for humans as AI develops, which would increase worries about data privacy and allow for more inadvertent data collection. Patient consent is crucial for addressing data privacy issues, as without sufficient individual consent, healthcare organizations may permit extensive utilization of patient data for AI training [[→32](#), [→33](#)].

Extensive datasets are a fundamental element of AI-driven systems and technologies, prompting privacy concerns related to data sharing and gathering. Due to the inclusion of personal information in patient records, the interchange and management of illness-related data across several

databases is exceedingly challenging [[→ 34](#), [→ 35](#)]. It indicates that software development organizations must adhere to secrecy regulations, potentially obstructing AI development. AI technology relies on machine learning to make conclusions based on available information, rather than taking into account the unique circumstances of each patient, hence raising moral, ethical, and legal concerns. AI technology should conform to the moral standards, legal obligations, and personal convictions that govern individuals' conduct in public areas. Numerous industries have transitioned from isolated silos in the digital era. It is also applicable to the medical domain [[→ 36](#)]. The healthcare sector was previously perceived as involving medical professionals, including doctors and nurses, providing patient care. In addition to access to superior medical treatment, contemporary health is also achieved via the adoption of healthy lifestyle choices, including a balanced diet, regular exercise, and daily self-care practices. It has rendered preventive medicine increasingly vital, complicating the differentiation between the roles of a physician, the utilization of technology, and the maintenance of a healthy lifestyle. AI is not limited by temporal or spatial parameters. Consequently, the concept of a closed healthcare system has become obsolete [[→ 37](#)].

The advancement of AI technology presents the next obstacle. Over fitting can happen when the system picks up on irrelevant correlations between patient characteristics and results. The algorithm predicts using unsuitable features as a result of having excessive variable parameters about outcomes [[→ 38](#)]. As a result, the algorithm may work correctly when applied to the

training data set, but it may yield unreliable results when applied to future events. The leakage of data is another issue. The algorithm's usefulness for predicting outcomes beyond the training dataset may be questioned if a covariate in the dataset unintentionally revealed the result, as would occur if the algorithm's optimistic accuracy was too high [[→ 39](#), [→ 40](#)]. One prevalent criticism leveled at AI is the "black-box" problem. It is often difficult for deep learning algorithms to provide comprehensive explanations for the predictions they make. The inability of the system to safeguard its conclusions creates a legal challenge if the recommendations turn out to be incorrect. It also makes it more difficult for scientists to understand how the data and the predictions relate to one another. More significantly, patients may become less trusting of the system as a result of the "black box" [[→ 41](#), [→ 42](#)].

Since the advent of AI, it has been accompanied with objections and ethical dilemmas. In addition to the previously addressed concerns around data security and confidentiality, accountability remains the primary apprehension. In the healthcare sector, poor decisions can yield significant consequences, and there is a consensus that accountability must be established. Occasionally, individuals perceive AI as a "black box," rendering the underlying algorithmic logic of a particular advice or prediction inscrutable [[→ 43](#)]. It can be contended that algorithms employed in lower-risk applications, which emphasize efficiency or enhanced operations rather than patient-centeredness, should not be apprehensive about the "black-box" problem. Therefore, if the system is defective, it

remains ambiguous who should assume responsibility. Attributing responsibility to the doctor may appear unjust, as they lacked involvement or authority in the algorithm’s creation; meanwhile, assigning accountability to the developer looks overly disconnected from the clinical environment. AI cannot legally make any healthcare decisions in China; all decisions must involve human decision-making and accountability [[→ 44](#)].

A primary concern with AI in healthcare is safety, as illustrated in [→ Fig. 12.1](#). IBM Watson for Oncology exemplifies the application of AI algorithms to analyze medical record data and support physicians in providing cancer therapy alternatives to patients. Recently, it has encountered fire for purportedly endorsing “unsafe and incorrect” cancer therapies [[→ 45](#)].

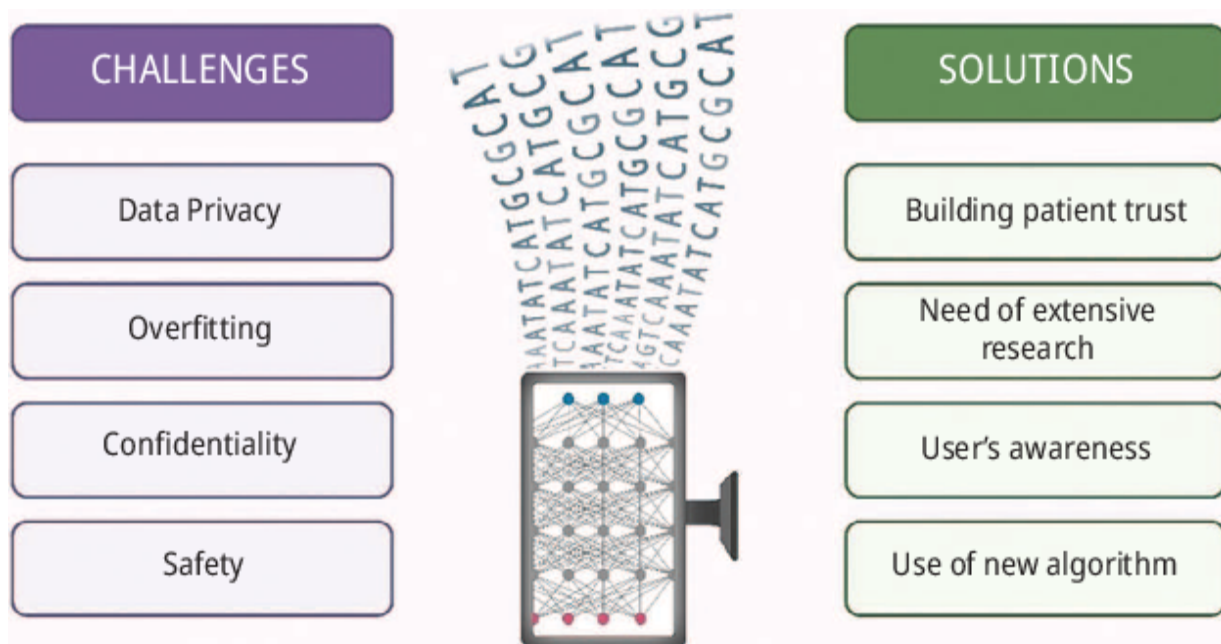


Fig. 12.1: Schematic representation of challenges and potential solutions of AI in healthcare.

Moreover, upholding a requisite level of transparency is crucial for safeguarding patient trust and security. In an ideal scenario, all data and algorithms would be accessible for public scrutiny; nonetheless, there are valid concerns around the protection of investments and intellectual property, as well as the mitigation of cyber security risks [[→46](#)]. The healthcare sector encounters significant ethical and privacy challenges that require strict compliance with laws, norms, and regulations. The implementation of AI in medicine raises several ethical and privacy concerns, including the confidentiality of patient information, the moral implications of innovation, and the tangible effects of technical advancements on both physicians and patients. To effectively leverage AI in healthcare, strategic decisions must be taken that balance competing interests and values, including privacy, reliability, intellectual property rights, and openness [[→47](#), [→48](#), [→49](#), [→50](#), [→51](#)]. Establishing patient trust is essential in the healthcare industry. It is especially applicable to technology based on AI. Healthcare experts and decision-makers acknowledge that, due to the historical significance of in-person consultations with physicians, the public may be reluctant to depend exclusively on AI-generated advice. Moreover, healthcare personnel may have difficulties in utilizing these technologies due to the public's pervasive misconceptions regarding the potential benefits and limitations of AI in medicine [[→52](#), [→53](#), [→54](#)].

Moreover, certain healthcare professionals such as physicians, nurses, and nursing assistants may be inadequately attending to digital technology,

thereby obstructing the implementation of AI in healthcare. The interpretability and opacity of AI algorithms pose significant challenges, particularly in the context of unstructured data. AI systems encounter difficulties as hospital administrators and physicians often grapple with technology apprehensions. Interoperability standards are essential for employing AI models and enabling the integration of data from various electronic health records [[→ 55](#), [→ 56](#)].

12.3 Ethical consideration in AI-driven healthcare

To ensure the responsible and beneficial deployment XAI in healthcare, it is essential to rigorously assess its ethical ramifications. Researchers emphasize the significance of AI efficacy, interpretability, and explicability in health technologies. Their findings indicate that organizations assessing healthcare technology must acquire substantial knowledge regarding the functionality of machine learning algorithms and their predictive capabilities [[→ 57](#)]. The authors advocate for the development and application of tools to assess these factors to foster a sense of accountability among stakeholders. Certain researchers employ AI for diagnostic purposes and examine the potential benefits and drawbacks associated with the interpretability and reliability of AI. While AI has the capacity to enhance diagnostic output and accuracy, it poses a risk to the skill sets of the clinical profession. The research advocates for the utilization of AI as an adjunct to

human expertise instead of attempting to supplant it. Potential biases in AI and big data applications utilized in the healthcare sector have been highlighted by certain sources [[→ 58](#), [→ 59](#)]. They offer recommendations for improving the equity of AI algorithms and caution against training these algorithms with non-representative samples. The authors contribute to the policy discussion on data oversight in the AI era by highlighting the necessity of incentivizing patients to disclose their health data. The authors contend that the revenue-generating nature of digital copyright content, such as music, can impart important knowledge and provide a GDPR-based remedy that leaves data flow unhindered [[→ 60](#)]. They contend that the strategy may provide a workable and proven way to encourage data sharing without restricting access to it, which is essential for the advancement and efficacy of healthcare AI. Academics talk about how urgently human-centered AI is needed in healthcare, especially in the so-called Global South a region mostly made up of low-resource areas [[→ 61](#), [→ 62](#)]. According to Okolo, biased AI systems may inadvertently widen the gaps that currently exist in these fields, so their use should be avoided. The author emphasizes how crucial it is to consider local requirements and conditions when creating AI systems to provide more equitable and effective healthcare solutions. Similarly, researchers discuss the moral implications of integrating AI and big data into mental health services a notion the author refers to as “iHealth.” In his thorough ethical examination of the possibilities of iHealth, Rubeis identifies three double-edged swords: data

mining, ecological momentary assessment (EMA), and self-monitoring [[→ 63](#)].

Moreover, transparency should be given top priority, and XAI should give clear justifications for its judgments as shown in [→ Fig. 12.2](#). It promotes accountability by enabling consumers and medical professionals to comprehend the reasoning behind AI-generated recommendations or diagnoses. Transparent AI systems help consumers validate as well as trust the results by avoiding the “black box” problem. Preexisting biases may be reinforced and amplified by AI systems that were trained on biased data, which might have discriminatory impacts on healthcare outcomes. When implementing XAI ethically, biases must be actively and thoughtfully taken into account during the phases of data collection, preliminary processing, and model training [[→ 64](#)]. XAI system bias monitoring and remediation can lead to more reasonable healthcare practices. Patients have a right to know if AI, including XAI, is used in their treatment. To gain informed consent, it is imperative to have clear communication about the purpose, constraints, and possible hazards associated with AI systems. It should be possible for patients to obtain human explanations and guidance, guaranteeing that their choices and autonomy are honored. Important elements of ethical XAI include putting patient welfare first and making sure AI systems make choices that are in the best interests of the patient [[→ 65](#)]. When making decisions, healthcare professionals should be assisted by XAI, which should be built with their specific needs and preferences in mind. Healthcare professionals must comprehend and apply XAI systems

correctly. It entails upholding their clinical expertise, critically assessing AI outputs, and being aware of the constraints and potential biases of AI systems. Healthcare workers should receive sufficient XAI education and training to guarantee its responsible and efficient use. XAI development and application in healthcare can be guided by adhering to accepted ethical frameworks, including the principles of kindness, nonmaleficence, independence, and justice. Healthcare-specific ethical guidelines, like those offered by governmental or professional medical organizations can help navigate the particular ethical challenges that XAI presents [[→66](#), [→67](#)].

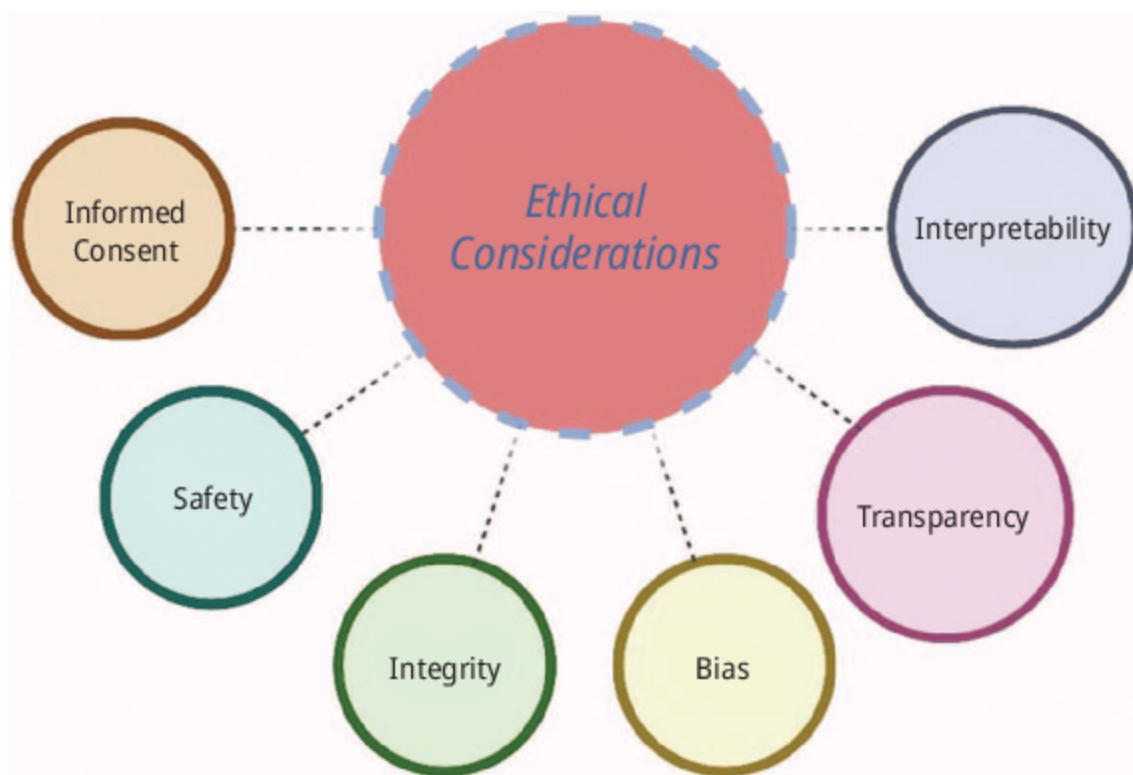


Fig. 12.2: Schematic representation of ethical considerations of AI in healthcare.

Another researcher explains that privacy is a key ethical factor to consider when using AI. Concerns about how AI systems gather, store, and use personal data are known as privacy issues in AI. AI systems can gather and analyze substantial amounts of personal data, and improper management of the data may lead to significant privacy concerns [[→68](#)]. AI systems often collect substantial amounts of personal data from individuals, which are subsequently utilized to make determinations about them. It prompts enquiries on the methodologies employed for data collection, storage, and utilization. The implementation of AI in surveillance systems may infringe upon privacy rights. Four principal ethical issues must be addressed for AI in healthcare to achieve its maximum potential: (1) informed permission for data utilization; (2) safety and integrity; (3) fairness in algorithms alongside biases; and (4) protection of personal data privacy. The majority of legal discourse on AI has been on the constraints of algorithmic transparency. The application of AI in high-risk situations has necessitated transparent, equitable, and accountable AI design and governance. Accessibility and comprehensibility of information are the two essential elements of transparency. It is standard practice to deliberately obscure the mechanisms of algorithmic operations [[→69](#), [→70](#)]. The potential for bias in AI algorithms, which may lead to disparities in healthcare results, represents another ethical problem. An unrepresentative training dataset can result in biased AI models that disproportionately affect specific demographic groups. It may result in inequitable access to healthcare and intensify pre-existing disparities in medical outcomes. To ensure fairness and equity,

developers must actively combat bias through inclusive and diverse data sets, constant monitoring, and iterative algorithm improvement [[→71](#)]. An important ethical consideration in AI decision-making is interpretability, particularly in the medical field where decisions affect patients' quality of life. Deep learning networks and other complex AI models can be hard to interpret, resulting in it being hard to understand the way the system comes to a particular conclusion [[→72](#)].

Moreover, another ethical cornerstone in AI applications for healthcare that needs particular consideration is informed consent. Patients ought to be made aware of how AI technologies are being used in their medical treatment and given the choice to accept or reject AI-driven diagnoses or treatments. It entails outlining the advantages of AI in healthcare as well as their drawbacks and possible hazards [[→73](#)]. AI's ability to automate some tasks in the healthcare industry could result in job displacement, which raises ethical questions about how it will affect the workforce. Maintaining human roles while taking advantage of AI's efficiency gains requires careful consideration. Prioritizing reskilling and up skilling initiatives are necessary for healthcare organizations to guarantee that their workforce is capable of adjusting to and cooperating with AI technologies [[→74](#), [→75](#)].

12.4 Impact on healthcare professionals

AI has applications in healthcare that include task automation, anomaly detection, correlation analysis, unstructured data analysis, and human

worker support. These are accomplished by using DL, ML, and NLP algorithms. NLP algorithms identify key terms and phrases in unprocessed electronic medical record data to assist with decision-making. The way AI can transform data into outcomes that could enhance clinical practices and the caliber of care given by medical professionals in innovative ways is fascinating to watch [[→76](#), [→77](#), [→78](#), [→79](#), [→80](#)]. Research has examined whether or not AI will be accepted or used, depending on how beneficial the technology is viewed to be and how to make it work better. Research on AI has mostly concentrated on the technology – where to connect it and how to best utilize it. Nonetheless, the human being is the other essential element in all of it. It is now necessary to take into account how the technology will affect the user, the healthcare professional – and how it will influence their capacity to provide care [[→81](#), [→82](#), [→83](#)]. The use and capability of AI will vary amongst healthcare professionals and disciplines. Since AI in health has no widely accepted definition, healthcare professionals might not be aware that they are utilizing AI in their work. It is crucial to comprehend how clinicians view AI in healthcare delivery to guarantee a long-term healthcare workforce prepared to manage its quickly evolving digital healthcare landscape. More research is required to determine the interactions between clinicians and technology [[→84](#), [→85](#), [→86](#)]. Researchers investigating the underlying factors influencing healthcare professionals' endorsement of an AIMDSS met the criteria. Medical practitioners are less inclined to utilize AI in their routine practice when they harbor skepticism about the technology or lack awareness of its

capacity to enhance patient outcomes or care delivery, as indicated by the researcher's limited observational study. The conclusion pertains specifically to the healthcare sector, highlighting similarities in aspects of digital health technology that do not incorporate AI [[→ 87](#), [→ 88](#), [→ 89](#), [→ 90](#)].

Researchers investigated how patients and healthcare providers interacted with digital technology used in the delivery of healthcare. The findings demonstrated that healthcare professionals were less likely to adopt digital technology if they did not know what it was used for or did not think it would enhance patient-provider relationships or the standard of care. Prior studies have indicated that healthcare practitioners possess a restricted comprehension of pertinent digital systems; consequently, a more profound understanding of the benefits and drawbacks of these systems is imperative, especially regarding their application in the healthcare sector [[→ 91](#), [→ 92](#)]. Researchers looked at the use of telemedicine in the treatment of diabetic foot ulcer patients and concluded that the context of technology use will be informed by the experiences of medical professionals. Others claimed that medical technology will have an impact on how doctors perceive different options for therapy and treatment, as well as how they comprehend patients' illnesses and interpret their complaints. Consequently, more study is required to determine how AI will affect the job and responsibilities of healthcare professionals [[→ 93](#), [→ 94](#), [→ 95](#)]. Medical professionals seemed to be optimistic about the benefits AI tools could provide for their patients. According to a survey conducted among healthcare professionals, AI tools

created by industrialists have the potential to save doctors' time, perform vigilant and watchful tasks, enhance population monitoring, mitigate certain issues associated with medical deserts, and even enhance management challenges within the healthcare industry, particularly in hospitals.

Therefore, AI may be a way to usher in a more effective era of medicine, enhancing patient safety while lowering costs and improving care [[→ 96](#), [→ 97](#)]. Thus, AI would signify a revolution in treating patients and medical practices for certain doctors, but it would also signify a continuation of the field's ongoing developments in medicine for other doctors. Nonetheless, every AI tool scenario discussed in the interviews was based on ongoing research. Healthcare experts then informed the interviewees that AI is still in its early stages; pointing out that only a few of these initiatives have yet shown real-world success. Physicians also noted a discrepancy between scientific findings and the sensationalized media announcements [[→ 98](#)].

For the concerned healthcare professionals, doctors' primary interest in AI is not just a wishful dream but also a result of the overwhelming volume of data that makes it more difficult for doctors to make diagnoses in the modern world. They thus stated that to evaluate and categorize such data, they require tools. Furthermore, rather than relying solely on AI tools, some doctors think that they ought to first inquire about themselves and what the needs and potential changes are. Physicians expressed their lack of readiness to face criminal charges if an AI tool is found to have made a medical error, should these tools become increasingly important in medical decision-making [[→ 99](#), [→ 100](#)].

12.5 Patient perspective and engagement

It is well-known how the public feels about sharing health data for research purposes [[→101](#), [→102](#), [→103](#)]. While worries about data sharing's security, confidentiality, and commercial implications are often raised, people generally favor the research when they believe that it will benefit society or the public at large and when they have faith in the institutions carrying it out [[→104](#)]. Nonetheless, the perceptions of the public and patients on the dissemination of medical information for AI research are inadequately articulated [[→105](#)]. Conditional public support for data sharing cannot be presumed to apply to the subject of study, as data sharing for AI research is a contentious issue. The medical literature in the field is rapidly expanding, and AI has the capacity to significantly transform patient interactions with healthcare practitioners. A recent journal issue analyzed the merits and drawbacks of AI technology, including enhancements in healthcare efficacy and quality, as well as threats to patient autonomy, informed consent, and privacy. However, there is a lack of depth and clarity in the present knowledge of how patients view AI and its use in healthcare [[→106](#)]. According to other researchers, people will likely lose faith in AI systems' abilities. People might not trust AI's diagnostic and predictive capabilities, for example, when it comes to medical interventions. According to a different study, users' perceptions of AI systems' trustworthiness are influenced by their autonomy [[→107](#), [→108](#), [→109](#)]. Furthermore, fewer participants signed up for an AI diagnosis in a survey

some researchers conducted to learn about consumer views of AI in medicine than for a diagnosis from a healthcare provider. The majority of respondents in the same study preferred a human provider over AI, even if there was a greater chance of a misdiagnosis. Additionally, they showed that patients are more inclined to select AI if the doctor makes the final treatment decision rather than just the AI [[→ 110](#)].

Historically, medical practitioners predominantly determined the health decisions of their patients. It constrains the opportunities for collaboration and collective decision-making between patients and healthcare providers. Studies have shown that digital technology has transformed the relationship between patients and medical personnel [[→ 111](#), [→ 112](#), [→ 113](#), [→ 114](#)]. The implementation of these technologies has fostered more collaborative ties between patients and doctors, enabling people to more readily express their concerns over their care. Patients with more awareness of their condition are those that commence self-management. It is characterized as the systematic distribution of knowledge and supportive measures by healthcare professionals to improve individuals' abilities and confidence in managing their health issues [[→ 115](#)]. In an ideal scenario, patients would possess the confidence to address their social, emotional, and physical challenges. Digital technology is enhancing patient engagement and preventive medicine within the healthcare sector. Research indicates that the incorporation of digital technology in healthcare enhances patients' awareness of their lifestyle choices and encourages them to engage more proactively in their health management. Researchers found that veterans

exhibited increased motivation to manage their health when they had access to their health data. Moreover, personalization is modifying a digital platform, product, or service's user experience to align with an individual's specific needs, interests, behaviors, and characteristics [[→ 116](#), [→ 117](#), [→ 118](#)]. It entails leveraging data and algorithms to produce interactions, recommendations, and user-specific content that are pertinent and helpful. It facilitates the change in the power dynamic from physicians making decisions for the sake of their patients to a collaborative process in which the needs of the patients are taken into account. Developing a user-friendly online interface for patients may be essential to boosting their engagement with their care [[→ 119](#)].

12.6 Conclusion

AI is a potent instrument in the field of healthcare, especially when it comes to ML, which helps to monitor the efficacy of interventions and understand illnesses. Healthcare organizations may be resistant to implementing AI due to issues with data accessibility and organizational resistance, so a shift toward overall patient outcomes is required. For data privacy, patient consent is essential, particularly since AI relies heavily on massive datasets. Responsible AI implementation in healthcare requires ethical considerations that prioritize patient trust, accountability, and transparency. Healthcare outcomes trained on biased data by AI systems can be discriminatory, highlighting the need to address biases in data collection as well as model

training. Limitations on algorithmic transparency have been the focus of legal discussions; however, in high-risk scenarios, transparent, equitable, and ethical AI design is essential. Creating reskilling programs as a priority is essential to ensuring that medical professionals can use AI technologies. The need for more education on the benefits and drawbacks of AI is highlighted by healthcare professionals' limited grasp of digital systems. Although worries about criminal penalties for AI errors remain, healthcare professionals are becoming more and more interested in using AI to handle the massive amount of data needed for diagnosis. Ultimately, AI in healthcare possesses the capacity to improve patient outcomes and the delivery of care; yet, factors such as transparency, data privacy, and ethical concerns must be addressed. The successful use of AI necessitates collaboration among medical experts. Further research is necessary to comprehensively understand the impact of AI on patients and healthcare professionals for the appropriate and effective integration of AI in the healthcare sector.

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Chapter 13 Future trends and innovations of AI in healthcare

Abstract

Artificial intelligence (AI) is an important technological advancement that has allowed humans to replace manual labor at greater psychological and logical capacities in various industries, beyond the physical labor promoted by machines. The chapter emphasizes the future trends and other innovations present in AI in the healthcare system. AI has advanced quickly in recent years and is now being applied to offer many benefits in the crucial healthcare industry. An overview of AI's current status in the medical and healthcare fields is given in the chapter, with an emphasis on the technology's future. AI is now being used to provide numerous advantages in the vital healthcare sector, particularly in the areas of early detection and diagnostics. With future health care delivered through the H-IoT system being a reality owing to the development of vital methods that the scientific community has assisted in paving the way for, the state of the technology today offers countless opportunities for enhancement and innovation. AI is now being used to provide numerous advantages in the vital healthcare sector, especially in the areas of early detection and diagnostics. AI will be utilized in smarter hospitals and patient-physician interactions in the future of smart-connected healthcare (SCH).

Keywords: innovation, Healthify, Dehealth, HintVR, bump galaxy, health blocks, AccuVein,

13.1 Introduction

Technology has advanced significantly since the Industrial Revolution and continues to be the primary driver of growth and production. Technological developments in machines that have substituted manual labor-intensive jobs have led to significant progress in human development. Artificial intelligence (AI) is an important advancement in technology that has allowed humans to take on physical labor at bigger mental and reasonable capacities across industries, beyond the manual support provided by machines [[→1](#)]. AI is a field of research and technological advances that allows intelligent machines and software applications to perform tasks that typically require human intelligence. Consequently, one of AI's primary advantages is its capacity to replicate human behavior in a variety of contexts, learn from errors, and adapt to new ones. Big Data and other relevant data sources are used by AI to improve performance on tasks. Among the numerous sectors that have profited from AI recent rapid development is the crucial healthcare sector [[→2](#)]. AI has digitally replaced manual health systems with automated ones in many areas. In some applications, managing patients and medical resources basic tasks is now all that is required of humans, with AI components handling more complex procedures or leaving them to their own devices. AI-powered healthcare systems are evolving swiftly, particularly in the fields of diagnosis and early detection as shown in [→Fig. 13.1](#) [[→3](#)].

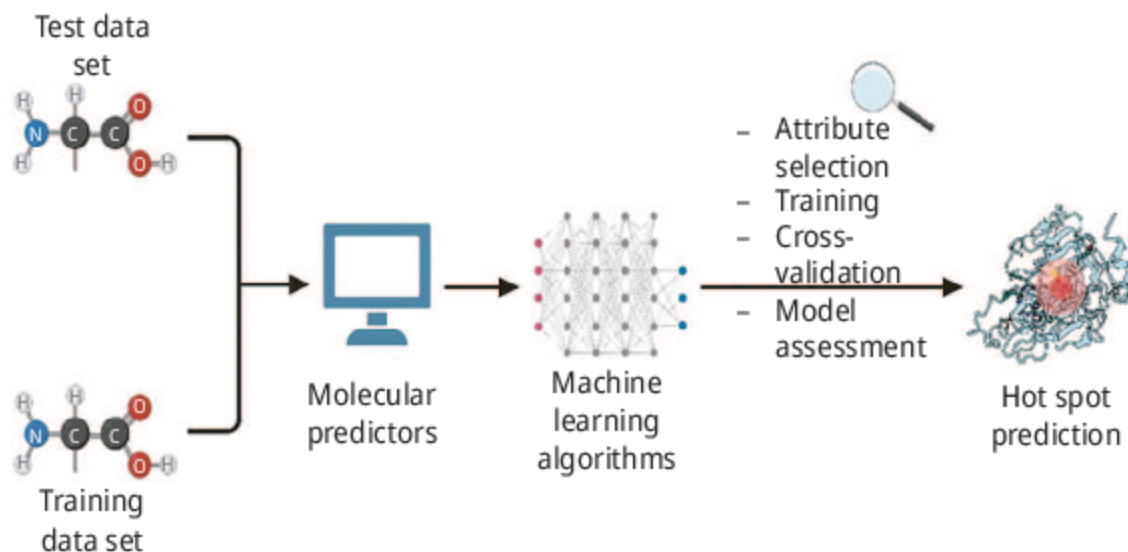


Fig. 13.1: Schematic representation of ML algorithm for hot spot prediction.

AI improves performance on a variety of tasks by utilizing relevant information sources. AI has advanced quickly in the past few years and has been used to great effect across numerous industries, including the vital healthcare sector. The healthcare industry is one of those that has benefited greatly from AI. In many fields, AI has already electronically transformed the healthcare industry into a computerized version [[→4](#), [→5](#)]. As a result, in certain programs, human intervention is limited to basic medical tasks like managing patients and medical resources; complex operations are either fully performed by humans or by AI components. Healthcare systems driven by AI are developing swiftly, particularly in the areas of early detection and diagnosis. As a result, AI is now able to complete tasks that are challenging for humans to complete with the same effectiveness, dependability, meticulousness, and speed as AI at a lower cost [[→6](#), [→7](#), [→8](#), [→9](#)]. If AI developers are successful in creating systems that can carry

out particular tasks, the technological advances enabled by the digitization of healthcare may be able to assist in overcoming other obstacles. AI, for instance, has a chance to dramatically reduce healthcare expenses while also enhancing patient care. The increasing number of people on the planet is predicted to result in a greater need for healthcare to be delivered more quickly. Because of this, the medical sector requires innovative AI solutions to increase productivity and effectiveness without driving up expenses [[→ 10](#)].

By using relevant data sources, AI enhances performance on a variety of tasks. AI has advanced quickly in the past few years and has been used to great effect across numerous industries, including the vital healthcare sector. The healthcare sector has benefited immensely from AI [[→ 11](#)]. AI has electronically transformed many areas of the healthcare system into a computerized version. Therefore, complex processes are either managed by AI components in some programs, or they are handled by humans for the most part, with humans only being required to handle basic medical practice tasks like managing clients along with medical resources. Healthcare systems driven by AI are rapidly developing, particularly in early detection and diagnostics [[→ 12](#)].

More creative and ingenious solutions will be used to improve patient-physician interactions as a result of smart-connected healthcare, or SCH. Mobile apps will enable the network to hyperlink geographically to the closest hospital in the event of an accident and transfer data before

admission, potentially saving lives [[→13](#)]. It is anticipated that new self-care methods made possible by technological advancements in the digital realm will be adopted in the future. Continuous as well as flash glucose monitoring systems, for instance, allow the sensors to serve as display devices, without wires transmit glucose data to a reader or mobile application, and generate hypo- and hyper-glycemia alerts [[→14](#)]. Eye-tracking can be used to guide patient-focused product development of new systems for drug delivery (self-injection systems) through the procedure of usability testing. The development could greatly improve how the various parts of the linked system like the sensors, pumps for infusion, handheld devices, and mobile apps interact with one another, increasing the product's overall efficacy [[→15](#)]. Cutting-edge technologies like virtual reality (VR) and augmented reality (AR) can raise the bar for medical and healthcare practices by enabling users to interact with and completely lose themselves in these worlds. These technologies include head-mounted displays, haptic devices, AR glasses, software, and tracking sensors. In plastic surgery, virtual reality (VR) has become popular and has potential uses in navigation, training, and surgical preparation. Training using VR has been well-studied. Currently, the market is filled with several commercial VR goods for medical education [[→16](#)].

13.2 Future trends and innovation of AI in drug discovery

Drug development is still a lengthy, costly, ineffective, and difficult process with a low rate of new-effective therapeutic discovery, especially in light of the complexity of disorders. Researchers have overcome obstacles in drug discovery with the help of AI/ML-based techniques, and these approaches have a wide range of applications in drug discovery programs for different diseases. In general, researchers from academia and business have expressed a great deal of interest in the application of AI/ML to pharmaceutical development [[→17](#)]. Furthermore, a large number of pharmaceutical companies have collaborated with colleges and universities and AI start-ups, as well as invested in their AI-based research initiatives. In reaction to the COVID-19 pandemic, academia and industry have recently shown an impressive level of cooperation. Due to open-sourced platforms, data and pipelines to AI labs and businesses can now be shared. For instance, the public can now view the 3D models of SARS-CoV-2 proteins that Google Deepmind's AlphaFold system predicted. The application of AI technology to enhance drug discovery is still in its infancy, even though AI has become a vital instrument for revolutionizing the medical development process [[→18](#)]. Because most next-generation AI structures are "black boxes," finding new drugs becomes more challenging. Because AI/ML results are difficult to interpret, their applications are restricted. It isn't true for ML models that are simpler (like XGBoost, TensorFlow, Lasso, Ridge, and Elastic Net) or more complex (like DNN). As a result, scientists are unable to understand the biological processes that underlie the model's output. Further complicating troubleshooting is the

unpredictable malfunctions of these models. As a result, creating strategies for unlocking the DL black boxes is crucial [[→19](#)].

Furthermore, since sufficient huge quantities of data to train them are necessary for the successful development of algorithms, the amount of data that is available directly impacts the effectiveness of AI/ML models. Finding novel targets or medications to treat a variety of neurological disorders for which there is no known treatment presents a unique challenge. Transfer learning technology, which applies knowledge from one task to another, can address the lack of training examples for a given drug discovery task. However, information sharing within the field of science appears to be the most promising long-term solution to data scarcity [[→20](#)]. The process of developing drugs would greatly benefit from such widespread data sharing, and shortly, faster devices could result from hardware advancements like quantum computing. The difficulty in identifying targets is a particular limitation of using AI/ML for drug discovery, as many neurological disorders have unclear pathophysiology. More sophisticated AI and ML tools that can extract information from data sets created across the different biological layers – such as transcriptomics, proteomics, and metabolomics – are required to study the intricate disease mechanisms and identify the appropriate biological targets [[→21](#)]. Capsule networks are a futuristic AI construction that can help in the situation. It is a networked module that houses CNNs. The remarkable ability of capsule networks to predict the cardiovascular side effects of materials during initial application highlights their unique potential in drug discovery efforts. Due

to CNNs' modular design, capsule neural networks can learn from a variety of data sets while maintaining the data's ordered features. Because illness records have a highly adaptable structure, capsule networks can be used to study the way various layers of protein molecules, metabolites, and genes connect and shift in function concerning different diseases [[→22](#)].

The resurgence of AI in organic synthesis can be attributed to developments in big data, computational power, deep neural networks, and optimization algorithms. Their primary drawback is their dependence on unambiguous chemical changes and reactions [[→23](#), [→24](#), [→25](#)]. These have to be built and assembled manually. Transformer depictions and sequence-to-sequence models are two methods for natural language processing that have recently gained traction in the area. The line of investigation is driven by the discovery that the distribution sequence of organic molecule fragments is comparable to the English word sequence. When rule-free approaches are applied, the products are usually conceived as part of a text-based representation. After processing the products, encoder–decoder architecture is used to predict their corresponding artificial precursors at a single-step response distance [[→26](#)]. Moreover, data indicates that the application of AI to the research and development of pharmaceuticals is beginning to gain traction. These methods are gradually living up to some of the community's expectations and are demonstrating remarkable advancements in synthesis planning, de novo molecule design, and quantitative structure-activity relationship (QSAR) modeling, among other fields. It's unclear, though; if these methods will help scientists create and synthesize “better drugs more

quickly”. Techniques based on more “raw” chemical pictures (e.g., neural networks with graphs and SMILES-based recurrent neural networks) ought to work at least as well as standard descriptor-based models in the context of ligand-based property prediction. Furthermore, a greater range of chemical materials and modeling tasks can be easily applied with deep learning techniques, enabling more effective data utilization through online and multitask learning. Presently, conformation-aware deep learning is in its infancy, particularly when approaches that make use of three-dimensional imbalances in their construction are taken into account. Nevertheless, they should advance quickly in the field of drug discovery as well as related fields like material science and quantum mechanics, especially when used to replace more computationally intensive first-principle calculations [[→27](#), [→28](#)].

Using generative models and deep learning, scientists can produce novel substances with target-binding properties that increase drug efficacy and decrease side effects. With AI, individualized dose formulations for each patient will be feasible. AI algorithms will take into account patient-specific factors like age, weight, genetics, and illness status to optimize drug pieces and delivery techniques and improve treatment outcomes. The application of AI algorithms to predict the toxicity and side effects of suggested medications will revolutionize safety evaluation. Medication adherence and remote patient care will be made possible by AI-powered monitoring systems. Wearable technology and instruments will make it easier to collect data continuously, which will allow AI algorithms to create more

individualized treatment plans and improve adherence. AI enhances patient recruitment, patient selection, and clinical trial design. AI algorithms will identify eligible patients, reduce trial costs, and expedite approval by utilizing genetic profiles, biomarkers, and electronic health records [[→ 29](#), [→ 30](#)]. Continuous manufacturing operations will be optimized by AI models' real-time monitoring and oversight of critical variables. Through data analysis and feedback, AI algorithms will make pharmaceutical manufacturing consistent and effective. Massive data sets will be analyzed by AI to support regulatory decision-making. It will expedite the approval process for pharmaceuticals by regulatory agencies and enhance safety. More creative approaches and procedures must be employed since the creation and study of pharmaceuticals necessitates such a significant time and financial commitment [[→ 31](#)]. The medical field is seeing significant opportunities due to AI. These include extensive multivariate data analysis; complex problem-solving in the development of workable medication delivery systems; enhanced decision-making; disease classification; dosage ratio optimization; quick drug development; prediction of drug reactions and bioactivities; cellular response; the efficacy of combination medications; and a host of other topics. As each section has shown, AI and machine learning have enormous potential to transform the way that medications are administered and improve the efficacy of therapies for infectious diseases [[→ 32](#)].

13.3 Future trends and innovations of AI in drug manufacturing

Initially, AI-based manufacturing control methods were limited to services for which official or analytical structures were lacking because of the complexity of the processes involved or the high computational costs associated with developing adjustable process controllers. However, like other factory groups, these kinds of applications will continue to be made possible by the growing number of instruments, the capacity to gather enormous amounts of data, and the ability to extract crucial process signals and features. These applications do not call for an exact mathematical model or a structured physics-based demonstration of the system [[→ 33](#), [→ 34](#)]. Factories are getting smarter as companies utilize AI intensively to transform data from various production system components into useful insights. However, a lot of gaps still need to be closed to ensure that integrating AI into manufacturing processes is simple.

At every rung of the manufacturing chain hierarchy, ML has been employed exclusively. However, ML is not as popular at the system-wide level for decision-making as it once was, especially when compared to its successes in more specialized applications such as process monitoring, optimization, and population health management (PHM). The main reasons for this are the complex multi-stage processes, the stochastic and nonlinear dynamic characteristics of these processes, and the dependencies among the

enormous amounts of diverse information generated in manufacturing systems. Furthermore, even though ML has the potential to solve non-deterministic polynomial (NP)-complete problems, which are frequently present in intelligent optimization problems, it is still challenging to select the best techniques and algorithms. A complete understanding of the problem, its root causes, consequences, and the intended solution must be known or thoroughly researched to increase the likelihood of choosing an AI tool that will be efficient and facilitate the ensuing creation of models, data analysis, and interpretation. The root cause of all these issues is the requirement for sufficient domain knowledge during the problem definition phase. It is required to make sure that every facet of the issue is comprehended and that no important information or suppositions are missed. Numerous interrelated plant management structures that control the plant's overall performance are positioned at the top of the manufacturing hierarchy. No single AI tool, or even a combination of tools, has been capable to bridge and integrate any of the performance goals of these control systems, despite the development of standalone machine learning approaches. A technique known as reinforcement learning (RL) has been suggested [[→ 35](#)]. Furthermore, because it would be excessively costly and labor-intensive to gather data in an organized way from different manufacturing configurations as well as operating conditions, real-world applications typically find it impractical to do so. Furthermore, limitations on the exchange of information and accessibility via both private and public networks may result from security concerns. To address these challenges,

the field of science has investigated two strategies: data synthesis and transfer learning. The process of transferring an existing model using a range of methods from an initial data-rich domain of origin to an education data-poor target domain is known as transfer learning. The most well-known advantage of transferable learning for fault diagnosis prognosis and condition monitoring is that it allows experimental data from defect-free machines to be used for building models on real production lines if faulty conditions occur [[→36](#)].

One application of transfer learning is the recognition of human actions in the context of human-robot cooperation during assembly. The example shows how transfer learning can address model scalability through appropriate transformations that match particular manufacturing scenarios, potentially leading to the establishment of a general solution instead of a specific one for a particular manufacturing aspect. However, it is also observed that transfer learning is still restricted to ad hoc, trial-and-error methods at this point [[→37](#)]. To ensure that transfer learning-based approaches are widely applicable and accepted, more thorough theoretical research on subjects like data transferability is still necessary. By learning enough characteristics of the real data, we can create synthetic data that closely mimics the real data, allowing us to get around the restriction of data availability. The process is known as data synthesis [[→38](#)].

Conventional data synthesis methods, like the Synthetic Minority Over-sampling Technique (SMOTE), often rely on interpolation, which is insufficient to capture complex features in the data. Generative adversarial

networks (GANs) is a deep learning technique that made significant strides in synthesizing high-fidelity data and learning salient features. The fundamental idea behind a GAN is a competition between two components: a discriminator that separates synthetic data from real data and a generator that uses real data analysis to create synthetic data. There have been reports on early GAN implementation efforts [[→39](#)]. For instance, to develop the gearbox fault diagnostic model, researchers looked into the use of synthetic vibration signals. According to spectral analysis, the synthetic data successfully captured the distinctive frequencies and other fault-related signal features. Similar to other deep learning methods, a comprehensive analysis of the underlying theories of GAN is still required to demonstrate the technique's applicability in resolving data limitation issues [[→40](#)].

There are new challenges arising from the boosting availability of various data in manufacturing systems. First, there's a chance that a significant portion of the data is either completely missing or contains a lot of unnecessary and irrelevant information. Because the accuracy and availability of manufacturing data have a significant impact on the suitability and efficacy of AI algorithms about expected outcomes, these information curation issues pose a challenge to the practical application of ML algorithms. Second, the setting that is created throughout the data gathering and structuring process determines the quality of any information that can be gleaned from data analysis. Gathering and arranging data techniques must dynamically link various data sets, and information, along with models to take into account multiple viewpoints for multiple uses,

given the wide range of viewpoints in production. Prospects for aligned local correlation-based forecasts (e.g., tool wellness and person process status) might be made possible by improved data quality. It will enable a causal understanding of the intricate and dynamic manufacturing processes that occur in factories [[→41](#)].

Given the restrictions on the use of contemporary technology that affect the decisions that are made about people's lives, it may be difficult to replace every antiquated or traditional scientific technique that is part of the healthcare infrastructure of today. These issues won't be resolved until every element of the current infrastructure – including the medical personnel – supports improving the situation and the available resources. Furthermore, it is crucial to comprehend the potential benefits and drawbacks of utilizing the technology, which may require modifications to the current infrastructure. Ultimately, it would require a large amount of budgetary and financial support from the organization in charge of overseeing healthcare infrastructure. Even though there are certain limitations on the use of contemporary technology that would place those in charge of making crucial decisions about people's lives in a difficult position, there might be difficulties in replacing all of the antiquated or traditional technical techniques that are employed in the current medical system [[→42](#)]. In addition to their primary role in addressing current issues in their respective fields of study, AI and ML-based solutions can be implemented in several healthcare subfields. One such area, which contributes to the annual increase in global death rates, is the diagnosis of

common illnesses and diseases. A significant portion of this issue can be resolved with the aid of deep neural networks and other AI-based technologies. With the rapid rate of population growth, doctors and other staff members would also find it challenging to complete their duties effectively due to excessive workloads and an absence of free time. As a result, hospitals can use ML and AI to help them solve the issue. These technologies can save critical time and resources by learning to predict every possible outcome at every stage with little room for error, which speeds up subsequent procedures. However, to carry out such solutions, hospitals might have to make investments in training physicians to use the technology, which requires the preparation of large datasets. Despite this, the existing infrastructure can incorporate virtual intelligence with minimal upfront costs for setting up internal diagnostic centers. It is anticipated that the long-term benefits will exceed those initial costs [[→ 43](#)].

13.4 Future trends and innovations of AI in healthcare

AI in health research and initiatives has surged in the last few years, combining various clinical data elements, establishing links between bio repositories and clinical data, and creating links between clinical data and pharmaceutical research and development. More importantly, knowledge and data must be combined to create AI algorithms for health that work. Unlike other computational fields like sight and speech evaluation, where

enormous amounts of information can be obtained, patient data is often sparse and subject to substantial variability [[→ 44](#), [→ 45](#)]. Furthermore, health issues in the real world are frequently complicated. Rule-based systems, for example, are examples of conventional AI technologies with high interpretability. Even though deep-learning models and other recent AI technologies may achieve beneficial quantitative performance, they are largely disregarded. Interpretability of models is essential and has been the topic of many conversations lately. The model interpretability may not always be as important as it might be, especially in scenarios where AI algorithms can produce accurate results consistently and widely. However, it isn't the medical case – at most not with computer science currently dominating healthcare analytics [[→ 46](#), [→ 47](#), [→ 48](#)]. For instance, it demonstrated that when employing claims data and electronic health records for readmission to hospital tasks, deep learning models were able to match the effectiveness of logistic regression. While deep-learning algorithms have demonstrated cutting-edge efficacy for medical image analysis, their generalizability remains questionable. In other words, it is difficult to argue that a model that performs well on a medical image data set from one radiology center will perform well on a data set from another radiology center. Additionally, human clinicians typically make decisions in healthcare, with AI algorithms serving as tools to assist them. To reassure the clinician, it is crucial to present comprehensive explanations for the assumptions made by those AI algorithms. Additionally, integrating AI

algorithms into routine clinical workflows will enhance their clinical utility [[→49](#), [→50](#)].

While there are numerous AI models being developed for healthcare, there is a potential risk regarding their security. Researchers found that autonomous driving systems can be easily tricked by pollution on transportation signs. Researchers found that even small changes to a patient's electronic health record can significantly change the predictor's mortality estimate, even if the predictor is otherwise well-trained [[→51](#), [→52](#), [→53](#), [→54](#)]. A more thorough explanation of the possible worries regarding the “incentives for developed adversarial assaults” in the healthcare sector is given by another researcher. According to the authors of the study, (i) medical practitioners must understand the possible risk; (ii) AI researchers create strong defenses against medical adversarial assaults; and (iii) it is recommended that policymakers consider the possibility of model security risks when developing new regulatory frameworks. All AI models require samples of training data [[→55](#)]. Generally, patient shifts and the intricate nature of their health issues are not well captured by a training sample size that is derived from patient data. A model created with patients from a single facility frequently does not work with patients from another. One of the main problems AI still has in the healthcare industry is the problem, also known as bias in the data. The researcher claims that bias of the following kind can exacerbate health disparities. The collection of extensive and varied patient data sets is one strategy to lessen bias [[→56](#), [→57](#), [→58](#)].

Numerous prominent healthcare companies worldwide are involved in and have announced various ongoing and upcoming Metaverse projects as shown in [→ Fig. 13.2](#). The Blockchain-Metaverse project HealthLand.io, also referred to as Healthify, was unveiled to offer its users virtual clubs, health excursions, meetings with heroes, and physical and mental health. Additionally, the project will enable patients and customers to use online stores and medical professionals to open online gyms. The idea is to establish a virtual community where people from all over the world come together for a common cause, irrespective of their nationality, religion, or ethnic background [[→ 59](#)]. A decentralized Metaverse has been conceptualized by the British organization DeHealth as a way to simplify the healthcare ecosystem. Along with MR, the technology will expand VR and AR. With the help of the platform, patients and doctors will be able to collaborate virtually, communicate with one another, and make virtual income by selling their private medical records [[→ 60](#)]. Bump Galaxy is a prototype Metaverse for mental health services built on Minecraft. Through the use of gaming environments, the project seeks to address patients' mental health concerns. In addition to giving the patients a sense of safety and support from society, the “game world therapy” will help them get over trauma, depression, or anxiety. By using a bottom-up approach, the deep hypnotic visualizations will improve mental health and resilience within the community [[→ 61](#)].

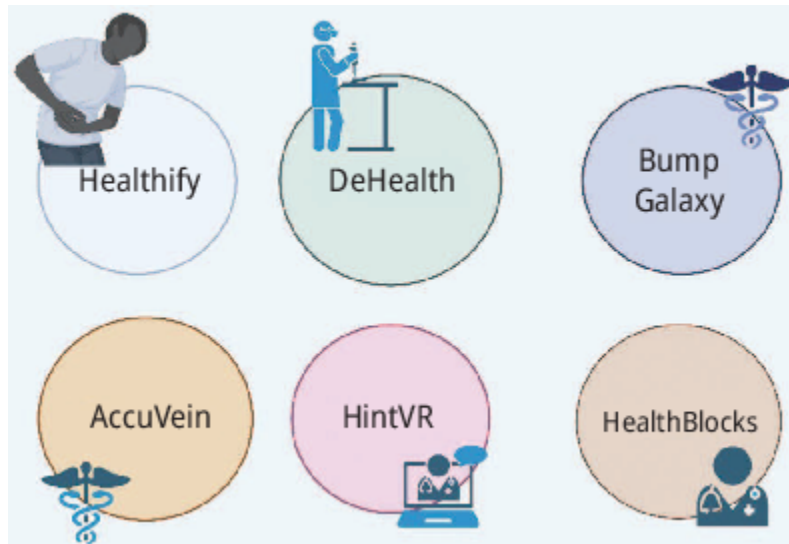


Fig. 13.2: Schematic representation of various ongoing and upcoming Metaverse projects in healthcare companies.

The vascular access rate will be greatly increased by the vein visualizing project AccuVein, which will be a very useful tool for physicians, neurosurgeons, and patients. It will raise the percentage of first-time injections that are successful and also raise the annual savings following clinical procedures. Following vein access, patients will not be concerned about bruises, and surgeons will be able to see a vast and intricate network of blood vessels. The project will undoubtedly enhance the security and caliber of treatments in critical medical applications [[→62](#)]. A new Metaverse structure called HintVR was created by the California-based technological advances startup 8Chili to help physicians, surgeons, patients, clinical staff, paramedical workers, medical students, and material curators. Using cutting-edge 3D visualizations for image-guided surgeries, surgeons can also have creative and engaging interactions with patients before, during, and after surgery. The platform therefore has the potential to greatly

assist the following parties: (i) physicians in making consultations more straightforward; (ii) surgeons in performing less invasive surgeries more skillfully; (iii) patients in understanding diagnosis, treatment options, and patient reviews; alongside (iv) medical/paramedical workers in competently adhering to all expert instructions [[→ 63](#)]. The goal of the Dutch startup project HealthBlocks, which is based on the IoTex Blockchain, is to eradicate healthcare inequality worldwide. It can help the patient in addition to providing remote patient monitoring. A new, highly secure, private ecosystem will be created by smart devices with decentralized identities. Users from all over the world can change their daily unhealthy habits and adopt a healthy lifestyle by completing the Web3 Metaverse application and earning rewards. The project aims to provide reduced transaction costs, quicker transaction times, and increased scalability [[→ 64](#)].

The development of a fully-connected society where smooth machine-to-machine along with machine-to-human communication anticipated is being fueled by the commercial launch of 5 G technology. The development of innovative biomaterials, intelligent materials, smart textiles, and sensors based on micro-electro-mechanical systems (MEMS), in addition to remote monitoring and illness diagnosis, is the focus of current research. Low-power, dependable, and safe network capabilities are needed for operating these sensor devices. Accurate real-time big data handling processing algorithms are required to handle the data collected by the sensors. The system's overall performance can be assessed using its quality of service (QoS). Future directions for H-IoT research can be seen in the possibility of

QoS improvements. Today's technological state presents a plethora of opportunities for creativity and advancement. The scientific community has contributed to the development of several critical technologies that have made it possible for future healthcare services delivered via the H-IoT system [[→ 65](#)].

13.5 Conclusion

Higher mental and intellectual capacities have replaced manual labor in many industries, including healthcare, due to AI. AI will be utilized in smarter hospitals and patient-physician interactions in the future of smart-connected healthcare. AI uses pertinent data sources to improve performance across various tasks. Technological innovations like VR alongside augmented reality (AR), which enable users to interact and immerse themselves fully in virtual environments, have the potential to raise the bar for medical and healthcare procedures. Initially, the application of AI techniques in manufacturing process control was limited to scenarios with a limited number of formal or analytical models. These applications are becoming simpler, though, due to the expansion of instrumentation, data collection, and the capacity to extract crucial process signals and features. Factories are getting smarter as AI is applied more and more in manufacturing processes. Making the best decisions about techniques and algorithms is still challenging. A comprehensive understanding of the problem, its causes, repercussions, and desired solution must be spoken or

thoroughly researched to increase the likelihood of choosing an AI tool that will be effective and facilitate the processes of creating models, assessing data, and interpreting it. Numerous well-known healthcare organizations, including HealthLand.io, DeHealth, Bump Galaxy, AccuVein, HintVR, and HealthBlocks, have disclosed their current and future Metaverse initiatives. These initiatives seek to eliminate healthcare disparities, enhance mental health services, increase vascular access rates, and simplify the healthcare system. The commercial launch of 5 G technology fuels the development of novel biomaterials, intelligent materials, smart fabrics, and sensors based on MEMS, remote monitoring, and disease diagnosis. The QoS of these systems can be used to evaluate their overall performance. The state of technology today offers numerous opportunities for improvement and innovation, with future healthcare services provided via the H-IoT system becoming a reality due to the development of critical technologies that has helped the research community.

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